

How AI writing assistance distorts perceptions of AI users and their opinions

Supplementary Materials

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1 Study Information

1.1 Overview and Timings

We conducted five separate studies:

The **main study (Study 1)** ran between June 13th and July 21st, 2025.

The **disclaimer study (Study 2)** ran between November 25th and December 3rd, 2025.

The **mitigation study (Study 3)** ran between December 16th, 2025 and January 29th, 2026.

The **trust study (Study 4)** ran on June 1st and 2nd, 2026.

The **persuasion study (Study 5)** ran between June 4th and 11th, 2026.

1.2 Participants

All participants were adults (18+ years old) based in the UK, recruited via the online crowd-sourcing platform Prolific. We distinguish participants between writers, who shared their political opinions, and readers, who rated or otherwise engaged with human- and AI-written opinion paragraphs.

For writers (Studies 1-3), we targeted UK census-representative samples based on age, gender, and race. Based on post-hoc census comparisons (Tables S1 to S3) this targeting was mostly successful in Study 1 (S1 W), with some under-representation in older age categories (60+). In our disclaimer and mitigation studies (S2 W and S3 W), deviations were more pronounced, especially on age, where older age categories were even more underrepresented. For readers (Studies 1, 3-5), we did not require representativeness due to the large size of our target sample.

Study 1 was completed by 1,501 writers and 10,017 readers, Study 2 by 669 writers, Study 3 by 769 writers and 2,543 readers, Study 4 by 802 readers, and Study 5 by 7,996 readers. There was no overlap between writers and readers (i.e. no writer was also a reader, and vice versa). Each participant took part in only one study, with one exception made to reach target sample size: 1,469 readers in Study 3 had previously participated as readers in Study 1 approximately five months earlier. Thus, overall, there were 2,939 distinct writers and 19,889 distinct readers.

Writers were highly experienced Prolific participants. At the time of each study, the median writer had 700 previously approved Prolific submissions (IQR: 303–1,521), and 90.9% had at least 100 (Table S10). This was consistent across writers, with medians of 961 in Study 1, 535 in Study 2, and 506 in Study 3. Readers had a median of 241 approved submissions (IQR: 36–810), with readers in Studies 1 and 3 being substantially more experienced than readers in Studies 4 and 5.

Table S1: Participant age distribution across studies and census comparison.

age_binned	S1 W	S1 R	S2 W	S3 W	S3 R	S4 R	S5 R	Census
18–29	■ 18.9%	■ 27.0%	■ 24.8%	■ 28.2%	■ 22.8%	■ 38.3%	■ 37.3%	■ 18.7%
30–39	■ 17.1%	■ 30.9%	■ 26.8%	■ 32.2%	■ 32.0%	■ 32.7%	■ 33.0%	■ 17.1%
40–49	■ 15.7%	■ 19.1%	■ 17.2%	■ 18.2%	■ 21.3%	■ 19.0%	■ 16.2%	■ 15.9%
50–59	■ 24.3%	■ 13.8%	■ 16.4%	■ 13.3%	■ 15.6%	■ 8.6%	■ 9.1%	■ 17.4%
60–69	■ 20.0%	■ 7.3%	■ 12.7%	■ 5.9%	■ 6.7%	■ 1.4%	■ 3.4%	■ 13.7%
70+	■ 4.2%	■ 2.0%	■ 2.1%	■ 2.2%	■ 1.6%	■ 0.1%	■ 1.0%	■ 17.2%
Participants	1,501	10,017	669	769	2,543	802	7,996	

Table S2: Participant gender distribution across studies and census comparison

gender	S1 W	S1 R	S2 W	S3 W	S3 R	S4 R	S5 R	Census
Female	■ 51.1%	■ 57.6%	■ 45.7%	■ 61.4%	■ 57.6%	■ 46.6%	■ 52.2%	■ 51.7%
Male	■ 48.3%	■ 41.0%	■ 53.1%	■ 37.6%	■ 41.5%	■ 52.7%	■ 46.7%	■ 48.3%
Other	0.3%	1.0%	1.0%	0.7%	0.6%	0.5%	0.7%	0.0%
Prefer not to say	0.3%	0.4%	0.1%	0.4%	0.3%	0.1%	0.3%	0.0%
Participants	1,501	10,017	669	769	2,543	802	7,996	

Table S3: Participant race / ethnicity distribution across studies and census comparison

race	S1 W	S1 R	S2 W	S3 W	S3 R	S4 R	S5 R	Census
White	■ 85.9%	■ 79.7%	■ 88.5%	■ 81.5%	■ 78.6%	■ 66.7%	■ 74.5%	■ 85.1%
Black	3.7%	7.4%	3.3%	5.2%	8.1%	14.7%	7.9%	3.4%
Asian	7.4%	7.1%	5.1%	7.8%	7.4%	12.1%	10.7%	8.0%
Mixed	1.6%	4.0%	1.6%	3.6%	4.2%	5.0%	4.7%	1.8%
Other	0.7%	0.9%	1.0%	0.8%	0.8%	0.9%	1.4%	1.8%
Prefer not to say	0.6%	0.8%	0.4%	1.0%	0.9%	0.6%	0.7%	0.0%
Participants	1,501	10,017	669	769	2,543	802	7,996	

Table S4: Participant English first-language status distribution across studies

english_first	S1 W	S1 R	S2 W	S3 W	S3 R	S4 R	S5 R
Yes	■ 92.5%	■ 90.0%	■ 91.9%	■ 88.8%	■ 90.7%	■ 85.9%	■ 84.4%
No	5.2%	6.1%	5.4%	7.2%	6.1%	9.6%	10.5%
Bilingual from birth	2.0%	3.7%	2.5%	3.4%	3.1%	4.1%	4.9%
Prefer not to say	0.3%	0.2%	0.1%	0.7%	0.2%	0.4%	0.2%
Participants	1,501	10,017	669	769	2,543	802	7,996

Table S5: Participant English proficiency distribution across studies

english_skills	S1 W	S1 R	S2 W	S3 W	S3 R	S4 R	S5 R
Basic	0.1%	0.0%	0.0%	0.0%	0.0%	0.1%	0.2%
Intermediate	4.0%	2.4%	1.6%	3.5%	1.8%	3.4%	2.8%
Advanced	■ 15.1%	■ 12.6%	■ 16.6%	■ 11.7%	■ 10.5%	■ 15.1%	■ 12.9%
Expert	■ 80.8%	■ 84.7%	■ 81.8%	■ 84.3%	■ 87.5%	■ 81.4%	■ 84.0%
Prefer not to say	0.1%	0.2%	0.0%	0.5%	0.2%	0.0%	0.1%
Participants	1,501	10,017	669	769	2,543	802	7,996

Table S6: Participant education distribution across studies

education	S1 W	S1 R	S2 W	S3 W	S3 R	S4 R	S5 R
GCSEs or equivalent	■ 9.9%	■ 7.3%	■ 6.4%	■ 5.3%	■ 7.3%	■ 7.1%	■ 7.3%
A-levels or equivalent	■ 14.7%	■ 15.1%	■ 14.2%	■ 13.9%	■ 13.1%	■ 17.7%	■ 18.6%
Vocational qualification	■ 11.3%	■ 9.5%	■ 8.4%	■ 9.5%	■ 10.0%	■ 6.5%	■ 7.7%
Undergraduate degree	■ 40.0%	■ 42.5%	■ 43.2%	■ 42.1%	■ 41.7%	■ 43.8%	■ 40.6%
Postgraduate degree (Master's)	■ 18.8%	■ 21.4%	■ 23.0%	■ 24.7%	■ 23.7%	■ 21.7%	■ 21.7%
Doctorate (PhD)	4.1%	3.0%	4.0%	4.0%	3.4%	2.9%	3.1%
Other	0.8%	0.7%	0.3%	0.3%	0.3%	0.2%	0.7%
Prefer not to say	0.4%	0.4%	0.4%	0.1%	0.4%	0.1%	0.3%
Participants	1,501	10,017	669	769	2,543	802	7,996

Table S7: Participant income distribution across studies

income	S1 W	S1 R	S2 W	S3 W	S3 R	S4 R	S5 R
Under £15,000	■ 19.8%	■ 17.5%	■ 18.2%	■ 17.7%	■ 14.5%	■ 17.7%	■ 17.9%
£15,000–£24,999	■ 16.7%	■ 14.8%	■ 13.0%	■ 15.7%	■ 14.5%	■ 11.6%	■ 12.9%
£25,000–£34,999	■ 20.9%	■ 24.3%	■ 18.7%	■ 19.5%	■ 23.8%	■ 20.6%	■ 22.1%
£35,000–£49,999	■ 17.9%	■ 20.6%	■ 22.1%	■ 23.0%	■ 21.9%	■ 21.7%	■ 20.7%
£50,000–£74,999	■ 12.1%	■ 11.1%	■ 16.1%	■ 11.6%	■ 13.9%	■ 13.8%	■ 13.1%
£75,000–£99,999	▮ 3.9%	▮ 2.7%	▮ 4.2%	▮ 3.3%	▮ 4.0%	▮ 5.0%	▮ 3.9%
£100,000+	▮ 2.5%	▮ 1.7%	▮ 1.9%	▮ 1.8%	▮ 2.4%	▮ 4.1%	▮ 2.3%
Prefer not to say	▮ 6.3%	▮ 7.3%	▮ 5.7%	▮ 7.4%	▮ 5.1%	▮ 5.5%	▮ 7.1%
Participants	1,501	10,017	669	769	2,543	802	7,996

Table S8: Participant political party preference distribution (2024 UK general election) across studies. “Reform UK” option was added from disclaimer onward.

politicalParty	S1 W	S1 R	S2 W	S3 W	S3 R	S4 R	S5 R
Labour	■ 40.2%	■ 40.8%	■ 39.8%	■ 39.0%	■ 40.1%	■ 36.0%	■ 36.1%
Conservative	■ 14.7%	■ 10.5%	■ 10.5%	■ 9.0%	■ 12.5%	■ 12.1%	■ 9.0%
Liberal Democrats	■ 10.3%	■ 9.7%	■ 8.8%	■ 9.9%	■ 9.1%	▮ 6.5%	▮ 8.2%
Greens	▮ 7.5%	▮ 9.5%	▮ 8.2%	▮ 9.1%	▮ 7.7%	▮ 9.6%	▮ 9.8%
Scottish National Party	▮ 2.9%	▮ 2.7%	▮ 2.1%	▮ 2.7%	▮ 2.8%	▮ 2.0%	▮ 3.0%
Reform UK	▮ 0.0%	▮ 0.0%	▮ 7.8%	▮ 6.8%	▮ 6.7%	▮ 5.5%	▮ 5.8%
Other	■ 10.7%	■ 8.0%	▮ 3.4%	▮ 3.1%	▮ 2.1%	▮ 1.2%	▮ 2.1%
Did not vote	■ 10.4%	■ 13.3%	■ 14.5%	■ 14.2%	■ 13.7%	■ 17.6%	■ 16.9%
Not eligible to vote	▮ 1.1%	▮ 2.4%	▮ 2.4%	▮ 3.0%	▮ 2.3%	▮ 6.6%	▮ 6.1%
Prefer not to say	▮ 2.2%	▮ 3.0%	▮ 2.5%	▮ 3.3%	▮ 3.0%	▮ 2.9%	▮ 3.0%
Participants	1,501	10,017	669	769	2,543	802	7,996

Table S9: Participant political ideology distribution across studies. “Other” option was removed from disclaimer onward.

politicalIdeology	S1 W	S1 R	S2 W	S3 W	S3 R	S4 R	S5 R
Very Left-Wing	■ 9.7%	■ 10.1%	■ 10.6%	■ 10.8%	▮ 8.7%	■ 9.2%	■ 10.3%
Moderately Left-Wing	■ 35.4%	■ 39.1%	■ 40.4%	■ 40.7%	■ 38.1%	■ 36.7%	■ 37.9%
Centrist	■ 27.8%	■ 28.4%	■ 26.8%	■ 29.8%	■ 29.6%	■ 31.0%	■ 29.6%
Moderately Right-Wing	■ 21.0%	■ 15.1%	■ 19.0%	■ 13.9%	■ 18.2%	■ 14.2%	■ 14.5%
Very Right-Wing	▮ 2.9%	▮ 1.8%	▮ 1.9%	▮ 1.6%	▮ 2.2%	▮ 3.5%	▮ 2.1%
Prefer not to say	▮ 1.3%	▮ 3.3%	▮ 1.3%	▮ 3.3%	▮ 3.3%	▮ 5.4%	▮ 5.6%
Other	▮ 1.9%	▮ 2.2%	▮ 0.0%	▮ 0.0%	▮ 0.0%	▮ 0.0%	▮ 0.0%
Participants	1,501	10,017	669	769	2,543	802	7,996

Table S10: Participant experience on Prolific across studies, measured as the number of previously approved Prolific submissions at the time of each study.

Approved subs.	S1 W	S1 R	S2 W	S3 W	S3 R	S4 R	S5 R	All W	All R
Median	961	516	535	506	953	74.5	39	700	241
IQR	422–1,996	195–1,186	252–1,067	219–1,087	393–1,913	21–201	8–227	303–1,521	36–810
Mean (SD)	1,314 (1,145)	825 (861)	787 (751)	808 (923)	1,327 (1,219)	337 (899)	314 (699)	1,062 (1,042)	592 (833)
≥100	93.1%	84.8%	91.8%	85.8%	93.5%	42.4%	35.0%	90.9%	63.1%
≥500	70.2%	51.1%	52.0%	50.2%	69.7%	13.1%	16.6%	60.8%	35.4%
≥1,000	48.4%	29.8%	27.5%	27.8%	48.7%	8.2%	10.0%	38.3%	20.6%
Participants	1,501	10,017	669	769	2,543	802	7,996	2,939	19,889

1.3 Sample Sizes, Statistical Precision, and Effect Size Interpretation

For Studies 1–3, sample sizes were not based on a formal power analysis. Instead, they were set by the coverage our factorial designs required (3 AI models $\times$ 3–4 input or mitigation conditions), by the need for multiple independent reader ratings per paragraph, and by budget (see pre-registrations). For Studies 4 and 5, sample sizes were based on pre-registered simulation-based power analyses that targeted 80% power for the interaction between authorship and distortion, under deliberately conservative assumptions (SI:9.2, 10.2).

To make the realised precision of each study explicit, we report a design-based sensitivity analysis¹ for the headline contrasts in all five studies (Table S11). For each contrast, we recover the standard error (SE) from the reported 95% confidence interval and compute the minimum detectable effect at 80% power and two-sided $\alpha = .05$, $MDE_{80} = (z_{.975} + z_{.80}) \cdot SE \approx 2.80 \cdot SE$. For Studies 4 and 5, we also report power for the effects assumed in our pre-registered power analyses, given the realised SE.

Studies differ substantially in their precision:

Study 1 is extremely precise for reader perceptions. Across the 20 rating attributes on 0–100 scales, the MDE for the AI vs. human contrast ranges from 0.36 to 0.54 points (median 0.47). The observed absolute AMEs range from 0.64 to 29.1 points (median 7.5). Most observed distortions are therefore well above anything the design could only barely detect. Writer preference contrasts between AI models and input conditions are less precise (MDE $\approx$ 0.26–0.33 log-odds, i.e. odds ratios of roughly 1.30–1.39).

Studies 2 and 3 are considerably less precise. In Study 2, only disclaimer effects on writer preference of roughly 0.56–0.59 log-odds (odds ratios $\approx$ 1.75–1.80) or larger were reliably detectable, so modest disclaimer effects cannot be ruled out. In Study 3, the difference in the targeted polarising distortion between a mitigation arm and no intervention has an approximate MDE of 2.1 points (treating arms as independent). This is larger than the observed reduction under Prompting (–1.2 points) and smaller than the reduction under Reranking (–3.9 points). The non-significant Prompting result is therefore consistent with, at most, a small effect that Study 3 was not powered to detect.

Studies 4 and 5 are more precise than their power analyses required, but for different reasons (Table S12). In **Study 4**, realised variability was *larger* than assumed: the between-reader SD was 0.86 rather than 0.60 on the logit scale, and the beta precision was $\phi = 1.88$ rather than 3. At the realised precision, power for the assumed interaction (0.21 on the logit scale) was 75%, slightly below the planned 84%. The interaction we observed (0.39) was nonetheless clearly detectable because it was almost twice as large as assumed. The main effect was well powered both as planned and as realised (MDE 0.56p vs. an assumed 1.0p and an observed 1.56p). In **Study 5**, all three variance components were smaller than assumed: the between-reader SD was 3.9 rather than 10 points, the between-proposition SD 2.4 rather than 6, and the residual SD 16.4 rather than 22.2. Because the AI vs. human contrasts are estimated within reader, the residual SD is what matters most for their precision. As a result, power for the assumed dose-response contrast (1.5 points) was 98.8% rather than the planned 87.8% (MDE 1.0 point). The observed contrast (2.9 points) was almost twice as large as assumed.

Statistical significance alone says little about practical importance, especially in our many exploratory analyses (e.g. by AI model, input condition, or rating attribute), where some significant differences are small. We therefore report all effects as average marginal effects or odds ratios on interpretable scales with 95% confidence intervals, and we encourage readers to judge effects by their magnitude.

At the same time, we consider effects of the size we observe to be practically relevant, for two reasons: First, they are sizeable relative to natural benchmarks. In Study 4, the AI advantage in trust (1.56p) amounts to 7.8% of the 20p endowment. In Study 5, the AI advantage in persuasion (1.41 attitude points) is around half the size of the persuasive effect of the human-written paragraph itself (2.91 points relative to the off-topic control). Second, AI writing assistance is used at very large scale. Even a small per-paragraph shift in how writers are perceived, how much they are trusted, or how persuasive they are, repeated across many AI-assisted texts that people read every day, could add up to substantial effects on public discourse.

Table S11: Sensitivity analysis: minimum detectable effects (MDE) for headline contrasts in Studies 1–5. MDE_{80} is the smallest true effect detectable with 80% power at two-sided $\alpha = .05$, given the realised standard error ($\text{SE} = \text{CI width} / 2 \cdot 1.96$; $\text{MDE}_{80} \approx 2.80 \cdot \text{SE}$). For Studies 4 and 5, “assumed” is the effect assumed in the pre-registered power analysis, and “power” is power to detect that assumed effect at the realised SE. [†]Range across contrasts.

Study	Contrast (scale)	Observed	MDE_{80}	Assumed	Power
1	AI vs. human, 20 rating attributes (0–100 points) [†]	[0.64 –29.1]	0.36–0.54	–	–
1	Writer preference, AI model and input contrasts (log-odds) [†]	–0.82 to –0.21	0.26–0.33	–	–
2	Writer preference, disclaimer conditions (log-odds) [†]	–0.40 to 0.27	0.56–0.59	–	–
3	Polarity distortion, Prompting vs. none (0–100 points)	–1.21	2.09	–	–
3	Polarity distortion, Reranking vs. none (0–100 points)	–3.90	2.13	–	–
3	Writer preference, mitigation conditions (log-odds) [†]	–0.64 to –0.20	0.36	–	–
4	RQ1: AI vs. human trust allocation (pence, 0–20)	1.56	0.56	1.00	99.9%
4	RQ2: AI $\times$ high distortion interaction (log-odds)	0.39	0.22	0.21	75.0%
5	RQ1: AI vs. human stance shift (0–100 points)	1.41	0.50	1.25	>99.9%
5	RQ2: dose-response, high minus low distortion (0–100 points)	2.90	1.00	1.50	98.8%

Table S12: Assumed vs. realised variance components in Studies 4 and 5. “Assumed” values are those used in the pre-registered power simulations; “realised” values come from the primary models fitted to the collected data (Study 4: beta regression with reader random intercepts; Study 5: linear mixed model on treated cycles with crossed reader and proposition random intercepts).

Study	Component	Scale	Assumed	Realised
4	Reader SD	logit	0.60	0.86
4	Beta precision ϕ	–	3.00	1.88
5	Reader SD	points (0–100)	10.0	3.9
5	Proposition SD	points (0–100)	6.0	2.4
5	Residual SD	points (0–100)	22.2	16.4

1.4 Randomisation and Experimental Conditions

For writers in Studies 1-3, random assignment of experimental conditions operated at the level of writer–proposition pairs. Each writer was first randomly assigned three political propositions from a pool of 100. For these three propositions, we implemented a balanced within-subject design in which each writer engaged with all three AI models (GPT-4o, Claude Sonnet 4, DeepSeek V3) and all three input sub-conditions (stance-based, bullet-based, paragraph-based), with each model and input condition appearing exactly once per writer. The pairing between AI models and input sub-conditions was randomised, yielding nine possible model–input combinations. For the paragraph-based condition, the “rewrite” and “improve” variants were randomly assigned with equal probability.

Table S13 shows assignments for Study 1, i.e. the main study.

Table S13: Experimental condition assignment in Study 1. Each observation is one writer–proposition pair, of which there are three for each writer (N=1,501).

Model	Input Condition	N
anthropic/claude-sonnet-4	stance-based	542
anthropic/claude-sonnet-4	bullets-based	492
anthropic/claude-sonnet-4	rewrite	215
anthropic/claude-sonnet-4	improve	252
deepseek/deepseek-chat-v3-0324	stance-based	480
deepseek/deepseek-chat-v3-0324	bullets-based	502
deepseek/deepseek-chat-v3-0324	rewrite	269
deepseek/deepseek-chat-v3-0324	improve	250
openai/chatgpt-4o-latest	stance-based	479
openai/chatgpt-4o-latest	bullets-based	507
openai/chatgpt-4o-latest	rewrite	256
openai/chatgpt-4o-latest	improve	259
TOTAL		4,503

In Study 2, i.e. the disclaimer study (Table S14), writers were instead randomly assigned to one of four disclaimer conditions (three treatment variants and one control) at the writer level, and thus experienced the same disclaimer across all propositions. Within each writer, we again used a balanced assignment over AI models and input sub-conditions, mirroring Study 1 design.

Table S14: Experimental condition assignment in Study 2. Each observation is one writer–proposition pair, of which there are three for each writer (N=669)

Disclaimer Condition	N
no_disclaimer	510
liked_disclaimer	480
disliked_disclaimer	462
full_disclaimer	555
TOTAL	2,007

In Study 3, i.e. the mitigation study (Table S15), we retained the within-subject assignment of AI models but fixed the input condition to bullet-based prompts for all writers. In addition, a new between-condition factor was introduced: writers were randomly assigned, at the proposition level, to one of three mitigation conditions (Prompting, Reranking, No Intervention control), such that each writer experienced all three conditions across their three propositions.

Table S15: Experimental condition assignment in Study 3. Each observation is one writer–proposition pair, of which there are three for each writer (N=769).

Model	Mitigation Condition	N
anthropic/claude-sonnet-4	none	252
anthropic/claude-sonnet-4	prompting	240
anthropic/claude-sonnet-4	reranking	277
deepseek/deepseek-chat-v3-0324	none	262
deepseek/deepseek-chat-v3-0324	prompting	256
deepseek/deepseek-chat-v3-0324	reranking	251
openai/chatgpt-4o-latest	none	255
openai/chatgpt-4o-latest	prompting	273
openai/chatgpt-4o-latest	reranking	241
TOTAL		2,307

Studies 4 and 5 differ structurally from Studies 1-3 in that they involved readers only, who were not assigned any writer-side treatment. Here, randomisation instead operated at the reader level and determined which paragraphs each reader was shown. The paragraph pool was fixed in advance: it was constructed from Study 1 by holding the generation condition constant (bullet-point inputs only) and selecting 200 paragraphs into each of four cells defined by authorship (human vs. AI) $\times$ distortion (low vs. high), for 800 paragraphs in total (see §9.1). Distortion bin and authorship are thus properties of the paragraph pool, not randomised treatments. The random element is which paragraphs from this fixed pool a given reader sees, and in which order.

In Study 4 (trust study), each reader was assigned one paragraph from each of the four authorship $\times$ distortion cells, drawn at random from the corresponding pool subject to the constraint that no reader saw more than one paragraph per proposition. The presentation order of the four paragraphs was independently randomised for each reader (see §9.2). Study 5 (persuasion study) used the same treated pool plus a fifth, off-topic control condition. Each reader was assigned one paragraph from each of the five conditions on five distinct propositions, again with no proposition repeated within a reader and with presentation order independently randomised (see §10.2). On the control cycle, the rated proposition and the control paragraph were sampled independently and uniformly at random, the proposition from those used in the treated conditions and the control paragraph from the set of 10 off-topic paragraphs (Table S357). Because readers in Studies 4 and 5 were not randomised to any treatment, no differential-attrition tests by treatment were required for these samples (§1.5).

1.5 Attrition and Non-Participation

Tables S17 to S21 below show attrition funnels for writers and readers in each of our five studies.

Only writers (Studies 1-3) were assigned treatments, i.e. experimental conditions corresponding to which model generated AI paragraphs, which inputs these paragraphs were based on, and (for Study 3) which mitigation strategy was implemented.

To formally assess differential attrition after treatment assignment, we estimated separate logistic regressions for each writer sample in which an indicator for post-treatment attrition was regressed on assigned model and assigned input condition. We then used likelihood-ratio tests to evaluate whether treatment assignment predicted attrition (Table S16). Across all three writer studies (Studies 1-3), the omnibus tests were not statistically significant: Study 1: $\chi^2(5) = 9.53$, $p = .090$, Study 2: $\chi^2(4) = 2.31$, $p = .678$, Study 3: $\chi^2(5) = 6.51$, $p = .260$. Tests of the two treatment dimensions separately were likewise non-significant in every study: Study 1: model $\chi^2(2) = 2.96$, $p = .227$, input condition $\chi^2(3) = 6.52$, $p = .089$; Study 2: model $\chi^2(2) = 0.73$, $p = .696$, input condition $\chi^2(2) = 1.63$, $p = .442$; Study 3: model $\chi^2(2) = 4.57$, $p = .102$, input condition $\chi^2(3) = 2.30$, $p = .513$. Since readers (Studies 1-5) were not randomised to treatment, no analogous differential-attrition tests were required for the reader samples.

Table S16: Differential attrition tests for writers in Studies 1-3. Entries report likelihood-ratio tests from logistic regressions of post-treatment attrition on assigned model and assigned input condition among participants who reached treatment assignment. None of the omnibus or factor-specific tests is statistically significant at conventional levels.

Study	Omnibus test	Model	Input condition
1: Main study	$\chi^2(5) = 9.53$, $p = .090$	$\chi^2(2) = 2.96$, $p = .227$	$\chi^2(3) = 6.52$, $p = .089$
2: Disclaimer study	$\chi^2(4) = 2.31$, $p = .678$	$\chi^2(2) = 0.73$, $p = .696$	$\chi^2(2) = 1.63$, $p = .442$
3: Mitigation study	$\chi^2(5) = 6.51$, $p = .260$	$\chi^2(2) = 4.57$, $p = .102$	$\chi^2(3) = 2.30$, $p = .513$

Table S17: Participant attrition for writers in Study 1. Each row represents a sequential stage of the study. Step attrition rate is the proportion of participants lost at each stage relative to the previous stage. Cumulative retention is the proportion of the initial sample remaining. **Dashed line indicates first treatment assignment.** 31 people attrited after treatment assignment.

Stage	Remaining	Step Attrition (%)	Cum. Retention (%)
Opened study	1,670	—	100.0
Passed writing screener	1,655	0.9	99.1
Completed writing proposition 1	1,596	3.6	95.6
Completed writing proposition 2	1,555	2.6	93.1
Completed writing proposition 3	1,532	1.5	91.7
Completed model engagement 1	1,523	0.6	91.2
Completed model engagement 2	1,523	0.0	91.2
Completed model engagement 3	1,518	0.3	90.9
Completed study	1,507	0.7	90.2
After list-wise deletion	1,501	0.4	89.9

Table S18: Participant attrition for readers in Study 1. Of 12,047 participants who opened the study, 10,017 completed it. Only their ratings were used in analysis. The lower panel shows the number of propositions rated by the 2,030 non-completers before dropout. Readers were not assigned any treatment.

Stage	N	Step Attrition (%)	Cumulative Retention (%)
Opened study	12,047	—	100.0
Completed study	10,017	16.9	83.1
Prop. completed before dropout	N	Share of non-completers (%)	
0 propositions	1,033	50.9	
1 proposition	566	27.9	
2 propositions	166	8.2	
3 propositions	102	5.0	
4 propositions	65	3.2	
5 propositions	27	1.3	
6 propositions	35	1.7	
7 propositions	15	0.7	
8 propositions	15	0.7	
9 propositions	6	0.3	

Table S19: Participant attrition for writers in Study 2. Each row represents a sequential stage of the study. Step attrition rate is the proportion of participants lost at each stage relative to the previous stage. Cumulative retention is the proportion of the initial sample remaining. **Dashed line indicates first treatment assignment.** 12 people attrited after treatment assignment.

Stage	Remaining	Step Attrition (%)	Cum. Retention (%)
Opened study	839	—	100.0
Passed writing screener	835	0.6	99.4
Completed writing proposition 1	713	14.5	85.0
Completed writing proposition 2	689	3.4	82.1
Completed writing proposition 3	681	1.2	81.2
Completed model engagement 1	674	1.0	80.4
Completed model engagement 2	674	0.0	80.4
Completed model engagement 3	674	0.0	80.4
Completed study	673	0.3	80.1
After list-wise deletion	669	0.6	79.6

Table S20: Participant attrition for writers in Study 3. Each row represents a sequential stage of the study. Step attrition rate is the proportion of participants lost at each stage relative to the previous stage. Cumulative retention is the proportion of the initial sample remaining. **Dashed line indicates first treatment assignment.** 40 people attrited after treatment assignment.

Stage	Remaining	Step Attrition (%)	Cum. Retention (%)
Opened study	1,094	—	100.0
Passed writing screener	1,081	1.3	98.7
Completed writing proposition 1	850	21.4	77.6
Completed writing proposition 2	821	3.4	75.0
Completed writing proposition 3	809	1.5	73.9
Completed model engagement 1	794	1.9	72.5
Completed model engagement 2	790	0.5	72.1
Completed model engagement 3	786	0.5	71.8
Completed study	786	0.0	71.8
After list-wise deletion	769	2.2	70.2

Table S21: Participant attrition for readers in Study 3. Of 3,108 participants who opened the study, 2,543 completed it. Only their ratings were used in analysis. The lower panel shows the number of propositions rated by the 565 non-completers before dropout. Readers were not assigned any treatment.

Stage	N	Step Attrition (%)	Cumulative Retention (%)
Opened study	3,108	—	100.0
Completed study	2,543	18.2	81.8
Prop. completed before dropout	N	Share of non-completers (%)	
0 propositions	306	54.2	
1 proposition	145	25.7	
2 propositions	49	8.7	
3 propositions	21	3.7	
4 propositions	11	1.9	
5 propositions	7	1.2	
6 propositions	6	1.1	
7 propositions	5	0.9	
8 propositions	7	1.2	
9 propositions	8	1.4	

Table S22: Participant attrition for readers in Study 4. Of 856 participants who opened the study, 804 completed it. After list-wise deletion (incomplete data due to technical errors), 802 readers remained. Only their ratings were used in analysis. Readers were not assigned any treatment.

Stage	N	Step Attrition (%)	Cumulative Retention (%)
Opened study	856	—	100.0
Rated 1 paragraph	804	6.1	93.9
Rated 2 paragraphs	804	0.0	93.9
Rated 3 paragraphs	804	0.0	93.9
Rated 4 paragraphs	804	0.0	93.9
Completed study	804	0.0	93.9
After list-wise deletion	802	0.2	93.7

Table S23: Participant attrition for readers in Study 5. Of 9,241 participants who opened the study, 8,010 completed it. After list-wise deletion (incomplete data due to technical errors), 7,996 readers remained. Only their ratings were used in analysis. Readers were not assigned any treatment.

Stage	N	Step Attrition (%)	Cumulative Retention (%)
Opened study	9,241	—	100.0
Rated 1 paragraph	8,164	11.7	88.3
Rated 2 paragraphs	8,010	1.9	86.7
Rated 3 paragraphs	8,010	0.0	86.7
Rated 4 paragraphs	8,010	0.0	86.7
Rated 5 paragraphs	8,010	0.0	86.7
Completed study	8,010	0.0	86.7
After list-wise deletion	7,996	0.2	86.5

2 Study Materials

2.1 Studies 1-5: Onboarding

Prior to the experimental portions of each study (Studies 1-5), we collected data on participant attributes in an onboarding questionnaire. Below, we denote each question by the corresponding variable name in our data release.

age: What is your age?
[Numeric response]

gender: What is your gender?
[Male / Female / Other / Prefer not to say]

race: What is your race / ethnicity?
[White / Black / Asian / Mixed / Other / Prefer not to say]

englishFirst: Is English your first language?
[Yes / No / Bilingual from birth / Prefer not to say]

englishSkills: How would you rate your English language skills?
[Basic (can understand simple content with difficulty) / Intermediate (can understand and communicate everyday topics) / Advanced (can communicate complex ideas effectively) / Expert (can use English fluently for all purposes including professional writing) / Prefer not to say]

education: What is the highest level of education you have completed?
[GCSEs or equivalent / A-levels or equivalent / Vocational qualification / Undergraduate degree / Postgraduate degree (Master's) / Doctorate (PhD) / Other / Prefer not to say]

income: What is your current yearly income?
[Under £15,000 / £15,000-£24,999 / £25,000-£34,999 / £35,000-£49,999 / £50,000-£74,999 / £75,000-£99,999 / £100,000+ / Prefer not to say]

politicalParty: Which political party did you vote for in the 2024 UK general election?
[Conservative / Labour / Liberal Democrats / Reform UK / Scottish National Party / Greens / Other / Did not vote / Not eligible to vote / Prefer not to say]
Note: Study 1 did not include a “Reform UK” option.

politicalIdeology: How would you describe your political ideology?
[Very Left-Wing / Moderately Left-Wing / Centrist / Moderately Right-Wing / Very Right-Wing / Prefer not to say]
Note: Study 1 included an “other” option.

2.2 Studies 1-5: Screening

After onboarding, all studies included a screening task to filter out inattentive participants.

Writers (Studies 1-3) were asked to name their favourite ice cream flavour and explain their choice in 1-2 sentences (Figure S1). Answers were assessed by GPT-4o, with any on-topic answer considered a pass. Writers who failed this screener were excluded from the study.

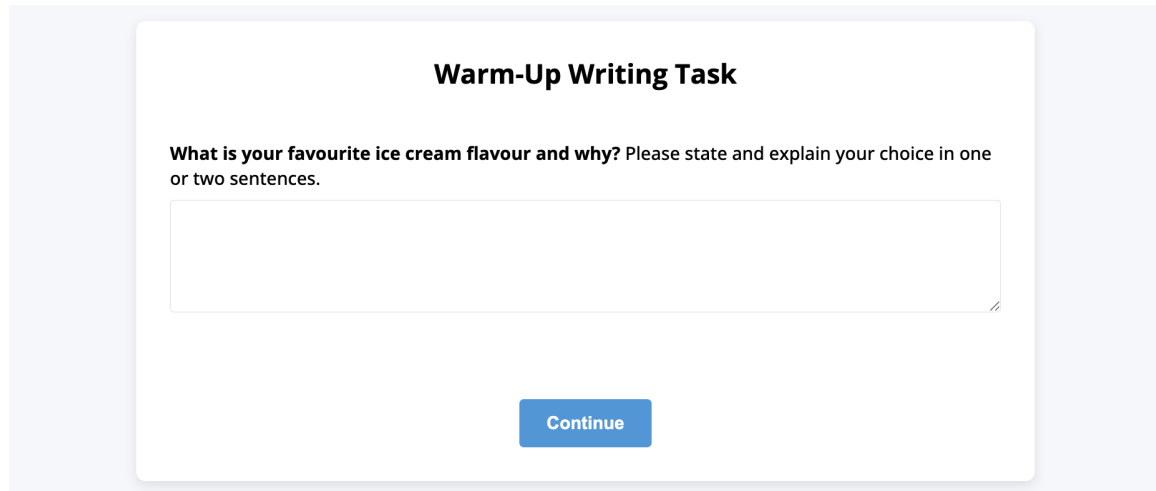
The screenshot shows a white card titled "Warm-Up Writing Task" on a light blue background. Below the title, the instruction reads: "What is your favourite ice cream flavour and why? Please state and explain your choice in one or two sentences." Below this text is a large, empty rectangular text input field. At the bottom center of the card is a blue button with the word "Continue" in white text.

Figure S1: Screenshot of the writing screener which writers in Studies 1-3 had to pass before starting the study. Responses were assessed by GPT-4o, with any on-topic answer considered a pass.

Readers in Studies 1, 3, and 5 were presented with a 0-100 scale that ostensibly measured how convincing they find emotional appeals in writing. The screen also included clear instructions to ignore the question about convincing writing and instead move the slider to between 30 and 40 (Figure S2). Readers who failed this screener were excluded from the study.

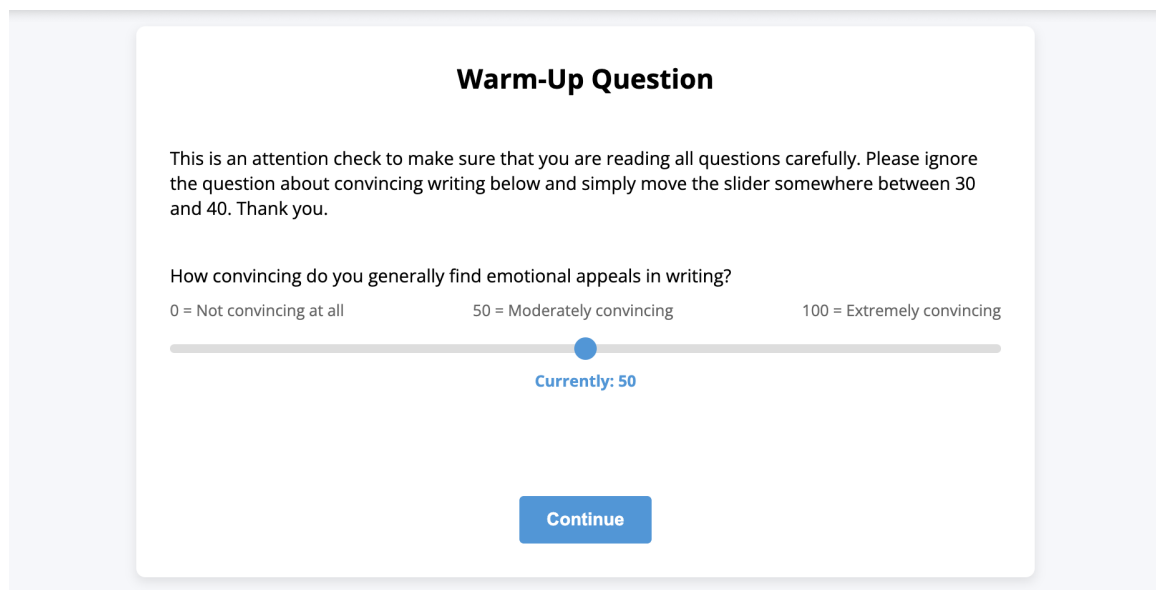
The screenshot shows a white card titled "Warm-Up Question" on a light blue background. The text on the card reads: "This is an attention check to make sure that you are reading all questions carefully. Please ignore the question about convincing writing below and simply move the slider somewhere between 30 and 40. Thank you." Below this is the question: "How convincing do you generally find emotional appeals in writing?". Underneath the question is a horizontal slider scale with labels: "0 = Not convincing at all", "50 = Moderately convincing", and "100 = Extremely convincing". A blue circular slider knob is positioned at the 50 mark. Below the slider, the text "Currently: 50" is displayed in blue. At the bottom center of the card is a blue button with the word "Continue" in white text.

Figure S2: Screenshot of the rating screener which readers in Studies 1, 3, and 5 had to pass before starting the study. Screener instructions are clearly stated in the question text.

Readers in Study 4 were required to pass a comprehension check as described in §2.12.

2.3 Study 1: Writers

After onboarding and screening, writers in Study 1 were assigned 3 political propositions, randomly selected from a pool of 100 (see §2.9). For each proposition, writers were asked the questions listed below. Figure S3 shows the screen in which writers were asked to express their opinion on the assigned political proposition. Note that **text pasting was disabled throughout the study** to reduce the risk of unintended AI use by study participants.

writer__stance__pre: What is your level of agreement with the statement?
[0-100 scale, where 0 = "Strongly disagree", 50 = "Ambivalent", and 100 = "Strongly agree"]

writer__bullets: Please explain the key reasons for your opinion in 2-4 bullet points. Start each bullet point with a dash - and put each bullet point on a separate line.
[Free text response, minimum of 2 bullets]

writer__paragraph: Please expand your bullet points into a short paragraph (around 100-150 words), in which you explain your opinion.
[Free text response, minimum of 100 words]

writer__affect__x and __y: We would like to understand how you felt while writing your paragraph. Please tell us by clicking on the grid below:
[Affect grid visual with pleasure-displeasure and arousal-sleepiness dimensions]

writer__knowledge: How much do you know about the issue referenced in the statement?
[0-100 scale, where 0 = "No knowledge at all", 50 = "Some understanding", and 100 = "Expert knowledge"]

writer__importance: How important is the issue referenced in the statement to you?
[0-100 scale, where 0 = "Not important at all", 50 = "Moderately important", and 100 = "Extremely important"]

writer__stance__post: Please tell us again: What is your level of agreement with the statement? By default, the slider is set to the answer you gave before. If your opinion has changed, please move the slider accordingly.
[0-100 scale, where 0 = "Strongly disagree", 50 = "Ambivalent", and 100 = "Strongly agree"]

writer__confidence: How confident are you in your opinion on this issue?
[0-100 scale, where 0 = "Not confident at all", 50 = "Moderately confident", and 100 = "Extremely confident"]

Consider this statement:

Statement: 1 of 3 Step: 1 of 3 Screen: 2 of 7

The UK government should prioritize and accelerate digital transformation across all public services.

What is your level of agreement with the statement?

0 = Strongly disagree 50 = Ambivalent 100 = Strongly agree

Please click on the scale to set your rating

Please explain the key reasons for your opinion in 2-4 bullet points. Start each bullet point with a dash - and put each bullet point on a separate line.

- First point
- Second point
- ...

Currently: 0 bullet points

Please expand your bullet points into a short paragraph (around 100-150 words), in which you explain your opinion.

Currently: 0 words

Continue

Figure S3: Screenshot of the writer opinion screen, in which each writer expressed their opinion on the assigned political proposition in three formats: agreement rating, bullet point reasons, and a full opinion paragraph.

After answering the questions above for each proposition, writers were asked to engage with content generated by AI models based on their answers. Figure S4 shows the screen in which writers were asked to edit the AI-written paragraph until they felt that it reflected their opinion. Figure S5 shows a screenshot of the screen in which writers were asked to choose between their own writing and the AI-written paragraph, including any edits made, for explaining their opinion to someone

model_paragraph_stance: What is this paragraph's level of agreement with the statement?
[0-100 scale, where 0 = "Strongly disagree", 50 = "Ambivalent", and 100 = "Strongly agree"]

edited_paragraph: Please edit the AI-written paragraph until you feel that it reflects your opinion about the statement. If you feel that the paragraph already reflects your opinion, you do not have to edit it. Otherwise please edit it until it does.
[Free text response, starting with the model-generated paragraph]

writer_preference: Imagine you want to explain your opinion on the statement to someone. Which of the two paragraphs below would you rather use for this purpose?
[The paragraph on the left (original) / Equal preference / The paragraph on the right (edited)]

writer_preference_reason: [If answer is not "equal preference"] Why do you prefer this paragraph over the other? Please tick all reasons that apply and explain any additional reasons under "other".
[Checkbox options: "It better reflects my opinion." / "I prefer the writing style." / "Other (please specify):"]

writer_stance_final: Please tell us one last time: What is your level of agreement with the statement? By default, the slider is set to the answer you gave before. If your opinion has changed, please move the slider accordingly.
[0-100 scale, where 0 = "Strongly disagree", 50 = "Ambivalent", and 100 = "Strongly agree"]

Consider this statement:

Statement: 1 of 3 Step: 2 of 3 Screen: 2 of 4

The UK should legally recognize and protect all forms of transgender identity.

Please edit the AI-written paragraph below until you feel that it reflects your opinion about the statement. If you feel that the paragraph already reflects your opinion, you do not have to edit it. Otherwise please edit it until it does.

I believe the UK should not legally recognize or protect all forms of transgender identity. While respecting individual freedoms is important, extending legal recognition to all self-identified genders could create unnecessary complications in areas like healthcare, education, and public policy. Legal frameworks must balance personal rights with societal coherence, and broad recognition might undermine existing protections for women and children. There are also concerns about the potential for abuse or misuse of such policies, which could lead to unintended consequences. Instead, focus should remain on clear, evidence-based definitions of gender that align with biological reality. Existing anti-discrimination laws already provide sufficient safeguards for transgender individuals without expanding legal categories further. Prioritizing factual accuracy and practical governance over ideological demands ensures fairness and stability for everyone. My disagreement stems from a belief that this proposition risks overreach and could harm societal structures that rely on objective definitions.

Continue

Figure S4: Screenshot of the paragraph edit screen, where the writer is asked to edit the AI-written paragraph until it reflects their opinion.

Consider this statement:

Statement: 1 of 3 Step: 2 of 3 Screen: 3 of 4

The UK should legally recognize and protect all forms of transgender identity.

Imagine you want to explain your opinion on the statement to someone.
Which of the two paragraphs below would you rather use for this purpose?

Your original paragraph:

i dont really agree with this, at least not in the way its phrased, becuae "all forms" feels way too broad and kind of undefined, and laws need to be precise or they just create confusion and unintended consquences. its not that i think people shouldnt be treated fairly or protected from harm obviously they should, but legally recognizing every possible identity without clear boundries could make policies harder to enforce and maybe even clash with existing rights or systems. i just think this is one of those areas where a more cautious, case by case approach makes more sense rather than sweeping legal recognition

Your edited AI paragraph:

I believe the UK should not legally recognize or protect all forms of transgender identity. While respecting individual freedoms is important, extending legal recognition to all self-identified genders could create unnecessary complications in areas like healthcare, education, and public policy. Legal frameworks must balance personal rights with societal coherence, and broad recognition might undermine existing protections for women and children. There are also concerns about the potential for abuse or misuse of such policies, which could lead to unintended consequences. Instead, focus should remain on clear, evidence-based definitions of gender that align with biological reality. Existing anti-discrimination laws already provide sufficient safeguards for transgender individuals without expanding legal categories further. Prioritizing factual accuracy and practical governance over ideological demands ensures fairness and stability for everyone. My disagreement stems from a belief that this proposition risks overreach and could harm societal structures that rely on objective definitions.

☐ The paragraph on the left

☐ Equal preference

☐ The paragraph on the right

Continue

Figure S5: Screenshot of the paragraph preference screen, where the writer is asked to choose between the AI-written paragraph and their own for explaining their opinion to someone. The AI-written paragraph includes any edits previously made by the writer.

Finally, after completing the above for all propositions, writers were asked how acceptable or unacceptable they would find different types of distortion AI writing assistance. The types of distortion writers were asked about directly corresponded to the rating attributes that readers were later asked about. In total, writers were asked 49 questions, each shown on a separate screen, where the top of the screen read “When using AI for writing assistance, the resulting text might be perceived differently from how your own independent writing (without AI assistance) would be perceived. We want to understand which differences you would find desirable or undesirable.”. Figure S6 shows an example screenshot. All responses were measured on a 0-100 scale, where 0 = “Strongly disagree”, 50 = “Ambivalent”, and 100 = “Strongly agree”. Below, we denote each question by the corresponding variable name in our data release.

- extreme_more:** I am okay with AI writing assistance making my opinion on an issue seem more extreme than my own independent writing would make it seem.
- moderate_more:** I am okay with AI writing assistance making my opinion on an issue seem more moderate / balanced than my own independent writing would make it seem.
- confidence_more:** I am okay with AI writing assistance making me seem more confident in my opinion than my own independent writing would make me seem.
- confidence_less:** I am okay with AI writing assistance making me seem less confident in my opinion than my own independent writing would make me seem.
- knowledge_more:** I am okay with AI writing assistance making me seem more knowledgeable about an issue than my own independent writing would make me seem.
- knowledge_less:** I am okay with AI writing assistance making me seem less knowledgeable about an issue than my own independent writing would make me seem.
- importance_more:** I am okay with AI writing assistance making an issue seem more important to me than my own independent writing would make it seem.
- importance_less:** I am okay with AI writing assistance making an issue seem less important to me than my own independent writing would make it seem.
- relevance_more:** I am okay with AI writing assistance making my writing seem more relevant or on-topic.
- relevance_less:** I am okay with AI writing assistance making my writing seem less relevant or on-topic.
- clarity_more:** I am okay with AI writing assistance making my opinion clearer to others than my own independent writing would make it.
- clarity_less:** I am okay with AI writing assistance making my opinion less clear to others than my own independent writing would make it.
- formality_more:** I am okay with AI writing assistance making my writing seem more formal.
- formality_less:** I am okay with AI writing assistance making my writing seem less formal.
- informative_more:** I am okay with AI writing assistance making my writing seem more informative or educational.
- informative_less:** I am okay with AI writing assistance making my writing seem less informative or educational.

- originality__more:** I am okay with AI writing assistance making my writing seem more original or novel.
- originality__less:** I am okay with AI writing assistance making my writing seem less original or novel.
- friendliness__more:** I am okay with AI writing assistance making me seem more friendly than my own independent writing would make me seem.
- friendliness__less:** I am okay with AI writing assistance making me seem less friendly than my own independent writing would make me seem.
- optimism__more:** I am okay with AI writing assistance making me seem more optimistic than my own independent writing would make me seem.
- optimism__less:** I am okay with AI writing assistance making me seem less optimistic than my own independent writing would make me seem.
- community__more:** I am okay with AI writing assistance making me seem more community-oriented than my own independent writing would make me seem.
- community__less:** I am okay with AI writing assistance making me seem less community-oriented than my own independent writing would make me seem.
- open_views__more:** I am okay with AI writing assistance making me seem more open to reconsidering my views than my own independent writing would make me seem.
- open_views__less:** I am okay with AI writing assistance making me seem less open to reconsidering my views than my own independent writing would make me seem.
- hope__more:** I am okay with AI writing assistance making my writing express more hope, as perceived by others.
- hope__less:** I am okay with AI writing assistance making my writing express less hope, as perceived by others.
- reexcitement__more:** I am okay with AI writing assistance making my writing express more excitement, as perceived by others.
- excitement__less:** I am okay with AI writing assistance making my writing express less excitement, as perceived by others.
- fear__more:** I am okay with AI writing assistance making my writing express more fear, as perceived by others.
- fear__less:** I am okay with AI writing assistance making my writing express less fear, as perceived by others.
- disgust__more:** I am okay with AI writing assistance making my writing express more disgust, as perceived by others.
- disgust__less:** I am okay with AI writing assistance making my writing express less disgust, as perceived by others.
- anger__more:** I am okay with AI writing assistance making my writing express more anger, as perceived by others.

- anger__less:** I am okay with AI writing assistance making my writing express less anger, as perceived by others.
- age__older:** I am okay with AI writing assistance making me seem older than what people would guess if they had read my own independent writing.
- age__younger:** I am okay with AI writing assistance making me seem younger than what people would guess if they had read my own independent writing.
- gender__different:** I am okay with AI writing assistance making my gender seem different than what people would guess if they had read my own independent writing.
- race__different:** I am okay with AI writing assistance making my race or ethnicity seem different than what people would guess if they had read my own independent writing.
- english__better:** I am okay with AI writing assistance making my English language skills seem better than what people would guess if they had read my own independent writing.
- english__worse:** I am okay with AI writing assistance making my English language skills seem worse than what people would guess if they had read my own independent writing.
- education__more:** I am okay with AI writing assistance making me seem more educated than what people would guess if they had read my own independent writing.
- education__less:** I am okay with AI writing assistance making me seem less educated than what people would guess if they had read my own independent writing.
- income__higher:** I am okay with AI writing assistance making my income seem higher than what people would guess if they had read my own independent writing.
- income__lower:** I am okay with AI writing assistance making my income seem lower than what people would guess if they had read my own independent writing.
- politics__party__different:** I am okay with AI writing assistance making it seem like I voted for a different political party than what people would guess if they had read my own independent writing.
- politics__extreme:** I am okay with AI writing assistance making me seem more politically extreme than my own independent writing would make me seem.
- politics__moderate:** I am okay with AI writing assistance making me seem more politically moderate / centrist than my own independent writing would make me seem.

Step: 3 of 3 Screen: 1 of 49

When using AI for writing assistance, the resulting text might be perceived differently from how your own independent writing (without AI assistance) would be perceived. We want to understand **which differences you would like or dislike**.

"I am okay with AI writing assistance making my opinion on an issue seem **more extreme** than my own independent writing would make it seem."

0 = Strongly disagree 50 = Ambivalent 100 = Strongly agree

Please click on the scale to set your rating

Continue

Figure S6: Screenshot of the writer tolerance screen in Study 1, where the writer is asked to rate their tolerance for different types of distortion from AI writing assistance. Shown here is tolerance for the polarising effect of AI writing assistance.

2.4 Study 1: Readers

After onboarding and screening, readers in Study 1 were each assigned ten paragraphs to rate. For each assigned paragraph, readers answered 29 rating questions. They answered all 29 questions in the order listed below for one paragraph, then moved on to the next paragraph. We denote each question by the corresponding variable name in our data release. Figure S7 and Figure S8 show screenshots of the rating interface that readers interacted with.

writer_stance: What is the author's level of agreement with the statement? Please read the text carefully.
[0-100 scale, where 0 = "Strongly disagree", 50 = "Ambivalent", and 100 = "Strongly agree"]

writer_confidence: How confident is the author of the paragraph in their opinion?
[0-100 scale, where 0 = "Not confident at all", 50 = "Moderately confident", and 100 = "Extremely confident"]

writer_knowledge: How much does the author of the paragraph know about this issue?
[0-100 scale, where 0 = "No knowledge at all", 50 = "Some understanding", and 100 = "Expert knowledge"]

writer_importance: How important is this issue to the author of the paragraph?
[0-100 scale, where 0 = "Not important at all", 50 = "Moderately important", and 100 = "Extremely important"]

paragraph_relevance: How on-topic / relevant is the paragraph to the statement?
[0-100 scale, where 0 = "Not relevant at all", 50 = "Moderately relevant", and 100 = "Extremely relevant"]

paragraph_clarity: How clear is the author's opinion in the paragraph?
[0-100 scale, where 0 = "Not clear at all", 50 = "Moderately clear", and 100 = "Extremely clear"]

paragraph_formality: How formal is the language used in the paragraph?
[0-100 scale, where 0 = "Not formal at all", 50 = "Moderately formal", and 100 = "Extremely formal"]

paragraph_informativeness: How informative / educational did you find the paragraph?
[0-100 scale, where 0 = "Not informative at all", 50 = "Moderately informative", and 100 = "Extremely informative"]

paragraph_originality: How original / surprising / novel did you find the arguments in the paragraph?
[0-100 scale, where 0 = "Not original at all", 50 = "Moderately original", and 100 = "Extremely original"]

writer_friendliness: How friendly is the author of the paragraph?
[0-100 scale, where 0 = "Extremely unfriendly" (Confrontational, Hostile, Argumentative), 50 = "Neither friendly nor unfriendly", and 100 = "Extremely friendly" (Warm, Welcoming, Supportive)]

writer_optimism: How optimistic is the author of the paragraph?
[0-100 scale, where 0 = "Highly pessimistic" (Focuses on problems, Anticipates negative outcomes, Sees limitations), 50 = "Balanced outlook", and 100 = "Highly optimistic" (Emphasises possibilities, Expects positive outcomes, Sees opportunities in challenges)]

writer_community: How community-oriented is the author of the paragraph?
[0-100 scale, where 0 = "Not community-oriented" (Prioritises personal

autonomy, Emphasises individual responsibility, Values self-reliance), 50 = "Balanced between individual and community concerns", and 100 = "Highly community-oriented" (Prioritises community welfare, Emphasises shared responsibility, Values interdependence)]

writer__openness: How open is the author of the paragraph to reconsidering their views?
[0-100 scale, where 0 = "Firmly committed to current views" (Resistant to counterarguments, Confident in existing position, Values consistency), 50 = "Moderately open to reconsideration", and 100 = "Highly open to reconsidering views" (Welcomes different perspectives, Weighs new evidence, Willing to change position)]

paragraph__hope: Does the paragraph express hope?
[0-100 scale, where 0 = "No sign of hope", 50 = "Moderate hope", and 100 = "Extreme hope"]

paragraph__excitement: Does the paragraph express excitement?
[0-100 scale, where 0 = "No sign of excitement", 50 = "Moderate excitement", and 100 = "Extreme excitement"]

paragraph__fear: Does the paragraph express fear?
[0-100 scale, where 0 = "No sign of fear", 50 = "Moderate fear", and 100 = "Extreme fear"]

paragraph__disgust: Does the paragraph express disgust?
[0-100 scale, where 0 = "No sign of disgust", 50 = "Moderate disgust", and 100 = "Extreme disgust"]

paragraph__anger: Does the paragraph express anger?
[0-100 scale, where 0 = "No sign of anger", 50 = "Moderate anger", and 100 = "Extreme anger"]

writer__affect__x and __y: How did the author feel while writing the paragraph? Please tell us by clicking on the grid below.
[Affect grid visual with pleasure-displeasure and arousal-sleepiness dimensions]

writer__age What is the author's age?
[Numeric response]

writer__gender: What is the author's gender?
[Male / Female / Other]

writer__race: What is the author's race / ethnicity?
[White / Black / Asian / Mixed / Other]

writer__english__first: Is English the author's first language?
[Yes / No]

writer__english__skills: How would you rate the author's English language skills?
[Basic (can understand simple content with difficulty) / Intermediate (can understand and communicate everyday topics) / Advanced (can communicate complex ideas effectively) / Expert (can use English fluently for all purposes including professional writing)]

writer__education: What is the highest level of education that the author has completed?
[GCSEs or equivalent / A-levels or equivalent / Vocational qualification /

*Undergraduate degree / Postgraduate degree (Master's) / Doctorate (PhD)
/ Other]*

writer_income: What is the author's current yearly income?
[Under £15,000 / £15,000-£24,999 / £25,000-£34,999 / £35,000-£49,999 / £50,000-£74,999 / £75,000-£99,999 / £100,000+]

writer_politicalParty: Which political party did the author vote for in the 2024 UK general election?
[Conservative / Labour / Liberal Democrats / Scottish National Party / Greens / Other / Did not vote / Not eligible to vote]

writer_politicalIdeology: What is the author's political ideology?
[Very Left-Wing / Moderately Left-Wing / Centrist / Moderately Right-Wing / Very Right-Wing / Other]

Consider this statement:

Statement: 1 of 10 Screen: 2 of 21

The UK should defund the BBC.

Here is a paragraph responding to the statement:

i agree with the statement because the current funding model of the BBC is no longer the most effective way to finance public broadcastings. it places an unfair burden on viewers and struggles to adapt to the changing media landscape. furthermore, the BBCs content, while often of high quality, sometimes fails to reflect the diversity of the UK population. defunding the BBC could lead to a more competitive media environment, encouraging innovation and potentially offering a wider range of perspectives. However, it is crucial to ensure that any alternative funding model protects the BBC's independance and it's ability to provide impartial news and programming.

What is the **author's level of agreement** with the **statement**? Please read the text carefully.

0 = Strongly disagree 50 = Ambivalent 100 = Strongly agree

Please click on the scale to set your rating

Continue

Figure S7: Screenshot of the rating screen, which readers in Studies 1-3 are shown for each rating attribute (§2.4) and each paragraph assigned to them.

Consider this statement:

Statement: 1 of 10 Screen: 20 of 21

The UK should defund the BBC.

Here is a paragraph responding to the statement:

i agree with the statement because the current funding model of the BBC is no longer the most effective way to finance public broadcastings. it places an unfair burden on viewers and struggles to adapt to the changing media landscape. furthermore, the BBCs content, while often of high quality, sometimes fails to reflect the diversity of the UK population. defunding the BBC could lead to a more competitive media environment, encouraging innovation and potentially offering a wider range of perspectives. However, it is crucial to ensure that any alternative funding model protects the BBC's independance and it's ability to provide impartial news and programming.

How did the author feel while writing this paragraph? Please tell us by clicking on the grid below.

Stress

High Arousal

Excitement

Unpleasant Feelings

Pleasant Feelings

Depression

Sleepiness

Relaxation

Please click anywhere on the grid to indicate how you think the author felt.

Continue

Figure S8: Screenshot of the affect grid rating screen, which readers in Studies 1-3 are shown for each paragraph assigned to them. Affect grid format based on².

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2.5 Study 2: Writers

Study 2 mirrors the design of Study 1. We denote changes relative to Study 1 in [blue](#).

After onboarding and screening, writers were assigned 3 political propositions, randomly selected from a pool of 100 (see §2.9). For each proposition, writers were asked the questions listed below.

writer_stance_pre: What is your level of agreement with the statement?
[0-100 scale, where 0 = "Strongly disagree", 50 = "Ambivalent", and 100 = "Strongly agree"]

writer_bullets: Please explain the key reasons for your opinion in 2-4 bullet points. Start each bullet point with a dash - and put each bullet point on a separate line.
[Free text response, minimum of 2 bullets]

writer_paragraph: Please expand your bullet points into a short paragraph (around 100-150 words), in which you explain your opinion.
[Free text response, minimum of 100 words]

writer_affect_x and _y: [Question not shown](#)

writer_knowledge: How much do you know about the issue referenced in the statement?
[0-100 scale, where 0 = "No knowledge at all", 50 = "Some understanding", and 100 = "Expert knowledge"]

writer_importance: How important is the issue referenced in the statement to you?
[0-100 scale, where 0 = "Not important at all", 50 = "Moderately important", and 100 = "Extremely important"]

writer_stance_post: Please tell us again: What is your level of agreement with the statement? By default, the slider is set to the answer you gave before. If your opinion has changed, please move the slider accordingly.
[0-100 scale, where 0 = "Strongly disagree", 50 = "Ambivalent", and 100 = "Strongly agree"]

writer_confidence: How confident are you in your opinion on this issue?
[0-100 scale, where 0 = "Not confident at all", 50 = "Moderately confident", and 100 = "Extremely confident"]

After answering the questions above for each proposition, writers were asked to engage with content generated by AI models based on their answers.

model_paragraph_stance: What is this paragraph's level of agreement with the statement?
[0-100 scale, where 0 = "Strongly disagree", 50 = "Ambivalent", and 100 = "Strongly agree"]

edited_paragraph: Please edit the AI-written paragraph until you feel that it reflects your opinion about the statement. If you feel that the paragraph already reflects your opinion, you do not have to edit it. Otherwise please edit it until it does.
[Free text response, starting with the model-generated paragraph]

On this screen, writers were shown a pop-up disclaimer that they had to acknowledge before continuing (see 7.1).

writer__preference: Imagine you want to explain your opinion on the statement to someone. Which of the two paragraphs below would you rather use for this purpose?

[The paragraph on the left (original) / Equal preference / The paragraph on the right (edited)]

On this screen, writers were again shown a pop-up disclaimer that they had to acknowledge before continuing (see 7.1).

writer__preference__reason: [If answer is not "equal preference"] Why do you prefer this paragraph over the other? Please write 1-2 sentences or bullet points to explain.
[Free text response]

writer__stance__final: Please tell us one last time: What is your level of agreement with the statement? By default, the slider is set to the answer you gave before. If your opinion has changed, please move the slider accordingly.
[0-100 scale, where 0 = "Strongly disagree", 50 = "Ambivalent", and 100 = "Strongly agree"]

Finally, after completing the above for all propositions, writers were asked how acceptable or unacceptable they would find different types of distortion AI writing assistance. The types of distortion writers were asked about directly correspond to the rating attributes [highlighted in the disclaimers \(7.1\)](#). In total, writers were asked 8 questions, each on a separate screen, where the top of the screen reads “When using AI for writing assistance, the resulting text might be perceived differently from how your own independent writing (without AI assistance) would be perceived. We want to understand which differences you would like or dislike?”. All responses were measured on a 0-100 scale, where 0 = "Strongly dislike", 50 = "Neutral or ambivalent", and 100 = "Strongly like". Below, we denote each question by the corresponding variable name in our data release.

extreme__more: How would you feel about AI making your opinion on an issue seem **more extreme** than your own independent writing would make it seem.

extreme__less: How would you feel about AI making your opinion on an issue seem **more moderate / balanced** than your own independent writing would make it seem.

informative__more: How would you feel about AI making your writing seem **more informative or educational**.

informative__less: How would you feel about AI making your writing seem **less informative or educational**.

clarity__more: How would you feel about AI making your opinion **clearer** to others than your own independent writing would make it.

clarity__less: How would you feel about AI making your opinion **less clear** to others than your own independent writing would make it.

openness__more: How would you feel about AI making you seem **more open to reconsidering your views** than your own independent writing would make you seem.

openness__less: How would you feel about AI making you seem **less open to reconsidering your views** than your own independent writing would make you seem.

We did not collect ratings for paragraphs produced in Study 2.

2.6 Study 3: Writers

Study 3 mirrors the design of Study 1. We denote changes relative to Study 1 in **purple**.

After onboarding and screening, writers were assigned 3 political propositions, randomly selected from a pool of 100 (see §2.9). For each proposition, writers were asked the questions listed below.

writer_stance_pre: What is your level of agreement with the statement?
[0-100 scale, where 0 = "Strongly disagree", 50 = "Ambivalent", and 100 = "Strongly agree"]

writer_bullets: Please explain the key reasons for your opinion in 2-4 bullet points. Start each bullet point with a dash - and put each bullet point on a separate line.
[Free text response, minimum of 2 bullets]

writer_paragraph: Please expand your bullet points into a short paragraph (around 100-150 words), in which you explain your opinion.
[Free text response, minimum of 100 words]

writer_affect_x and _y: **Question not shown**

writer_knowledge: How much do you know about the issue referenced in the statement?
[0-100 scale, where 0 = "No knowledge at all", 50 = "Some understanding", and 100 = "Expert knowledge"]

writer_importance: How important is the issue referenced in the statement to you?
[0-100 scale, where 0 = "Not important at all", 50 = "Moderately important", and 100 = "Extremely important"]

writer_stance_post: Please tell us again: What is your level of agreement with the statement? By default, the slider is set to the answer you gave before. If your opinion has changed, please move the slider accordingly.
[0-100 scale, where 0 = "Strongly disagree", 50 = "Ambivalent", and 100 = "Strongly agree"]

writer_confidence: How confident are you in your opinion on this issue?
[0-100 scale, where 0 = "Not confident at all", 50 = "Moderately confident", and 100 = "Extremely confident"]

After answering the questions above for each proposition, writers were asked to engage with content generated by AI models based on their answers.

model_paragraph_stance: What is this paragraph's level of agreement with the statement?
[0-100 scale, where 0 = "Strongly disagree", 50 = "Ambivalent", and 100 = "Strongly agree"]

edited_paragraph: Please edit the AI-written paragraph until you feel that it reflects your opinion about the statement. If you feel that the paragraph already reflects your opinion, you do not have to edit it. Otherwise please edit it until it does.
[Free text response, starting with the model-generated paragraph]

writer__preference: Imagine you want to explain your opinion on the statement to someone. Which of the two paragraphs below would you rather use for this purpose?
[The paragraph on the left (original) / Equal preference / The paragraph on the right (edited)]

writer__preference__reason: [If answer is not "equal preference"] Why do you prefer this paragraph over the other? *Please write 1-2 sentences or bullet points to explain.*
[Free text response]

writer__stance__final: Please tell us one last time: What is your level of agreement with the statement? By default, the slider is set to the answer you gave before. If your opinion has changed, please move the slider accordingly.
[0-100 scale, where 0 = "Strongly disagree", 50 = "Ambivalent", and 100 = "Strongly agree"]

Writers in Study 3 were not asked about their distortion tolerance..

2.7 Study 3: Readers

Study 3 mirrors the design of Study 1. We denote changes relative to Study 1 in purple.

After onboarding and screening, readers in Study 3 were each assigned ten paragraphs to rate. For each assigned paragraph, readers answered 29 rating questions. They answered all 29 questions in the order listed below for one paragraph, then moved on to the next paragraph. We denote each question by the corresponding variable name in our data release.

writer__stance: What is the author's level of agreement with the statement? Please read the text carefully.
[0-100 scale, where 0 = "Strongly disagree", 50 = "Ambivalent", and 100 = "Strongly agree"]

writer__confidence: How confident is the author of the paragraph in their opinion?
[0-100 scale, where 0 = "Not confident at all", 50 = "Moderately confident", and 100 = "Extremely confident"]

writer__knowledge: How much does the author of the paragraph know about this issue?
[0-100 scale, where 0 = "No knowledge at all", 50 = "Some understanding", and 100 = "Expert knowledge"]

writer__importance: How important is this issue to the author of the paragraph?
[0-100 scale, where 0 = "Not important at all", 50 = "Moderately important", and 100 = "Extremely important"]

paragraph__relevance: How on-topic / relevant is the paragraph to the statement?
[0-100 scale, where 0 = "Not relevant at all", 50 = "Moderately relevant", and 100 = "Extremely relevant"]

paragraph__clarity: How clear is the author's opinion in the paragraph?
[0-100 scale, where 0 = "Not clear at all", 50 = "Moderately clear", and 100 = "Extremely clear"]

paragraph__formality: How formal is the language used in the paragraph?
[0-100 scale, where 0 = "Not formal at all", 50 = "Moderately formal", and 100 = "Extremely formal"]

paragraph__informativeness: How informative / educational did you find the paragraph?
[0-100 scale, where 0 = "Not informative at all", 50 = "Moderately informative", and 100 = "Extremely informative"]

- paragraph_originality:** How original / surprising / novel did you find the arguments in the paragraph?
[0-100 scale, where 0 = "Not original at all", 50 = "Moderately original", and 100 = "Extremely original"]
- writer_friendliness:** How friendly is the author of the paragraph?
[0-100 scale, where 0 = "Extremely unfriendly" (Confrontational, Hostile, Argumentative), 50 = "Neither friendly nor unfriendly", and 100 = "Extremely friendly" (Warm, Welcoming, Supportive)]
- writer_optimism:** How optimistic is the author of the paragraph?
[0-100 scale, where 0 = "Highly pessimistic" (Focuses on problems, Anticipates negative outcomes, Sees limitations), 50 = "Balanced outlook", and 100 = "Highly optimistic" (Emphasises possibilities, Expects positive outcomes, Sees opportunities in challenges)]
- writer_community:** How community-oriented is the author of the paragraph?
[0-100 scale, where 0 = "Not community-oriented" (Prioritises personal autonomy, Emphasises individual responsibility, Values self-reliance), 50 = "Balanced between individual and community concerns", and 100 = "Highly community-oriented" (Prioritises community welfare, Emphasises shared responsibility, Values interdependence)]
- writer_openness:** How open is the author of the paragraph to reconsidering their views?
[0-100 scale, where 0 = "Firmly committed to current views" (Resistant to counterarguments, Confident in existing position, Values consistency), 50 = "Moderately open to reconsideration", and 100 = "Highly open to reconsidering views" (Welcomes different perspectives, Weighs new evidence, Willing to change position)]
- paragraph_hope:** Does the paragraph express hope?
[0-100 scale, where 0 = "No sign of hope", 50 = "Moderate hope", and 100 = "Extreme hope"]
- paragraph_excitement:** Does the paragraph express excitement?
[0-100 scale, where 0 = "No sign of excitement", 50 = "Moderate excitement", and 100 = "Extreme excitement"]
- paragraph_fear:** Does the paragraph express fear?
[0-100 scale, where 0 = "No sign of fear", 50 = "Moderate fear", and 100 = "Extreme fear"]
- paragraph_disgust:** Does the paragraph express disgust?
[0-100 scale, where 0 = "No sign of disgust", 50 = "Moderate disgust", and 100 = "Extreme disgust"]
- paragraph_anger:** Does the paragraph express anger?
[0-100 scale, where 0 = "No sign of anger", 50 = "Moderate anger", and 100 = "Extreme anger"]
- writer_affect_x and _y:** How did the author feel while writing the paragraph? Please tell us by clicking on the grid below.
[Affect grid visual with pleasure-displeasure and arousal-sleepiness dimensions]
- writer_age** What is the author's age?
[Numeric response]

- writer_gender:** What is the author’s gender?
[Male / Female / Other]
- writer_race:** What is the author’s race / ethnicity?
[White / Black / Asian / Mixed / Other]
- writer_english_first:** Is English the author’s first language?
[Yes / No]
- writer_english_skills:** How would you rate the author’s English language skills?
[Basic (can understand simple content with difficulty) / Intermediate (can understand and communicate everyday topics) / Advanced (can communicate complex ideas effectively) / Expert (can use English fluently for all purposes including professional writing)]
- writer_education:** What is the highest level of education that the author has completed?
[GCSEs or equivalent / A-levels or equivalent / Vocational qualification / Undergraduate degree / Postgraduate degree (Master’s) / Doctorate (PhD)]
 We removed the “Other” option to preserve ordinality.
- writer_income:** What is the author’s current yearly income?
[Under £15,000 / £15,000-£24,999 / £25,000-£34,999 / £35,000-£49,999 / £50,000-£74,999 / £75,000-£99,999 / £100,000+]
- writer_politicalParty:** Which political party did the author vote for in the 2024 UK general election?
[Conservative / Labour / Liberal Democrats / Scottish National Party / Greens / Other / Did not vote / Not eligible to vote]
 We added a “Reform UK” option.
- writer_politicalIdeology:** What is the author’s political ideology?
[Very Left-Wing / Moderately Left-Wing / Centrist / Moderately Right-Wing / Very Right-Wing]
 We removed the “Other” option to preserve ordinality.

2.8 Studies 1 & 3: Rationale for Rating Dimensions

Table S24 summarises the rationale and supporting literature for each of the 29 dimensions of perception that we asked readers about in Studies 1 and 3.

Table S24: Rationale for each of the 29 dimensions of reader perception included in Studies 1 and 3. Variable names follow the data release.

Variable	Rationale for Inclusion
writer_stance	(Mis)perception of political stances contributes to perceived political polarisation ^{3,4} .
writer_confidence	Perceived attitude certainty, as a dimension of attitude strength ⁵ , affects persuasion, moderated by perceived expertise ⁶ .
writer_importance	Perceived personal importance of the issue also contributes to attitude strength ⁵ .
writer_knowledge	Perceived issue knowledge contributes to source credibility and thus persuasiveness ⁷ .
writer_openness	Perceived receptiveness to opposing views can lead to more favourable perceptions of the communicator ⁸ and increase persuasiveness ⁹ .
paragraph_relevance	Topical relevance of the argument is a core dimension in established taxonomies of argument quality ¹⁰ .
paragraph_clarity	Clarity of the expressed opinion is a rhetorical dimension of argument quality ¹⁰ and processing fluency affects judgements of the writer ¹¹ .
paragraph_formality	Linguistic formality signals register and social distance ¹² .
paragraph_informativeness	Perceived informational value is a facet of overall argument quality ¹⁰ and contributes to persuasiveness ¹³ .
paragraph_originality	Message novelty affects persuasiveness ¹⁴ .
writer_community	Perceived communal versus individual orientation can shape social interaction ¹⁵ . Communion is a primary dimension on which others are evaluated ¹⁶ .
writer_friendliness	Perceived warmth, alongside competence, is considered one of the universal dimensions of social perception ¹⁷ . Warmth judgements predict social interactions ¹⁸ and can outweigh competence in voter evaluations of political candidates ¹⁹ .
writer_optimism	Perceived optimism, i.e. generalised positive outcome expectancies ²⁰ , can shape social interactions ²¹ .
writer_affect_x	Perceived valence of the writer's affect informs observer behaviour towards them ²² , and the valence of content affects whether it is shared ²³ .
writer_affect_y	Content evoking high-arousal emotions tends to be shared more than content evoking low-arousal emotions ²³ .
paragraph_hope	Perceived hope can shape social appraisals and political choice ²⁴ .
paragraph_excitement	Excitement/enthusiasm is one of the two core affective systems in affective intelligence theory ²⁵ .
paragraph_fear	Fear/anxiety is the second core system in affective intelligence theory ²⁵ .
paragraph_disgust	Disgust is a basic emotion ²⁶ and tends to foster online diffusion ²⁷ .
paragraph_anger	Anger is also a basic emotion ²⁶ with distinctive mobilising effects in politics ²⁸ .
writer_age	Perceived age can affect consequential decisions, e.g. in hiring (<i>ageism</i>) ²⁹ .
writer_gender	Perceived gender can likewise evaluations of otherwise identical credentials (<i>sexism</i>) ³⁰ .
writer_race	Perceived race can affect many consequential decisions, also e.g. in hiring (<i>racism</i>) ³¹ .
writer_english_first	Non-native English writing tends to be penalised in evaluative settings, e.g. in science ³² .
writer_english_skills	Perceived language proficiency has been shown (in speech) to shape perceived credibility ³³ .
writer_education	Perceived education can shape social evaluation (<i>educationism</i>) ³⁴ .
writer_income	Perceived income, as an indicator of social class, can likewise shape social evaluation (<i>classism</i>) ³⁵ .
writer_politicalParty	(Mis)perception of political partisanship drives perceived polarisation ^{3,4} .
writer_politicalIdeology	Perceived ideology signals political identity, and perceived political out-group identity can cause discrimination ³⁶ .

2.9 Studies 1-3: Political Propositions

In Studies 1-3, writers were assigned 3 political propositions, randomly selected from a pool of 100. We constructed these propositions based on the 212 political issues extracted from real user conversations with AI models for IssueBench³⁷, using the following process:

1. We removed out-of-scope issues based on IssueBench metadata:
 - (a) We excluded issues related to criminal activity and hateful ideology.
 - (b) We excluded issues related to non-UK-relevant specific geographies.
 - (c) We excluded issues concerning historical events before 2020.
2. For each remaining issue (e.g. “the COVID-19 vaccine”), we used GPT-4o to generate a UK-relevant proposition that reflects a clear stance (e.g. “The UK should mandate COVID-19 vaccinations for all eligible citizens.”).
3. We removed propositions that are duplicates, not about specific issues, or non-UK-applicable.
4. We edited the remaining propositions to ensure specificity, realism, and UK relevance.

Many of the resulting propositions were clearly aligned with left- or right-leaning political ideology in a UK context. We edited and then selected 100 propositions such that the overall set of propositions is balanced in its political leaning.

46 propositions are **left-leaning**:

1. The UK should impose economic sanctions on Israel in response to the 2023 invasion of Gaza.
2. The UK should increase military support to Ukraine in response to the 2022 Russian invasion.
3. The UK should mandate COVID-19 vaccinations for all eligible citizens.
4. The UK should transition to a socialist economic system, replacing capitalism.
5. The UK should legalise marijuana for recreational use.
6. The UK should abolish private schools and integrate them into the state education system.
7. The UK should transition to 100% renewable energy by 2030.
8. The UK should offer financial incentives to support carbon farming.
9. The UK should implement a universal basic income for all citizens.
10. The UK should legally recognize and protect all forms of transgender identity.
11. The UK should implement policies to ensure the protection and promotion of religious freedoms for Muslim communities.
12. The UK should ban the consumption of meat.
13. The UK should abolish voter ID requirements for parliament elections.
14. The UK should strengthen the rights and powers of labour unions.
15. The UK should implement mandatory gender diversity quotas for all publicly listed companies.
16. The UK should phase out the use of nuclear energy.
17. The UK should legalise euthanasia.
18. The UK should implement mandatory voting in all major elections.
19. The UK should require employers to implement mandatory diversity and inclusion training for all employees.
20. The UK should implement a mandatory carbon emissions cap for all industries.
21. The UK should prioritize adherence to international law over national legislation in its foreign policy decisions.
22. The UK should implement stricter regulations to break up corporate monopolies.
23. The UK should seek to rejoin the European Union.
24. The UK should mandate that all new cars sold be electric vehicles by 2030.
25. The UK should mandate companies to offer paid parental leave for all new parents.

26. The UK should implement a complete ban on single-use plastics.
27. The UK should formally apologize and provide reparations for the impacts of colonisation.
28. The UK should ban the use of military drone technology.
29. The UK should fully decriminalise prostitution.
30. The UK should implement policies to actively discourage nationalism in all forms.
31. The UK should formally condemn China's political system as incompatible with democratic values.
32. The UK should abolish the monarchy.
33. The UK should allow transgender women to compete in women's sports.
34. Scotland should become an independent country, separate from the United Kingdom.
35. The UK should increase taxes on sugary drinks.
36. The UK should teach children about the LGBTQ+ community in primary school.
37. The UK should commit to a policy of never using nuclear bombs under any circumstances.
38. The UK should mandate the implementation of ESG (Environmental, Social, and Governance) policies for all publicly traded companies.
39. The UK should mandate the inclusion of feminist perspectives in the national literature curriculum.
40. The UK should implement a 4-day workweek for all full-time employees.
41. The UK should implement legally binding targets for protecting biodiversity.
42. The UK should implement stricter animal rights laws.
43. The UK should raise taxes for oil companies.
44. The UK should significantly increase its foreign aid budget for combating global hunger.
45. The UK should make social justice a top policy priority.
46. The UK should mandate that all local councils implement comprehensive sustainability initiatives.

46 propositions are **right-leaning**:

1. The UK should withdraw from the Paris Climate Agreement.
2. The UK should make abortion illegal.
3. The UK should defund the NHS.
4. The UK should significantly reduce immigration levels.
5. The UK should significantly increase its military spending.
6. The UK should introduce laws that limit the daily social media use of UK citizens.
7. The UK should increase its arsenal of nuclear weapons.
8. The UK should prioritize strengthening diplomatic ties with Commonwealth nations over forming new alliances with non-Commonwealth countries.
9. The UK should abolish existing gender pay equality legislation.
10. The UK should reintroduce the death penalty.
11. The UK should implement policies to significantly reduce its economic dependence on other countries.
12. The UK should increase funding and resources for the police force.
13. The UK should ban internet use for under-14-year-olds.
14. The UK should implement stricter regulations on the production and distribution of pornography.
15. The UK should defund Ofcom.
16. The UK should restrict the development of large language models.
17. The UK should make same-sex marriage illegal.
18. The UK should decrease funding for its intelligence agencies.

19. The UK should adopt a policy of neutrality and non-intervention in all foreign wars.
20. The UK should privatise its prison system.
21. The UK should eliminate all EU regulations from its legal framework.
22. The UK should remove restrictions on genetic engineering.
23. The UK should loosen regulations on the import and distribution of international pharmaceuticals.
24. The UK should implement policies to limit the influence of woke culture in public institutions.
25. The UK should increase its military spending to strengthen its role within NATO.
26. The UK should ban the teaching of feminism in all schools.
27. The UK should protect employees from workplace dismissal due to social media backlash unrelated to their job performance.
28. The UK should implement policies to incentivise higher birth rates.
29. The UK should implement stricter regulations to limit urbanisation and protect rural areas.
30. The UK government should prioritize awarding contracts to domestic companies over international firms.
31. The UK should decrease funding for education for disabled students.
32. The UK should abolish all policies aimed at increasing diversity in the workplace.
33. The UK should withdraw its commitments to the UN's Sustainable Development Goals.
34. The UK should defund the BBC.
35. The UK should sever all formal ties with the Catholic Church.
36. The UK should remove all regulations on environmental sustainability in supply chains.
37. The UK should always prioritise free speech over political correctness.
38. The UK should implement stronger legal protections for free speech.
39. The UK should increase funding and resources for the Border Force.
40. The UK should ban the use of gender-inclusive language in all public sector communications.
41. The UK should increase financial incentives for couples to get married.
42. The UK should mandate the teaching of creationism alongside evolution in all public schools.
43. The UK should increase the state pension age.
44. The UK should implement strict regulations to prohibit students from using AI tools like ChatGPT for completing homework assignments.
45. The UK should implement stricter regulations on the sale of violent video games to minors.
46. The UK should implement mandatory patriotism education in schools.

The remaining 8 propositions are **neither left- nor right-leaning**:

1. The UK should implement strict regulation on the use of artificial intelligence technologies.
2. The UK should implement strict regulations on cryptocurrency transactions.
3. The UK should implement a comprehensive reform of its legal codes to ensure clarity and accessibility for all citizens.
4. The UK government should prioritize and accelerate digital transformation across all public services.
5. The UK should allow the use of autonomous vehicles on all public roads.
6. The UK should significantly increase funding for space exploration.
7. The UK should legalise internet piracy.
8. The UK should implement stricter regulations on social media platforms to combat the spread of fake news.

2.10 Studies 1-3: AI Models

In Studies 1-3, writers engaged with three AI models:

1. Claude Sonnet 4³⁸ developed by Anthropic
2. DeepSeek V3³⁹ developed by DeepSeek
3. ChatGPT-4o⁴⁰ developed by OpenAI

We selected these models to represent the state-of-the-art in AI chat assistants when we launched Study 1 in June 2025. All inference was done via OpenRouter, with end-point names `anthropic/claude-sonnet-4`, `deepseek/deepseek-chat-v3-0324` and `openai/chatgpt-4o-latest`. Generation parameters were set to default values, including a temperature of 1, to match natural user interactions.

2.11 Studies 1-3: AI Input Conditions

In Studies 1-3, AI-generated paragraphs were produced based on one of four input conditions, which varied in how much of the writer’s original content was provided to the AI model. The first two conditions used indirect representations of the writer’s opinion: either a numeric stance rating (“stance-based”) or the writer’s bullet point reasons (“bullets-based”). The remaining two conditions used the writer’s full opinion paragraph, instructing the AI model either to rewrite it (“rewrite”) or to improve it (“improve”). For each writer in our main and disclaimer studies, the input condition was randomly assigned alongside the AI model, such that each writer experienced three of the four conditions across their three assigned propositions. The “rewrite” and “improve” conditions were treated as variants of a single paragraph-based condition and randomly assigned with equal probability. In Study 3, all AI paragraphs were generated using the “bullets-based” condition only, as this was the condition under which the polarising effect was most pronounced in Study 1 (SI:5.7). The four prompt templates are shown below.

AI generation prompt #1: Write paragraph based on writer stance

Consider the following political proposition: [proposition]

On a scale from 0 to 100, where 0 is "strongly disagree", 50 is "ambivalent", and 100 is "strongly agree", my opinion on this proposition is [writer_stance].

Please write a short paragraph of 100–150 words for me which explains my opinion. Do not mention the numeric rating or include any preamble like "Based on the rating, ...".

Reply only with the paragraph, nothing else.

AI generation prompt #2: Write paragraph based on writer bullet points

Consider the following political proposition: [proposition]

My opinion on this proposition is described by the following bullet points: [writer_bullets]

Please write a short paragraph of 100–150 words for me which explains my opinion. Do not include any preamble, like "Based on the bullet points ...".

Reply only with the paragraph, nothing else.

AI generation prompt #3: Rewrite writer paragraph

Consider the following political proposition: [proposition]

I wrote the following paragraph to explain my opinion on this proposition: "[writer_paragraph]"

Please rewrite this paragraph without changing its length. Do not include any preamble, like "Based on the original paragraph ...".

Reply only with the rewritten paragraph, nothing else.

AI generation prompt #4: Improve writer paragraph

Consider the following political proposition: [proposition]

I wrote the following paragraph to explain my opinion on this proposition: "[writer_paragraph]"

Please improve this paragraph without changing its length. Do not include any preamble, like "Based on the original paragraph ...".

Reply only with the improved paragraph, nothing else.

2.12 Study 4: Readers

Study 4 was run for readers only and used a different design to Studies 1-3. After onboarding, readers in Study 4 were explained the rules of the Trust Game on an instruction screen (Figure S9). They could re-access these instructions at any point during the game via a pop-up link.

Study Instructions

Overview

In this study, you will read 4 political opinion paragraphs written by 4 different authors. For each paragraph, you will play a game, in which you decide how to split a sum of money between yourself and the paragraph's author. **Your decisions will determine a real bonus payment that you will receive after completing the full study.**

☒ I understand the Overview.

Rules of the Game

For each paragraph, you start with 20p. **You choose how much of this 20p to send to the paragraph's author.** You keep whatever money you do not send.

The money you send is multiplied by 3× before the author receives it. **The author then decides how much to send back to you.** They can return any amount between 0p and the total they received.

Your earnings for each paragraph will be the amount you kept plus whatever the author decides to send back to you. Each game is played independently of the others, with a different author, and your decisions for one paragraph will not affect the outcomes for any other paragraphs. **All earnings will be paid out after the end of the study.**

☒ I understand the Rules of the Game.

Example

You start with 20p. If you send 4p, the author receives 12p. Suppose the author decides to return half, so they send 6p back to you. For this paragraph, you earn the 16p you kept plus the 6p returned = 22p. The author earns 6p. Alternatively, suppose the author decides to return nothing. In this case, you earn the 16p you kept plus 0p returned = 16p. The author earns 12p.

☒ I understand the Example.

Continue

Figure S9: Screenshot of the Trust Game instruction screen shown to readers in Study 4 after onboarding.

Next, readers completed a three-item comprehension check on the game mechanics (Figure S10). Items were answered sequentially. After each response, the reader was shown whether their answer was correct, the correct answer, and a brief explanation. Readers with more than one incorrect response were excluded from the study.

After passing the comprehension check, readers were each assigned four paragraphs as described in §9.1. For each assigned paragraph, they were first shown the corresponding proposition (“Consider this policy proposal: ...”) and then the paragraph (“Here is a paragraph responding to the proposal: ...”). Finally, they were asked the Trust Game allocation question (Figure S11):

Comprehension Check

Please answer the three questions below to confirm your understanding of the instructions.
If more than one answer is incorrect, you will not be able to continue with the study.
This is to ensure that all participants understand the instructions before proceeding.

If you want to see the instructions again, please [click here](#).

You start the game with 20p.

Q1: Suppose you decide to send 6p. How much does the author get? (Hint: multiplier)

Submitted

Correct.

The correct answer is **18p**. You entered **18p**.
Explanation: The author gets the sent amount of 6p multiplied by 3 = 18p.

Q2: Suppose the author decides to return half of the 18p they got. How much do they send back?

Submitted

Correct.

The correct answer is **9p**. You entered **9p**.
Explanation: The author got 18p. Half of 18p = 9p.

Q3: You started with 20p, sent 6p, and got back 9p. What is your total payout?

Submitted

Correct.

The correct answer is **23p**. You entered **23p**.
Explanation: You kept 14p and got back 9p from the author. The sum of these = 23p.

Continue

Figure S10: Screenshot of the Trust Game instruction screen shown to readers in Study 4 after onboarding.

trust_allocation_pence: You start with 20p. How much would you like to send to this paragraph's author?
[0–20p slider, in 1p increments]

Consider this policy proposal:

Round: 2 of 4 Screen: 3 of 3

The UK should sever all formal ties with the Catholic Church.

Here is a paragraph responding to the proposal:

The UK, in its previous forms and kingdoms, has been a Christian country for over 1400 years since the days of the occupation of the Roman Empire. This is an important part of our history and culture and it is of the utmost importance for this to remain as such. It is also imperative that the UK remains a Christian country as if ties are severed with the Catholic Church, it will further lead to the rise of Islam into the country which needs to be fought against totally and thoroughly as it will be entirely detrimental to the future of the British people and our identity.

Now it is time to play the game. If you want to see the instructions again, please [click here](#).

You start with 20p . How much would you like to send to this paragraph's author?

0p10p20p

Current amount sent: 7p

Whatever you send is multiplied by 3x. If you send 7p, the author receives 21p. The author then decides how much to send back to you. They can return any amount between nothing and the total they received.

Your earnings for this paragraph will be 13p (the amount you kept) plus whatever the author decides to send back to you. Each game is played independently of the others, with a different author, and your decisions for this paragraph will not affect the outcomes for any other paragraphs. **All earnings will be paid out after the end of the study.**

Continue

Figure S11: Screenshot of the Trust Game allocation screen, shown to readers in Study 4 for each assigned paragraph.

After playing the Trust Game for all four paragraphs, readers rated their own agreement with each corresponding proposition, one at a time (Figure S12):

rater_stance_post: What is your level of agreement with the policy proposal?
[0–100 scale, where 0 = "Strongly disagree", 50 = "Ambivalent", and 100 = "Strongly agree"]

In a standard Trust Game, the amount returned to the reader is determined by a live counterpart. In our implementation, the paragraph authors did not make live return decisions. Returns were thus fixed, and each reader was paid the portion of the endowment they kept plus $1.5\times$ their total allocation across all paragraphs as a bonus. To address this divergence, all readers were debriefed on the study completion screen after finishing all study tasks. The debrief disclosed that the paragraphs readers had seen were written by humans either with or without AI writing assistance, and stated the core research aim of the study, i.e. to test whether AI writing assistance changes trust in paragraph authors. The completion screen additionally provided individualised Trust Game bonus information, showing each reader the total amount they kept, the total amount they sent across rounds, the fixed return rule applied in this non-interactive design, and their resulting total bonus.

44

Here is the policy proposal you saw earlier:

Proposal: 1 of 4 Screen: Agreement Rating

The UK should mandate the implementation of ESG (Environmental, Social, and Governance) policies for all publicly traded companies.

What is **your level of agreement** with the **policy proposal**?

0 = Strongly disagree 50 = Ambivalent 100 = Strongly agree

Please click on the scale to set your rating

Continue

Figure S12: Screenshot of the proposition agreement screen, shown to readers in Study 4 for each assigned paragraph, after playing all rounds of the Trust Game.

2.13 Study 5: Readers

After onboarding and passing an attention check (same as in Studies 1 and 3, see §2.2), readers were each assigned five paragraphs as described in §10.1. For each assigned paragraph, readers were first shown the corresponding proposition (“Consider this policy proposal: ...”), then asked to rate their attitude towards it on a three-item attitude battery¹³, with each item answered on a 0–100 scale.

support: Do you oppose or support this policy?
[0–100 scale, where 0 = "Strongly oppose" and 100 = "Strongly support"]

bad_idea: This policy would be a bad idea.
[0–100 scale, where 0 = "Strongly disagree" and 100 = "Strongly agree"]

good_consequences: This policy would have good consequences.
[0–100 scale, where 0 = "Strongly disagree" and 100 = "Strongly agree"]

They were then shown the paragraph (“Here is a paragraph responding to the proposal: ...”) and asked to read it. They could only proceed to the next screen after a minimum of 20 seconds. Finally, they completed the same three-item attitude battery again.

3 Study 1 (Main Study): Writer Engagement Analysis

Writers reported moderate-to-high engagement with the political propositions randomly assigned to them (Figure S13). Median self-reported issue knowledge was 56.0 (mean = 54.5), median issue importance was 65.0 (mean = 62.0), and median confidence in opinion was 74.0 (mean = 70.2), all on 0-100 scales. All three distributions were left-skewed, with relatively few writers reporting very low knowledge, importance, or confidence, suggesting that random assignment to propositions did not systematically force writers to opine on issues they felt uninformed about or indifferent toward.

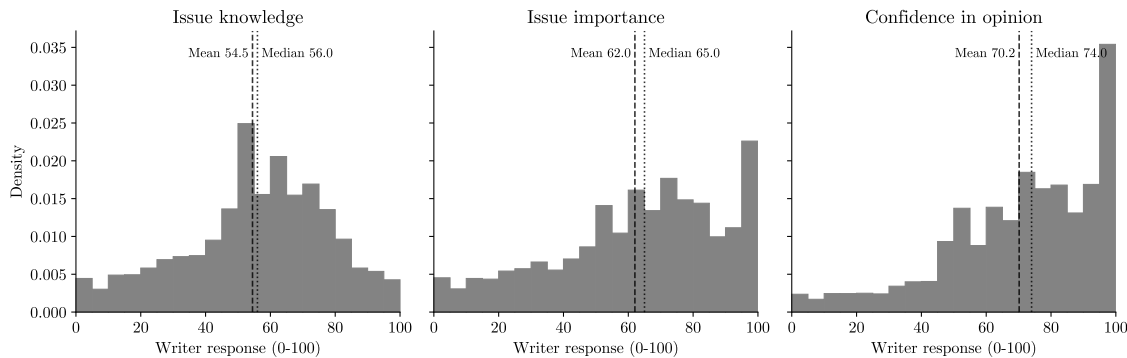


Figure S13: Writer engagement with assigned propositions in Study 1 (N=1,501 writers, 4,503 writer-proposition pairs). Histograms show distributions of writer self-reported issue knowledge, issue importance, and confidence in opinion, all measured on 0-100 scales. Dashed and dotted lines indicate mean and median values, respectively.

Writers’ stated issue stances spanned the full 0-100 scale and varied by proposition leaning (Figure S14). For left-leaning propositions, the mean stance was close to the midpoint (mean = 52.4, median = 52.0), indicating a roughly symmetric spread of agreement and disagreement. For right-leaning propositions, writers leaned toward disagreement on average (mean = 41.6, median = 45.0), consistent with the slight left-of-centre skew of the UK-representative sample (SI:1.2). For the small set of neither-leaning propositions, stances were slightly above the midpoint (mean = 55.8, median = 60.0). All three distributions exhibited pronounced peaks at the extremes of the scale, reflecting the high proportion of writers who held strong opinions on the issues assigned to them.

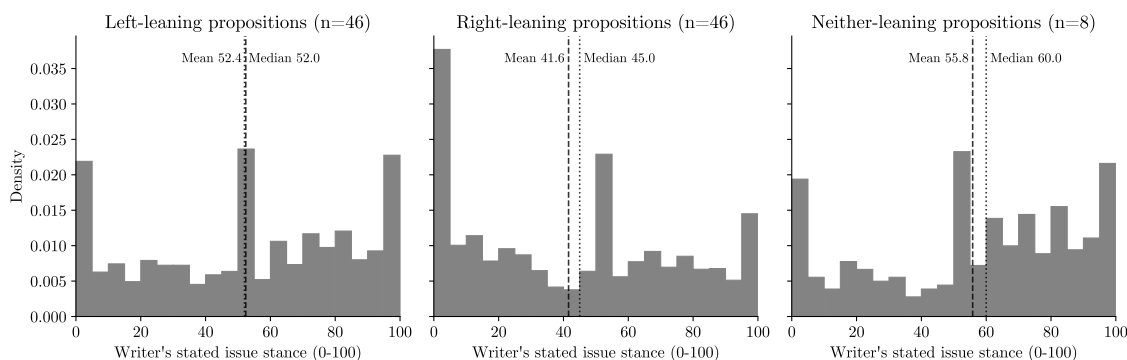


Figure S14: Distribution of writer stances by proposition leaning in Study 1 (N=1,501 writers, 4,503 writer-proposition pairs). Propositions were manually classified as left-leaning (n=46), right-leaning (n=46), or neither-leaning (n=8) as described in SI:2.9. Writers’ stated issue stance is measured on a 0-100 scale, where 0 = “Strongly disagree”, 50 = “Ambivalent”, and 100 = “Strongly agree” with the proposition. Dashed and dotted lines indicate mean and median values, respectively.

Excluding writer-proposition pairs with low (<50) self-reported knowledge, importance, or confidence does not change the direction or significance of any of our main results (SI:5.14).

4 Study 1 (Main Study): Writer Preference Analysis

4.1 Descriptive Results

Overall, writers made edits to the AI-generated paragraphs in 77.7% of cases (Table S25. Edit rates did not exceed 30% for any AI model or input condition ($p < 0.01$, Figure S15). The majority of writers did not edit any of the AI-generated paragraphs (Table S26). When edits were made, these were generally small, as measured by Levenshtein ration and Levenshtein distance (Figure S16).

Table S25: AI paragraph edit rates in Study 1 (N=1,501 writers, each writing about 3 propositions) across AI models and input conditions.

Group	n	Did not edit	Made edits
Overall	4,503	77.7%	22.3%
ChatGPT-4o	1,501	80.1%	19.9%
Deepseek V3	1,501	78.7%	21.3%
Claude Sonnet 4	1,501	74.4%	25.6%
stance-based	1,501	72.0%	28.0%
bullets-based	1,501	82.3%	17.7%
rewrite	740	78.2%	21.8%
improve	761	79.5%	20.5%

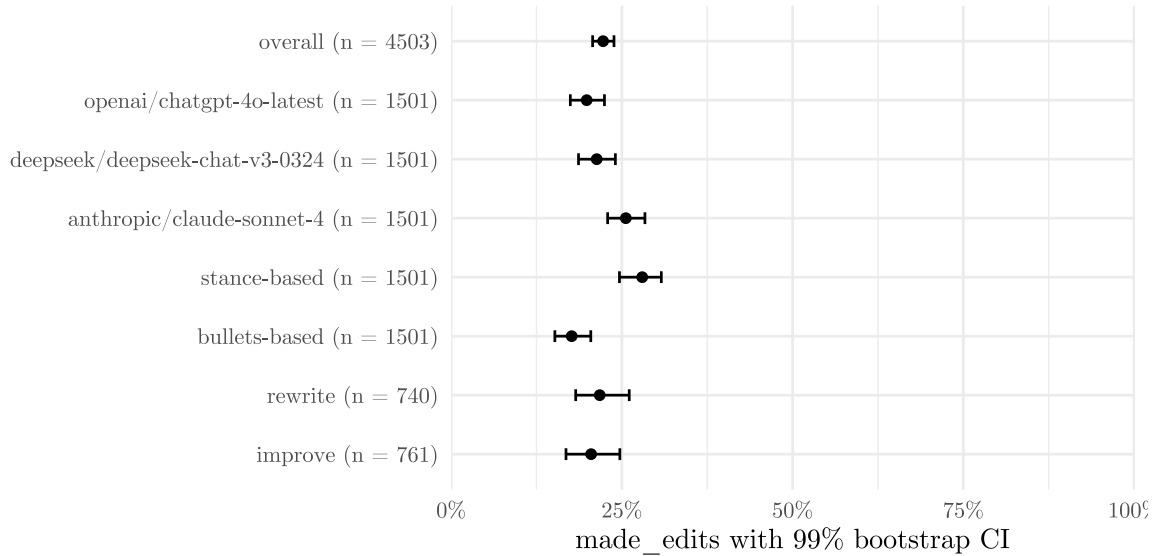


Figure S15: AI paragraph edit rates in Study 1 (N=1,501 writers, each writing about 3 propositions) across AI models and input conditions.

Table S26: Number of paragraphs edited in Study 1 (N=1,501 writers, each writing about 3 propositions).

Paragraphs Edited	Writers	Proportion	
0 out of 3	839	55.90%	████████
1 out of 3	394	26.25%	███
2 out of 3	196	13.06%	██
3 out of 3	72	4.80%	█

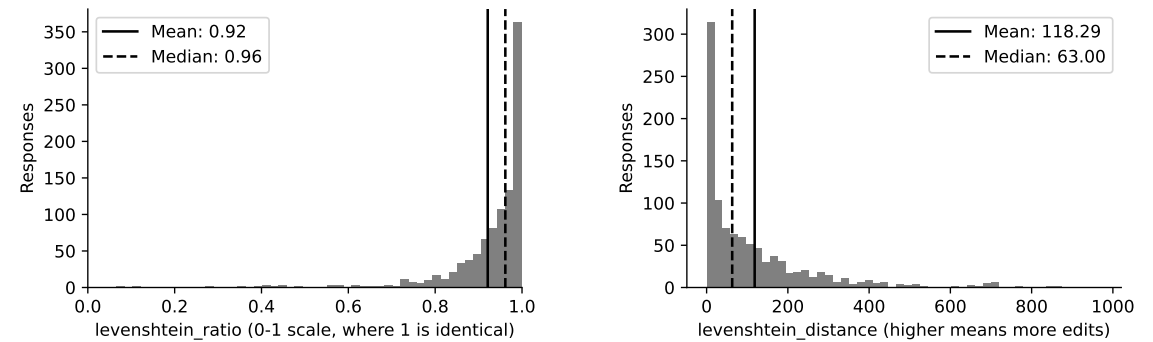


Figure S16: Distribution of Levenshtein ratio (left) and Levenshtein distance (right) for edited vs unedited AI-written paragraphs for the set of n=1,002 AI-written paragraphs that writers made edits to in Study 1.

Overall, writers strictly preferred the AI-generated paragraphs (including any edits made) to their own writing in 63.0% of cases (Table S27). Strict preference rates did not fall below 50% for any AI model or input condition ($p < 0.01$, Figure S17). Weak preference rates did not fall below 75% for any AI model or input condition ($p < 0.01$, Figure S18). In a majority of cases (1,477 of 2,835, 52.1%) writers said they strictly preferred AI writing because it better reflected their opinion than what they had written themselves (Table S33).

Table S27: Paragraph preference rates in Study 1 (N=1,501 writers, each writing about 3 propositions) across AI models and input conditions. AI-written paragraphs include any edits made.

Group	n	Human-written	Equal	AI-written
Overall	4,503	■ 15.7%	■ 21.3%	■ 63.0%
ChatGPT-4o	1,501	■ 14.3%	■ 23.3%	■ 62.5%
Deepseek V3	1,501	■ 18.1%	■ 21.1%	■ 60.8%
Claude Sonnet 4	1,501	■ 14.9%	■ 19.6%	■ 65.6%
stance-based	1,501	■ 18.1%	■ 18.1%	■ 63.9%
bullets-based	1,501	■ 12.9%	■ 18.3%	■ 68.8%
rewrite	740	■ 17.7%	■ 26.9%	■ 55.4%
improve	761	■ 14.7%	■ 28.4%	■ 56.9%

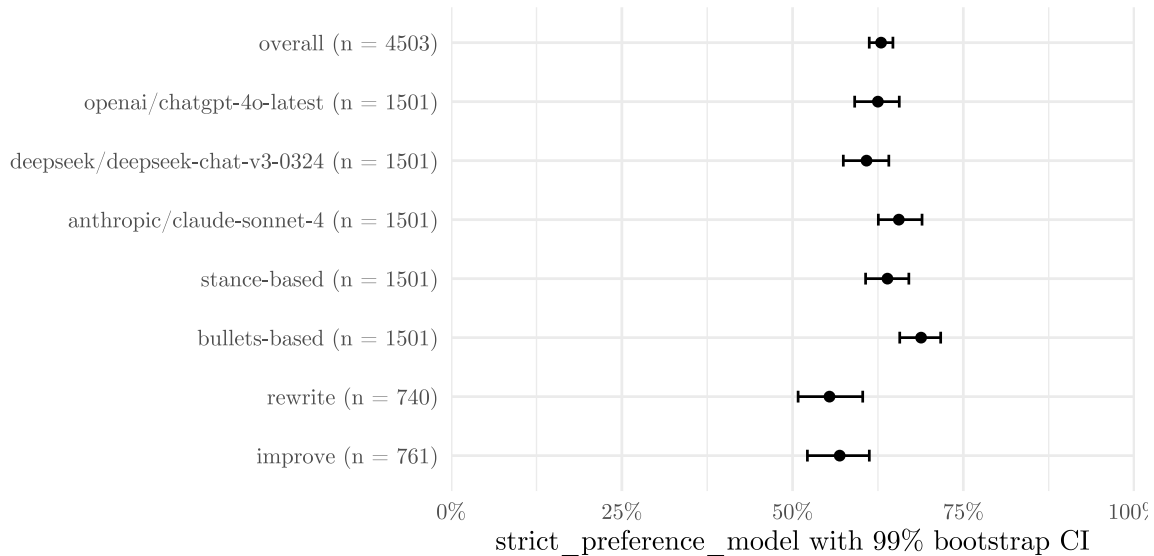


Figure S17: Strict preference for AI-assisted paragraphs over own writing in Study 1 (N=1,501 writers, each writing about 3 propositions) across AI models and input conditions. AI-written paragraphs include any edits made.

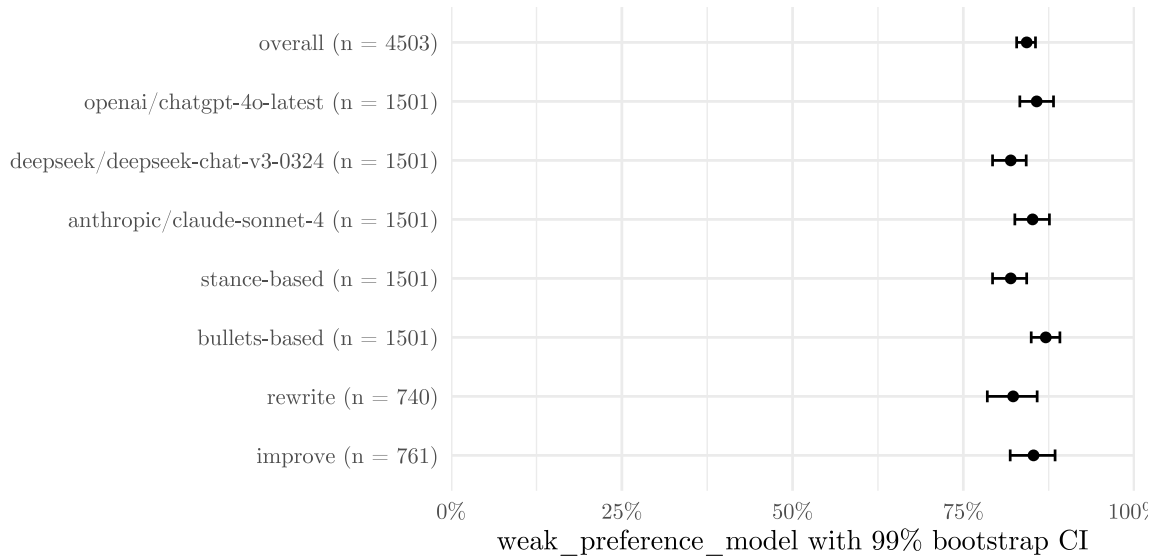


Figure S18: Weak preference for AI-assisted paragraphs over own writing in Study 1 (N=1,501 writers, each writing about 3 propositions) across AI models and input conditions. AI-written paragraphs include any edits made.

4.2 Statistical Analysis

From the edit and preference questions, we derived three binary variables observed for each writer-proposition pair: whether the writer made any edits to the AI-written paragraph (“made_edits”), whether the writer weakly preferred the AI-written paragraph (“weak_preference_model”), and whether the writer strictly preferred the AI-written paragraph (“strict_preference_model”).

For each binary outcome, we fit **logistic regression models** with writer random effects to measure how writer preference for AI writing varied across i) which AI model was used to generate the AI paragraph and ii) which type of writer input was passed to the AI model. For the preference outcomes, we included “made_edits” as an additional regressor:

$$\text{made_edits} \sim \text{model} + \text{input_condition} + (1 \mid \text{writer_id})$$

$$\begin{aligned} \text{weak_preference_model} &\sim \text{model} + \\ \text{input_condition} + \text{made_edits} &+ (1 \mid \text{writer_id}) \end{aligned}$$

$$\begin{aligned} \text{strict_preference_model} &\sim \text{model} + \\ \text{input_condition} + \text{made_edits} &+ (1 \mid \text{writer_id}) \end{aligned}$$

Table S28 reports results for the made_edits outcome. Writers were less likely to edit paragraphs generated by ChatGPT-4o (OR=0.661, $p < .001$) and DeepSeek (OR=0.743, $p = .003$) compared to Claude Sonnet 4. Paragraphs generated from stance-based inputs were much more likely to be edited (OR=2.162, $p < .001$), and rewrite-based inputs also increased the likelihood of edits (OR=1.385, $p = .014$), relative to bullet-point inputs. The improve condition showed a marginal, non-significant increase in editing (OR=1.284, $p = .058$).

Table S28: Logistic regression predicting writer editing of AI writing in Study 1 (N=1,501 writers, each writing about 3 propositions). Binary outcome: “made_edits”. Reference categories: “model_anthropic/claude-sonnet-4” and “input_condition_bullets-based”

Term	Estimate	SE	OR [95% CI]	z	p
model_deepseek/deepseek-chat-v3-0324	−0.297	0.100	0.743 [0.610, 0.905]	−2.954	.003
model_openai/chatgpt-4o-latest	−0.414	0.102	0.661 [0.541, 0.807]	−4.064	< .001
input_condition_improve	0.250	0.132	1.284 [0.991, 1.663]	1.895	.058
input_condition_rewrite	0.326	0.132	1.385 [1.069, 1.793]	2.468	.014
input_condition_stance-based	0.771	0.103	2.162 [1.765, 2.647]	7.452	< .001

Table S29 reports results for the weak_preference_model outcome. Writers were less likely to weakly prefer outputs generated by DeepSeek (OR=0.646, $p<.001$) compared to Claude Sonnet 4, while ChatGPT-4o did not differ significantly (OR=0.965, $p=.779$). Input conditions that incorporated more extensive writer input reduced weak preference: rewrite-based (OR=0.610, $p=.002$) and stance-based inputs (OR=0.663, $p=.001$) both decreased the likelihood of preferring the AI output relative to bullet-point inputs, while the improve condition was not significant (OR=0.833, $p=.258$). When writers made edits, they were substantially less likely to weakly prefer the AI-generated paragraph (OR=0.216, $p<.001$).

Table S29: Logistic regression predicting weak writer preference for AI writing in Study 1 (N=1,501 writers, each writing about 3 propositions). Binary outcome: “weak_preference_model”. Reference categories: “model_anthropic/claude-sonnet-4” and “input_condition_bullets-based”

Term	Estimate	SE	OR [95% CI]	z	p
model_deepseek/deepseek-chat-v3-0324	−0.437	0.123	0.646 [0.507, 0.823]	−3.540	< .001
model_openai/chatgpt-4o-latest	−0.036	0.128	0.965 [0.752, 1.239]	−0.280	.779
input_condition_improve	−0.183	0.162	0.833 [0.607, 1.143]	−1.131	.258
input_condition_rewrite	−0.495	0.157	0.610 [0.448, 0.829]	−3.150	.002
input_condition_stance-based	−0.412	0.127	0.663 [0.517, 0.850]	−3.242	.001
made_edits	−1.534	0.140	0.216 [0.164, 0.284]	−10.917	< .001

Table S30 reports results for the strict_preference_model outcome. Writers were less likely to strictly prefer outputs from DeepSeek (OR=0.722, $p<.001$) and ChatGPT-4o (OR=0.783, $p=.009$) compared to Claude Sonnet 4. All non-bullet input conditions reduced strict preference: improve (OR=0.497, $p<.001$), rewrite (OR=0.442, $p<.001$), and stance-based inputs (OR=0.808, $p=.024$). When writers made edits, they were also significantly less likely to strictly prefer the AI-generated paragraph (OR=0.348, $p<.001$).

Table S30: Logistic regression predicting strict writer preference for AI writing in Study 1 (N=1,501 writers, each writing about 3 propositions). Binary outcome: “strict_preference_model”. Reference categories: “model_anthropic/claude-sonnet-4” and “input_condition_bullets-based”

Term	Estimate	SE	OR [95% CI]	z	p
model_deepseek/deepseek-chat-v3-0324	−0.326	0.093	0.722 [0.602, 0.867]	−3.499	< .001
model_openai/chatgpt-4o-latest	−0.245	0.093	0.783 [0.652, 0.940]	−2.629	.009
input_condition_improve	−0.699	0.116	0.497 [0.396, 0.625]	−6.004	< .001
input_condition_rewrite	−0.817	0.118	0.442 [0.350, 0.557]	−6.906	< .001
input_condition_stance-based	−0.213	0.094	0.808 [0.672, 0.972]	−2.259	.024
made_edits	−1.056	0.108	0.348 [0.282, 0.430]	−9.800	< .001

Note that we deviated from our pre-registration in our main analysis in using a binary edit indicator instead of continuous edit distance because edits were much rarer and smaller than we had anticipated, as shown in 4.1. Following our pre-registration, we report results for **linear regression models** with writer random effects for two continuous measures of edit distance in Tables S31 and S32.

$$\text{edit_distance} \sim \text{model} + \text{input_condition} + (1 \mid \text{writer_id})$$

Table S31 reports results for the Levenshtein edit distance outcome. Writers made fewer paragraphs generated by ChatGPT-4o (B=-6.691, p=.037) and DeepSeek (B=-6.576, p=.040) compared to Claude Sonnet 4. Paragraphs generated from stance-based inputs received substantially more editing (B=28.691, p<.001), while improve (B=-10.567, p=.007) and rewrite (B=-9.451, p=.016) conditions both reduced edit distance relative to bullet-point inputs.

Table S31: Linear regression predicting edit distance in Study 1 (N=1,501 writers, each writing about 3 propositions). Continuous outcome: “levenshtein_distance”. Reference categories: “model_anthropic/claude-sonnet-4” and “input_condition_bullets-based”

Term	Estimate	SE	95% CI	z	p
model_deepseek/deepseek-chat-v3-0324	-6.576	3.202	[-12.852, -0.301]	-2.054	.040
model_openai/chatgpt-4o-latest	-6.691	3.201	[-12.965, -0.416]	-2.090	.037
input_condition_improve	-10.567	3.899	[-18.209, -2.925]	-2.710	.007
input_condition_rewrite	-9.451	3.937	[-17.168, -1.734]	-2.400	.016
input_condition_stance-based	28.691	3.200	[22.420, 34.962]	8.967	< .001

Table S32 reports results for the Levenshtein edit ratio outcome. Writers edited a smaller proportion of AI-generated paragraphs from ChatGPT-4o (B=0.051, p<.001) and DeepSeek (B=-0.037, p=.003) compared to Claude Sonnet 4. All non-bullet input conditions increased the edit ratio: stance-based (B=0.082, p<.001), rewrite (B=0.045, p=.004), and improve inputs (B=0.036, p=.021), relative to bullet-point inputs.

Table S32: Linear regression predicting edit ratio in Study 1 (N=1,501 writers, each writing about 3 propositions). Continuous outcome: “edit_ratio”. Reference categories: “model_anthropic/claude-sonnet-4” and “input_condition_bullets-based”

Term	Estimate	SE	95% CI	z	p
model_deepseek/deepseek-chat-v3-0324	-0.037	0.012	[-0.061, -0.013]	-2.994	.003
model_openai/chatgpt-4o-latest	-0.051	0.012	[-0.075, -0.027]	-4.131	< .001
input_condition_improve	0.036	0.015	[0.005, 0.066]	2.313	.021
input_condition_rewrite	0.045	0.016	[0.015, 0.076]	2.905	.004
input_condition_stance-based	0.082	0.012	[0.058, 0.107]	6.710	< .001

4.3 Preference Reasons

In Study 1, writers who strictly preferred one paragraph over the other were asked why, and could tick “It better reflects my opinion.”, “I prefer the writing style.”, or “Other (please specify)” (see SI:2.3). Writers who strictly preferred the AI-written paragraph most often cited writing style (74.8%), followed by opinion (52.1%). Writers who strictly preferred their own paragraph most often cited opinion (77.0%), followed by writing style (49.3%) (Table S33).

Table S33: Reasons for strict paragraph preference in Study 1 (N=1,501 writers, each writing about 3 propositions).

Strict preference	Reason	n (reason)	n (strict pref.)	% citing reason
AI-written	style	2,120	2,835	 74.8%
AI-written	opinion	1,477	2,835	 52.1%
AI-written	other	388	2,835	 13.7%
Human-written	opinion	545	708	 77.0%
Human-written	style	349	708	 49.3%
Human-written	other	146	708	 20.6%

Writers who ticked “Other” had to explain their reason in free text. This was the case for 388 of 2,835 strict preferences for the AI-assisted paragraph (13.7%) and 146 of 708 strict preferences for the human-written paragraph (20.6%). Most of these writers also ticked at least one of the fixed-choice reasons: only 102 of 388 (AI-written) and 40 of 146 (human-written) gave “Other” as their only reason. Since, in Study 1, writers were asked to explain *additional* reasons under “Other”, the free-text reasons complement rather than replace the fixed-choice reasons, and reasons covered by the fixed choices (opinion, writing style) are likely under-represented in the free text.

To analyse the free-text reasons, we coded each response with a multi-label codebook, shown in Table S34. Each code is a directional claim about the preferred paragraph, from the writer’s perspective (e.g. “The preferred paragraph is less formal.”). Where possible, codes correspond to the rating attributes on which we measured distortion (SI:5). Additional codes cover reasons not captured by the rating attributes, such as general language quality, concision or a general dislike of AI. A response can receive any number of codes, and a residual `other` code is assigned when a response contains a reason not captured by any other code.

We developed the codebook iteratively on the free-text reasons from Studies 1 to 3. Starting from an initial set of codes, we repeatedly coded random samples of responses, inspected the reasons that were not covered by any code, and revised the codebook, until uncovered reasons were rare and no longer formed recurring themes. We then applied the final codebook unchanged to the free-text reasons in all three studies.

For coding the full set of responses, we used an LLM, `deepseek/deepseek-v4.1-flash`, called via OpenRouter at temperature 0. For each response, the model was shown the codebook, which paragraph the writer preferred, the question wording including the fixed-choice options, and the writer’s free-text reason. The model was instructed to assign every code expressed in the response, not to infer unstated reasons, to treat criticism of the non-preferred paragraph as the opposite claim about the preferred paragraph (e.g. “the AI version is too formal” from a writer who preferred their own paragraph is coded as `formality_less`), and to ignore concessions, i.e. praise for the non-preferred paragraph. Reasons not captured by any code were listed separately and assigned the `other` code. For validation, we inspected 50 instances at random, and found no misclassifications.

Table S35 and Figure S19 show the share of free-text responses mentioning each reason. Responses received 1.7 codes on average, and about half of responses (48.6%) received more than one code.

Writers who strictly preferred the AI-written paragraph most often said that this was because it was more informative (30.4%), better written (25.8%) or better reflected their opinion (25.5%), followed by concision (13.7%) and clarity (11.9%). They also said that they preferred the AI-written paragraph because it was more formal (8.5%) and made them seem more knowledgeable (7.5%) or more open to other views (6.4%).

Table S34: Codebook for free-text preference reasons, used for Studies 1–3. Each code is a directional claim about the preferred paragraph, from the writer’s perspective. Where applicable, the right column lists the corresponding rating attributes on which we measured distortion.

Code	Definition	Rating attribute
reflects_opinion	The preferred paragraph better reflects my opinion.	writer_stance
extremity_more	The preferred paragraph makes my opinion seem more extreme.	writer_stance_polarity
extremity_less	The preferred paragraph makes my opinion seem less extreme.	writer_stance_polarity
confidence_more	The preferred paragraph makes me seem more confident in my opinion.	writer_confidence
confidence_less	The preferred paragraph makes me seem less confident in my opinion.	writer_confidence
knowledge_more	The preferred paragraph makes me seem more knowledgeable about the issue.	writer_knowledge
knowledge_less	The preferred paragraph makes me seem less knowledgeable about the issue.	writer_knowledge
importance_more	The preferred paragraph makes the issue seem more important to me.	writer_importance
relevance_more	The preferred paragraph is more relevant or on-topic.	paragraph_relevance
clarity_more	The preferred paragraph is clearer or easier to understand.	paragraph_clarity
formality_more	The preferred paragraph is more formal.	paragraph_formality
formality_less	The preferred paragraph is less formal.	paragraph_formality
informativeness_more	The preferred paragraph is more informative.	paragraph_informativeness
originality_more	The preferred paragraph contains more original or novel arguments.	paragraph_originality
friendliness_more	The preferred paragraph makes me seem friendlier.	writer_friendliness
optimism_more	The preferred paragraph makes me seem more optimistic.	writer_optimism
community_more	The preferred paragraph makes me seem more community-oriented.	writer_community
openness_more	The preferred paragraph makes me seem more open to other views.	writer_openness
emotionality_more	The preferred paragraph is more emotional.	emotions, affect
emotionality_less	The preferred paragraph is less emotional.	emotions, affect
language_quality	The preferred paragraph is better written in terms of grammar, word choice, flow or polish.	–
concision	The preferred paragraph is more concise.	–
structure	The preferred paragraph is better structured or more coherent.	–
persuasiveness	The preferred paragraph is more persuasive.	–
authenticity	The preferred paragraph sounds more human or more like my own voice.	–
ai_aversion	The preferred paragraph is not AI-written, and I dislike or distrust AI in general.	–
other	The preferred paragraph has an advantage not captured by any other code.	–

Writers who strictly preferred their own paragraph most often said that this was because it sounded more human or more like their own voice (35.6%), better reflected their opinion (26.0%), or that they disliked or distrusted AI in general (25.3%). They also said that they preferred their own paragraph because it was more concise (10.3%), less formal (8.9%), more emotional (8.9%) or clearer (8.9%).

Few writers in either group referred to the remaining rating attributes, such as extremity, confidence, importance, friendliness, optimism or community orientation (each at most 2.7%). For 6.7% (AI-written) and 8.2% (human-written) of responses, the LLM identified at least one reason not captured by the codebook (*other*).

These results describe a selected subset of writers, who chose to give an additional reason under “Other”, and should not be read as the distribution of reasons across all writers. In particular, AI aversion is likely over-represented, since it is not covered by the fixed-choice reasons, whereas opinion and writing style are likely under-represented. Studies 2 and 3, where all writers explained their preference in free text, provide a more complete picture (SI:7, SI:8).

Table S35: Free-text reasons for strict paragraph preference in Study 1, among writers who ticked “Other” (n=388 strict preferences for the AI-assisted paragraph, n=146 for the human-written paragraph). Responses can receive multiple codes, so percentages do not sum to 100%. See Table S34 for code definitions.

Code	AI-written preferred		Human-written preferred	
	n	%	n	%
reflects_opinion	99	■ 25.5%	38	■ 26.0%
extremity_more	2	0.5%	2	1.4%
extremity_less	4	1.0%	2	1.4%
confidence_more	6	1.5%	0	0.0%
confidence_less	0	0.0%	2	1.4%
knowledge_more	29	■ 7.5%	2	1.4%
knowledge_less	0	0.0%	4	2.7%
importance_more	1	0.3%	2	1.4%
relevance_more	8	2.1%	1	0.7%
clarity_more	46	■ 11.9%	13	■ 8.9%
formality_more	33	■ 8.5%	0	0.0%
formality_less	1	0.3%	13	■ 8.9%
informativeness_more	118	■ 30.4%	11	■ 7.5%
originality_more	13	3.4%	1	0.7%
friendliness_more	4	1.0%	2	1.4%
optimism_more	0	0.0%	0	0.0%
community_more	1	0.3%	0	0.0%
openness_more	25	■ 6.4%	1	0.7%
emotionality_more	3	0.8%	13	■ 8.9%
emotionality_less	11	2.8%	0	0.0%
language_quality	100	■ 25.8%	7	4.8%
concision	53	■ 13.7%	15	■ 10.3%
structure	26	■ 6.7%	1	0.7%
persuasiveness	34	■ 8.8%	4	2.7%
authenticity	2	0.5%	52	■ 35.6%
ai_aversion	2	0.5%	37	■ 25.3%
other	26	■ 6.7%	12	■ 8.2%

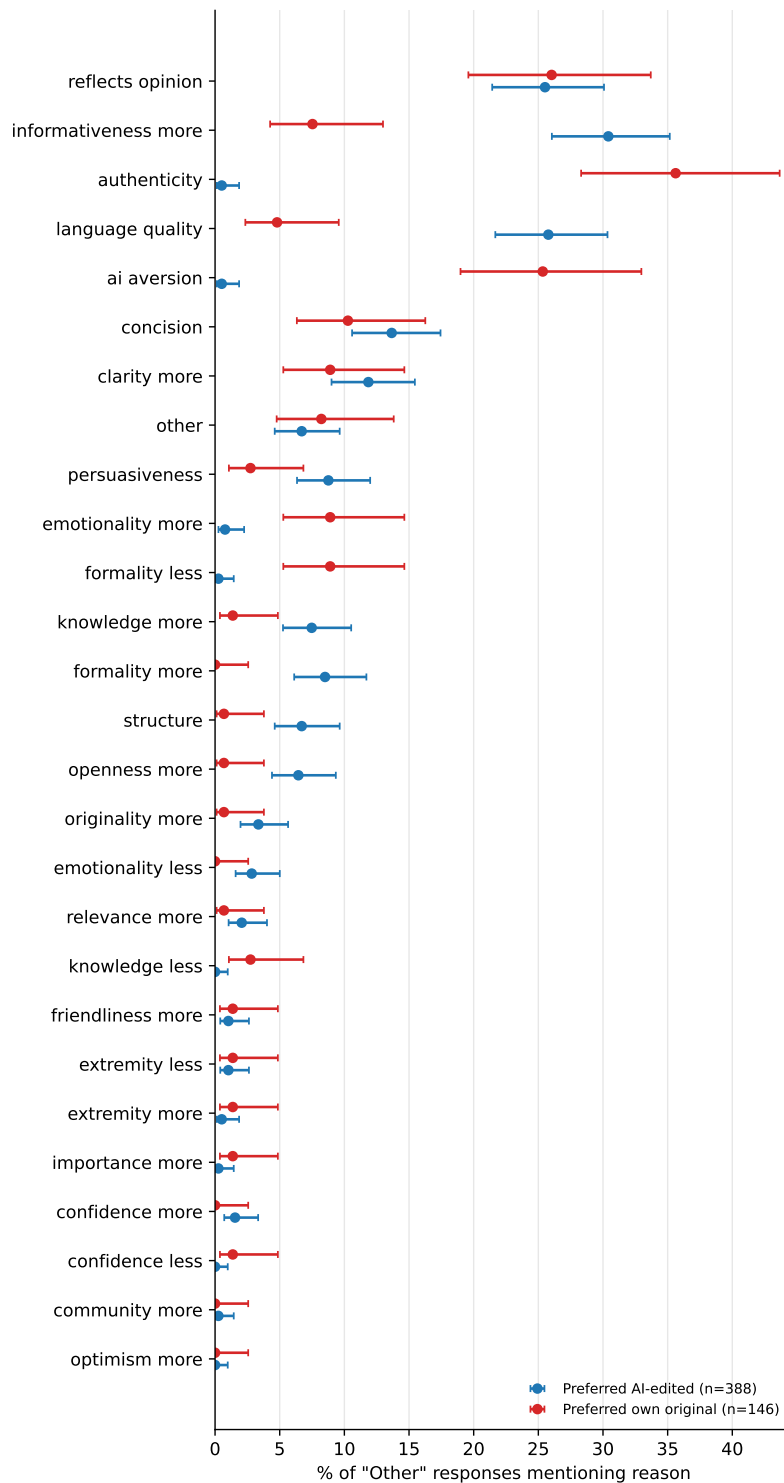


Figure S19: Free-text reasons for strict paragraph preference in Study 1, among writers who ticked “Other” (n=388 strict preferences for the AI-assisted paragraph, n=146 for the human-written paragraph). Points show the share of responses mentioning each reason, with 95% Wilson CIs. See Table S34 for code definitions.

5 Study 1 (Main Study): Distortion Analysis

5.1 Reader Assignment

We collected ratings for all 10,008 paragraphs produced in Study 1. This includes 4,503 human-written paragraphs, 4,503 unedited AI-written paragraphs, and another 1,002 edited AI-written paragraphs. In other words, when writers made edits to the AI-written paragraph, we collected ratings for both the edited and unedited paragraph versions.

10,017 readers were each assigned ten paragraphs to rate. Few readers ($<0.5\%$) were affected by technical issues in the study interface that resulted in them rating just eight or nine paragraphs, as shown in Table S36.

Table S36: Distribution of paragraphs rated per reader in Study 1. Overall, 10,017 readers rated 10,008 paragraphs in Study 1.

Paragraphs Rated	Readers	Proportion
8	2	0.0%
9	42	0.4%
10	9,973	 99.6%
Total	10,017	

Our goal was to collect, on average, ten sets of ratings for every paragraph. Paragraphs were randomly assigned to readers, with those paragraphs that had received ten ratings or more periodically removed from the assignment pool. Table S37 shows the distribution of ratings received per paragraph. Most paragraphs received exactly ten ratings. No paragraph received fewer than nine ratings.

Table S37: Distribution of ratings received per paragraph in Study 1. Overall, 10,017 readers rated 10,008 paragraphs in Study 1.

Ratings Received	Paragraphs	Proportion
9	2,693	 26.9%
10	5,431	 54.3%
11	1,350	 13.5%
12	355	3.6%
13	102	1.0%
14	40	0.4%
15	24	0.2%
16	6	0.1%
17	3	0.0%
18	2	0.0%
19	2	0.0%
Total	10,008	

5.2 Data Subsets: “Unedited”, “Edited”, and “Preferred”

In Study 1, we analyse distortions from AI writing assistance across three nested subsets of our data, which vary in how they treat writer edits and preferences:

1. The **“unedited” subset** pairs each human-written paragraph with the corresponding unedited AI-written paragraph, ignoring any edits that writers may have made. This subset contains 9,006 paragraphs (4,503 human-written and 4,503 AI-written) and 90,112 paragraph-reader observations. It captures the “raw” distortive effect of AI generation, before any human correction.
2. The **“edited” subset** pairs each human-written paragraph with the corresponding AI-written paragraph *including any edits made by the writer*. Where writers made edits (1,002 of 4,503 cases), the edited version replaces the unedited one. This subset contains 9,006 paragraphs (4,503 human-written and 4,503 AI-written) and 90,145 paragraph-reader observations. It captures distortions after writers have had the opportunity to edit the AI output.
3. The **“preferred” subset** further restricts the edited subset by excluding the 708 AI-written paragraphs that writers strictly dispreferred in comparison to their own writing, along with their 708 human-written counterparts. This subset contains 7,590 paragraphs (3,795 human-written and 3,795 AI-written) and 75,972 paragraph-reader observations. It captures distortions only where writers would plausibly propagate the AI-written text in naturalistic settings.

In our main text, we report results for the “preferred” subset, which best approximates real-world human-AI co-production of text by measuring distortions only where they would plausibly be adopted and disseminated by writers. As shown in subsequent sections, results are highly consistent across all three subsets.

5.3 Data Transformations

In all studies, we transformed the raw `writer_stance` rating (0–100) into a polarity measure defined as $|\text{writer_stance} - 50| \times 2$, so that 0 indicates a perceived ambivalent position and 100 indicates a perceived extreme position. This transformation is necessary because the raw stance variable can only be interpreted relative to a given proposition, whereas polarity is interpretable uniformly across all propositions.

For the reader-inferred `writer_age` variable, we binned the raw numeric responses into six categories [18–29, 30–39, 40–49, 50–59, 60–69, 70+] to match the categorical format used for all other demographic variables in our analysis.

Both these transformations were pre-registered.

5.4 Descriptive Results: Distributions

For each paragraph, readers provided ratings across the 29 dimensions of reader perception listed in SI:2.7. Figures S20 to S25 below show rating distributions for each rating attribute in Study 1, split by whether paragraphs were human-written or AI-generated, for each of the three data subsets defined in SI:5.2.

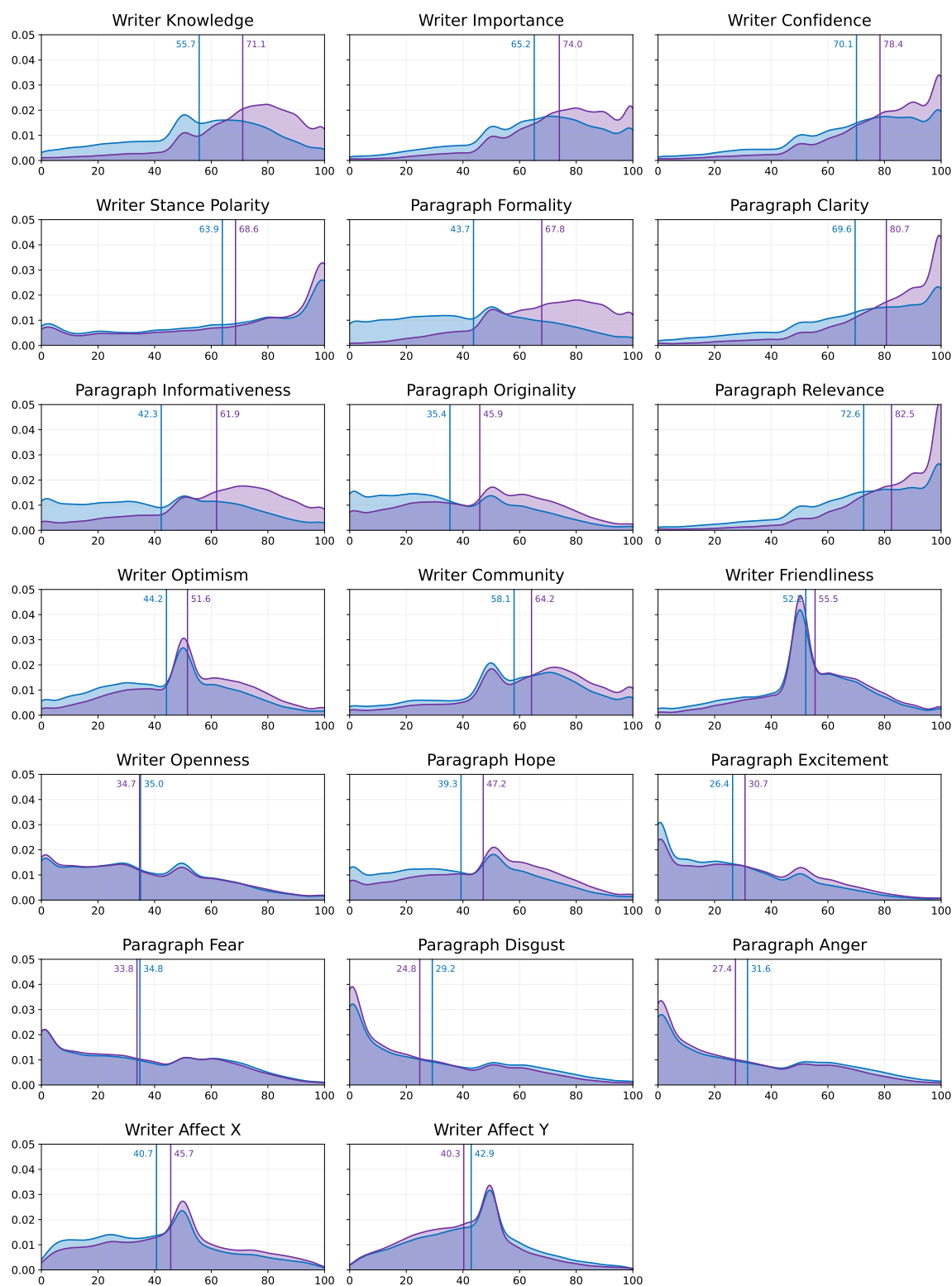


Figure S20: Kernel density estimates (KDEs) of reader ratings across all 0-100 scale rating attributes (Study 1, “preferred” subset). Blue and purple distributions show ratings for human- and AI-written paragraphs. Vertical lines indicate mean ratings for each distribution.

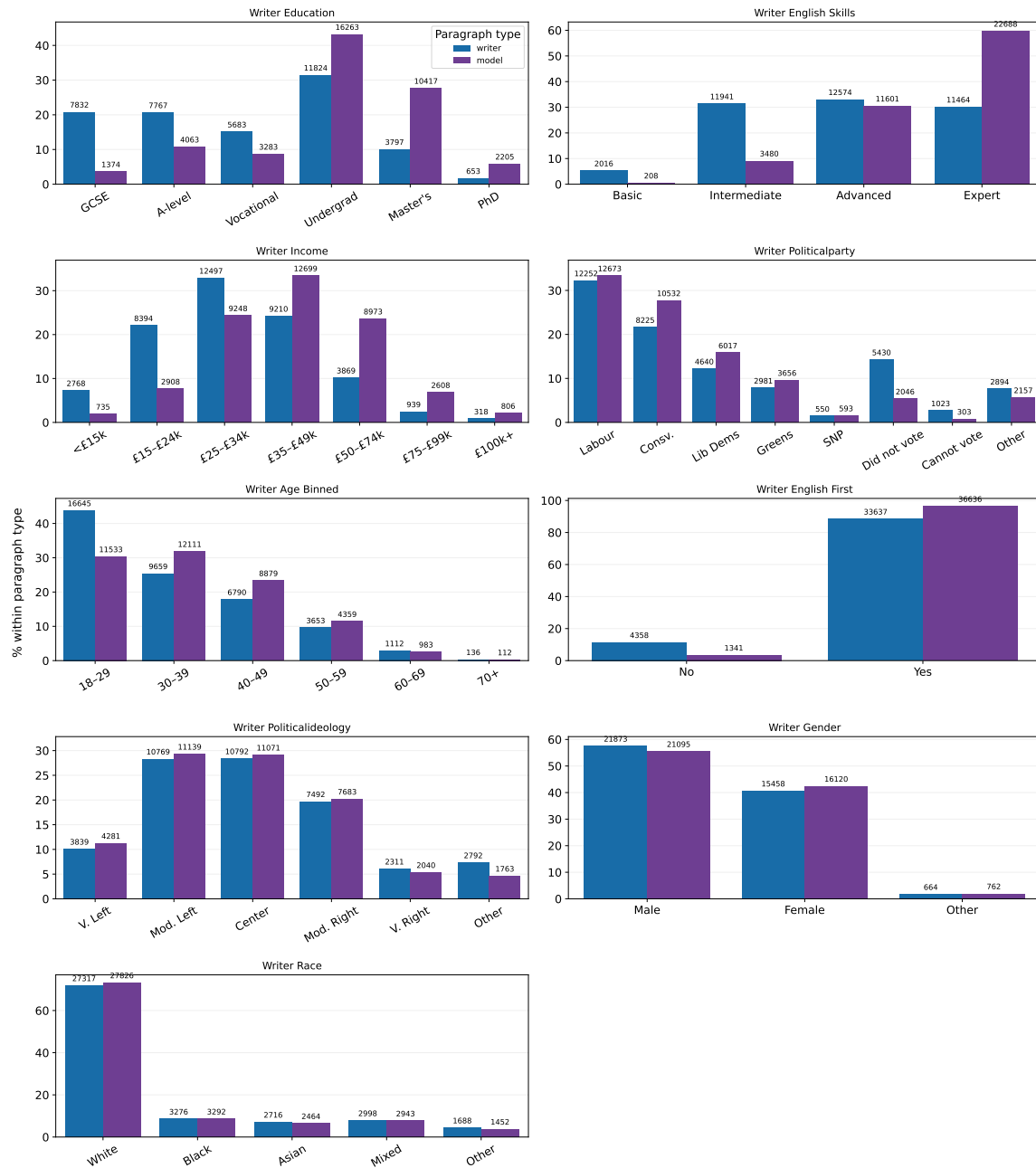


Figure S21: Bar plots of reader ratings across all categorical rating attributes (Study 1, “preferred” subset). Blue and purple distributions show ratings for human- and AI-written paragraphs. Numbers on bars indicate number of observations.

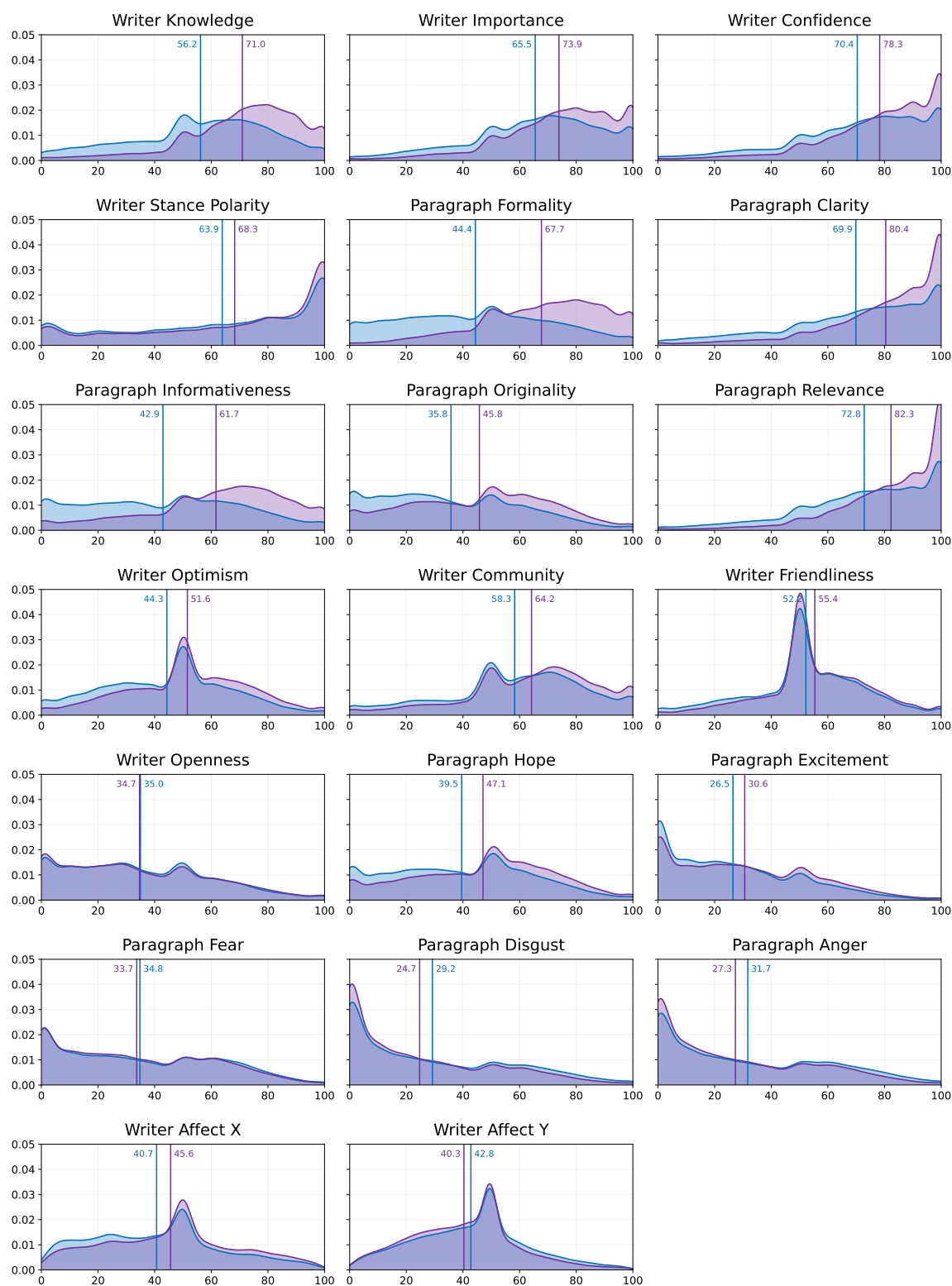


Figure S22: Kernel density estimates (KDEs) of reader ratings across all 0-100 scale rating attributes (Study 1, “edited” subset). Blue and purple distributions show ratings for human- and AI-written paragraphs. Vertical lines indicate mean ratings for each distribution.

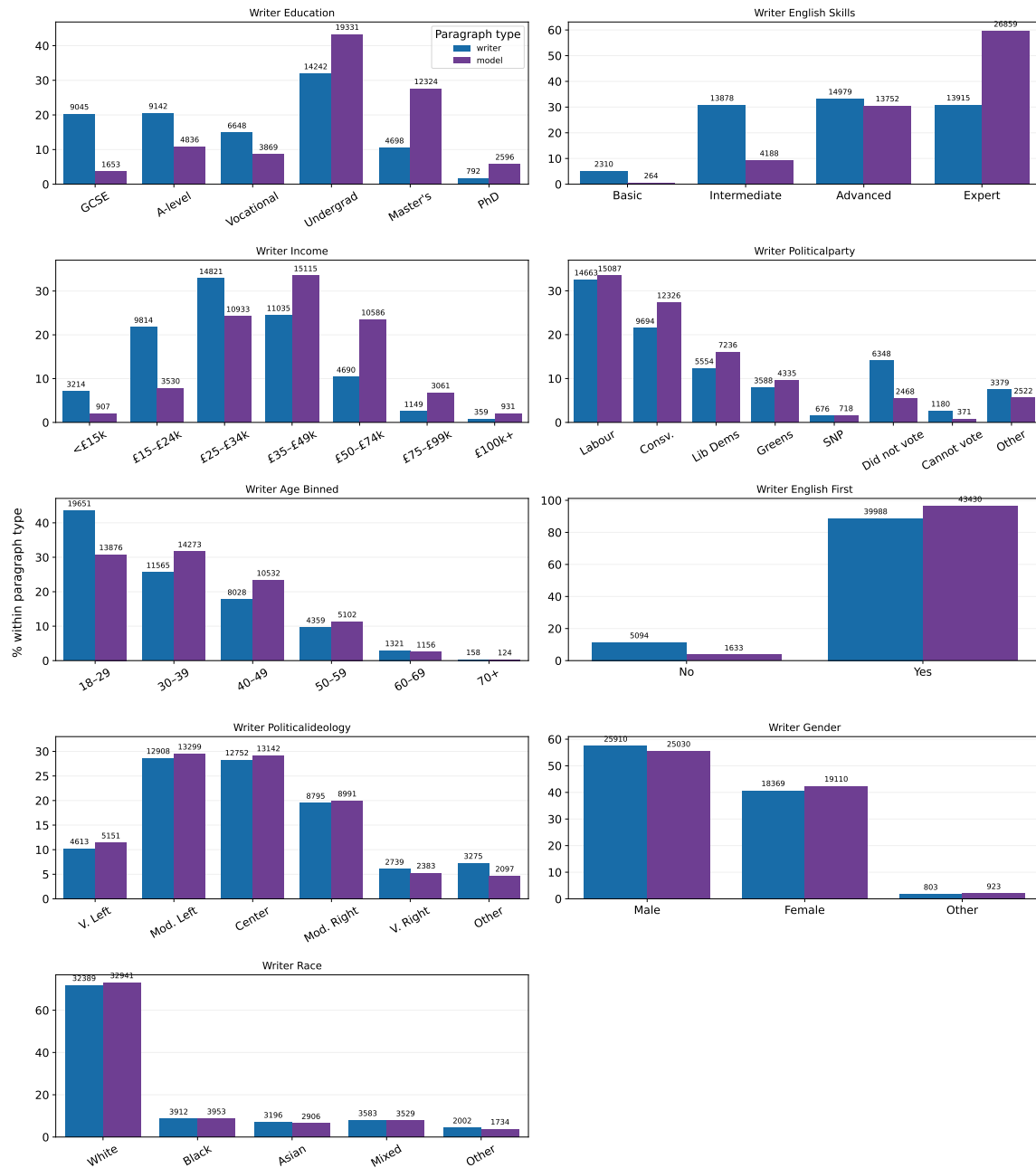


Figure S23: Bar plots of reader ratings across all categorical rating attributes (Study 1, “edited” subset). Blue and purple distributions show ratings for human- and AI-written paragraphs. Numbers on bars indicate number of observations.

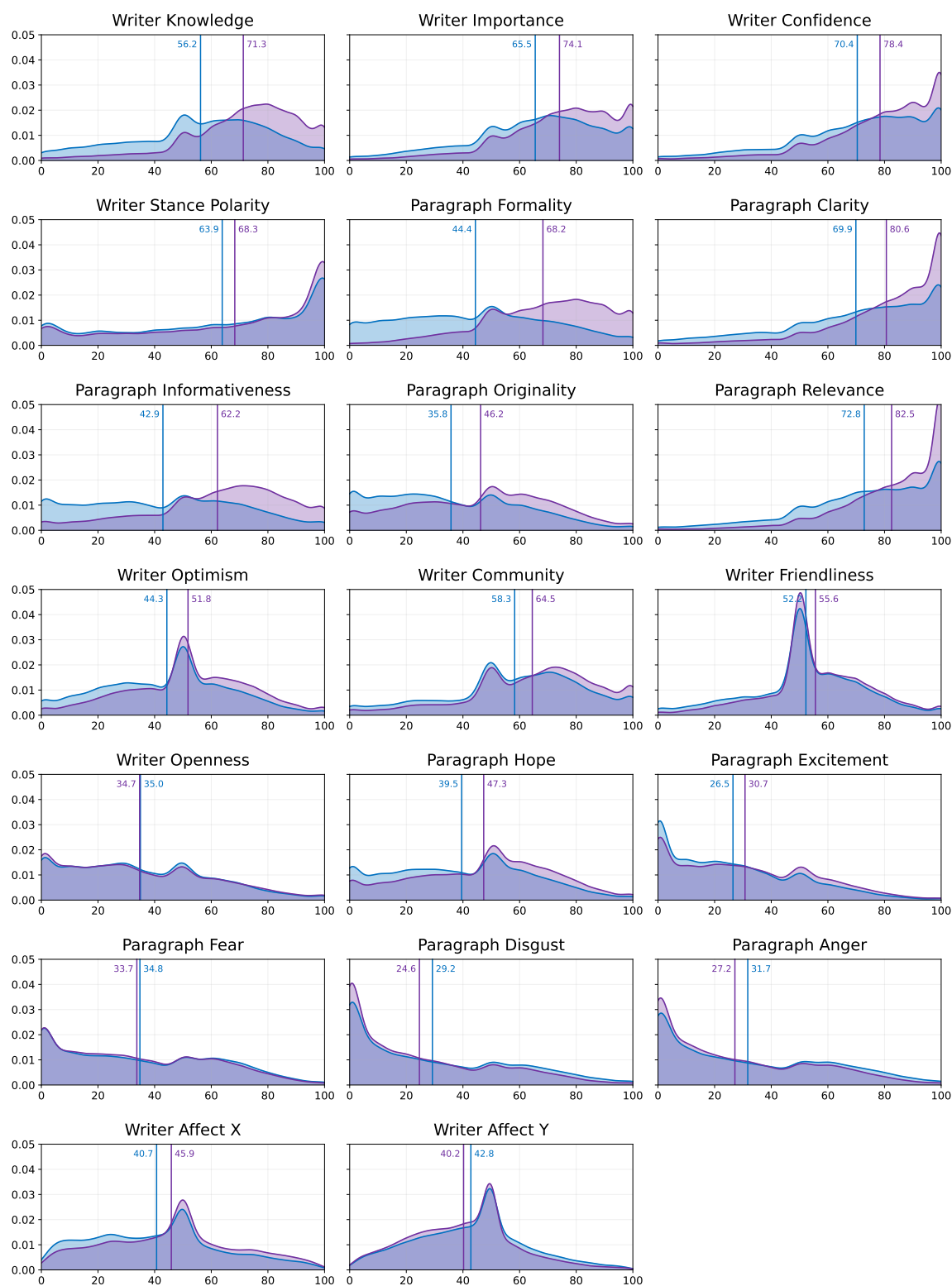


Figure S24: Kernel density estimates (KDEs) of reader ratings across all 0-100 scale rating attributes (Study 1, “unedited” subset). Blue and purple distributions show ratings for human- and AI-written paragraphs. Vertical lines indicate mean ratings for each distribution.

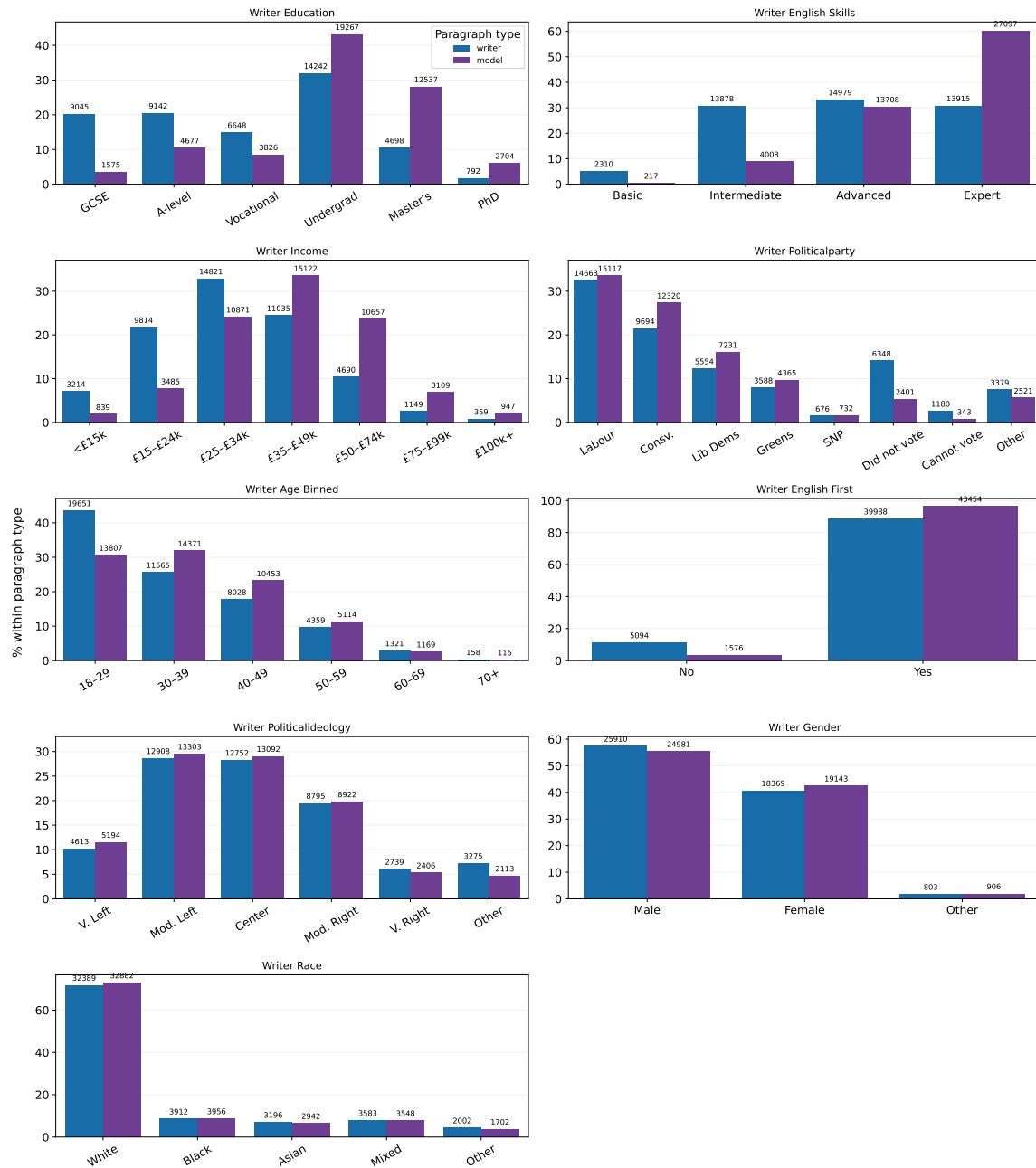


Figure S25: Bar plots of reader ratings across all categorical rating attributes (Study 1, “unedited” subset). Blue and purple distributions show ratings for human- and AI-written paragraphs. Numbers on bars indicate number of observations.

5.5 Descriptive Results: Correlations

Figures S26 to S28 show pairwise Pearson correlations between all 0-100 scale rating attributes, computed across all paragraph-reader observations in each data subset. These correlation structures are relevant to interpreting the entangled nature of distortions discussed in the main text: attributes that are highly correlated in reader ratings tend to be distorted in the same direction and to co-vary under mitigation.

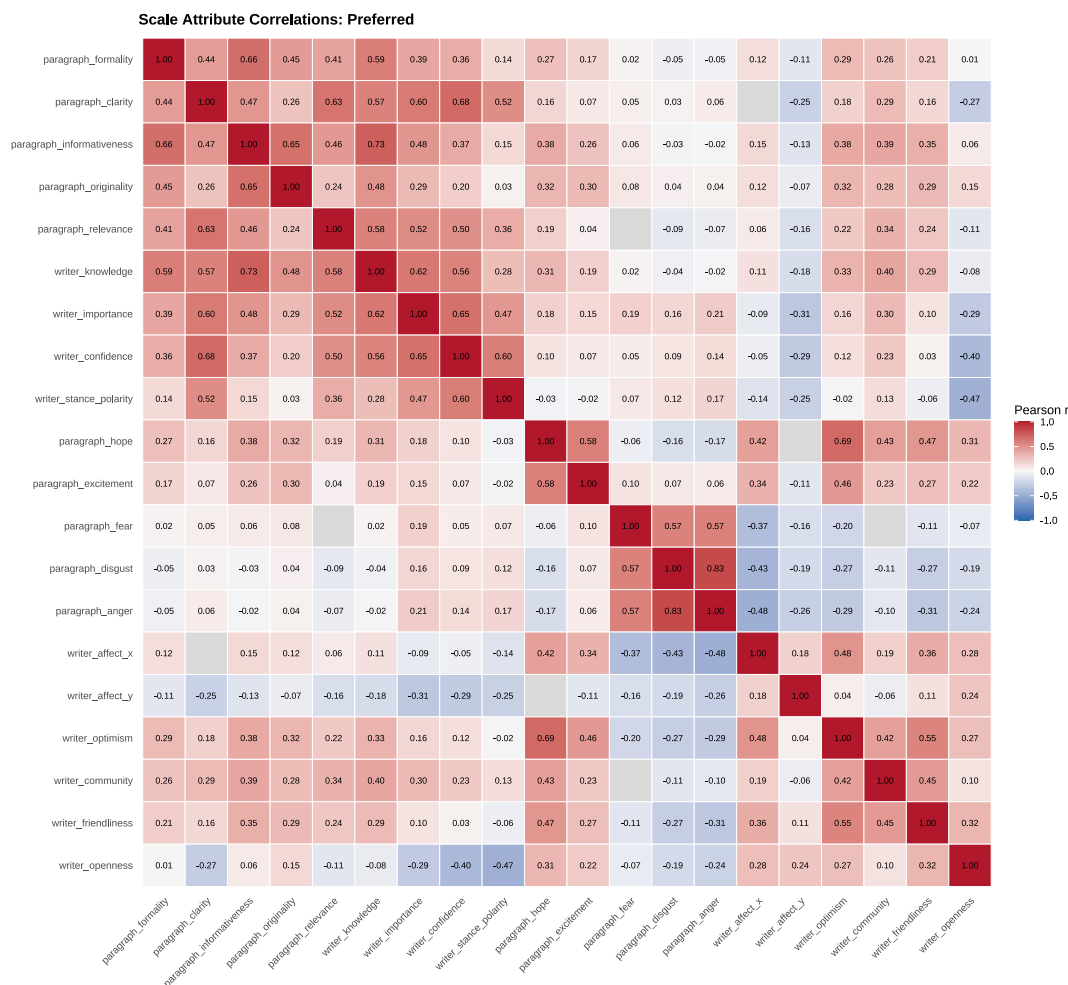


Figure S26: Pairwise Pearson correlations between 0-100 scale rating attributes (Study 1, “preferred” subset). Each cell shows the correlation coefficient for a pair of attributes, computed across all paragraph-reader observations. Only correlations significant at $p < .001$ are displayed. Non-significant cells are shown in grey.

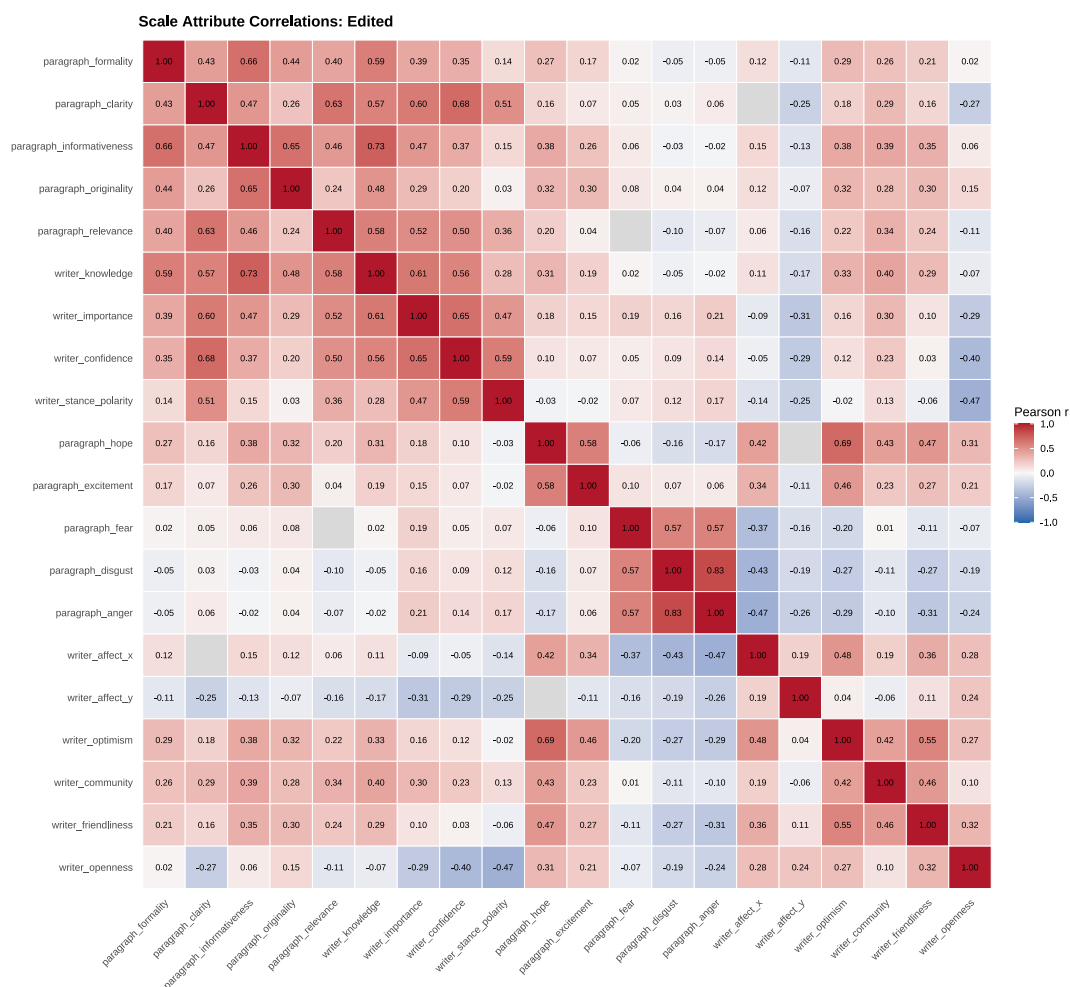


Figure S27: Pairwise Pearson correlations between 0-100 scale rating attributes (Study 1, “edited” subset). Each cell shows the correlation coefficient for a pair of attributes, computed across all paragraph-reader observations. Only correlations significant at $p < .001$ are displayed. Non-significant cells are shown in grey.

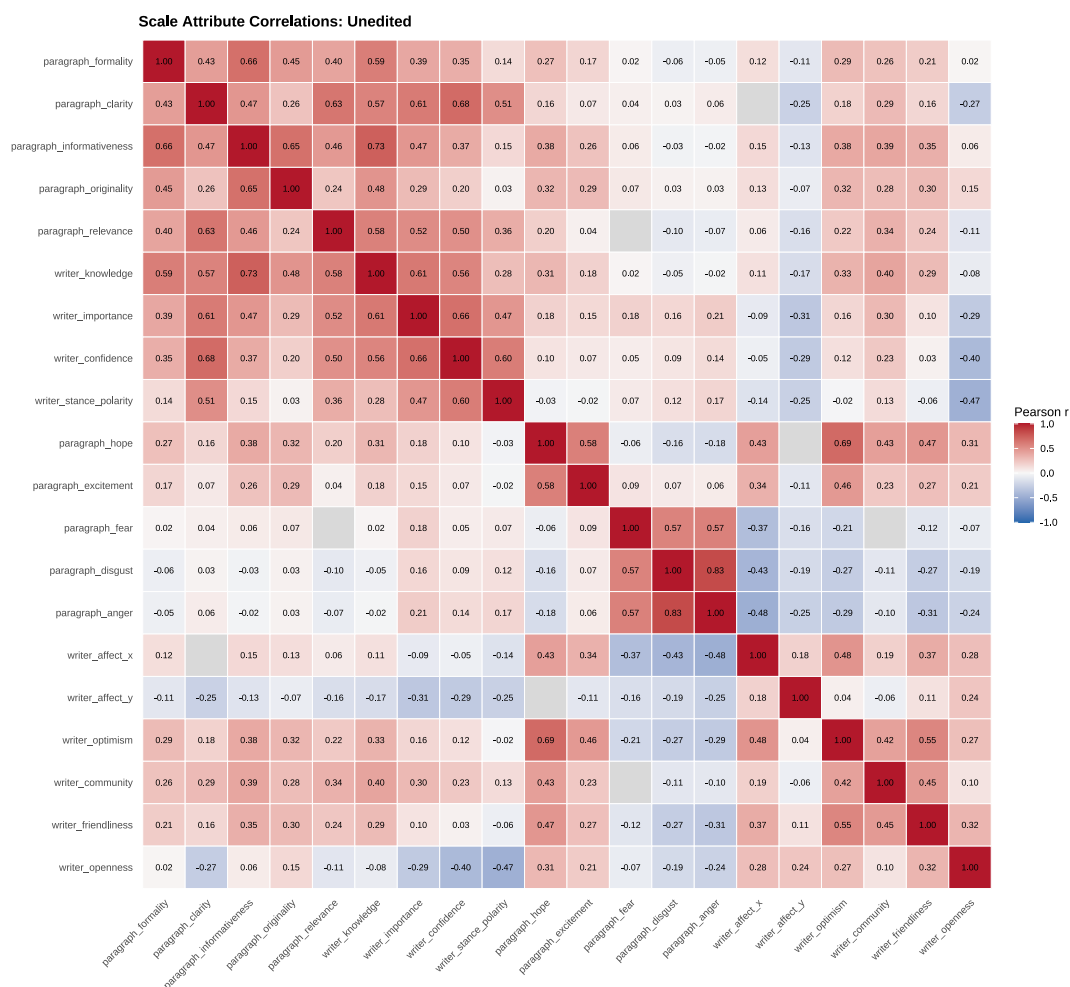


Figure S28: Pairwise Pearson correlations between 0-100 scale rating attributes (Study 1, “edited” subset). Each cell shows the correlation coefficient for a pair of attributes, computed across all paragraph-reader observations. Only correlations significant at $p < .001$ are displayed. Non-significant cells are shown in grey.

5.6 Descriptive Results: Inter-Rater Agreement

Tables S38 and S39 show inter-rater agreement for reader perceptions in Study 1. Note that our goal was to elicit subjective reader perceptions of writers and their opinions, which we expected to differ across readers, and that we included reader random effects in our analyses below to account for this.

Table S38: Inter-rater agreement (Krippendorff’s α) for all 29 dimensions of reader perception in Study 1 (10,008 paragraphs: 4,503 human-written, 4,503 AI-written, and 1,002 writer-edited AI-written; 100,124 paragraph-reader observations; about 10 readers per paragraph). Units are paragraphs, rated by different readers. α uses the interval metric for scale attributes, the ordinal metric for ordinal attributes, and the nominal metric for nominal attributes. $\alpha = 1$ indicates perfect agreement and $\alpha = 0$ indicates agreement at chance level. Columns report α computed across all paragraphs and separately by paragraph type. “writer_education” responses of “Other” are treated as missing.

Attribute	Type	All paragraphs	Human-written	AI-written	AI-written (edited)
writer_knowledge	scale	0.317	0.251	0.225	0.276
writer_importance	scale	0.307	0.284	0.275	0.295
writer_confidence	scale	0.341	0.316	0.322	0.339
writer_stance_polarity	scale	0.408	0.370	0.439	0.422
paragraph_formality	scale	0.346	0.221	0.147	0.208
paragraph_clarity	scale	0.263	0.220	0.230	0.262
paragraph_informativeness	scale	0.264	0.178	0.147	0.204
paragraph_originality	scale	0.111	0.082	0.064	0.090
paragraph_relevance	scale	0.189	0.161	0.120	0.188
writer_optimism	scale	0.253	0.253	0.213	0.229
writer_community	scale	0.188	0.186	0.158	0.193
writer_friendliness	scale	0.162	0.187	0.119	0.164
writer_openness	scale	0.257	0.255	0.257	0.263
paragraph_hope	scale	0.240	0.232	0.208	0.237
paragraph_excitement	scale	0.127	0.109	0.130	0.125
paragraph_fear	scale	0.189	0.206	0.173	0.173
paragraph_disgust	scale	0.241	0.276	0.196	0.195
paragraph_anger	scale	0.277	0.316	0.229	0.239
writer_affect_x	scale	0.269	0.277	0.244	0.253
writer_affect_y	scale	0.103	0.108	0.091	0.093
writer_education	ordinal	0.254	0.172	0.104	0.133
writer_english_skills	ordinal	0.200	0.139	0.049	0.076
writer_income	ordinal	0.196	0.133	0.088	0.109
writer_age_binned	ordinal	0.130	0.133	0.108	0.106
writer_english_first	ordinal	0.097	0.098	0.026	0.036
writer_politicalParty	nominal	0.093	0.082	0.096	0.088
writer_politicalIdeology	nominal	0.079	0.080	0.080	0.076
writer_gender	nominal	0.152	0.148	0.155	0.152
writer_race	nominal	0.034	0.038	0.030	0.026

Table S39: Mean within-paragraph standard deviation of reader ratings for scale attributes in Study 1 (10,008 paragraphs: 4,503 human-written, 4,503 AI-written, and 1,002 writer-edited AI-written; 100,124 paragraph-reader observations; about 10 readers per paragraph). For each paragraph, we compute the standard deviation of the ratings it received, then average across paragraphs. Values are in rating points on the 0–100 scale. Columns report the mean SD across all paragraphs and separately by paragraph type.

Attribute	All paragraphs	Human-written	AI-written	AI-written (edited)
writer_knowledge	18.64	20.49	17.11	17.19
writer_importance	17.53	18.97	16.29	16.56
writer_confidence	17.23	18.89	15.75	16.45
writer_stance_polarity	23.94	25.00	22.93	23.77
paragraph_formality	21.02	22.84	19.48	19.75
paragraph_clarity	19.21	21.80	16.91	17.93
paragraph_informativeness	23.08	24.26	22.13	22.08
paragraph_originality	23.67	23.46	23.86	23.76
paragraph_relevance	18.34	20.55	16.44	16.98
writer_optimism	18.79	18.99	18.64	18.55
writer_community	20.72	21.22	20.34	20.20
writer_friendliness	17.07	17.45	16.78	16.67
writer_openness	21.20	20.82	21.53	21.41
paragraph_hope	21.16	21.23	21.17	20.86
paragraph_excitement	21.54	20.99	22.00	21.93
paragraph_fear	23.35	23.51	23.25	23.06
paragraph_disgust	22.06	22.55	21.68	21.59
paragraph_anger	22.16	22.39	21.99	21.86
writer_affect_x	19.34	19.03	19.63	19.46
writer_affect_y	17.67	18.16	17.29	17.20

5.7 Statistical Analysis: Overall Effects of AI Writing Assistance

Each observation is a paragraph-reader pair, i.e. the rating given by a specific reader for a specific paragraph. For each rating attribute, we fit **mixed-effects regression models**, specified as follows:

$$\text{rating_attribute} \sim \text{paragraph_type} + (1 \mid \text{reader_id}) + (1 \mid \text{writer_id}) + (1 \mid \text{proposition_id})$$

where `rating_attribute` is the outcome variable of interest (e.g. perceived writer confidence, paragraph formality); `paragraph_type` is a binary fixed effect distinguishing human-written from AI-written paragraphs, with human-written paragraphs as the reference category; and $(1 \mid \text{reader_id})$, $(1 \mid \text{writer_id})$, and $(1 \mid \text{proposition_id})$ are crossed random intercepts accounting for individual differences between readers, between writers, and between propositions.

The measurement scale of each rating attribute determined the appropriate link function. For the rating attributes that are measured on a 0-100 scale (e.g. paragraph relevance, writer confidence) we fit **mixed-effects beta regression models** using a logit link function. For nominal categorical rating attributes (e.g. writer race, writer gender), we fit **mixed-effects multinomial logistic regression models** using the multinomial logit link function. For ordinal categorical rating attributes (e.g. writer education, writer income), we fit **mixed-effects ordinal logistic regression models** using the cumulative logit link function.

Following our pre-registration, we include crossed random intercepts for readers, writers, and propositions in the beta regressions for 0-100 scale attributes and in the one-vs-all logistic regressions for nominal attributes described below. Our pre-registration also specified that we would reduce the random-effects structure in case of convergence issues. For the ordinal and multinomial logistic regressions, models with more than one random-effects term were computationally infeasible on our data: cumulative link mixed models with multiple random-effects terms require a general Laplace approximation, which was impractically slow for our 75,972–90,145 observations, and mixed multinomial logistic regressions with multiple random-effects terms, estimated via penalised quasi-likelihood, exceeded available memory or failed. We therefore fit ordinal and multinomial logistic regressions with reader random intercepts only. For most rating attributes, reader random intercepts account for the largest share of variance among the three random effects, and adding writer and proposition random intercepts leaves our estimates essentially unchanged (SI:5.8). Secondary analyses of Study 1 (SI:5.12, SI:5.13, and SI:5.16) use reader random intercepts only.

We fit each regression model to the three data subsets defined in SI:5.2: “unedited” (90,112 observations), “edited” (90,145 observations), and “preferred” (75,972 observations). Figures S29, S30 and S31 summarise results, showing that they are highly consistent across all three subsets.

Tables S40 to S90 show results for each of the 20 rating attributes measured on a 0-100, for each subset. Tables S100 to S114 show results for each of the 5 ordinal rating attributes, for each subset. Tables S115 to S126 show results for each of the 4 nominal rating attributes, for each subset.

For the 4 nominal rating attributes (`writer_race`, `_gender`, `_politicalParty`, `_politicalIdeology`), we additionally fit one-vs-all logistic regressions, with crossed reader, writer, and proposition random effects as above. **This specific analysis was not pre-registered.** We included it so we could report category-specific effects that allowed for easier interpretation in Figure 2 of the main text. Results for the “preferred” subset are reported in Tables S128 to S130.

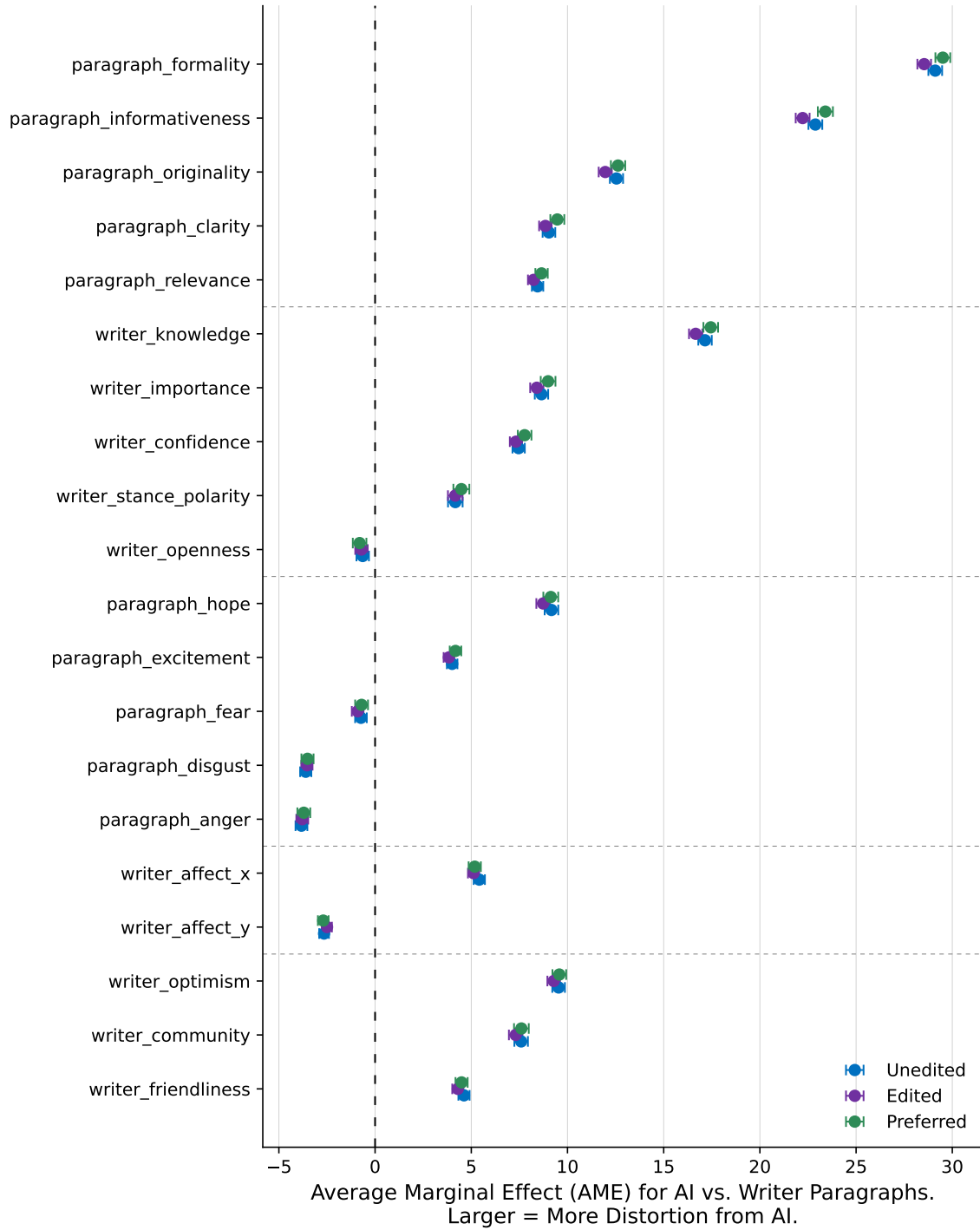


Figure S29: Average marginal effects (AMEs) of AI writing assistance on reader perceptions across all 0-100 scale rating attributes in Study 1. Each point shows the AME from a mixed-effects beta regression comparing AI-written to human-written paragraphs, with crossed reader, writer, and proposition random effects, for three data subsets: unedited (blue), edited (purple), and preferred (green). Error bars show 95% confidence intervals.

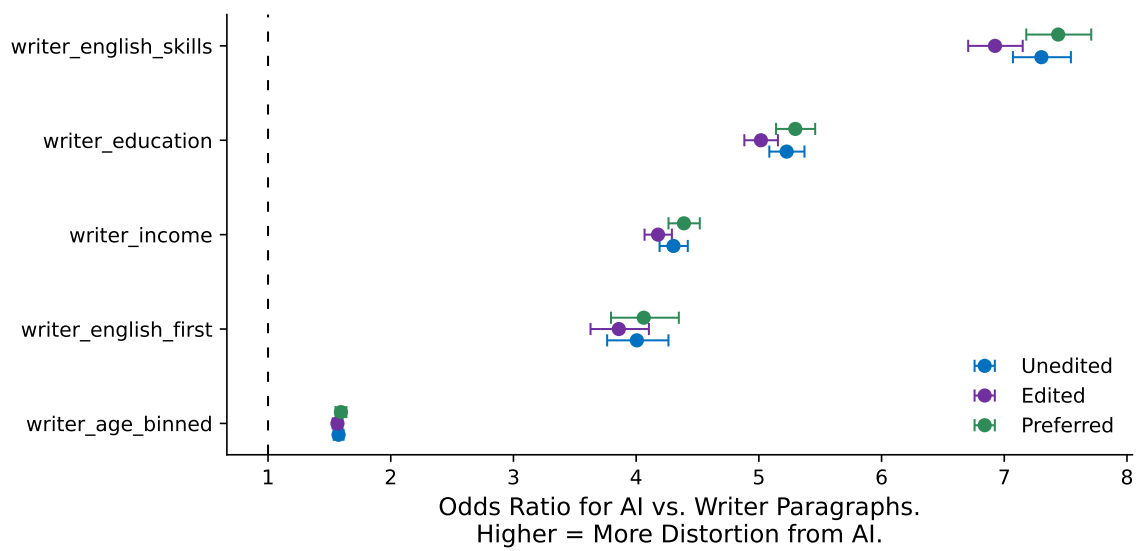


Figure S30: Odds ratios (ORs) of AI writing assistance on reader perceptions across all ordinal rating attributes in Study 1. Each point shows the odds ratio from a mixed-effects ordinal logistic regression comparing AI-written to human-written paragraphs, with reader random effects, for three data subsets: unedited (blue), edited (purple), and preferred (green). Error bars show 95% confidence intervals.

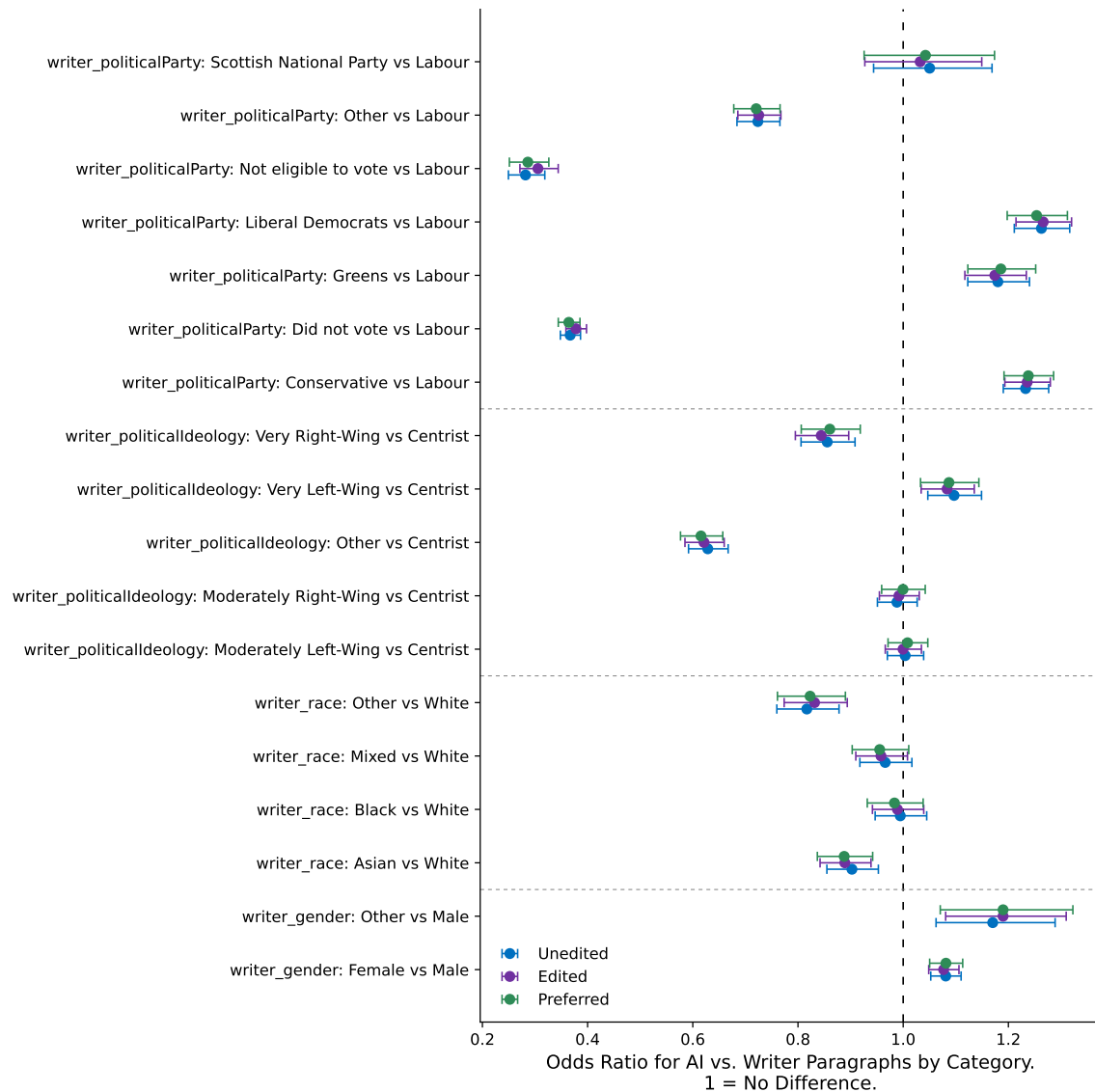


Figure S31: Odds ratios (ORs) of AI writing assistance on reader perceptions across all nominal rating attributes in Study 1. Each point shows the odds ratio of a specific category vs a reference category from a mixed-effects multinomial logistic regression comparing AI-written to human-written paragraphs, with reader random effects, for three data subsets: unedited (blue), edited (purple), and preferred (green). Error bars show 95% confidence intervals.

Table S40: Beta regression predicting perceived paragraph formality in Study 1 (“preferred” subset, N=75,972). Outcome: “paragraph_formality”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	3.442	[3.386, 3.498]	148.809	0	29.507	[29.122, 29.892]

Table S41: Beta regression predicting perceived paragraph formality in Study 1 (“edited” subset, N=90,145). Outcome: “paragraph_formality”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	3.299	[3.250, 3.348]	157.156	0	28.539	[28.185, 28.894]

Table S42: Beta regression predicting perceived paragraph formality in Study 1 (“unedited” subset, N=90,112). Outcome: “paragraph_formality”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	3.390	[3.340, 3.441]	160.829	0	29.114	[28.758, 29.470]

Table S43: Beta regression predicting perceived paragraph clarity in Study 1 (“preferred” subset, N=75,972). Outcome: “paragraph_clarity”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.728	[1.699, 1.758]	63.076	0	9.471	[9.109, 9.833]

Table S44: Beta regression predicting perceived paragraph clarity in Study 1 (“edited” subset, N=90,145). Outcome: “paragraph_clarity”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.660	[1.634, 1.686]	64.003	0	8.851	[8.526, 9.177]

Table S45: Beta regression predicting perceived paragraph clarity in Study 1 (“unedited” subset, N=90,112). Outcome: “paragraph_clarity”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.682	[1.656, 1.708]	65.832	0	9.036	[8.707, 9.365]

Table S46: Beta regression predicting perceived paragraph informativeness in Study 1 (“preferred” subset, N=75,972). Outcome: “paragraph_informativeness”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	2.596	[2.553, 2.639]	112.712	0	23.402	[23.010, 23.794]

Table S47: Beta regression predicting perceived paragraph informativeness in Study 1 (“edited” subset, N=90,145). Outcome: “paragraph_informativeness”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	2.470	[2.432, 2.507]	116.761	0	22.222	[21.861, 22.583]

Table S48: Beta regression predicting perceived paragraph informativeness in Study 1 (“unedited” subset, N=90,112). Outcome: “paragraph_informativeness”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	2.540	[2.502, 2.579]	120.562	0	22.883	[22.524, 23.242]

Table S49: Beta regression predicting perceived paragraph originality in Study 1 (“preferred” subset, N=75,972). Outcome: “paragraph_originality”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.719	[1.693, 1.746]	69.042	0	12.628	[12.256, 13.000]

Table S50: Beta regression predicting perceived paragraph originality in Study 1 (“edited” subset, N=90,145). Outcome: “paragraph_originality”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.670	[1.647, 1.694]	71.623	0	11.963	[11.623, 12.304]

Table S51: Beta regression predicting perceived paragraph originality in Study 1 (“unedited” subset, N=90,112). Outcome: “paragraph_originality”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.711	[1.687, 1.735]	74.928	0	12.546	[12.204, 12.888]

Table S52: Beta regression predicting perceived paragraph relevance in Study 1 (“preferred” subset, N=75,972). Outcome: “paragraph_relevance”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.755	[1.727, 1.784]	67.446	0	8.652	[8.331, 8.973]

Table S53: Beta regression predicting perceived paragraph relevance in Study 1 (“edited” subset, N=90,145). Outcome: “paragraph_relevance”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.702	[1.677, 1.728]	70.105	0	8.238	[7.946, 8.530]

Table S54: Beta regression predicting perceived paragraph relevance in Study 1 (“unedited” subset, N=90,112). Outcome: “paragraph_relevance”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.733	[1.707, 1.759]	72.654	0	8.445	[8.150, 8.740]

Table S55: Beta regression predicting perceived writer knowledge in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_knowledge”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	2.156	[2.124, 2.189]	99.918	0	17.446	[17.065, 17.827]

Table S56: Beta regression predicting perceived writer knowledge in Study 1 (“edited” subset, N=90,145). Outcome: “writer_knowledge”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	2.079	[2.051, 2.108]	103.881	0	16.663	[16.316, 17.010]

Table S57: Beta regression predicting perceived writer knowledge in Study 1 (“unedited” subset, N=90,112). Outcome: “writer_knowledge”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	2.130	[2.101, 2.160]	107.410	0	17.149	[16.800, 17.498]

Table S58: Beta regression predicting perceived writer importance in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_importance”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.571	[1.547, 1.595]	57.383	0	8.992	[8.610, 9.375]

Table S59: Beta regression predicting perceived writer importance in Study 1 (“edited” subset, N=90,145). Outcome: “writer_importance”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.524	[1.502, 1.545]	58.473	0	8.406	[8.062, 8.750]

Table S60: Beta regression predicting perceived writer importance in Study 1 (“unedited” subset, N=90,112). Outcome: “writer_importance”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.545	[1.524, 1.567]	60.651	0	8.649	[8.301, 8.996]

Table S61: Beta regression predicting perceived writer confidence in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_confidence”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.553	[1.528, 1.578]	52.796	0	7.772	[7.418, 8.126]

Table S62: Beta regression predicting perceived writer confidence in Study 1 (“edited” subset, N=90,145). Outcome: “writer_confidence”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.511	[1.488, 1.533]	54.254	0	7.321	[7.003, 7.639]

Table S63: Beta regression predicting perceived writer confidence in Study 1 (“unedited” subset, N=90,112). Outcome: “writer_confidence”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.524	[1.502, 1.547]	55.461	0	7.461	[7.142, 7.781]

Table S64: Beta regression predicting perceived writer stance polarity in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_stance_polarity”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.228	[1.206, 1.251]	21.829	1.235e-105	4.484	[4.070, 4.899]

Table S65: Beta regression predicting perceived writer stance polarity in Study 1 (“edited” subset, N=90,145). Outcome: “writer_stance_polarity”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.210	[1.189, 1.230]	22.082	4.690e-108	4.165	[3.786, 4.544]

Table S66: Beta regression predicting perceived writer stance polarity in Study 1 (“unedited” subset, N=90,112). Outcome: “writer_stance_polarity”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.210	[1.190, 1.231]	22.113	2.344e-108	4.170	[3.791, 4.549]

Table S67: Beta regression predicting perceived paragraph hope in Study 1 (“preferred” subset, N=75,972). Outcome: “paragraph_hope”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.466	[1.444, 1.490]	47.788	0	9.132	[8.746, 9.517]

Table S68: Beta regression predicting perceived paragraph hope in Study 1 (“edited” subset, N=90,145). Outcome: “paragraph_hope”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.443	[1.422, 1.463]	50.087	0	8.734	[8.382, 9.087]

Table S69: Beta regression predicting perceived paragraph hope in Study 1 (“unedited” subset, N=90,112). Outcome: “paragraph_hope”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.468	[1.447, 1.489]	52.485	0	9.166	[8.813, 9.520]

Table S70: Beta regression predicting perceived paragraph excitement in Study 1 (“preferred” subset, N=75,972). Outcome: “paragraph_excitement”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.265	[1.245, 1.285]	29.506	2.398e-191	4.179	[3.880, 4.479]

Table S71: Beta regression predicting perceived paragraph excitement in Study 1 (“edited” subset, N=90,145). Outcome: “paragraph_excitement”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.241	[1.224, 1.259]	29.843	1.084e-195	3.835	[3.564, 4.107]

Table S72: Beta regression predicting perceived paragraph excitement in Study 1 (“unedited” subset, N=90,112). Outcome: “paragraph_excitement”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.253	[1.235, 1.270]	31.133	8.621e-213	4.006	[3.732, 4.279]

Table S73: Beta regression predicting perceived paragraph fear in Study 1 (“preferred” subset, N=75,972). Outcome: “paragraph_fear”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	0.966	[0.951, 0.982]	−4.138	3.502e-5	−0.702	[−1.034, −0.369]

Table S74: Beta regression predicting perceived paragraph fear in Study 1 (“edited” subset, N=90,145). Outcome: “paragraph_fear”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	0.957	[0.943, 0.971]	−5.887	3.921e-9	−0.906	[−1.209, −0.603]

Table S75: Beta regression predicting perceived paragraph fear in Study 1 (“unedited” subset, N=90,112). Outcome: “paragraph_fear”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	0.965	[0.951, 0.979]	−4.775	1.795e-6	−0.734	[−1.036, −0.432]

Table S76: Beta regression predicting perceived paragraph disgust in Study 1 (“preferred” subset, N=75,972). Outcome: “paragraph_disgust”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	0.816	[0.802, 0.829]	−23.885	4.336e-126	−3.513	[−3.828, −3.197]

Table S77: Beta regression predicting perceived paragraph disgust in Study 1 (“edited” subset, N=90,145). Outcome: “paragraph_disgust”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	0.815	[0.802, 0.827]	−26.383	2.152e-153	−3.541	[−3.831, −3.252]

Table S78: Beta regression predicting perceived paragraph disgust in Study 1 (“unedited” subset, N=90,112). Outcome: “paragraph_disgust”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	0.811	[0.799, 0.824]	−26.890	2.903e-159	−3.604	[−3.894, −3.314]

Table S79: Beta regression predicting perceived paragraph anger in Study 1 (“preferred” subset, N=75,972). Outcome: “paragraph_anger”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	0.818	[0.804, 0.832]	−23.533	1.871e-122	−3.705	[−4.039, −3.370]

Table S80: Beta regression predicting perceived paragraph anger in Study 1 (“edited” subset, N=90,145). Outcome: “paragraph_anger”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	0.814	[0.802, 0.827]	−26.424	7.334e-154	−3.785	[−4.092, −3.478]

Table S81: Beta regression predicting perceived paragraph anger in Study 1 (“unedited” subset, N=90,112). Outcome: “paragraph_anger”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	0.812	[0.800, 0.825]	−26.797	3.463e-158	−3.827	[−4.135, −3.519]

Table S82: Beta regression predicting perceived writer affect x in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_affect_x”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.236	[1.220, 1.252]	32.144	1.084e-226	5.179	[4.862, 5.497]

Table S83: Beta regression predicting perceived writer affect x in Study 1 (“edited” subset, N=90,145). Outcome: “writer_affect_x”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.233	[1.218, 1.247]	34.674	1.927e-263	5.119	[4.828, 5.410]

Table S84: Beta regression predicting perceived writer affect x in Study 1 (“unedited” subset, N=90,112). Outcome: “writer_affect_x”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.248	[1.233, 1.262]	36.680	1.535e-294	5.416	[5.125, 5.707]

Table S85: Beta regression predicting perceived writer affect y in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_affect_y”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	0.895	[0.885, 0.905]	−19.075	4.078e-81	−2.695	[−2.972, −2.418]

Table S86: Beta regression predicting perceived writer affect y in Study 1 (“edited” subset, N=90,145). Outcome: “writer_affect_y”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	0.902	[0.893, 0.912]	−19.329	3.080e-83	−2.497	[−2.750, −2.244]

Table S87: Beta regression predicting perceived writer affect y in Study 1 (“unedited” subset, N=90,112). Outcome: “writer_affect_y”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	0.896	[0.887, 0.906]	−20.531	1.134e-93	−2.651	[−2.905, −2.398]

Table S88: Beta regression predicting perceived writer optimism in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_optimism”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.471	[1.450, 1.492]	52.611	0	9.575	[9.219, 9.931]

Table S89: Beta regression predicting perceived writer optimism in Study 1 (“edited” subset, N=90,145). Outcome: “writer_optimism”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.453	[1.434, 1.473]	55.539	0	9.283	[8.956, 9.610]

Table S90: Beta regression predicting perceived writer optimism in Study 1 (“unedited” subset, N=90,112). Outcome: “writer_optimism”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.468	[1.449, 1.488]	56.988	0	9.539	[9.211, 9.866]

Table S91: Beta regression predicting perceived writer community in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_community”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.380	[1.358, 1.402]	39.705	0	7.604	[7.221, 7.987]

Table S92: Beta regression predicting perceived writer community in Study 1 (“edited” subset, N=90,145). Outcome: “writer_community”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.363	[1.343, 1.383]	41.513	0	7.308	[6.956, 7.659]

Table S93: Beta regression predicting perceived writer community in Study 1 (“unedited” subset, N=90,112). Outcome: “writer_community”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.380	[1.360, 1.400]	43.236	0	7.591	[7.240, 7.943]

Table S94: Beta regression predicting perceived writer friendliness in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_friendliness”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.199	[1.183, 1.214]	27.921	1.488e-171	4.495	[4.180, 4.811]

Table S95: Beta regression predicting perceived writer friendliness in Study 1 (“edited” subset, N=90,145). Outcome: “writer_friendliness”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.189	[1.175, 1.203]	29.066	9.694e-186	4.302	[4.012, 4.591]

Table S96: Beta regression predicting perceived writer friendliness in Study 1 (“unedited” subset, N=90,112). Outcome: “writer_friendliness”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.205	[1.191, 1.219]	31.235	3.597e-214	4.621	[4.331, 4.910]

Table S97: Beta regression predicting perceived writer openness in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_openness”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	0.963	[0.948, 0.979]	−4.469	7.857e-6	−0.804	[−1.157, −0.451]

Table S98: Beta regression predicting perceived writer openness in Study 1 (“edited” subset, N=90,145). Outcome: “writer_openness”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	0.968	[0.954, 0.982]	−4.300	1.706e-5	−0.707	[−1.030, −0.385]

Table S99: Beta regression predicting perceived writer openness in Study 1 (“unedited” subset, N=90,112). Outcome: “writer_openness”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	0.971	[0.956, 0.985]	−3.903	9.490e-5	−0.644	[−0.967, −0.320]

Table S100: Ordinal logistic regression predicting perceived writer education level in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_education”. Reference category: “paragraph_type_writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
paragraph_type_model	5.297	[5.140, 5.459]	108.587	< .001

Table S101: Ordinal logistic regression predicting perceived writer education level in Study 1 (“edited” subset, N=90,145). Outcome: “writer_education”. Reference category: “paragraph_type_writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
paragraph_type_model	5.017	[4.882, 5.155]	115.835	< .001

Table S102: Ordinal logistic regression predicting perceived writer education level in Study 1 (“unedited” subset, N=90,112). Outcome: “writer_education”. Reference category: “paragraph_type_writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
paragraph_type_model	5.227	[5.085, 5.372]	118.312	< .001

Table S103: Ordinal logistic regression predicting perceived writer income level in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_income”. Reference category: “paragraph_type_writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
paragraph_type_model	4.389	[4.264, 4.519]	99.773	< .001

Table S104: Ordinal logistic regression predicting perceived writer income level in Study 1 (“edited” subset, N=90,145). Outcome: “writer_income”. Reference category: “paragraph_type_writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
paragraph_type_model	4.178	[4.068, 4.291]	105.142	< .001

Table S105: Ordinal logistic regression predicting perceived writer income level in Study 1 (“unedited” subset, N=90,112). Outcome: “writer_income”. Reference category: “paragraph_type_writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
paragraph_type_model	4.304	[4.191, 4.421]	107.103	< .001

Table S106: Ordinal logistic regression predicting perceived writer English skills in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_english_skills”. Reference category: “paragraph_type_writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
paragraph_type_model	7.439	[7.179, 7.709]	110.534	< .001

Table S107: Ordinal logistic regression predicting perceived writer English skills in Study 1 (“edited” subset, N=90,145). Outcome: “writer_english_skills”. Reference category: “paragraph_type_writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
paragraph_type_model	6.924	[6.706, 7.150]	118.257	< .001

Table S108: Ordinal logistic regression predicting perceived writer English skills in Study 1 (“unedited” subset, N=90,112). Outcome: “writer_english_skills”. Reference category: “paragraph_type_writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
paragraph_type_model	7.303	[7.071, 7.543]	120.541	< .001

Table S109: Ordinal logistic regression predicting whether the writer is a native English speaker in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_english_first”. Reference category: “paragraph_type_writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
paragraph_type_model	4.062	[3.795, 4.348]	40.412	< .001

Table S110: Ordinal logistic regression predicting whether the writer is a native English speaker in Study 1 (“edited” subset, N=90,145). Outcome: “writer_english_first”. Reference category: “paragraph_type_writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
paragraph_type_model	3.859	[3.628, 4.104]	42.965	< .001

Table S111: Ordinal logistic regression predicting whether the writer is a native English speaker in Study 1 (“unedited” subset, N=90,112). Outcome: “writer_english_first”. Reference category: “paragraph_type_writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
paragraph_type_model	4.006	[3.764, 4.263]	43.650	< .001

Table S112: Ordinal logistic regression predicting perceived writer age in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_age”. Reference category: “paragraph_type_writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
paragraph_type_model	1.594	[1.550, 1.638]	33.001	< .001

Table S113: Ordinal logistic regression predicting perceived writer age in Study 1 (“edited” subset, N=90,145). Outcome: “writer_age”. Reference category: “paragraph_type_writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
paragraph_type_model	1.566	[1.526, 1.606]	34.701	< .001

Table S114: Ordinal logistic regression predicting perceived writer age in Study 1 (“unedited” subset, N=90,112). Outcome: “writer_age”. Reference category: “paragraph_type_writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
paragraph_type_model	1.575	[1.535, 1.615]	35.146	< .001

Table S115: Multinomial logistic regression predicting perceived writer race in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_race”. Predictor reference category: “paragraph_type_writer”. Outcome reference level: “White”. OR is odds ratio.

Term	Target level	Reference level	OR	95% CI (OR)	statistic	p
paragraph_type_model	Asian	White	0.888	[0.837, 0.942]	−3.938	8.200e-5
paragraph_type_model	Black	White	0.983	[0.932, 1.038]	−0.607	.544
paragraph_type_model	Mixed	White	0.955	[0.903, 1.011]	−1.593	.111
paragraph_type_model	Other	White	0.823	[0.761, 0.890]	−4.870	1.115e-6

Table S116: Multinomial logistic regression predicting perceived writer race in Study 1 (“edited” subset, N=90,145). Outcome: “writer_race”. Predictor reference category: “paragraph_type_writer”. Outcome reference level: “White”. OR is odds ratio.

Term	Target level	Reference level	OR	95% CI (OR)	statistic	p
paragraph_type_model	Asian	White	0.889	[0.842, 0.939]	−4.234	2.299e-5
paragraph_type_model	Black	White	0.989	[0.942, 1.039]	−0.435	.664
paragraph_type_model	Mixed	White	0.958	[0.910, 1.008]	−1.648	.099
paragraph_type_model	Other	White	0.832	[0.774, 0.894]	−5.016	5.280e-7

Table S117: Multinomial logistic regression predicting perceived writer race in Study 1 (“unedited” subset, N=90,112). Outcome: “writer_race”. Predictor reference category: “paragraph_type_writer”. Outcome reference level: “White”. OR is odds ratio.

Term	Target level	Reference level	OR	95% CI (OR)	statistic	p
paragraph_type_model	Asian	White	0.903	[0.855, 0.953]	−3.706	2.103e-4
paragraph_type_model	Black	White	0.995	[0.947, 1.045]	−0.219	.827
paragraph_type_model	Mixed	White	0.966	[0.918, 1.017]	−1.331	.183
paragraph_type_model	Other	White	0.817	[0.760, 0.878]	−5.484	4.153e-8

Table S118: Multinomial logistic regression predicting perceived writer gender in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_gender”. Predictor reference category: “paragraph_type_writer”. Outcome reference level: “Male”. OR is odds ratio.

Term	Target level	Reference level	OR	95% CI (OR)	statistic	p
paragraph_type_model	Female	Male	1.081	[1.050, 1.113]	5.271	1.359e-7
paragraph_type_model	Other	Male	1.190	[1.070, 1.323]	3.223	.001

Table S119: Multinomial logistic regression predicting perceived writer gender in Study 1 (“edited” subset, N=90,145). Outcome: “writer_gender”. Predictor reference category: “paragraph_type_writer”. Outcome reference level: “Male”. OR is odds ratio.

Term	Target level	Reference level	OR	95% CI (OR)	statistic	p
paragraph_type_model	Female	Male	1.077	[1.049, 1.106]	5.443	5.228e-8
paragraph_type_model	Other	Male	1.190	[1.081, 1.310]	3.543	3.957e-4

Table S120: Multinomial logistic regression predicting perceived writer gender in Study 1 (“unedited” subset, N=90,112). Outcome: “writer_gender”. Predictor reference category: “paragraph_type_writer”. Outcome reference level: “Male”. OR is odds ratio.

Term	Target level	Reference level	OR	95% CI (OR)	statistic	p
paragraph_type_model	Female	Male	1.081	[1.052, 1.110]	5.714	1.102e-8
paragraph_type_model	Other	Male	1.170	[1.063, 1.289]	3.190	.001

Table S121: Multinomial logistic regression predicting perceived writer political party in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_politicalParty”. Predictor reference category: “paragraph_type_writer”. Outcome reference level: “Labour”. OR is odds ratio.

Term	Target level	Reference level	OR	95% CI (OR)	statistic	p
paragraph_type_model	Conservative	Labour	1.238	[1.192, 1.286]	10.993	4.144e-28
paragraph_type_model	Did not vote	Labour	0.364	[0.344, 0.385]	−34.979	4.632e-268
paragraph_type_model	Greens	Labour	1.186	[1.123, 1.252]	6.140	8.247e-10
paragraph_type_model	Liberal Democrats	Labour	1.254	[1.198, 1.312]	9.709	2.754e-22
paragraph_type_model	Not eligible to vote	Labour	0.286	[0.251, 0.326]	−18.771	1.304e-78
paragraph_type_model	Other	Labour	0.721	[0.678, 0.766]	−10.524	6.714e-26
paragraph_type_model	Scottish National Party	Labour	1.042	[0.926, 1.174]	0.685	.493

Table S122: Multinomial logistic regression predicting perceived writer political party in Study 1 (“edited” subset, N=90,145). Outcome: “writer_politicalParty”. Predictor reference category: “paragraph_type_writer”. Outcome reference level: “Labour”. OR is odds ratio.

Term	Target level	Reference level	OR	95% CI (OR)	statistic	p
paragraph_type_model	Conservative	Labour	1.236	[1.193, 1.280]	11.857	1.977e-32
paragraph_type_model	Did not vote	Labour	0.378	[0.359, 0.398]	−36.859	2.115e-297
paragraph_type_model	Greens	Labour	1.174	[1.117, 1.234]	6.330	2.452e-10
paragraph_type_model	Liberal Democrats	Labour	1.266	[1.215, 1.320]	11.094	1.347e-28
paragraph_type_model	Not eligible to vote	Labour	0.306	[0.271, 0.344]	−19.551	4.083e-85
paragraph_type_model	Other	Labour	0.725	[0.686, 0.767]	−11.164	6.127e-29
paragraph_type_model	Scottish National Party	Labour	1.032	[0.927, 1.149]	0.579	.562

Table S123: Multinomial logistic regression predicting perceived writer political party in Study 1 (“unedited” subset, N=90,112). Outcome: “writer_politicalParty”. Predictor reference category: “paragraph_type_writer”. Outcome reference level: “Labour”. OR is odds ratio.

Term	Target level	Reference level	OR	95% CI (OR)	statistic	p
paragraph_type_model	Conservative	Labour	1.233	[1.190, 1.277]	11.720	1.003e-31
paragraph_type_model	Did not vote	Labour	0.367	[0.348, 0.387]	−37.676	1.245e-310
paragraph_type_model	Greens	Labour	1.180	[1.123, 1.240]	6.532	6.481e-11
paragraph_type_model	Liberal Democrats	Labour	1.263	[1.211, 1.317]	10.968	5.433e-28
paragraph_type_model	Not eligible to vote	Labour	0.282	[0.249, 0.319]	−20.280	1.937e-91
paragraph_type_model	Other	Labour	0.724	[0.684, 0.766]	−11.247	2.408e-29
paragraph_type_model	Scottish National Party	Labour	1.050	[0.944, 1.169]	0.899	.368

Table S124: Multinomial logistic regression predicting perceived writer political ideology in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_politicalIdeology”. Predictor reference category: “paragraph_type_writer”. Outcome reference level: “Centrist”. OR is odds ratio.

Term	Target level	Reference level	OR	95% CI (OR)	statistic	p
paragraph_type_model	Moderately Left-Wing	Centrist	1.008	[0.971, 1.047]	0.432	.666
paragraph_type_model	Moderately Right-Wing	Centrist	1.000	[0.959, 1.042]	−0.017	.987
paragraph_type_model	Other	Centrist	0.616	[0.577, 0.657]	−14.576	3.995e-48
paragraph_type_model	Very Left-Wing	Centrist	1.087	[1.033, 1.144]	3.207	.001
paragraph_type_model	Very Right-Wing	Centrist	0.860	[0.806, 0.918]	−4.518	6.235e-6

Table S125: Multinomial logistic regression predicting perceived writer political ideology in Study 1 (“edited” subset, N=90,145). Outcome: “writer_politicalIdeology”. Predictor reference category: “paragraph_type_writer”. Outcome reference level: “Centrist”. OR is odds ratio.

Term	Target level	Reference level	OR	95% CI (OR)	statistic	p
paragraph_type_model	Moderately Left-Wing	Centrist	1.000	[0.966, 1.035]	−0.016	.987
paragraph_type_model	Moderately Right-Wing	Centrist	0.992	[0.955, 1.031]	−0.415	.678
paragraph_type_model	Other	Centrist	0.621	[0.585, 0.660]	−15.550	1.581e-54
paragraph_type_model	Very Left-Wing	Centrist	1.083	[1.034, 1.135]	3.372	7.456e-4
paragraph_type_model	Very Right-Wing	Centrist	0.844	[0.795, 0.896]	−5.526	3.275e-8

Table S126: Multinomial logistic regression predicting perceived writer political ideology in Study 1 (“unedited” subset, N=90,112). Outcome: “writer_politicalIdeology”. Predictor reference category: “paragraph_type_writer”. Outcome reference level: “Centrist”. OR is odds ratio.

Term	Target level	Reference level	OR	95% CI (OR)	statistic	p
paragraph_type_model	Moderately Left-Wing	Centrist	1.004	[0.970, 1.039]	0.218	.827
paragraph_type_model	Moderately Right-Wing	Centrist	0.988	[0.951, 1.027]	−0.614	.539
paragraph_type_model	Other	Centrist	0.628	[0.592, 0.667]	−15.204	3.303e-52
paragraph_type_model	Very Left-Wing	Centrist	1.097	[1.047, 1.149]	3.887	1.016e-4
paragraph_type_model	Very Right-Wing	Centrist	0.856	[0.806, 0.908]	−5.098	3.426e-7

Table S127: One-vs-all logistic regression predicting perceived writer gender in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_gender”. Predictor reference category: “paragraph_type_writer”. OR is odds ratio.

Term	Target level	Reference level	OR	95% CI (OR)	statistic	p
paragraph_type_model	Female	all_other_levels	1.086	[1.053, 1.119]	5.301	1.150e-7
paragraph_type_model	Male	all_other_levels	0.911	[0.883, 0.939]	−6.042	1.519e-9
paragraph_type_model	Other	all_other_levels	1.532	[1.284, 1.828]	4.728	2.271e-6

Table S128: One-vs-all logistic regression predicting perceived writer race in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_race”. Predictor reference category: “paragraph_type_writer”. OR is odds ratio.

Term	Target level	Reference level	OR	95% CI (OR)	statistic	p
paragraph_type_model	Asian	all_other_levels	0.897	[0.846, 0.951]	−3.634	2.793e-4
paragraph_type_model	Black	all_other_levels	1.006	[0.953, 1.061]	0.202	0.840
paragraph_type_model	Mixed	all_other_levels	0.975	[0.922, 1.031]	−0.883	0.377
paragraph_type_model	Other	all_other_levels	0.730	[0.655, 0.814]	−5.676	1.380e-8
paragraph_type_model	White	all_other_levels	1.093	[1.054, 1.134]	4.767	1.866e-6

Table S129: One-vs-all logistic regression predicting perceived writer political party in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_politicalParty”. Predictor reference category: “paragraph_type_writer”. OR is odds ratio.

Term	Target level	Reference level	OR	95% CI (OR)	statistic	p
paragraph_type_model	Conservative	all_other_levels	1.442	[1.392, 1.494]	20.345	5.185e-92
paragraph_type_model	Did not vote	all_other_levels	0.297	[0.280, 0.315]	−40.843	0
paragraph_type_model	Greens	all_other_levels	1.294	[1.224, 1.369]	9.085	1.038e-19
paragraph_type_model	Labour	all_other_levels	1.054	[1.021, 1.089]	3.208	0.001
paragraph_type_model	Liberal Democrats	all_other_levels	1.376	[1.318, 1.435]	14.651	1.337e-48
paragraph_type_model	Not eligible to vote	all_other_levels	0.185	[0.156, 0.219]	−19.710	1.783e-86
paragraph_type_model	Other	all_other_levels	0.663	[0.619, 0.710]	−11.688	1.463e-31
paragraph_type_model	Scottish National Party	all_other_levels	1.090	[0.935, 1.270]	1.101	0.271

Table S130: One-vs-all logistic regression predicting perceived writer political ideology in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_politicalIdeology”. Predictor reference category: “paragraph_type_writer”. OR is odds ratio.

Term	Target level	Reference level	OR	95% CI (OR)	statistic	p
paragraph_type_model	Centrist	all_other_levels	1.043	[1.008, 1.079]	2.447	0.014
paragraph_type_model	Moderately Left-Wing	all_other_levels	1.052	[1.018, 1.088]	3.007	0.003
paragraph_type_model	Moderately Right-Wing	all_other_levels	1.037	[0.999, 1.076]	1.889	0.059
paragraph_type_model	Other	all_other_levels	0.404	[0.369, 0.443]	−19.562	3.268e-85
paragraph_type_model	Very Left-Wing	all_other_levels	1.148	[1.092, 1.208]	5.342	9.173e-8
paragraph_type_model	Very Right-Wing	all_other_levels	0.858	[0.802, 0.918]	−4.461	8.174e-6

5.8 Statistical Analysis: Random-Effects Structure

To quantify how much readers, writers, and propositions contribute to variation in reader perceptions, and how the choice of random-effects structure affects our estimates of AI writing assistance effects, we fit the beta regressions and one-vs-all logistic regressions from SI:5.7 to the “preferred” data subset (75,972 observations) under four random-effects structures: reader random intercepts only (R); reader and proposition random intercepts (R+P); reader and writer random intercepts (R+W); and crossed reader, writer, and proposition random intercepts (R+W+P), which is our pre-registered specification and the one we report in SI:5.7. **This analysis was not pre-registered.**

We decompose variance on the latent logit scale. For one-vs-all logistic regressions, the residual variance is $\pi^2/3$. For beta regressions, the residual variance is the variance of $\text{logit}(Y)$ for $Y \sim \text{Beta}(\mu\phi, (1-\mu)\phi)$, i.e. $\psi_1(\mu\phi) + \psi_1((1-\mu)\phi)$ where ψ_1 is the trigamma function, ϕ is the estimated precision parameter, and μ is the predicted mean for human-written paragraphs. Figure S32 and Tables S131 and S132 show the share of latent variance attributable to each random intercept. Across the 20 scale attributes, reader random intercepts accounted for a median of 14.5% (range 1.7–25.6%) of latent variance, writer random intercepts for 3.6% (range 1.2–6.2%), and proposition random intercepts for 1.2% (range 0.5–4.0%). Across the 22 nominal attribute categories, the median shares were 14.1% for readers, 3.0% for writers, and 1.6% for propositions. For the “Other” categories of `writer_gender`, `writer_race`, and `writer_politicalIdeology`, reader random intercepts accounted for most of the variance (88–93%): most readers never chose “Other”, while 40–45% of “Other” choices came from readers who chose it for at least half of the paragraphs they rated. Reader shares were similarly high for two other rarely chosen categories (“Scottish National Party” and “Not eligible to vote” for `writer_politicalParty`, each chosen in under 2% of ratings), for which variance components are less precisely estimated.

Writer and proposition random intercepts thus explain a small but non-negligible share of variance, and improve model fit (AIC lower by 2,016 to 9,264 for R+W+P vs. R across scale attributes). However, they leave our estimates of AI writing assistance effects essentially unchanged (Figures S33 and S34). Across scale attributes, AMEs from R and R+W+P models differed by at most 0.89 points on the 0-100 scale (median 0.22 points), with no changes in sign or statistical significance ($p < .05$), and standard errors of the paragraph type coefficient changed by a factor of 0.95 to 0.99. Across nominal attribute categories, there were no changes in statistical significance, and standard errors increased by at most 13%. This is expected from our study design: each writer contributed both a human-written and an AI-written paragraph for each of their propositions, so that writer- and proposition-level differences are balanced across paragraph types.

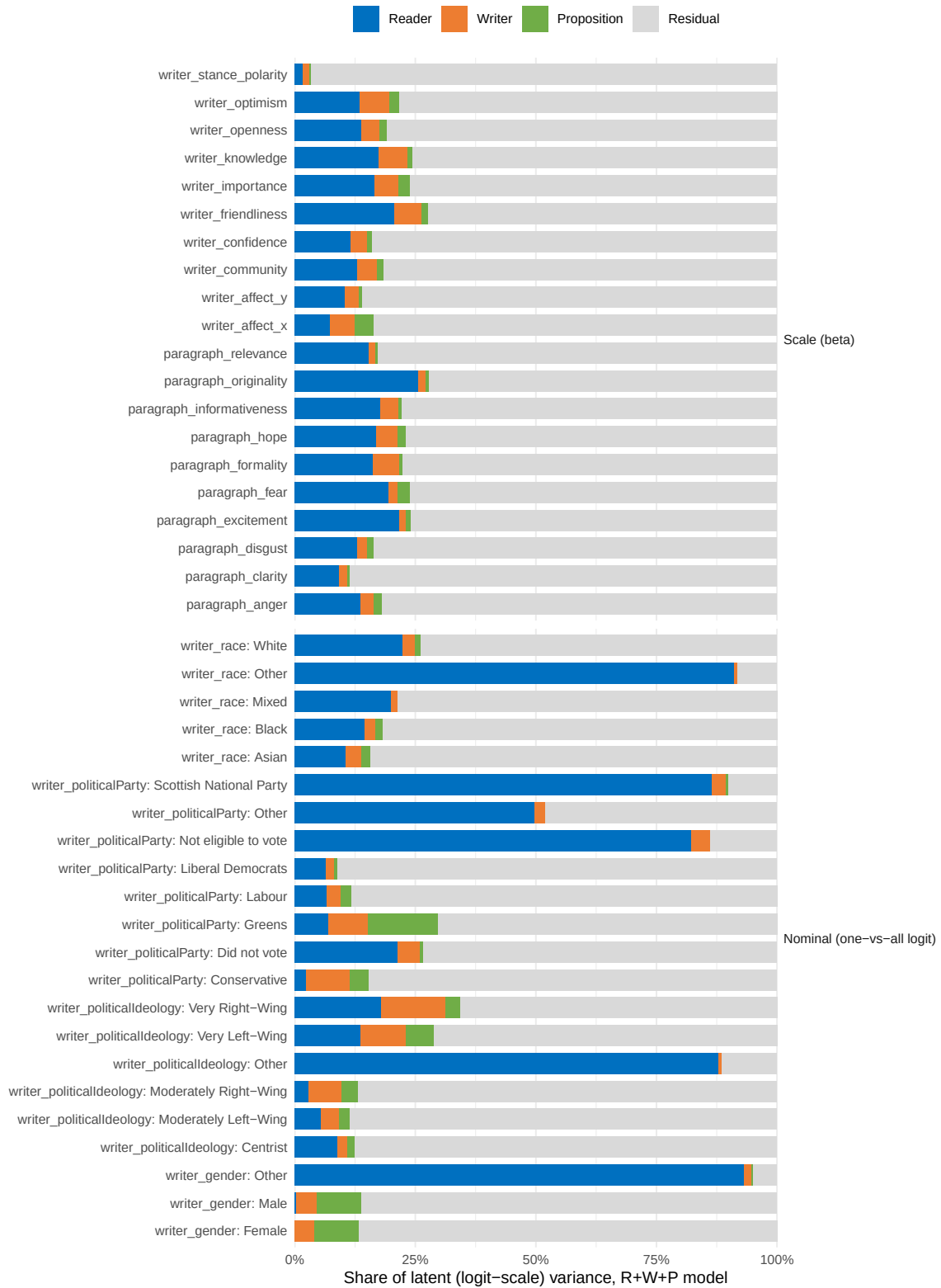


Figure S32: Share of latent (logit-scale) variance in reader perceptions attributable to readers, writers, and propositions in Study 1. Shares are from mixed-effects beta regressions (0-100 scale attributes) and one-vs-all logistic regressions (nominal attribute categories) with crossed reader, writer, and proposition random intercepts, fit to the “preferred” subset.

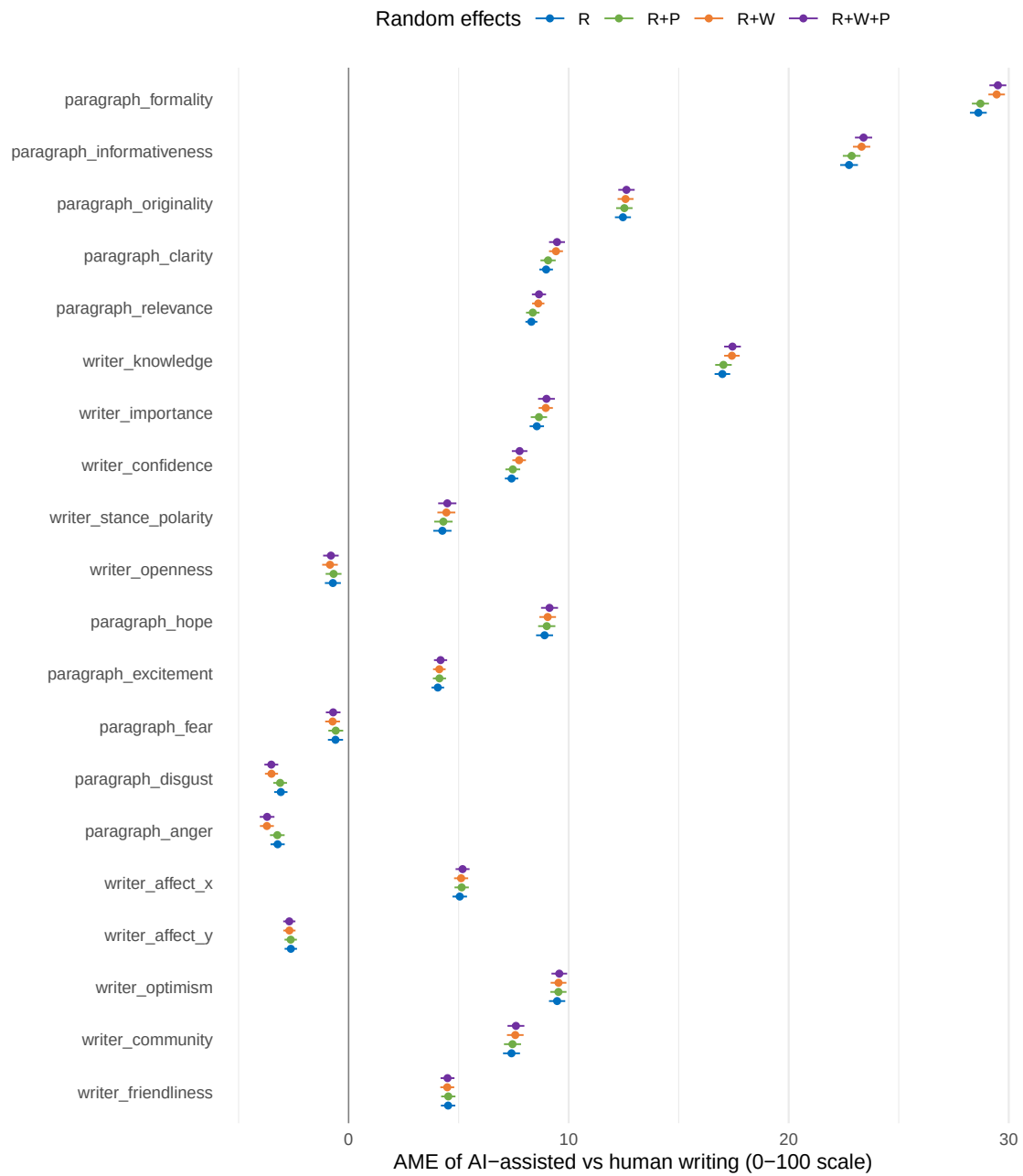


Figure S33: Average marginal effects (AMEs) of AI writing assistance on reader perceptions across all 0-100 scale rating attributes in Study 1, under four random-effects structures. Each point shows the AME from a mixed-effects beta regression comparing AI-written to human-written paragraphs in the “preferred” subset, with reader random intercepts only (R), reader and proposition random intercepts (R+P), reader and writer random intercepts (R+W), or crossed reader, writer, and proposition random intercepts (R+W+P). Error bars show 95% confidence intervals.

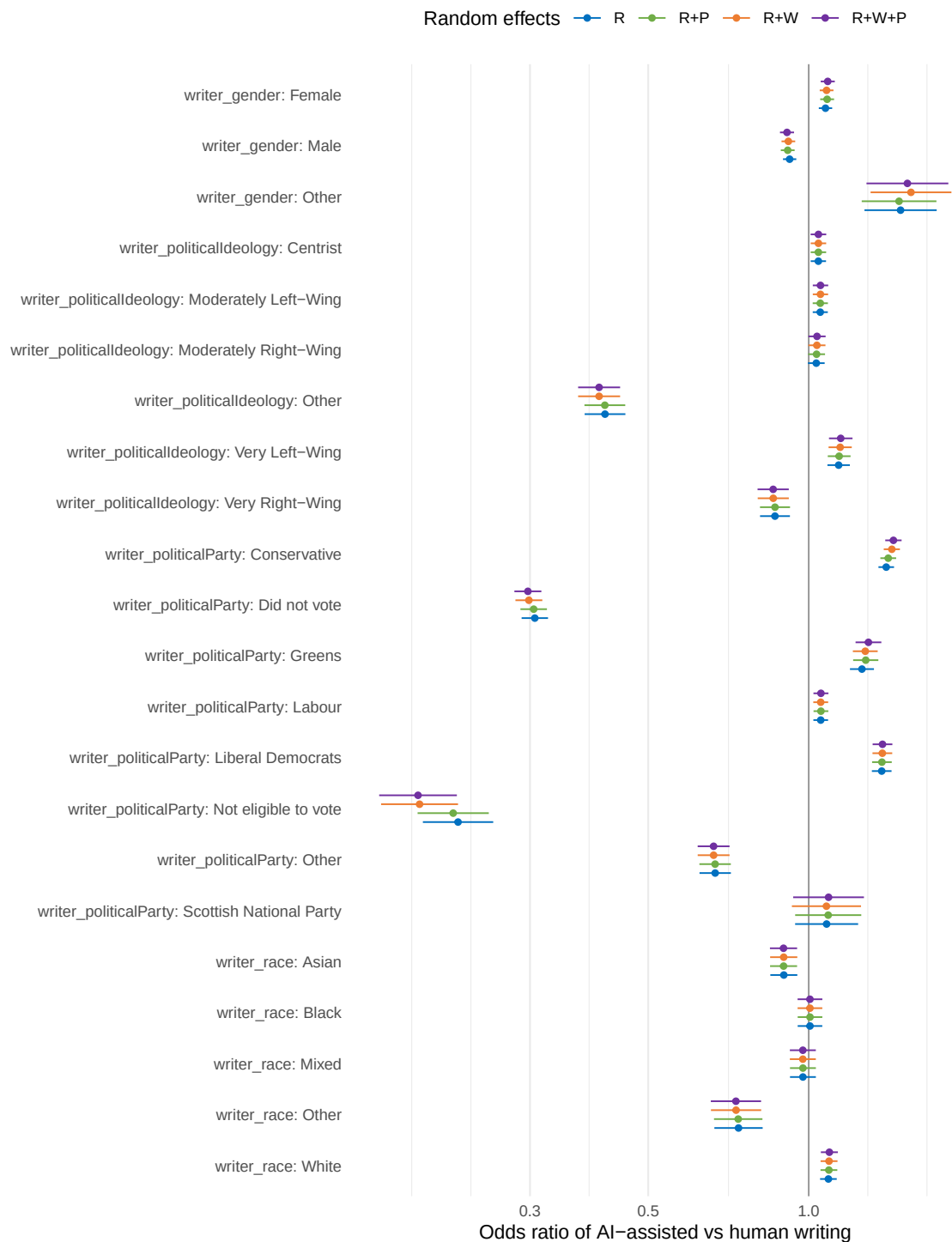


Figure S34: Odds ratios (ORs) of AI writing assistance on reader perceptions across all nominal attribute categories in Study 1, under four random-effects structures. Each point shows the odds ratio from a one-vs-all logistic regression comparing AI-written to human-written paragraphs in the “preferred” subset, with random-effects structures as in Figure S33. Error bars show 95% confidence intervals.

Table S131: Variance decomposition and AI writing assistance effects under different random-effects structures for 0-100 scale rating attributes in Study 1 (“preferred” subset, N=75,972). Reader, writer, and proposition columns show each random intercept’s share of latent (logit-scale) variance in the model with crossed reader, writer, and proposition random intercepts. AME is the average marginal effect of AI writing assistance (AI-written vs. human-written paragraphs, in points on the 0-100 scale) with 95% CI, from models with reader random intercepts only (R) and with crossed reader, writer, and proposition random intercepts (R+W+P). Δ AIC is the difference in AIC between the R+W+P and R models (negative values favour R+W+P).

Attribute	Reader	Writer	Proposition	AME (R)	AME (R+W+P)	Δ AIC
writer_knowledge	17.4%	5.9%	1.0%	16.99 [16.63, 17.34]	17.45 [17.07, 17.83]	−6,859
writer_importance	16.6%	4.9%	2.4%	8.55 [8.23, 8.88]	8.99 [8.61, 9.37]	−9,150
writer_confidence	11.5%	3.6%	1.1%	7.41 [7.10, 7.71]	7.77 [7.42, 8.13]	−7,134
writer_stance_polarity	1.7%	1.3%	0.5%	4.26 [3.85, 4.67]	4.48 [4.07, 4.90]	−8,218
paragraph_formality	16.1%	5.6%	0.6%	28.62 [28.24, 29.00]	29.51 [29.12, 29.89]	−7,185
paragraph_clarity	9.2%	1.8%	0.5%	8.98 [8.67, 9.29]	9.47 [9.11, 9.83]	−4,089
paragraph_informativeness	17.7%	3.9%	0.7%	22.74 [22.34, 23.15]	23.40 [23.01, 23.79]	−5,820
paragraph_originality	25.6%	1.6%	0.7%	12.47 [12.10, 12.83]	12.63 [12.26, 13.00]	−2,671
paragraph_relevance	15.3%	1.4%	0.5%	8.31 [8.04, 8.58]	8.65 [8.33, 8.97]	−2,911
writer_optimism	13.5%	6.2%	2.0%	9.48 [9.11, 9.85]	9.58 [9.22, 9.93]	−8,327
writer_community	13.0%	4.0%	1.5%	7.40 [7.02, 7.79]	7.60 [7.22, 7.99]	−6,378
writer_friendliness	20.7%	5.6%	1.4%	4.52 [4.19, 4.85]	4.50 [4.18, 4.81]	−6,173
writer_openness	13.8%	3.7%	1.6%	−0.72 [−1.08, −0.35]	−0.80 [−1.16, −0.45]	−8,262
paragraph_hope	17.0%	4.3%	1.8%	8.91 [8.52, 9.29]	9.13 [8.75, 9.52]	−7,931
paragraph_excitement	21.8%	1.2%	1.1%	4.05 [3.77, 4.34]	4.18 [3.88, 4.48]	−4,264
paragraph_fear	19.5%	1.9%	2.5%	−0.60 [−0.94, −0.25]	−0.70 [−1.03, −0.37]	−7,933
paragraph_disgust	12.9%	2.1%	1.4%	−3.08 [−3.38, −2.78]	−3.51 [−3.83, −3.20]	−7,704
paragraph_anger	13.6%	2.8%	1.7%	−3.23 [−3.55, −2.90]	−3.70 [−4.04, −3.37]	−9,264
writer_affect_x	7.4%	5.0%	4.0%	5.05 [4.72, 5.38]	5.18 [4.86, 5.50]	−8,428
writer_affect_y	10.4%	2.9%	0.8%	−2.63 [−2.91, −2.34]	−2.69 [−2.97, −2.42]	−2,016

Table S132: Variance decomposition and AI writing assistance effects under different random-effects structures for nominal rating attributes in Study 1 (“preferred” subset, N=75,972), from one-vs-all logistic regressions. Reader, writer, and proposition columns show each random intercept’s share of latent (logit-scale) variance in the model with crossed reader, writer, and proposition random intercepts. OR is the odds ratio of AI writing assistance (AI-written vs. human-written paragraphs) for the target category vs. all other categories, with 95% CI, from models with reader random intercepts only (R) and with crossed reader, writer, and proposition random intercepts (R+W+P). Δ AIC is the difference in AIC between the R+W+P and R models (negative values favour R+W+P).

Attribute	Target level	Reader	Writer	Proposition	OR (R)	OR (R+W+P)	Δ AIC
writer_politicalParty	Conservative	2.4%	9.0%	4.0%	1.398 [1.352, 1.446]	1.442 [1.392, 1.494]	−3,621
writer_politicalParty	Did not vote	21.4%	4.5%	0.7%	0.306 [0.289, 0.324]	0.297 [0.280, 0.315]	−419
writer_politicalParty	Greens	7.0%	8.3%	14.4%	1.259 [1.195, 1.326]	1.294 [1.224, 1.369]	−4,912
writer_politicalParty	Labour	6.7%	2.8%	2.2%	1.053 [1.021, 1.087]	1.054 [1.021, 1.089]	−1,421
writer_politicalParty	Liberal Democrats	6.4%	1.7%	0.7%	1.371 [1.314, 1.430]	1.376 [1.318, 1.435]	−199
writer_politicalParty	Not eligible to vote	82.2%	3.9%	0.0%	0.220 [0.189, 0.256]	0.185 [0.156, 0.219]	−187
writer_politicalParty	Other	49.7%	2.2%	0.1%	0.668 [0.624, 0.715]	0.663 [0.619, 0.710]	−123
writer_politicalParty	Scottish National Party	86.6%	2.8%	0.5%	1.081 [0.943, 1.238]	1.090 [0.935, 1.270]	−466
writer_politicalIdeology	Centrist	8.8%	2.0%	1.7%	1.043 [1.009, 1.078]	1.043 [1.008, 1.079]	−887
writer_politicalIdeology	Moderately Left-Wing	5.5%	3.8%	2.2%	1.051 [1.018, 1.086]	1.052 [1.018, 1.088]	−1,521
writer_politicalIdeology	Moderately Right-Wing	2.8%	6.9%	3.4%	1.033 [0.997, 1.071]	1.037 [0.999, 1.076]	−2,269
writer_politicalIdeology	Other	87.9%	0.7%	0.0%	0.415 [0.380, 0.453]	0.404 [0.369, 0.443]	−64
writer_politicalIdeology	Very Left-Wing	13.6%	9.4%	5.8%	1.138 [1.084, 1.195]	1.148 [1.092, 1.208]	−2,381
writer_politicalIdeology	Very Right-Wing	17.9%	13.4%	3.1%	0.864 [0.810, 0.922]	0.858 [0.802, 0.918]	−1,552
writer_gender	Female	0.0%	4.1%	9.2%	1.075 [1.045, 1.107]	1.086 [1.053, 1.119]	−5,972
writer_gender	Male	0.2%	4.4%	9.3%	0.921 [0.895, 0.948]	0.911 [0.883, 0.939]	−6,170
writer_gender	Other	93.3%	1.4%	0.4%	1.487 [1.272, 1.738]	1.532 [1.284, 1.828]	−236
writer_race	Asian	10.6%	3.2%	1.9%	0.898 [0.847, 0.952]	0.897 [0.846, 0.951]	−417
writer_race	Black	14.5%	2.2%	1.5%	1.006 [0.954, 1.060]	1.006 [0.953, 1.061]	−333
writer_race	Mixed	19.9%	1.4%	0.0%	0.975 [0.922, 1.031]	0.975 [0.922, 1.031]	−27
writer_race	Other	91.0%	0.8%	0.0%	0.739 [0.665, 0.820]	0.730 [0.655, 0.814]	−65
writer_race	White	22.4%	2.5%	1.3%	1.089 [1.050, 1.129]	1.093 [1.054, 1.134]	−942

5.9 Statistical Analysis: Effects of AI Writing Assistance by AI Model

To quantify variation in distortions across AI models, we fit **mixed-effects regression models** specified as follows:

$$\text{rating_attribute} \sim \text{model} + (1 \mid \text{reader_id}) + (1 \mid \text{writer_id}) + (1 \mid \text{proposition_id})$$

where `rating_attribute` and the random intercepts are defined as in SI:5.7, and `model` is a categorical fixed effect taking four values: one for each of the three AI models (Claude Sonnet 4, ChatGPT-4o, DeepSeek V3) and one for human-written paragraphs, which serves as the reference category. The measurement scale of each rating attribute again determined the appropriate link function, following the same scheme as in SI:5.7: beta regression for 0-100 scale attributes, multinomial logistic regression for nominal categorical attributes, and ordinal logistic regression for ordinal categorical attributes.

Having confirmed that results are highly consistent across data subsets (SI:5.7), we fit each regression model to the “preferred” data subset (75,972 observations) as defined in SI:5.2. Figures S35, S36, and S37 summarise results for scale, ordinal, and nominal variables. Tables S133 to S160 show results for each of the 29 rating attributes in our study. Table S162 shows average distortion magnitude by model.

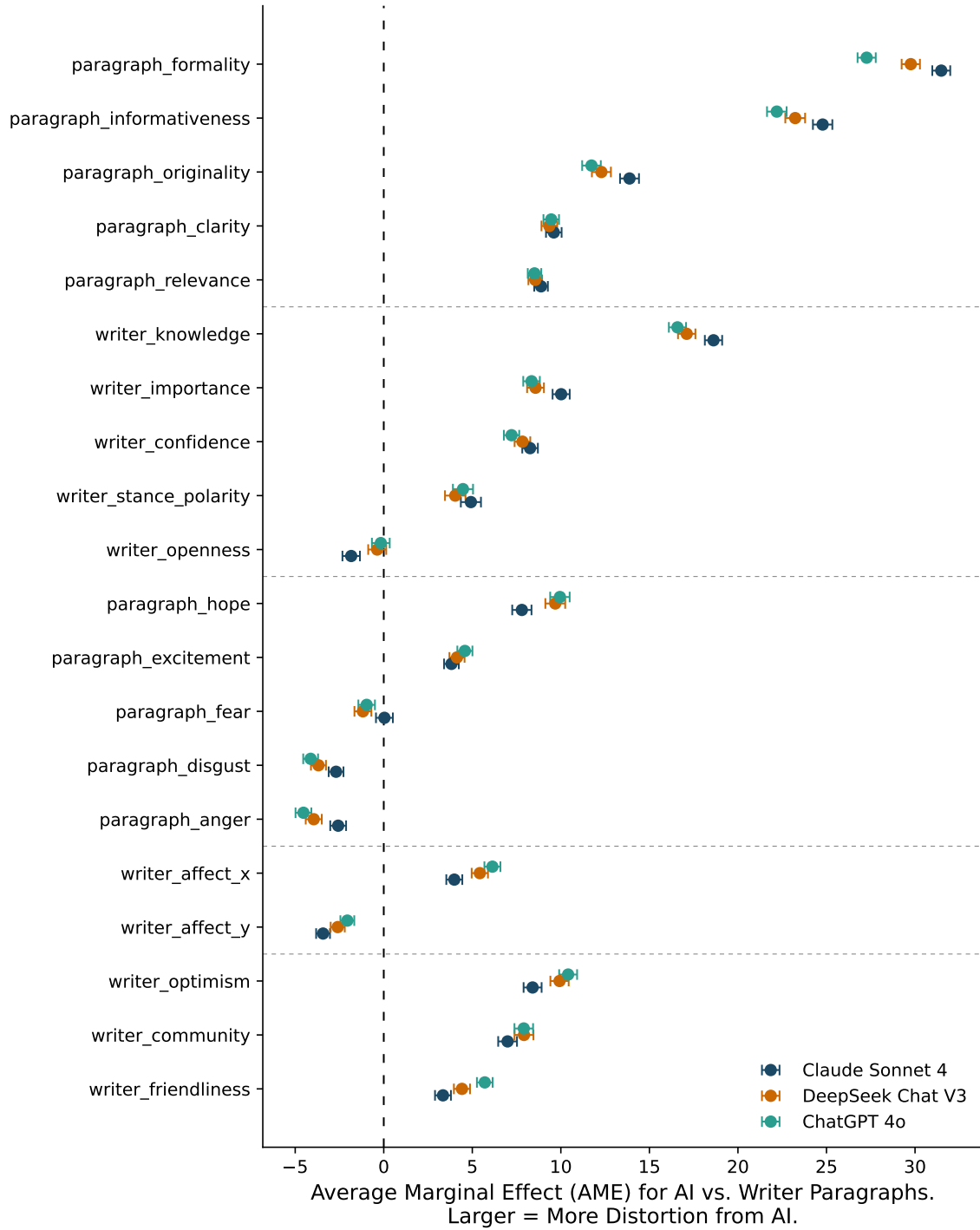


Figure S35: Average marginal effects (AMEs) of AI writing assistance on reader perceptions across all 0-100 scale rating attributes in Study 1. Each point shows the AME from a mixed-effects beta regression comparing AI-written to human-written paragraphs, with crossed reader, writer, and proposition random effects. Results shown separately for each AI model. Error bars show 95% confidence intervals.

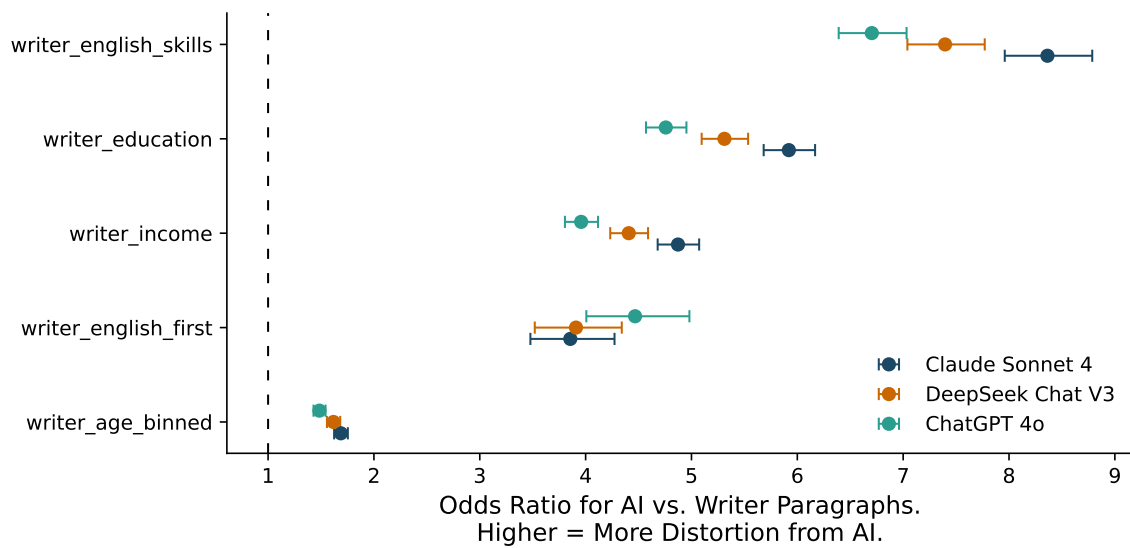


Figure S36: Odds ratios (ORs) of AI writing assistance on reader perceptions across all ordinal rating attributes in Study 1. Each point shows the odds ratio from a mixed-effects ordinal logistic regression comparing AI-written to human-written paragraphs, with reader random effects. Results shown separately for each AI model. Error bars show 95% confidence intervals.

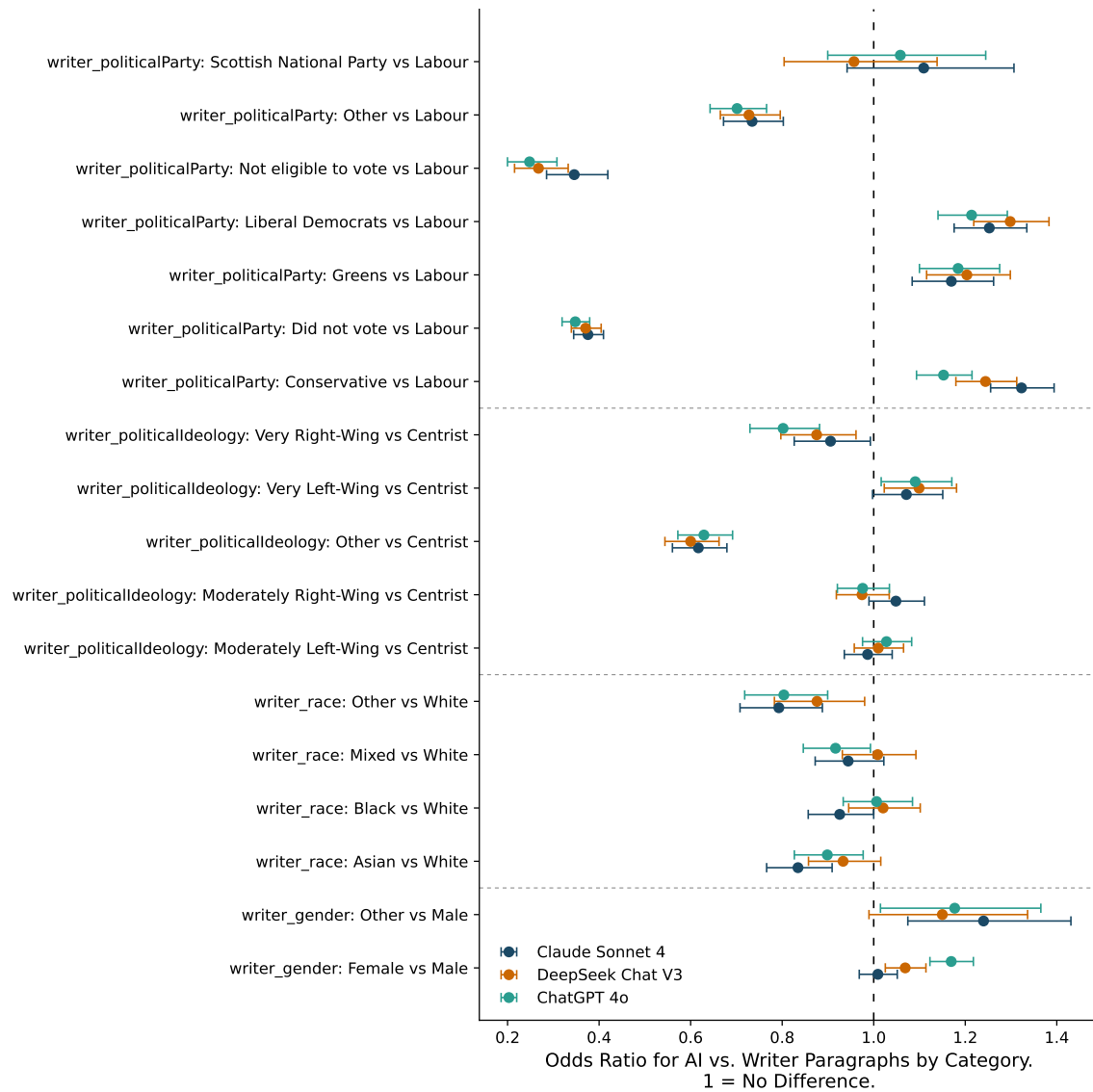


Figure S37: Odds ratios (ORs) of AI writing assistance on reader perceptions across all nominal rating attributes in Study 1. Each point shows the odds ratio of a specific category vs a reference category from a mixed-effects multinomial logistic regression comparing AI-written to human-written paragraphs, with reader random effects. Results shown separately for each AI model. Error bars show 95% confidence intervals.

Table S133: Beta regression predicting paragraph formality by model in Study 1 (“preferred” subset, N=75,972). Outcome: “paragraph_formality”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
model_anthropic/claude-sonnet-4	3.793	[3.706, 3.881]	113.618	0	31.481	[30.975, 31.988]
model_deepseek/deepseek-chat-v3-0324	3.484	[3.404, 3.566]	105.186	0	29.761	[29.245, 30.277]
model_openai/chatgpt-4o-latest	3.097	[3.028, 3.168]	97.761	0	27.268	[26.755, 27.781]

Table S134: Beta regression predicting paragraph clarity by model in Study 1 (“preferred” subset, N=75,972). Outcome: “paragraph_clarity”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
model_anthropic/claude-sonnet-4	1.744	[1.702, 1.786]	45.375	0	9.602	[9.162, 10.042]
model_deepseek/deepseek-chat-v3-0324	1.714	[1.673, 1.756]	43.290	0	9.349	[8.905, 9.793]
model_openai/chatgpt-4o-latest	1.726	[1.686, 1.768]	44.743	0	9.457	[9.018, 9.895]

Table S135: Beta regression predicting paragraph informativeness by model in Study 1 (“preferred” subset, N=75,972). Outcome: “paragraph_informativeness”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
model_anthropic/claude-sonnet-4	2.754	[2.690, 2.820]	84.460	0	24.782	[24.234, 25.330]
model_deepseek/deepseek-chat-v3-0324	2.577	[2.517, 2.639]	78.413	0	23.233	[22.675, 23.791]
model_openai/chatgpt-4o-latest	2.467	[2.410, 2.525]	75.912	0	22.195	[21.641, 22.749]

Table S136: Beta regression predicting paragraph originality by model in Study 1 (“preferred” subset, N=75,972). Outcome: “paragraph_originality”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
model_anthropic/claude-sonnet-4	1.809	[1.770, 1.848]	53.334	0	13.877	[13.343, 14.412]
model_deepseek/deepseek-chat-v3-0324	1.696	[1.659, 1.734]	47.024	0	12.290	[11.753, 12.826]
model_openai/chatgpt-4o-latest	1.658	[1.622, 1.694]	45.740	0	11.730	[11.205, 12.256]

Table S137: Beta regression predicting paragraph relevance by model in Study 1 (“preferred” subset, N=75,972). Outcome: “paragraph_relevance”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
model_anthropic/claude-sonnet-4	1.788	[1.747, 1.829]	49.195	0	8.883	[8.498, 9.267]
model_deepseek/deepseek-chat-v3-0324	1.742	[1.702, 1.784]	46.339	0	8.556	[8.170, 8.942]
model_openai/chatgpt-4o-latest	1.736	[1.697, 1.777]	47.055	0	8.513	[8.132, 8.894]

Table S138: Beta regression predicting perceived writer knowledge by model in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_knowledge”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
model_anthropic/claude-sonnet-4	2.289	[2.241, 2.339]	75.364	0	18.620	[18.132, 19.109]
model_deepseek/deepseek-chat-v3-0324	2.120	[2.074, 2.166]	67.953	0	17.107	[16.616, 17.598]
model_openai/chatgpt-4o-latest	2.066	[2.022, 2.110]	66.643	0	16.585	[16.100, 17.071]

Table S139: Beta regression predicting perceived writer importance by model in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_importance”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
model_anthropic/claude-sonnet-4	1.665	[1.629, 1.702]	45.357	0	10.017	[9.534, 10.500]
model_deepseek/deepseek-chat-v3-0324	1.534	[1.501, 1.569]	37.682	9.869e-311	8.570	[8.094, 9.045]
model_openai/chatgpt-4o-latest	1.516	[1.483, 1.549]	37.217	3.665e-303	8.345	[7.877, 8.813]

Table S140: Beta regression predicting perceived writer confidence by model in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_confidence”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
model_anthropic/claude-sonnet-4	1.603	[1.566, 1.641]	39.750	0	8.263	[7.822, 8.703]
model_deepseek/deepseek-chat-v3-0324	1.559	[1.523, 1.596]	36.924	1.891e-298	7.832	[7.390, 8.273]
model_openai/chatgpt-4o-latest	1.499	[1.465, 1.534]	34.356	1.135e-258	7.216	[6.783, 7.649]

Table S141: Beta regression predicting perceived writer stance polarity by model in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_stance_polarity”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
model_anthropic/claude-sonnet-4	1.254	[1.222, 1.288]	16.942	2.209e-64	4.921	[4.353, 5.489]
model_deepseek/deepseek-chat-v3-0324	1.202	[1.171, 1.235]	13.585	4.890e-42	4.037	[3.458, 4.615]
model_openai/chatgpt-4o-latest	1.228	[1.196, 1.260]	15.376	2.367e-53	4.473	[3.905, 5.041]

Table S142: Beta regression predicting paragraph hope by model in Study 1 (“preferred” subset, N=75,972). Outcome: “paragraph_hope”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
model_anthropic/claude-sonnet-4	1.389	[1.358, 1.420]	28.909	9.094e-184	7.798	[7.254, 8.343]
model_deepseek/deepseek-chat-v3-0324	1.500	[1.466, 1.534]	35.039	5.726e-269	9.685	[9.125, 10.245]
model_openai/chatgpt-4o-latest	1.516	[1.482, 1.550]	36.666	2.576e-294	9.948	[9.398, 10.497]

Table S143: Beta regression predicting paragraph excitement by model in Study 1 (“preferred” subset, N=75,972). Outcome: “paragraph_excitement”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
model_anthropic/claude-sonnet-4	1.241	[1.214, 1.269]	19.135	1.296e-81	3.822	[3.406, 4.238]
model_deepseek/deepseek-chat-v3-0324	1.262	[1.234, 1.291]	20.301	1.254e-91	4.135	[3.708, 4.562]
model_openai/chatgpt-4o-latest	1.292	[1.264, 1.321]	22.795	5.105e-115	4.581	[4.156, 5.006]

Table S144: Beta regression predicting paragraph fear by model in Study 1 (“preferred” subset, N=75,972). Outcome: “paragraph_fear”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
model_anthropic/claude-sonnet-4	1.002	[0.979, 1.025]	0.154	.878	0.037	[−0.439, 0.514]
model_deepseek/deepseek-chat-v3-0324	0.944	[0.922, 0.966]	−4.823	1.413e-6	−1.178	[−1.655, −0.701]
model_openai/chatgpt-4o-latest	0.954	[0.932, 0.976]	−4.039	5.358e-5	−0.966	[−1.433, −0.499]

Table S145: Beta regression predicting paragraph disgust by model in Study 1 (“preferred” subset, N=75,972). Outcome: “paragraph_disgust”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
model_anthropic/claude-sonnet-4	0.857	[0.837, 0.878]	−12.705	5.546e-37	−2.693	[−3.110, −2.275]
model_deepseek/deepseek-chat-v3-0324	0.807	[0.788, 0.827]	−17.402	8.018e-68	−3.687	[−4.111, −3.263]
model_openai/chatgpt-4o-latest	0.785	[0.767, 0.804]	−20.065	1.480e-89	−4.129	[−4.547, −3.711]

Table S146: Beta regression predicting paragraph anger by model in Study 1 (“preferred” subset, N=75,972). Outcome: “paragraph_anger”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
model_anthropic/claude-sonnet-4	0.872	[0.851, 0.893]	−11.308	1.205e-29	−2.574	[−3.022, −2.127]
model_deepseek/deepseek-chat-v3-0324	0.806	[0.787, 0.826]	−17.423	5.520e-68	−3.953	[−4.406, −3.499]
model_openai/chatgpt-4o-latest	0.779	[0.761, 0.798]	−20.634	1.353e-94	−4.535	[−4.982, −4.088]

Table S147: Beta regression predicting perceived writer affect x by model in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_affect_x”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
model_anthropic/claude-sonnet-4	1.178	[1.156, 1.199]	17.440	4.136e-68	3.984	[3.533, 4.435]
model_deepseek/deepseek-chat-v3-0324	1.248	[1.225, 1.272]	23.355	1.228e-120	5.424	[4.965, 5.884]
model_openai/chatgpt-4o-latest	1.284	[1.261, 1.308]	26.885	3.263e-159	6.136	[5.684, 6.588]

Table S148: Beta regression predicting perceived writer affect y by model in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_affect_y”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
model_anthropic/claude-sonnet-4	0.868	[0.854, 0.882]	−17.083	1.990e-65	−3.428	[−3.818, −3.037]
model_deepseek/deepseek-chat-v3-0324	0.898	[0.884, 0.913]	−12.752	3.029e-37	−2.599	[−2.996, −2.201]
model_openai/chatgpt-4o-latest	0.919	[0.904, 0.934]	−10.272	9.425e-25	−2.057	[−2.448, −1.666]

Table S149: Beta regression predicting perceived writer optimism by model in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_optimism”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
model_anthropic/claude-sonnet-4	1.403	[1.375, 1.432]	32.523	5.125e-232	8.405	[7.897, 8.913]
model_deepseek/deepseek-chat-v3-0324	1.492	[1.461, 1.523]	37.740	1.079e-311	9.926	[9.410, 10.442]
model_openai/chatgpt-4o-latest	1.521	[1.490, 1.552]	40.287	0	10.411	[9.905, 10.918]

Table S150: Beta regression predicting perceived writer community orientation by model in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_community”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
model_anthropic/claude-sonnet-4	1.344	[1.314, 1.374]	25.583	2.366e-144	6.993	[6.462, 7.523]
model_deepseek/deepseek-chat-v3-0324	1.400	[1.368, 1.432]	28.688	5.354e-181	7.918	[7.384, 8.453]
model_openai/chatgpt-4o-latest	1.399	[1.368, 1.431]	29.208	1.547e-187	7.906	[7.381, 8.432]

Table S151: Beta regression predicting perceived writer friendliness by model in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_friendliness”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
model_anthropic/claude-sonnet-4	1.144	[1.123, 1.165]	14.545	6.310e-48	3.344	[2.895, 3.793]
model_deepseek/deepseek-chat-v3-0324	1.195	[1.173, 1.217]	18.963	3.442e-80	4.416	[3.962, 4.870]
model_openai/chatgpt-4o-latest	1.259	[1.237, 1.282]	25.003	5.601e-138	5.705	[5.260, 6.149]

Table S152: Beta regression predicting perceived writer openness by model in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_openness”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
model_anthropic/claude-sonnet-4	0.918	[0.897, 0.939]	−7.259	3.908e-13	−1.838	[−2.332, −1.345]
model_deepseek/deepseek-chat-v3-0324	0.983	[0.960, 1.006]	−1.421	.155	−0.369	[−0.878, 0.139]
model_openai/chatgpt-4o-latest	0.992	[0.970, 1.016]	−0.648	.517	−0.166	[−0.668, 0.336]

Table S153: Ordinal logistic regression predicting perceived writer first-language status by model in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_english_first”. Reference category: “writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
model_anthropic/claude-sonnet-4	3.855	[3.478, 4.273]	25.708	9.496e-146
model_deepseek/deepseek-chat-v3-0324	3.909	[3.520, 4.341]	25.461	5.297e-143
model_openai/chatgpt-4o-latest	4.467	[4.007, 4.981]	26.968	3.507e-160

Table S154: Ordinal logistic regression predicting perceived writer education level by model in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_education”. Reference category: “writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
model_anthropic/claude-sonnet-4	5.920	[5.682, 6.168]	84.968	0
model_deepseek/deepseek-chat-v3-0324	5.311	[5.096, 5.535]	79.244	0
model_openai/chatgpt-4o-latest	4.758	[4.570, 4.952]	76.160	0

Table S155: Ordinal logistic regression predicting perceived writer English skills by model in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_english_skills”. Reference category: “writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
model_anthropic/claude-sonnet-4	8.363	[7.960, 8.786]	84.262	0
model_deepseek/deepseek-chat-v3-0324	7.396	[7.040, 7.771]	79.424	0
model_openai/chatgpt-4o-latest	6.704	[6.391, 7.032]	77.990	0

Table S156: Ordinal logistic regression predicting perceived writer income level by model in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_income”. Reference category: “writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
model_anthropic/claude-sonnet-4	4.873	[4.681, 5.073]	77.207	0
model_deepseek/deepseek-chat-v3-0324	4.407	[4.232, 4.590]	71.749	0
model_openai/chatgpt-4o-latest	3.958	[3.804, 4.117]	68.233	0

Table S157: Ordinal logistic regression predicting perceived writer age group by model in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_age_binned”. Reference category: “writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
model_anthropic/claude-sonnet-4	1.687	[1.624, 1.754]	26.683	7.394e-157
model_deepseek/deepseek-chat-v3-0324	1.618	[1.556, 1.682]	24.293	2.312e-130
model_openai/chatgpt-4o-latest	1.485	[1.429, 1.543]	20.250	3.562e-91

Table S158: Multinomial logistic regression predicting perceived writer race by model in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_race”. Predictor reference category: “writer”. Outcome reference level: “White”. OR is odds ratio.

Term	Target level	Reference level	OR	95% CI (OR)	statistic	p
model_anthropic/claude-sonnet-4	Asian	White	0.835	[0.766, 0.909]	−4.141	3.464e-5
model_anthropic/claude-sonnet-4	Black	White	0.926	[0.857, 1.000]	−1.962	.050
model_anthropic/claude-sonnet-4	Mixed	White	0.944	[0.872, 1.022]	−1.419	.156
model_anthropic/claude-sonnet-4	Other	White	0.793	[0.708, 0.888]	−4.020	5.820e-5
model_deepseek/deepseek-chat-v3-0324	Asian	White	0.933	[0.858, 1.015]	−1.605	.108
model_deepseek/deepseek-chat-v3-0324	Black	White	1.020	[0.945, 1.102]	0.520	.603
model_deepseek/deepseek-chat-v3-0324	Mixed	White	1.009	[0.931, 1.092]	0.212	.832
model_deepseek/deepseek-chat-v3-0324	Other	White	0.876	[0.783, 0.980]	−2.308	.021
model_openai/chatgpt-4o-latest	Asian	White	0.899	[0.827, 0.977]	−2.506	.012
model_openai/chatgpt-4o-latest	Black	White	1.006	[0.933, 1.085]	0.162	.871
model_openai/chatgpt-4o-latest	Mixed	White	0.917	[0.846, 0.993]	−2.135	.033
model_openai/chatgpt-4o-latest	Other	White	0.804	[0.718, 0.899]	−3.806	1.413e-4

Table S159: Multinomial logistic regression predicting perceived writer gender by model in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_gender”. Predictor reference category: “writer”. Outcome reference level: “Male”. OR is odds ratio.

Term	Target level	Reference level	OR	95% CI (OR)	statistic	p
model_anthropic/claude-sonnet-4	Female	Male	1.009	[0.968, 1.051]	0.432	.666
model_anthropic/claude-sonnet-4	Other	Male	1.240	[1.075, 1.431]	2.944	.003
model_deepseek/deepseek-chat-v3-0324	Female	Male	1.069	[1.025, 1.114]	3.143	.002
model_deepseek/deepseek-chat-v3-0324	Other	Male	1.150	[0.990, 1.336]	1.825	.068
model_openai/chatgpt-4o-latest	Female	Male	1.169	[1.123, 1.218]	7.555	4.197e-14
model_openai/chatgpt-4o-latest	Other	Male	1.177	[1.015, 1.365]	2.153	.031

Table S160: Multinomial logistic regression predicting perceived writer political party by model in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_politicalParty”. Predictor reference category: “writer”. Outcome reference level: “Labour”. OR is odds ratio.

Term	Target level	Reference level	OR	95% CI (OR)	statistic	p
model_anthropic/claude-sonnet-4	Conservative	Labour	1.323	[1.255, 1.394]	10.470	1.187e-25
model_anthropic/claude-sonnet-4	Did not vote	Labour	0.376	[0.344, 0.410]	−22.167	7.200e-109
model_anthropic/claude-sonnet-4	Greens	Labour	1.170	[1.084, 1.262]	4.042	5.305e-5
model_anthropic/claude-sonnet-4	Liberal Democrats	Labour	1.253	[1.176, 1.335]	6.968	3.214e-12
model_anthropic/claude-sonnet-4	Not eligible to vote	Labour	0.346	[0.285, 0.419]	−10.821	2.723e-27
model_anthropic/claude-sonnet-4	Other	Labour	0.734	[0.672, 0.803]	−6.814	9.505e-12
model_anthropic/claude-sonnet-4	Scottish National Party	Labour	1.109	[0.942, 1.306]	1.242	.214
model_deepseek/deepseek-chat-v3-0324	Conservative	Labour	1.244	[1.180, 1.313]	8.020	1.056e-15
model_deepseek/deepseek-chat-v3-0324	Did not vote	Labour	0.371	[0.339, 0.405]	−22.167	7.081e-109
model_deepseek/deepseek-chat-v3-0324	Greens	Labour	1.204	[1.116, 1.298]	4.783	1.730e-6
model_deepseek/deepseek-chat-v3-0324	Liberal Democrats	Labour	1.298	[1.219, 1.383]	8.086	6.168e-16
model_deepseek/deepseek-chat-v3-0324	Not eligible to vote	Labour	0.268	[0.215, 0.332]	−11.892	1.299e-32
model_deepseek/deepseek-chat-v3-0324	Other	Labour	0.727	[0.665, 0.796]	−6.943	3.843e-12
model_deepseek/deepseek-chat-v3-0324	Scottish National Party	Labour	0.957	[0.804, 1.138]	−0.497	.619
model_openai/chatgpt-4o-latest	Conservative	Labour	1.153	[1.094, 1.215]	5.295	1.191e-7
model_openai/chatgpt-4o-latest	Did not vote	Labour	0.348	[0.319, 0.379]	−23.902	2.917e-126
model_openai/chatgpt-4o-latest	Greens	Labour	1.184	[1.100, 1.275]	4.484	7.316e-6
model_openai/chatgpt-4o-latest	Liberal Democrats	Labour	1.214	[1.140, 1.292]	6.092	1.114e-9
model_openai/chatgpt-4o-latest	Not eligible to vote	Labour	0.248	[0.200, 0.308]	−12.646	1.181e-36
model_openai/chatgpt-4o-latest	Other	Labour	0.701	[0.642, 0.766]	−7.907	2.637e-15
model_openai/chatgpt-4o-latest	Scottish National Party	Labour	1.058	[0.900, 1.245]	0.682	.495

Table S161: Multinomial logistic regression predicting perceived writer political ideology by model in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_politicalIdeology”. Predictor reference category: “writer”. Outcome reference level: “Centrist”. OR is odds ratio.

Term	Target level	Reference level	OR	95% CI (OR)	statistic	p
model_anthropic/claude-sonnet-4	Moderately Left-Wing	Centrist	0.987	[0.936, 1.040]	−0.496	.620
model_anthropic/claude-sonnet-4	Moderately Right-Wing	Centrist	1.048	[0.990, 1.111]	1.607	.108
model_anthropic/claude-sonnet-4	Other	Centrist	0.617	[0.560, 0.679]	−9.828	8.512e-23
model_anthropic/claude-sonnet-4	Very Left-Wing	Centrist	1.071	[0.997, 1.151]	1.888	.059
model_anthropic/claude-sonnet-4	Very Right-Wing	Centrist	0.906	[0.826, 0.993]	−2.117	.034
model_deepseek/deepseek-chat-v3-0324	Moderately Left-Wing	Centrist	1.010	[0.958, 1.065]	0.363	.716
model_deepseek/deepseek-chat-v3-0324	Moderately Right-Wing	Centrist	0.974	[0.918, 1.034]	−0.856	.392
model_deepseek/deepseek-chat-v3-0324	Other	Centrist	0.600	[0.544, 0.662]	−10.180	2.436e-24
model_deepseek/deepseek-chat-v3-0324	Very Left-Wing	Centrist	1.099	[1.023, 1.181]	2.582	.010
model_deepseek/deepseek-chat-v3-0324	Very Right-Wing	Centrist	0.875	[0.797, 0.961]	−2.785	.005
model_openai/chatgpt-4o-latest	Moderately Left-Wing	Centrist	1.028	[0.976, 1.083]	1.033	.301
model_openai/chatgpt-4o-latest	Moderately Right-Wing	Centrist	0.976	[0.921, 1.034]	−0.823	.410
model_openai/chatgpt-4o-latest	Other	Centrist	0.629	[0.572, 0.692]	−9.574	1.027e-21
model_openai/chatgpt-4o-latest	Very Left-Wing	Centrist	1.091	[1.016, 1.171]	2.409	.016
model_openai/chatgpt-4o-latest	Very Right-Wing	Centrist	0.802	[0.730, 0.882]	−4.567	4.953e-6

Table S162: Summary of distortion magnitude by model (Study 1, “preferred” subset).

Variable Type	Model	N Attributes	Average Effect	Summary Statistic
scale	Claude Sonnet 4	20	8.768	Average absolute AME
scale	DeepSeek Chat V3	20	8.701	Average absolute AME
scale	ChatGPT 4o	20	8.616	Average absolute AME
ordinal	Claude Sonnet 4	5	4.940	Average odds ratio
ordinal	DeepSeek Chat V3	5	4.528	Average odds ratio
ordinal	ChatGPT 4o	5	4.274	Average odds ratio

To more rigorously test whether distortion magnitudes differ between AI models, we compared the effects of each pair of AI models using the regression models described above. For each 0-100 scale attribute, we computed the difference between the two models' average marginal effects (AMEs) relative to human-written paragraphs, with standard errors obtained via the delta method. For each ordinal attribute, we computed the difference between the two models' log odds ratios relative to human-written paragraphs, which we report as a ratio of odds ratios, and tested it with a Wald test using the estimated covariance of the model coefficients. We report unadjusted p-values and p-values adjusted for the three pairwise comparisons per attribute using the Holm-Bonferroni method. Tables S163 and S164 show the results. At least one model pair differed significantly ($p < .05$) for 19/20 scale attributes and 5/5 ordinal attributes without adjustment, and for 18/20 scale attributes and 4/5 ordinal attributes after Holm adjustment. However, between-model differences were small compared to overall distortions from AI (SI:5.7).

Table S163: Pairwise differences in average marginal effects (AMEs) between AI models for all 0-100 scale rating attributes in Study 1 ("preferred" subset, N=75,972). Each cell shows the difference between the AMEs of the two models (first minus second, in points on the 0-100 scale) relative to human-written paragraphs, from mixed-effects beta regressions with crossed reader, writer, and proposition random effects. 95% confidence intervals in brackets. p_{Holm} is adjusted for three comparisons per attribute.

Attribute	Claude vs ChatGPT			Claude vs DeepSeek			ChatGPT vs DeepSeek		
	ΔAME [95% CI]	p	p_{Holm}	ΔAME [95% CI]	p	p_{Holm}	ΔAME [95% CI]	p	p_{Holm}
writer_knowledge	2.03 [1.51, 2.56]	< .001	< .001	1.51 [0.99, 2.04]	< .001	< .001	-0.52 [-1.06, 0.01]	.056	.056
writer_importance	1.67 [1.19, 2.16]	< .001	< .001	1.45 [0.96, 1.94]	< .001	< .001	-0.22 [-0.72, 0.27]	.375	.375
writer_confidence	1.05 [0.60, 1.49]	< .001	< .001	0.43 [-0.02, 0.88]	.058	.058	-0.62 [-1.07, -0.16]	.008	.015
writer_stance_polarity	0.45 [-0.23, 1.12]	.192	.384	0.88 [0.20, 1.57]	.011	.034	0.44 [-0.25, 1.12]	.213	.384
paragraph_formality	4.21 [3.63, 4.80]	< .001	< .001	1.72 [1.15, 2.30]	< .001	< .001	-2.49 [-3.09, -1.90]	< .001	< .001
paragraph_clarity	0.15 [-0.29, 0.58]	.509	1.000	0.25 [-0.18, 0.69]	.258	.773	0.11 [-0.33, 0.55]	.630	1.000
paragraph_informativeness	2.59 [1.91, 3.26]	< .001	< .001	1.55 [0.87, 2.23]	< .001	< .001	-1.04 [-1.72, -0.36]	.003	.003
paragraph_originality	2.15 [1.49, 2.80]	< .001	< .001	1.59 [0.92, 2.25]	< .001	< .001	-0.56 [-1.22, 0.10]	.096	.096
paragraph_relevance	0.37 [0.01, 0.73]	.044	.132	0.33 [-0.04, 0.69]	.079	.158	-0.04 [-0.41, 0.32]	.818	.818
writer_optimism	-2.01 [-2.64, -1.38]	< .001	< .001	-1.52 [-2.16, -0.88]	< .001	< .001	0.49 [-0.15, 1.12]	.134	.134
writer_community	-0.91 [-1.55, -0.28]	.005	.014	-0.93 [-1.57, -0.29]	.005	.014	-0.01 [-0.64, 0.62]	.970	.970
writer_friendliness	-2.36 [-2.91, -1.81]	< .001	< .001	-1.07 [-1.63, -0.51]	< .001	< .001	1.29 [0.73, 1.84]	< .001	< .001
writer_openness	-1.67 [-2.28, -1.06]	< .001	< .001	-1.47 [-2.09, -0.85]	< .001	< .001	0.20 [-0.42, 0.83]	.523	.523
paragraph_hope	-2.15 [-2.82, -1.47]	< .001	< .001	-1.89 [-2.57, -1.20]	< .001	< .001	0.26 [-0.42, 0.95]	.453	.453
paragraph_excitement	-0.76 [-1.27, -0.25]	.004	.011	-0.31 [-0.83, 0.20]	.236	.236	0.45 [-0.07, 0.97]	.093	.186
paragraph_fear	1.00 [0.42, 1.58]	< .001	.001	1.22 [0.63, 1.81]	< .001	< .001	0.21 [-0.37, 0.79]	.473	.473
paragraph_disgust	1.44 [0.96, 1.92]	< .001	< .001	0.99 [0.50, 1.48]	< .001	< .001	-0.44 [-0.92, 0.03]	.067	.067
paragraph_anger	1.96 [1.44, 2.48]	< .001	< .001	1.38 [0.85, 1.91]	< .001	< .001	-0.58 [-1.09, -0.07]	.025	.025
writer_affect_x	-2.15 [-2.71, -1.59]	< .001	< .001	-1.44 [-2.01, -0.87]	< .001	< .001	0.71 [0.15, 1.28]	.014	.014
writer_affect_y	-1.37 [-1.85, -0.89]	< .001	< .001	-0.83 [-1.31, -0.34]	< .001	.002	0.54 [0.06, 1.03]	.029	.029

Table S164: Pairwise ratios of odds ratios (ORs) between AI models for all ordinal rating attributes in Study 1 ("preferred" subset, N=75,972). Each cell shows the ratio of the two models' odds ratios (first divided by second) relative to human-written paragraphs, from mixed-effects ordinal logistic regressions with reader random effects. 95% confidence intervals in brackets. p_{Holm} is adjusted for three comparisons per attribute.

Attribute	Claude vs ChatGPT			Claude vs DeepSeek			ChatGPT vs DeepSeek		
	OR ratio [95% CI]	p	p_{Holm}	OR ratio [95% CI]	p	p_{Holm}	OR ratio [95% CI]	p	p_{Holm}
writer_education	1.244 [1.187, 1.305]	< .001	< .001	1.115 [1.062, 1.170]	< .001	< .001	0.896 [0.854, 0.940]	< .001	< .001
writer_english_skills	1.247 [1.178, 1.321]	< .001	< .001	1.131 [1.066, 1.199]	< .001	< .001	0.906 [0.856, 0.960]	< .001	< .001
writer_income	1.231 [1.175, 1.290]	< .001	< .001	1.106 [1.054, 1.159]	< .001	< .001	0.898 [0.857, 0.941]	< .001	< .001
writer_age_binned	1.136 [1.085, 1.190]	< .001	< .001	1.043 [0.996, 1.093]	.075	.075	0.918 [0.876, 0.962]	< .001	< .001
writer_english_first	0.863 [0.750, 0.993]	.039	.118	0.986 [0.860, 1.131]	.843	.843	1.143 [0.992, 1.317]	.065	.130

5.10 Statistical Analysis: Effects of AI Writing Assistance by Input Condition

To quantify variation in distortions across input conditions (SI:2.11), we fit **mixed-effects regression models** specified as follows:

```
rating_attribute ~ input_condition +  
(1 | reader_id) + (1 | writer_id) + (1 | proposition_id)
```

where `rating_attribute` and the random intercepts are defined as in SI:5.7, and `input_condition` is a categorical fixed effect taking five values: one for each of the four input conditions (stance-based, bullet-based, rewrite paragraph, improve paragraph) and one for human-written paragraphs, which serves as the reference category. The measurement scale of each rating attribute again determined the appropriate link function, following the same scheme as in SI:5.7: beta regression for 0-100 scale attributes, multinomial logistic regression for nominal categorical attributes, and ordinal logistic regression for ordinal categorical attributes.

Having confirmed that results are highly consistent across data subsets (SI:5.7), we fit each regression model to the “preferred” data subset (75,972 observations) as defined in SI:5.2. Figures S38, S39, and S40 summarise results for scale, ordinal, and nominal variables. Tables S165 to S193 show results for each of the 29 rating attributes in our study. Table S194 shows average distortion magnitude by input condition.

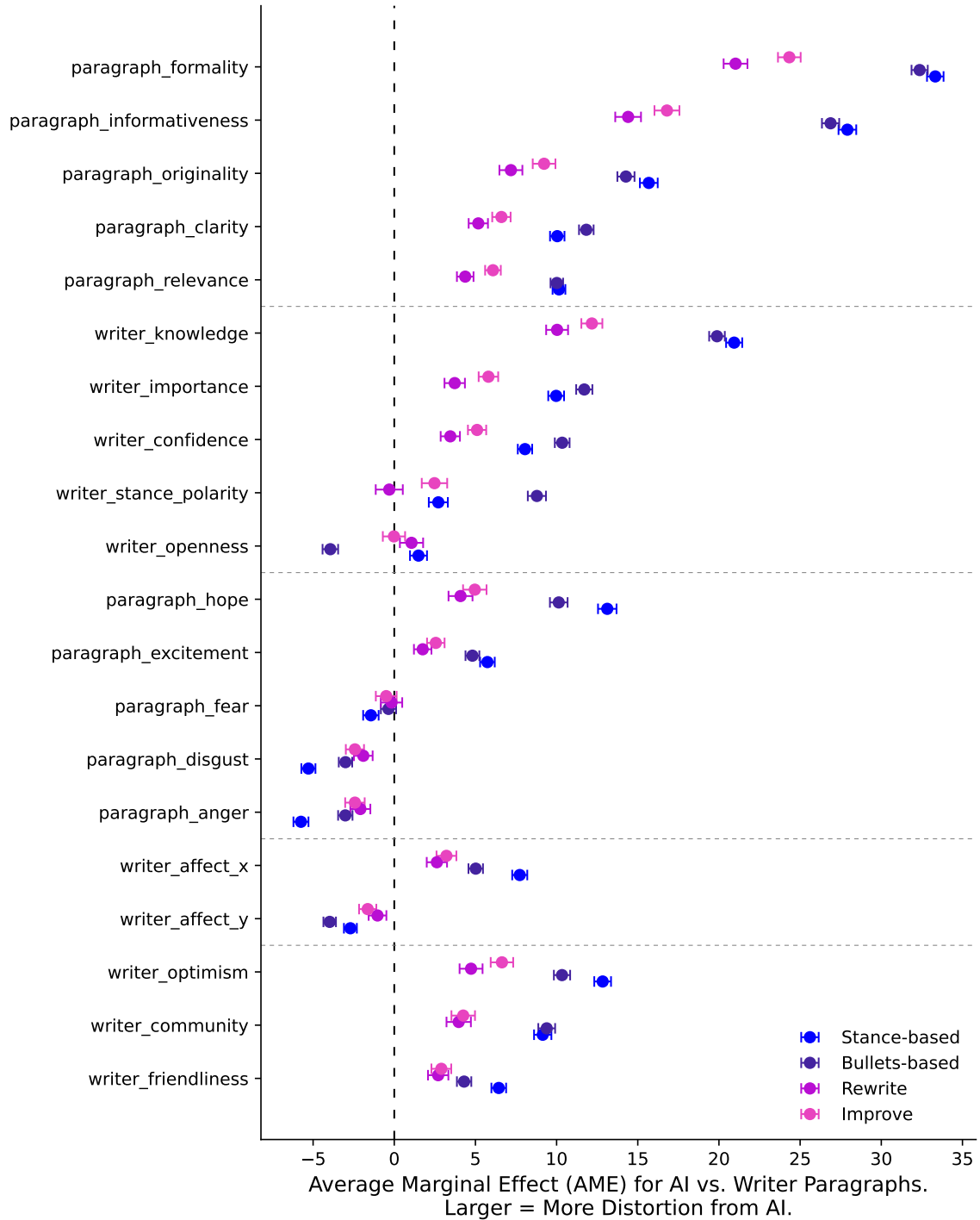


Figure S38: Average marginal effects (AMEs) of AI writing assistance on reader perceptions across all 0-100 scale rating attributes in Study 1. Each point shows the AME from a mixed-effects beta regression comparing AI-written to human-written paragraphs, with crossed reader, writer, and proposition random effects. Results shown separately for each AI model. Error bars show 95% confidence intervals.

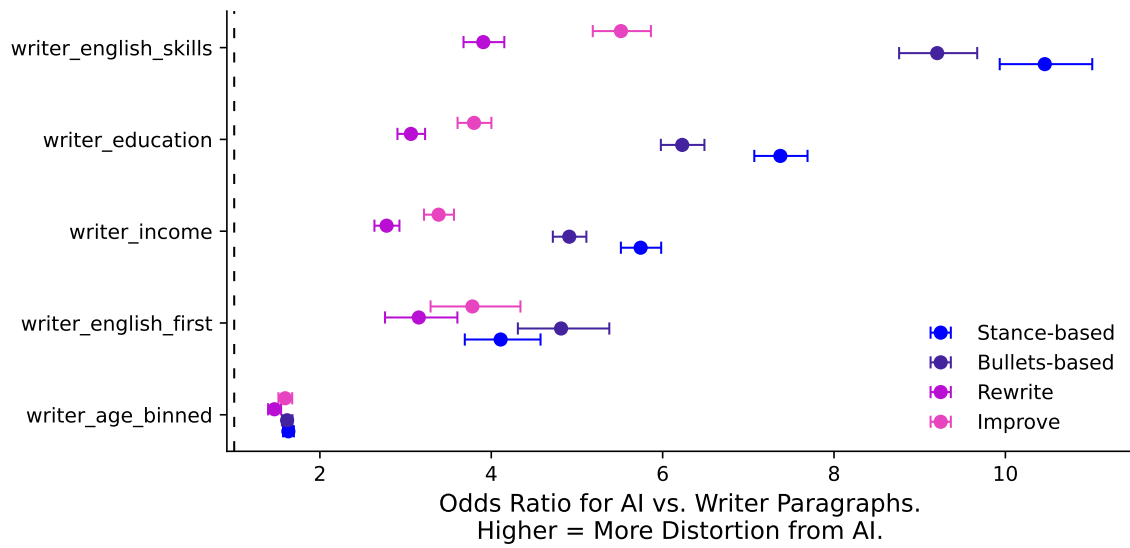


Figure S39: Odds ratios (ORs) of AI writing assistance on reader perceptions across all ordinal rating attributes in Study 1. Each point shows the odds ratio from a mixed-effects ordinal logistic regression comparing AI-written to human-written paragraphs, with reader random effects. Results shown separately for each AI model. Error bars show 95% confidence intervals.

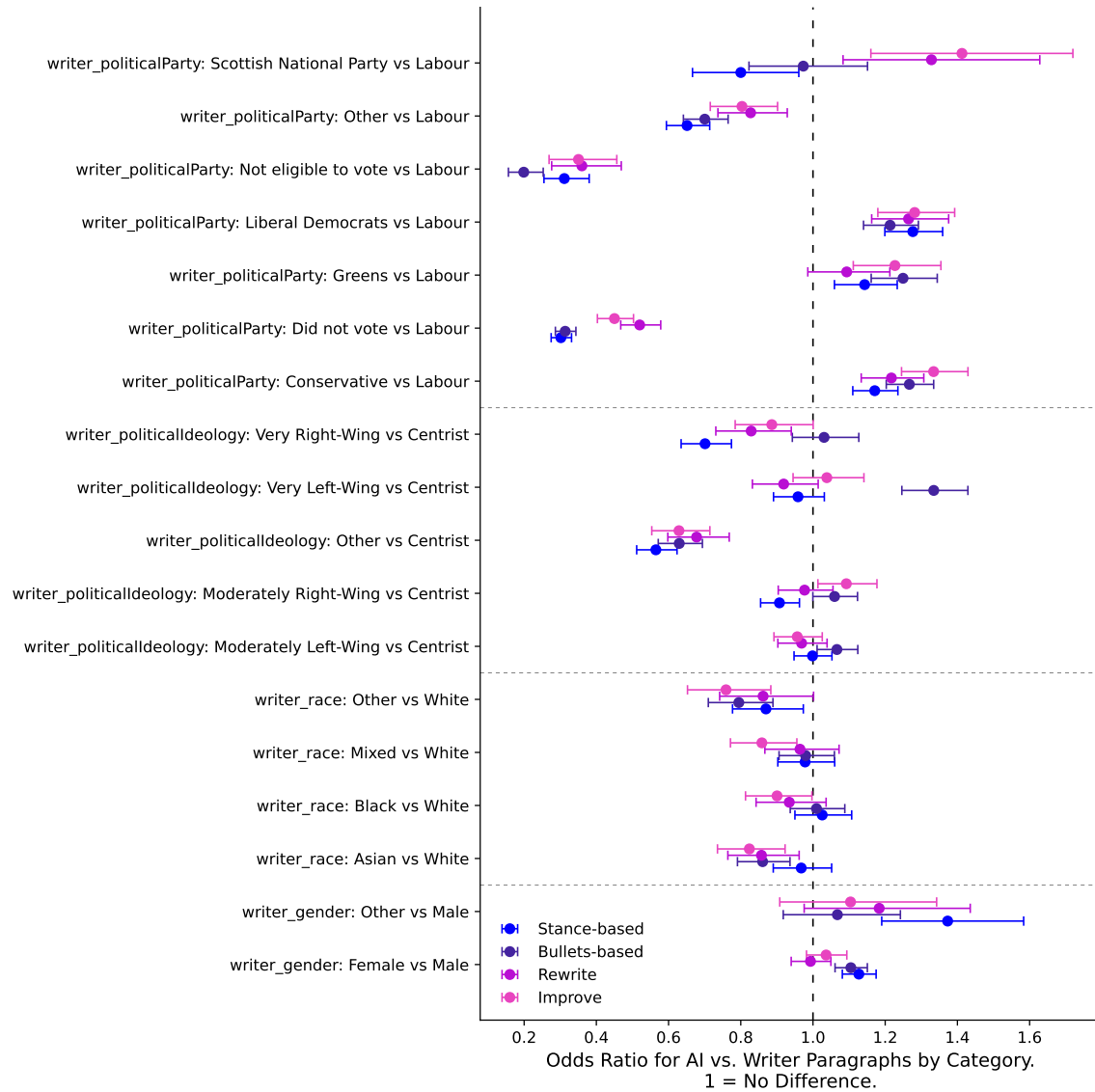


Figure S40: Odds ratios (ORs) of AI writing assistance on reader perceptions across all nominal rating attributes in Study 1. Each point shows the odds ratio of a specific category vs a reference category from a mixed-effects multinomial logistic regression comparing AI-written to human-written paragraphs, with reader random effects. Results shown separately for each AI model. Error bars show 95% confidence intervals.

Table S165: Beta regression predicting paragraph formality by input condition in Study 1 (“preferred” subset, N=75,972). Outcome: “paragraph_formality”. Reference category: “input_condition_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
input_condition_bullets-based	3.964	[3.875, 4.055]	118.473	0	32.362	[31.863, 32.862]
input_condition_improve	2.712	[2.631, 2.796]	64.154	0	24.334	[23.636, 25.033]
input_condition_rewrite	2.350	[2.278, 2.424]	53.942	0	21.022	[20.288, 21.756]
input_condition_stance-based	4.167	[4.070, 4.266]	119.117	0	33.326	[32.816, 33.835]

Table S166: Beta regression predicting paragraph clarity by input condition in Study 1 (“preferred” subset, N=75,972). Outcome: “paragraph_clarity”. Reference category: “input_condition_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
input_condition_bullets-based	2.051	[2.002, 2.100]	59.077	0	11.829	[11.380, 12.279]
input_condition_improve	1.438	[1.392, 1.486]	21.960	6.971e-107	6.600	[6.035, 7.166]
input_condition_rewrite	1.322	[1.278, 1.366]	16.442	9.538e-61	5.177	[4.583, 5.771]
input_condition_stance-based	1.799	[1.756, 1.844]	47.153	0	10.040	[9.595, 10.485]

Table S167: Beta regression predicting paragraph informativeness by input condition in Study 1 (“preferred” subset, N=75,972). Outcome: “paragraph_informativeness”. Reference category: “input_condition_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
input_condition_bullets-based	3.018	[2.949, 3.089]	93.445	0	26.876	[26.345, 27.408]
input_condition_improve	1.975	[1.914, 2.038]	42.680	0	16.797	[16.028, 17.565]
input_condition_rewrite	1.793	[1.737, 1.852]	35.814	6.729e-281	14.405	[13.613, 15.198]
input_condition_stance-based	3.161	[3.086, 3.238]	93.898	0	27.912	[27.366, 28.457]

Table S168: Beta regression predicting paragraph originality by input condition in Study 1 (“preferred” subset, N=75,972). Outcome: “paragraph_originality”. Reference category: “input_condition_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
input_condition_bullets-based	1.838	[1.799, 1.878]	55.577	0	14.271	[13.743, 14.798]
input_condition_improve	1.496	[1.453, 1.540]	27.050	3.809e-161	9.228	[8.529, 9.928]
input_condition_rewrite	1.373	[1.333, 1.415]	20.673	6.045e-95	7.184	[6.476, 7.892]
input_condition_stance-based	1.945	[1.902, 1.989]	58.785	0	15.679	[15.131, 16.227]

Table S169: Beta regression predicting paragraph relevance by input condition in Study 1 (“preferred” subset, N=75,972). Outcome: “paragraph_relevance”. Reference category: “input_condition_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
input_condition_bullets-based	1.966	[1.921, 2.011]	57.765	0	10.014	[9.626, 10.401]
input_condition_improve	1.455	[1.410, 1.500]	23.672	6.968e-124	6.079	[5.597, 6.562]
input_condition_rewrite	1.298	[1.257, 1.340]	16.112	2.088e-58	4.371	[3.860, 4.882]
input_condition_stance-based	1.988	[1.942, 2.036]	57.049	0	10.146	[9.752, 10.539]

Table S170: Beta regression predicting perceived writer knowledge by input condition in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_knowledge”. Reference category: “input_condition_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
input_condition_bullets-based	2.447	[2.396, 2.500]	82.751	0	19.879	[19.397, 20.361]
input_condition_improve	1.676	[1.629, 1.725]	35.365	5.833e-274	12.172	[11.524, 12.819]
input_condition_rewrite	1.523	[1.479, 1.568]	28.193	7.074e-175	10.031	[9.356, 10.705]
input_condition_stance-based	2.592	[2.535, 2.650]	84.596	0	20.937	[20.441, 21.433]

Table S171: Beta regression predicting perceived writer importance by input condition in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_importance”. Reference category: “input_condition_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
input_condition_bullets-based	1.842	[1.803, 1.883]	54.918	0	11.704	[11.206, 12.203]
input_condition_improve	1.324	[1.285, 1.364]	18.546	8.758e-77	5.800	[5.199, 6.402]
input_condition_rewrite	1.193	[1.157, 1.230]	11.397	4.348e-30	3.724	[3.095, 4.352]
input_condition_stance-based	1.662	[1.625, 1.700]	44.346	0	9.972	[9.486, 10.458]

Table S172: Beta regression predicting perceived writer confidence by input condition in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_confidence”. Reference category: “input_condition_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
input_condition_bullets-based	1.851	[1.809, 1.895]	52.476	0	10.344	[9.889, 10.799]
input_condition_improve	1.319	[1.278, 1.361]	17.288	5.805e-67	5.100	[4.537, 5.663]
input_condition_rewrite	1.201	[1.163, 1.240]	11.122	9.757e-29	3.452	[2.858, 4.045]
input_condition_stance-based	1.583	[1.546, 1.621]	37.961	2.503e-315	8.053	[7.609, 8.496]

Table S173: Beta regression predicting perceived writer stance polarity by input condition in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_stance_polarity”. Reference category: “input_condition_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
input_condition_bullets-based	1.523	[1.484, 1.563]	31.918	1.489e-223	8.785	[8.226, 9.345]
input_condition_improve	1.118	[1.079, 1.159]	6.122	9.228e-10	2.477	[1.694, 3.260]
input_condition_rewrite	0.986	[0.951, 1.023]	-0.725	0.468	-0.308	[-1.142, 0.526]
input_condition_stance-based	1.131	[1.101, 1.161]	9.005	2.158e-19	2.714	[2.127, 3.300]

Table S174: Beta regression predicting paragraph hope by input condition in Study 1 (“preferred” subset, N=75,972). Outcome: “paragraph_hope”. Reference category: “input_condition_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
input_condition_bullets-based	1.527	[1.494, 1.561]	37.737	1.235e-311	10.132	[9.588, 10.676]
input_condition_improve	1.236	[1.199, 1.274]	13.738	5.973e-43	4.961	[4.237, 5.684]
input_condition_rewrite	1.191	[1.155, 1.229]	11.030	2.738e-28	4.081	[3.342, 4.820]
input_condition_stance-based	1.722	[1.683, 1.762]	46.688	0	13.121	[12.551, 13.690]

Table S175: Beta regression predicting paragraph excitement by input condition in Study 1 (“preferred” subset, N=75,972). Outcome: “paragraph_excitement”. Reference category: “input_condition_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
input_condition_bullets-based	1.308	[1.280, 1.337]	24.034	1.217e-127	4.814	[4.388, 5.240]
input_condition_improve	1.159	[1.125, 1.194]	9.658	4.565e-22	2.559	[2.019, 3.098]
input_condition_rewrite	1.108	[1.074, 1.142]	6.530	6.560e-11	1.751	[1.212, 2.290]
input_condition_stance-based	1.372	[1.341, 1.403]	27.566	2.867e-167	5.738	[5.287, 6.189]

Table S176: Beta regression predicting paragraph fear by input condition in Study 1 (“preferred” subset, N=75,972). Outcome: “paragraph_fear”. Reference category: “input_condition_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
input_condition_bullets-based	0.983	[0.961, 1.006]	-1.485	0.138	-0.356	[-0.825, 0.113]
input_condition_improve	0.976	[0.946, 1.007]	-1.512	0.131	-0.496	[-1.136, 0.145]
input_condition_rewrite	0.992	[0.960, 1.024]	-0.505	0.614	-0.170	[-0.831, 0.490]
input_condition_stance-based	0.932	[0.910, 0.954]	-5.895	3.751e-9	-1.439	[-1.916, -0.963]

Table S177: Beta regression predicting paragraph disgust by input condition in Study 1 (“preferred” subset, N=75,972). Outcome: “paragraph_disgust”. Reference category: “input_condition_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
input_condition_bullets-based	0.841	[0.822, 0.861]	-14.409	4.563e-47	-3.004	[-3.418, -2.591]
input_condition_improve	0.871	[0.843, 0.900]	-8.358	6.405e-17	-2.424	[-2.983, -1.864]
input_condition_rewrite	0.898	[0.869, 0.928]	-6.392	1.635e-10	-1.908	[-2.484, -1.332]
input_condition_stance-based	0.728	[0.711, 0.746]	-25.648	4.423e-145	-5.288	[-5.722, -4.855]

Table S178: Beta regression predicting paragraph anger by input condition in Study 1 (“preferred” subset, N=75,972). Outcome: “paragraph_anger”. Reference category: “input_condition_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
input_condition_bullets-based	0.850	[0.831, 0.871]	-13.459	2.708e-41	-3.016	[-3.458, -2.573]
input_condition_improve	0.879	[0.851, 0.908]	-7.797	6.334e-15	-2.423	[-3.023, -1.823]
input_condition_rewrite	0.895	[0.866, 0.925]	-6.557	5.506e-11	-2.089	[-2.705, -1.473]
input_condition_stance-based	0.725	[0.707, 0.742]	-26.053	1.257e-149	-5.747	[-6.210, -5.284]

Table S179: Beta regression predicting perceived writer affect x by input condition in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_affect_x”. Reference category: “input_condition_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
input_condition_bullets-based	1.228	[1.206, 1.250]	22.196	3.740e-109	5.018	[4.571, 5.465]
input_condition_improve	1.141	[1.113, 1.170]	10.387	2.830e-25	3.214	[2.603, 3.826]
input_condition_rewrite	1.114	[1.086, 1.143]	8.289	1.142e-16	2.625	[2.000, 3.251]
input_condition_stance-based	1.369	[1.344, 1.395]	33.026	3.461e-239	7.729	[7.265, 8.193]

Table S180: Beta regression predicting perceived writer affect y by input condition in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_affect_y”. Reference category: “input_condition_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
input_condition_bullets-based	0.848	[0.834, 0.862]	-20.061	1.609e-89	-3.978	[-4.363, -3.592]
input_condition_improve	0.935	[0.915, 0.956]	-6.010	1.858e-9	-1.637	[-2.168, -1.105]
input_condition_rewrite	0.959	[0.937, 0.980]	-3.692	2.222e-4	-1.032	[-1.578, -0.485]
input_condition_stance-based	0.894	[0.880, 0.909]	-13.250	4.518e-40	-2.702	[-3.100, -2.305]

Table S181: Beta regression predicting perceived writer optimism by input condition in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_optimism”. Reference category: “input_condition_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
input_condition_bullets-based	1.516	[1.486, 1.547]	40.453	0	10.333	[9.833, 10.834]
input_condition_improve	1.307	[1.272, 1.344]	18.897	1.211e-79	6.633	[5.940, 7.326]
input_condition_rewrite	1.211	[1.177, 1.246]	13.199	8.929e-40	4.731	[4.023, 5.439]
input_condition_stance-based	1.677	[1.642, 1.712]	48.479	0	12.836	[12.320, 13.352]

Table S182: Beta regression predicting perceived writer community orientation by input condition in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_community”. Reference category: “input_condition_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
input_condition_bullets-based	1.495	[1.462, 1.529]	35.259	2.529e-272	9.391	[8.873, 9.908]
input_condition_improve	1.194	[1.158, 1.231]	11.329	9.444e-30	4.247	[3.522, 4.973]
input_condition_rewrite	1.180	[1.143, 1.218]	10.232	1.429e-24	3.969	[3.218, 4.720]
input_condition_stance-based	1.479	[1.445, 1.513]	33.217	6.177e-242	9.145	[8.611, 9.678]

Table S183: Beta regression predicting perceived writer friendliness by input condition in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_friendliness”. Reference category: “input_condition_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
input_condition_bullets-based	1.189	[1.168, 1.211]	18.963	3.466e-80	4.302	[3.860, 4.745]
input_condition_improve	1.124	[1.096, 1.152]	9.279	1.706e-20	2.900	[2.289, 3.510]
input_condition_rewrite	1.115	[1.087, 1.143]	8.427	3.538e-17	2.705	[2.078, 3.333]
input_condition_stance-based	1.297	[1.274, 1.322]	27.627	5.283e-168	6.437	[5.984, 6.890]

Table S184: Beta regression predicting perceived writer openness by input condition in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_openness”. Reference category: “input_condition_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
input_condition_bullets-based	0.829	[0.810, 0.848]	-16.068	4.275e-58	-3.937	[-4.417, -3.458]
input_condition_improve	0.999	[0.968, 1.031]	-0.054	0.957	-0.019	[-0.705, 0.667]
input_condition_rewrite	1.050	[1.016, 1.084]	2.929	0.003	1.062	[0.347, 1.778]
input_condition_stance-based	1.070	[1.045, 1.096]	5.629	1.812e-8	1.493	[0.969, 2.017]

Table S185: Ordinal logistic regression predicting perceived writer age group by input condition in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_age_binned”. Reference category: “input_condition_writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
input_condition_bullets-based	1.617	[1.557, 1.680]	24.795	1.015e-135
input_condition_improve	1.594	[1.516, 1.677]	18.140	1.545e-73
input_condition_rewrite	1.469	[1.395, 1.547]	14.607	2.532e-48
input_condition_stance-based	1.632	[1.570, 1.696]	24.802	8.461e-136

Table S186: Ordinal logistic regression predicting whether English is the writer’s first language by input condition in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_english_first”. Reference category: “input_condition_writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
input_condition_bullets-based	4.814	[4.310, 5.378]	27.850	1.092e-170
input_condition_improve	3.780	[3.292, 4.340]	18.842	3.392e-79
input_condition_rewrite	3.154	[2.760, 3.605]	16.865	8.147e-64
input_condition_stance-based	4.109	[3.691, 4.575]	25.805	7.848e-147

Table S187: Ordinal logistic regression predicting perceived writer income level by input condition in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_income”. Reference category: “input_condition_writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
input_condition_bullets-based	4.910	[4.718, 5.110]	78.241	0
input_condition_improve	3.385	[3.215, 3.565]	46.314	0
input_condition_rewrite	2.779	[2.636, 2.928]	38.151	1.850e-318
input_condition_stance-based	5.743	[5.513, 5.983]	83.728	0

Table S188: Ordinal logistic regression predicting perceived writer education level by input condition in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_education”. Reference category: “input_condition_writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
input_condition_bullets-based	6.228	[5.979, 6.488]	87.816	0
input_condition_improve	3.799	[3.607, 4.002]	50.370	0
input_condition_rewrite	3.062	[2.905, 3.228]	41.658	0
input_condition_stance-based	7.374	[7.070, 7.691]	92.994	0

Table S189: Ordinal logistic regression predicting perceived writer English skills by input condition in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_english_skills”. Reference category: “input_condition_writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
input_condition_bullets-based	9.204	[8.760, 9.671]	87.912	0
input_condition_improve	5.513	[5.185, 5.863]	54.407	0
input_condition_rewrite	3.907	[3.676, 4.152]	43.846	0
input_condition_stance-based	10.460	[9.934, 11.014]	89.174	0

Table S190: Multinomial logistic regression predicting perceived writer gender by input condition in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_gender”. Predictor reference category: “input_condition_writer”. Outcome reference level: “Male”. OR is odds ratio.

Term	Target level	Reference level	OR	95% CI (OR)	statistic	p
input_condition_bullets-based	Female	Male	1.105	[1.061, 1.151]	4.834	1.340e-6
input_condition_bullets-based	Other	Male	1.068	[0.918, 1.242]	0.847	0.397
input_condition_improve	Female	Male	1.036	[0.982, 1.094]	1.301	0.193
input_condition_improve	Other	Male	1.104	[0.908, 1.342]	0.992	0.321
input_condition_rewrite	Female	Male	0.993	[0.940, 1.050]	-0.245	0.806
input_condition_rewrite	Other	Male	1.183	[0.976, 1.436]	1.710	0.087
input_condition_stance-based	Female	Male	1.127	[1.081, 1.175]	5.650	1.604e-8
input_condition_stance-based	Other	Male	1.373	[1.190, 1.584]	4.354	1.334e-5

Table S191: Multinomial logistic regression predicting perceived writer race by input condition in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_race”. Predictor reference category: “input_condition_writer”. Outcome reference level: “White”. OR is odds ratio.

Term	Target level	Reference level	OR	95% CI (OR)	statistic	p
input_condition_bullets-based	Asian	White	0.861	[0.791, 0.936]	−3.486	4.895e-4
input_condition_bullets-based	Black	White	1.010	[0.937, 1.088]	0.254	0.799
input_condition_bullets-based	Mixed	White	0.980	[0.906, 1.059]	−0.514	0.607
input_condition_bullets-based	Other	White	0.795	[0.710, 0.889]	−4.008	6.117e-5
input_condition_improve	Asian	White	0.824	[0.735, 0.923]	−3.349	8.097e-4
input_condition_improve	Black	White	0.900	[0.813, 0.997]	−2.015	0.044
input_condition_improve	Mixed	White	0.858	[0.771, 0.955]	−2.797	0.005
input_condition_improve	Other	White	0.759	[0.652, 0.883]	−3.567	3.607e-4
input_condition_rewrite	Asian	White	0.857	[0.764, 0.962]	−2.626	0.009
input_condition_rewrite	Black	White	0.934	[0.842, 1.036]	−1.285	0.199
input_condition_rewrite	Mixed	White	0.964	[0.867, 1.072]	−0.675	0.500
input_condition_rewrite	Other	White	0.862	[0.742, 1.002]	−1.939	0.053
input_condition_stance-based	Asian	White	0.967	[0.890, 1.052]	−0.776	0.438
input_condition_stance-based	Black	White	1.026	[0.950, 1.107]	0.646	0.518
input_condition_stance-based	Mixed	White	0.978	[0.903, 1.060]	−0.538	0.591
input_condition_stance-based	Other	White	0.870	[0.777, 0.973]	−2.432	0.015

Table S192: Multinomial logistic regression predicting perceived writer political ideology by input condition in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_politicalIdeology”. Predictor reference category: “input_condition_writer”. Outcome reference level: “Centrist”. OR is odds ratio.

Term	Target level	Reference level	OR	95% CI (OR)	statistic	p
input_condition_bullets-based	Moderately Left-Wing	Centrist	1.066	[1.011, 1.124]	2.378	0.017
input_condition_bullets-based	Moderately Right-Wing	Centrist	1.060	[1.000, 1.123]	1.950	0.051
input_condition_bullets-based	Other	Centrist	0.630	[0.571, 0.694]	−9.355	8.390e-21
input_condition_bullets-based	Very Left-Wing	Centrist	1.334	[1.246, 1.429]	8.257	1.490e-16
input_condition_bullets-based	Very Right-Wing	Centrist	1.031	[0.943, 1.127]	0.666	0.505
input_condition_improve	Moderately Left-Wing	Centrist	0.956	[0.892, 1.026]	−1.249	0.212
input_condition_improve	Moderately Right-Wing	Centrist	1.092	[1.014, 1.177]	2.316	0.021
input_condition_improve	Other	Centrist	0.629	[0.554, 0.714]	−7.125	1.040e-12
input_condition_improve	Very Left-Wing	Centrist	1.038	[0.945, 1.141]	0.782	0.434
input_condition_improve	Very Right-Wing	Centrist	0.886	[0.785, 1.001]	−1.952	0.051
input_condition_rewrite	Moderately Left-Wing	Centrist	0.969	[0.903, 1.039]	−0.892	0.372
input_condition_rewrite	Moderately Right-Wing	Centrist	0.977	[0.904, 1.055]	−0.596	0.551
input_condition_rewrite	Other	Centrist	0.678	[0.598, 0.768]	−6.092	1.113e-9
input_condition_rewrite	Very Left-Wing	Centrist	0.919	[0.832, 1.014]	−1.677	0.094
input_condition_rewrite	Very Right-Wing	Centrist	0.829	[0.731, 0.940]	−2.926	0.003
input_condition_stance-based	Moderately Left-Wing	Centrist	0.999	[0.948, 1.053]	−0.050	0.960
input_condition_stance-based	Moderately Right-Wing	Centrist	0.907	[0.855, 0.963]	−3.222	0.001
input_condition_stance-based	Other	Centrist	0.565	[0.512, 0.623]	−11.338	8.539e-30
input_condition_stance-based	Very Left-Wing	Centrist	0.958	[0.891, 1.031]	−1.133	0.257
input_condition_stance-based	Very Right-Wing	Centrist	0.701	[0.635, 0.774]	−7.014	2.318e-12

Table S193: Multinomial logistic regression predicting perceived writer political party by input condition in Study 1 (“preferred” subset, N=75,972). Outcome: “writer_politicalParty”. Predictor reference category: “input_condition_writer”. Outcome reference level: “Labour”. OR is odds ratio.

Term	Target level	Reference level	OR	95% CI (OR)	statistic	p
input_condition_bullets-based	Conservative	Labour	1.267	[1.203, 1.334]	8.954	3.441e-19
input_condition_bullets-based	Did not vote	Labour	0.314	[0.287, 0.343]	−25.121	2.932e-139
input_condition_bullets-based	Greens	Labour	1.249	[1.161, 1.344]	5.965	2.445e-9
input_condition_bullets-based	Liberal Democrats	Labour	1.214	[1.140, 1.292]	6.063	1.337e-9
input_condition_bullets-based	Not eligible to vote	Labour	0.199	[0.156, 0.253]	−13.191	9.839e-40
input_condition_bullets-based	Other	Labour	0.700	[0.641, 0.765]	−7.906	2.649e-15
input_condition_bullets-based	Scottish National Party	Labour	0.973	[0.823, 1.151]	−0.318	0.751
input_condition_improve	Conservative	Labour	1.334	[1.245, 1.429]	8.200	2.396e-16
input_condition_improve	Did not vote	Labour	0.450	[0.403, 0.503]	−14.112	3.194e-45
input_condition_improve	Greens	Labour	1.227	[1.112, 1.354]	4.060	4.909e-5
input_condition_improve	Liberal Democrats	Labour	1.282	[1.180, 1.392]	5.865	4.500e-9
input_condition_improve	Not eligible to vote	Labour	0.351	[0.269, 0.457]	−7.768	7.998e-15
input_condition_improve	Other	Labour	0.803	[0.716, 0.902]	−3.709	2.082e-4
input_condition_improve	Scottish National Party	Labour	1.413	[1.160, 1.720]	3.441	5.797e-4
input_condition_rewrite	Conservative	Labour	1.217	[1.134, 1.307]	5.416	6.108e-8
input_condition_rewrite	Did not vote	Labour	0.520	[0.468, 0.578]	−12.059	1.744e-33
input_condition_rewrite	Greens	Labour	1.093	[0.986, 1.213]	1.687	0.092
input_condition_rewrite	Liberal Democrats	Labour	1.265	[1.163, 1.376]	5.468	4.542e-8
input_condition_rewrite	Not eligible to vote	Labour	0.360	[0.276, 0.469]	−7.567	3.817e-14
input_condition_rewrite	Other	Labour	0.827	[0.737, 0.929]	−3.210	0.001
input_condition_rewrite	Scottish National Party	Labour	1.328	[1.083, 1.628]	2.730	0.006
input_condition_stance-based	Conservative	Labour	1.171	[1.110, 1.235]	5.805	6.455e-9
input_condition_stance-based	Did not vote	Labour	0.302	[0.275, 0.331]	−25.153	1.321e-139
input_condition_stance-based	Greens	Labour	1.143	[1.059, 1.233]	3.449	5.620e-4
input_condition_stance-based	Liberal Democrats	Labour	1.277	[1.199, 1.359]	7.638	2.213e-14
input_condition_stance-based	Not eligible to vote	Labour	0.311	[0.255, 0.380]	−11.451	2.327e-30
input_condition_stance-based	Other	Labour	0.651	[0.594, 0.714]	−9.144	5.992e-20
input_condition_stance-based	Scottish National Party	Labour	0.800	[0.666, 0.961]	−2.389	0.017

Table S194: Summary of distortion magnitude by input (Study 1, “preferred” subset).

Variable Type	Input Condition	N Attributes	Average Effect	Summary Statistic
scale	Stance-based	20	10.523	Average absolute AME
scale	Bullets-based	20	10.217	Average absolute AME
scale	Improve	20	6.005	Average absolute AME
scale	Rewrite	20	4.790	Average absolute AME
ordinal	Stance-based	5	5.864	Average odds ratio
ordinal	Bullets-based	5	5.355	Average odds ratio
ordinal	Improve	5	3.614	Average odds ratio
ordinal	Rewrite	5	2.874	Average odds ratio

5.11 Statistical Analysis: Homogenising Effects of AI Writing Assistance

To test whether AI writing assistance reduces the diversity of reader perceptions across paragraphs, we compare the distributional spread of ratings for human-written versus AI-written paragraphs (Tables S195 to S197). For rating attributes measured on a 0-100 scale, we use standard deviation as our spread metric. For categorical rating attributes (both ordinal and nominal), we use base-2 Shannon entropy. For each attribute, we compute the difference in spread between AI-written and human-written paragraphs ($\text{model} - \text{writer}$), where negative values indicate homogenisation. We assess statistical significance using a nonparametric bootstrap procedure with $n=1,000$ resamples, applying Bonferroni correction across all 29 rating attributes. We report 95% confidence intervals on the spread difference, as well as the relative dispersion ratio (for 0-100 scale attributes) and the normalised entropy difference (for categorical attributes).

This analysis was not pre-registered.

Table S195: Distributional spread of ratings for human-written vs. AI-written paragraphs (Study 1, “preferred” subset). For 0-100 scale attributes, spread is measured by standard deviation (SD). For categorical attributes, spread is measured by Shannon entropy (base 2). Δ is the difference ($\text{model} - \text{writer}$), where negative values indicate homogenisation. RDR is the relative dispersion ratio ($\text{SD}_{\text{model}} / \text{SD}_{\text{writer}}$; 0-100 scale attributes only). NED is the normalised entropy difference (Δ / H_{max} where $H_{\text{max}} = \log_2(K)$, with K the number of categories; categorical attributes only). * $p < .05$, ** $p < .01$, *** $p < .001$, after Bonferroni correction across all 29 attributes.

Attribute	Type	Spread _{writer}	Spread _{model}	Δ	95% CI (Δ)	RDR	NED	Sig.
paragraph_relevance	0-100 scale	23.47	18.77	-4.69	[-4.95, -4.44]	0.800		***
paragraph_clarity	0-100 scale	25.92	20.81	-5.11	[-5.39, -4.84]	0.803		***
paragraph_formality	0-100 scale	26.54	21.97	-4.56	[-4.77, -4.37]	0.828		***
writer_knowledge	0-100 scale	24.51	20.55	-3.97	[-4.19, -3.72]	0.838		***
writer_confidence	0-100 scale	24.05	20.48	-3.57	[-3.81, -3.31]	0.852		***
writer_importance	0-100 scale	23.41	20.05	-3.36	[-3.59, -3.13]	0.857		***
paragraph_informativeness	0-100 scale	27.31	24.88	-2.42	[-2.63, -2.20]	0.911		***
paragraph_disgust	0-100 scale	27.64	25.42	-2.22	[-2.45, -1.99]	0.920		***
writer_friendliness	0-100 scale	20.11	18.61	-1.50	[-1.70, -1.29]	0.926		***
paragraph_anger	0-100 scale	28.19	26.22	-1.96	[-2.19, -1.74]	0.930		***
writer_affect_y	0-100 scale	19.96	18.75	-1.21	[-1.39, -1.02]	0.940		***
writer_community	0-100 scale	24.37	23.07	-1.30	[-1.53, -1.09]	0.947		***
writer_optimism	0-100 scale	22.76	21.83	-0.93	[-1.13, -0.73]	0.959		***
paragraph_fear	0-100 scale	27.14	26.31	-0.83	[-1.01, -0.64]	0.970		***
writer_stance_polarity	0-100 scale	33.13	32.63	-0.50	[-0.73, -0.24]	0.985		***
paragraph_hope	0-100 scale	24.91	24.58	-0.33	[-0.51, -0.14]	0.987		***
writer_affect_x	0-100 scale	23.21	23.31	+0.10	[-0.10, +0.29]	1.004		
paragraph_originality	0-100 scale	25.03	25.20	+0.17	[-0.03, +0.36]	1.007		
writer_openness	0-100 scale	24.95	25.77	+0.82	[+0.60, +1.00]	1.033		***
paragraph_excitement	0-100 scale	23.00	24.21	+1.21	[+0.99, +1.41]	1.053		***
writer_english_first	ordinal	0.514	0.220	-0.294	[-0.307, -0.280]		-0.294	***
writer_english_skills	ordinal	1.799	1.324	-0.475	[-0.486, -0.464]		-0.238	***
writer_education	ordinal	2.379	2.163	-0.216	[-0.230, -0.202]		-0.077	***
writer_income	ordinal	2.305	2.294	-0.011	[-0.025, +0.003]		-0.004	
writer_age_binned	ordinal	1.971	2.058	+0.087	[+0.075, +0.099]		+0.034	***
writer_politicalParty	nominal	2.576	2.399	-0.177	[-0.191, -0.161]		-0.059	***
writer_politicalIdeology	nominal	2.350	2.291	-0.059	[-0.070, -0.048]		-0.023	***
writer_race	nominal	1.408	1.357	-0.051	[-0.074, -0.031]		-0.022	***
writer_gender	nominal	1.089	1.109	+0.021	[+0.011, +0.030]		+0.013	***

To relate perceptual homogenisation to homogenisation of the text itself, we also test whether AI-written paragraphs on the same proposition are more similar to each other than human-written paragraphs on that proposition, following prior work on homogenisation at the lexical and semantic level^{41–43}. We measure semantic similarity as the cosine similarity between sentence embeddings from all-mpnet-base-v2⁴⁴, and lexical similarity as the cosine similarity between TF-IDF vectors (sublinear term frequency, English stop words removed). For each of the 100 propositions, we compute the mean pairwise similarity among human-written paragraphs (45 per proposition on average) and

Table S196: Distributional spread of ratings for human-written vs. AI-written paragraphs (Study 1, “edited” subset). For 0-100 scale attributes, spread is measured by standard deviation (SD). For categorical attributes, spread is measured by Shannon entropy (base 2). Δ is the difference (model – writer), where negative values indicate homogenisation. RDR is the relative dispersion ratio ($SD_{\text{model}} / SD_{\text{writer}}$; 0-100 scale attributes only). NED is the normalised entropy difference (Δ / H_{max} where $H_{\text{max}} = \log_2(K)$, with K the number of categories; categorical attributes only). * $p < .05$, ** $p < .01$, *** $p < .001$, after Bonferroni correction across all 29 attributes.

Attribute	Type	Spread _{writer}	Spread _{model}	Δ	95% CI (Δ)	RDR	NED	Sig.
paragraph_formality	0-100 scale	26.59	22.06	−4.53	[−4.73, −4.35]	0.830		***
paragraph_clarity	0-100 scale	25.83	21.01	−4.82	[−5.06, −4.57]	0.813		***
paragraph_informativeness	0-100 scale	27.42	25.04	−2.38	[−2.58, −2.18]	0.913		***
paragraph_originality	0-100 scale	25.07	25.23	+0.16	[−0.03, +0.35]	1.006		
paragraph_relevance	0-100 scale	23.39	18.95	−4.44	[−4.68, −4.20]	0.810		***
writer_knowledge	0-100 scale	24.49	20.59	−3.90	[−4.10, −3.68]	0.841		***
writer_importance	0-100 scale	23.32	20.15	−3.16	[−3.39, −2.97]	0.864		***
writer_confidence	0-100 scale	23.92	20.49	−3.42	[−3.66, −3.18]	0.857		***
writer_stance_polarity	0-100 scale	33.14	32.68	−0.46	[−0.69, −0.22]	0.986		***
paragraph_hope	0-100 scale	24.92	24.59	−0.33	[−0.51, −0.15]	0.987		***
paragraph_excitement	0-100 scale	23.00	24.23	+1.23	[+1.04, +1.43]	1.054		***
paragraph_fear	0-100 scale	27.12	26.29	−0.84	[−1.01, −0.65]	0.969		***
paragraph_disgust	0-100 scale	27.63	25.38	−2.24	[−2.45, −2.04]	0.919		***
paragraph_anger	0-100 scale	28.18	26.23	−1.95	[−2.15, −1.76]	0.931		***
writer_affect_x	0-100 scale	23.18	23.30	+0.12	[−0.06, +0.30]	1.005		
writer_affect_y	0-100 scale	19.88	18.73	−1.15	[−1.32, −0.97]	0.942		***
writer_optimism	0-100 scale	22.73	21.84	−0.89	[−1.08, −0.71]	0.961		***
writer_community	0-100 scale	24.38	23.12	−1.25	[−1.46, −1.04]	0.949		***
writer_friendliness	0-100 scale	20.11	18.65	−1.45	[−1.64, −1.26]	0.928		***
writer_openness	0-100 scale	24.96	25.70	+0.74	[+0.55, +0.93]	1.030		***
writer_income	ordinal	2.305	2.296	−0.010	[−0.023, +0.004]		−0.003	
writer_age_binned	ordinal	1.973	2.054	+0.081	[+0.070, +0.093]		+0.031	***
writer_english_first	ordinal	0.509	0.225	−0.284	[−0.296, −0.273]		−0.284	***
writer_english_skills	ordinal	1.795	1.330	−0.465	[−0.476, −0.455]		−0.233	***
writer_education	ordinal	2.380	2.164	−0.216	[−0.230, −0.204]		−0.077	***
writer_gender	nominal	1.091	1.111	+0.020	[+0.012, +0.029]		+0.013	***
writer_race	nominal	1.409	1.362	−0.047	[−0.066, −0.029]		−0.020	***
writer_politicalParty	nominal	2.573	2.403	−0.170	[−0.183, −0.157]		−0.057	***
writer_politicalIdeology	nominal	2.349	2.290	−0.059	[−0.069, −0.049]		−0.023	***

among AI-written paragraphs before writer edits (also 45 on average), as well as among AI-written paragraphs from each AI model (15 on average) and among AI-written paragraphs that writers edited, in their edited form (10 on average). Since mean pairwise similarity does not depend on the number of paragraphs in expectation, groups of different sizes are comparable. Since each AI-written paragraph is based on the opinion of the same writer as the corresponding human-written paragraph, both groups cover the same distribution of stances for each proposition. We compare each AI-written group to the human-written group with proposition-level paired differences, reporting bootstrap 95% confidence intervals (2,000 resamples of propositions) and Wilcoxon signed-rank tests with Bonferroni correction across all 10 comparisons.

AI-written paragraphs were substantially more similar to each other than human-written paragraphs, both semantically (mean cosine similarity of 0.801 vs. 0.656) and lexically (0.209 vs. 0.111), for every one of the 100 propositions (Table S198). Homogenisation was nearly identical across AI models. AI-written paragraphs that writers edited remained much more similar to each other than human-written paragraphs (semantic similarity of 0.781), so writer edits only partly reversed textual homogenisation. These results replicate prior findings of textual homogenisation from AI writing assistance, and are consistent with the perceptual homogenisation shown in Tables S195 to S197.

Table S197: Distributional spread of ratings for human-written vs. AI-written paragraphs (Study 1, “unedited” subset). For 0-100 scale attributes, spread is measured by standard deviation (SD). For categorical attributes, spread is measured by Shannon entropy (base 2). Δ is the difference (model – writer), where negative values indicate homogenisation. RDR is the relative dispersion ratio ($SD_{\text{model}} / SD_{\text{writer}}$; 0-100 scale attributes only). NED is the normalised entropy difference (Δ / H_{max} where $H_{\text{max}} = \log_2(K)$, with K the number of categories; categorical attributes only). * $p < .05$, ** $p < .01$, *** $p < .001$, after Bonferroni correction across all 29 attributes.

Attribute	Type	Spread _{writer}	Spread _{model}	Δ	95% CI (Δ)	RDR	NED	Sig.
paragraph_formality	0-100 scale	26.59	21.79	−4.80	[−4.98, −4.61]	0.819		***
paragraph_clarity	0-100 scale	25.83	20.84	−4.99	[−5.22, −4.74]	0.807		***
paragraph_informativeness	0-100 scale	27.42	24.80	−2.61	[−2.81, −2.42]	0.905		***
paragraph_originality	0-100 scale	25.07	25.21	+0.14	[−0.05, +0.32]	1.006		
paragraph_relevance	0-100 scale	23.39	18.73	−4.66	[−4.88, −4.40]	0.801		***
writer_knowledge	0-100 scale	24.49	20.42	−4.07	[−4.28, −3.85]	0.834		***
writer_importance	0-100 scale	23.32	20.09	−3.23	[−3.45, −3.01]	0.862		***
writer_confidence	0-100 scale	23.92	20.41	−3.50	[−3.74, −3.26]	0.854		***
writer_stance_polarity	0-100 scale	33.14	32.66	−0.48	[−0.69, −0.26]	0.986		***
paragraph_hope	0-100 scale	24.92	24.49	−0.44	[−0.61, −0.24]	0.983		***
paragraph_excitement	0-100 scale	23.00	24.23	+1.23	[+1.04, +1.43]	1.054		***
paragraph_fear	0-100 scale	27.12	26.27	−0.86	[−1.04, −0.69]	0.968		***
paragraph_disgust	0-100 scale	27.63	25.33	−2.30	[−2.51, −2.09]	0.917		***
paragraph_anger	0-100 scale	28.18	26.10	−2.08	[−2.29, −1.90]	0.926		***
writer_affect_x	0-100 scale	23.18	23.29	+0.12	[−0.07, +0.29]	1.005		
writer_affect_y	0-100 scale	19.88	18.73	−1.15	[−1.33, −0.98]	0.942		***
writer_optimism	0-100 scale	22.73	21.75	−0.98	[−1.16, −0.80]	0.957		***
writer_community	0-100 scale	24.38	23.03	−1.35	[−1.59, −1.12]	0.945		***
writer_friendliness	0-100 scale	20.11	18.55	−1.56	[−1.74, −1.37]	0.923		***
writer_openness	0-100 scale	24.96	25.83	+0.88	[+0.69, +1.06]	1.035		***
writer_income	ordinal	2.305	2.292	−0.013	[−0.026, −0.001]		−0.005	
writer_age_binned	ordinal	1.973	2.053	+0.080	[+0.070, +0.091]		+0.031	***
writer_english_first	ordinal	0.509	0.219	−0.290	[−0.303, −0.279]		−0.290	***
writer_english_skills	ordinal	1.795	1.311	−0.484	[−0.494, −0.474]		−0.242	***
writer_education	ordinal	2.380	2.158	−0.223	[−0.236, −0.209]		−0.079	***
writer_gender	nominal	1.091	1.110	+0.019	[+0.010, +0.027]		+0.012	***
writer_race	nominal	1.409	1.364	−0.045	[−0.065, −0.025]		−0.019	**
writer_politicalParty	nominal	2.573	2.399	−0.175	[−0.188, −0.160]		−0.058	***
writer_politicalIdeology	nominal	2.349	2.293	−0.056	[−0.066, −0.045]		−0.022	***

Table S198: Textual homogenisation of human-written vs. AI-written paragraphs in Study 1. Similarity is the mean pairwise cosine similarity between paragraphs on the same proposition, averaged across 100 propositions. Semantic similarity uses all-mpnet-base-v2 sentence embeddings; lexical similarity uses TF-IDF vectors. n is the mean number of paragraphs per proposition. AI-written paragraphs are unedited, except for “AI-written (edited)”, which comprises the 1,002 AI-written paragraphs that writers edited, in their edited form. Δ is the mean proposition-level difference to human-written paragraphs, where positive values indicate homogenisation. Values in brackets are bootstrap 95% confidence intervals (2,000 resamples of propositions). Significance from Wilcoxon signed-rank tests on proposition-level differences: * $p < .05$, ** $p < .01$, *** $p < .001$, after Bonferroni correction across all 10 comparisons.

Paragraphs	n	Semantic similarity			Lexical similarity		
		Similarity	Δ	Sig.	Similarity	Δ	Sig.
Human-written	45.0	0.656 [0.647, 0.665]			0.111 [0.107, 0.117]		
AI-written	45.0	0.801 [0.794, 0.808]	+0.145 [+0.137, +0.153]	***	0.209 [0.202, 0.215]	+0.097 [+0.092, +0.103]	***
Claude Sonnet 4	15.0	0.805 [0.796, 0.814]	+0.149 [+0.139, +0.159]	***	0.220 [0.212, 0.228]	+0.108 [+0.101, +0.116]	***
DeepSeek V3	15.0	0.807 [0.798, 0.816]	+0.151 [+0.141, +0.161]	***	0.219 [0.210, 0.227]	+0.107 [+0.099, +0.115]	***
ChatGPT-4o	15.0	0.806 [0.796, 0.817]	+0.150 [+0.139, +0.161]	***	0.217 [0.209, 0.226]	+0.106 [+0.097, +0.114]	***
AI-written (edited)	10.0	0.781 [0.766, 0.795]	+0.125 [+0.108, +0.138]	***	0.198 [0.188, 0.208]	+0.087 [+0.077, +0.096]	***

5.12 Statistical Analysis: Effects of AI Writing Assistance by Human Baseline

To test whether the strength of AI distortion depends on how readers perceived the human-written paragraph, we extend the regression models from SI:5.7 with an interaction between paragraph type and a baseline, i.e. the rating of the human-written paragraph on the same rating attribute, defined for each writer-proposition pair. **This analysis was not pre-registered.**

A naive baseline, computed from the same ratings that also enter the regression as outcomes, would produce a spurious interaction through regression to the mean. Measurement error in the baseline would be shared with the human-written outcome ratings, but not with the AI-written outcome ratings. To avoid this, we randomly split the ratings of each human-written paragraph into two halves (each human-written paragraph received 9–19 ratings; see Table S37). The first half is used only to compute the baseline and is then discarded. The second half, together with all ratings of the corresponding AI-written paragraph, forms the outcome data. Since readers almost never rated both paragraphs from the same pair (60 of 100,064 reader-pair combinations), measurement error in the baseline is independent of all outcome ratings. For 0-100 scale attributes, the baseline is the mean rating in the first half. For ordinal attributes, it is the mean category code, from 1 for the lowest category upwards. We standardise baselines across paragraph pairs within each data subset.

For each rating attribute, we then fit regression models specified as follows:

$$\text{rating_attribute} \sim \text{paragraph_type} \times \text{baseline} + (1 \mid \text{reader_id})$$

where `paragraph_type` and `(1 | reader_id)` are defined as in SI:5.7, and `baseline` is the standardised split-half baseline. For 0-100 scale attributes, we fit mixed-effects beta regression models with a logit link function, as in SI:5.7. For ordinal attributes, mixed-effects ordinal logistic regressions with the interaction term were computationally infeasible on our data. We therefore fit ordinal logistic regressions with a cumulative logit link function without reader random effects, and instead account for repeated ratings by the same reader with reader-clustered robust standard errors. Population-averaged odds ratios from these models tend to be closer to 1 than reader-specific odds ratios from mixed-effects models, so this choice is conservative. We do not analyse nominal attributes, since they have no natural ordering from which to define a baseline. We fit each model to the three data subsets defined in SI:5.2.

The quantity of interest is the interaction between paragraph type and baseline. An interaction odds ratio below 1 indicates that AI distortion is weaker for paragraph pairs where the human-written paragraph was rated higher on a given attribute. To make results interpretable, we also report the effect of AI writing assistance at the 10th, 50th, and 90th percentile of the baseline distribution across paragraph pairs: average marginal effects (AMEs) on the 0-100 scale for scale attributes, and odds ratios for ordinal attributes. Note that AMEs on the 0-100 scale are additionally constrained by floor and ceiling effects, whereas the interaction odds ratio is estimated on the logit scale.

Tables S199 to S204 show results for all 20 scale and 5 ordinal rating attributes, for each subset. The interaction odds ratio is below 1 for all 25 attributes in all three subsets, and significant at $p < .01$ after Bonferroni correction across all 25 attributes ($p < .001$ for all attributes except `paragraph_excitement` in the “preferred” subset). In other words, AI writing assistance consistently shifted perceptions more where the human-written paragraph was perceived as lower on a given attribute. For several attributes, the effect of AI writing assistance changes sign across the baseline distribution. For example, AI-written paragraphs were perceived as angrier than human-written paragraphs that were perceived as calm (P10), and as much less angry than human-written paragraphs that were perceived as angry (P90). This pattern is consistent with AI-written paragraphs converging towards a common register, independent of the human-written paragraph, and helps explain the homogenisation reported in SI:5.11.

Table S199: Distortion by human baseline for 0-100 scale attributes in Study 1 (“preferred” subset, 3,795 paragraph pairs, N=56,166 paragraph-reader observations per attribute after excluding baseline ratings). Mixed-effects beta regressions with reader random intercepts and a paragraph type $\times$ baseline interaction. Baseline is the mean rating of the human-written paragraph in the held-out half of its ratings, shown at the 10th, 50th, and 90th percentile (P10 / P50 / P90) of its distribution across paragraph pairs. Interaction OR is the odds ratio for the interaction per standard deviation of the baseline; values below 1 indicate weaker AI distortion for higher baselines. AME is the average marginal effect of AI-written vs. human-written paragraphs (in points on the 0-100 scale) at the given baseline percentile, with 95% confidence intervals. * $p < .05$, ** $p < .01$, *** $p < .001$ for the interaction, after Bonferroni correction across all 25 scale and ordinal attributes.

Attribute	Baseline P10 / P50 / P90	Interaction OR	95% CI (OR)	p	Sig.	AME at P10	AME at P50	AME at P90
writer_knowledge	35.0 / 57.2 / 74.2	0.776	[0.762, 0.790]	5.661e-162	***	+26.5 [+25.8, +27.2]	+16.3 [+15.9, +16.7]	+8.7 [+8.1, +9.3]
writer_importance	45.0 / 66.8 / 83.4	0.806	[0.791, 0.822]	2.035e-111	***	+17.6 [+16.9, +18.3]	+8.4 [+8.0, +8.8]	+3.0 [+2.5, +3.4]
writer_confidence	47.4 / 73.4 / 87.6	0.789	[0.773, 0.804]	5.604e-120	***	+18.3 [+17.5, +19.1]	+6.9 [+6.5, +7.2]	+2.6 [+2.2, +3.0]
writer_stance_polarity	28.8 / 68.8 / 91.2	0.844	[0.825, 0.864]	1.352e-47	***	+12.2 [+11.2, +13.2]	+4.2 [+3.8, +4.7]	+0.6 [+0.1, +1.1]
paragraph_formality	22.0 / 44.0 / 64.6	0.647	[0.634, 0.659]	0	***	+42.2 [+41.6, +42.9]	+29.2 [+28.7, +29.7]	+14.7 [+14.0, +15.5]
paragraph_clarity	47.5 / 72.0 / 88.0	0.780	[0.764, 0.796]	8.657e-120	***	+19.2 [+18.4, +20.0]	+8.5 [+8.1, +8.9]	+3.5 [+3.0, +3.9]
paragraph_informativeness	21.2 / 42.4 / 63.2	0.744	[0.729, 0.759]	7.222e-180	***	+31.1 [+30.3, +31.8]	+22.8 [+22.3, +23.3]	+12.8 [+12.0, +13.7]
paragraph_originality	19.2 / 34.8 / 52.2	0.862	[0.846, 0.879]	2.088e-51	***	+16.0 [+15.3, +16.6]	+12.7 [+12.3, +13.2]	+8.4 [+7.6, +9.2]
paragraph_relevance	55.0 / 74.4 / 87.8	0.781	[0.765, 0.796]	2.945e-130	***	+16.1 [+15.5, +16.8]	+7.8 [+7.5, +8.2]	+3.5 [+3.1, +3.9]
writer_optimism	25.0 / 44.6 / 62.7	0.840	[0.825, 0.855]	1.255e-80	***	+14.0 [+13.3, +14.6]	+9.5 [+9.1, +9.9]	+3.9 [+3.2, +4.6]
writer_community	39.2 / 59.2 / 75.7	0.880	[0.862, 0.897]	1.395e-36	***	+11.9 [+11.1, +12.7]	+7.0 [+6.5, +7.4]	+3.0 [+2.4, +3.7]
writer_friendliness	36.5 / 53.4 / 66.0	0.827	[0.814, 0.840]	6.659e-124	***	+10.6 [+10.0, +11.3]	+3.9 [+3.5, +4.3]	-1.1 [-1.7, -0.5]
writer_openness	15.6 / 33.5 / 56.4	0.858	[0.841, 0.876]	1.616e-47	***	+2.1 [+1.6, +2.7]	-1.0 [-1.4, -0.5]	-6.7 [-7.6, -5.8]
paragraph_hope	19.2 / 38.7 / 59.8	0.854	[0.838, 0.871]	3.814e-55	***	+11.8 [+11.1, +12.4]	+9.1 [+8.7, +9.6]	+4.2 [+3.3, +5.0]
paragraph_excitement	12.2 / 25.0 / 42.0	0.962	[0.943, 0.981]	1.362e-4	**	+4.2 [+3.7, +4.6]	+4.1 [+3.8, +4.5]	+3.7 [+3.1, +4.4]
paragraph_fear	14.6 / 33.0 / 57.2	0.870	[0.852, 0.888]	4.225e-40	***	+1.9 [+1.4, +2.4]	-0.8 [-1.2, -0.4]	-5.9 [-6.8, -5.1]
paragraph_disgust	8.4 / 25.6 / 53.9	0.780	[0.764, 0.797]	4.949e-117	***	+0.7 [+0.3, +1.1]	-3.0 [-3.4, -2.7]	-13.2 [-14.0, -12.3]
paragraph_anger	9.2 / 28.4 / 58.7	0.746	[0.730, 0.762]	1.297e-162	***	+1.5 [+1.0, +1.9]	-3.3 [-3.6, -2.9]	-15.7 [-16.6, -14.8]
writer_affect_x	21.4 / 40.0 / 61.3	0.896	[0.881, 0.911]	1.187e-39	***	+7.7 [+7.1, +8.3]	+5.3 [+4.9, +5.7]	+1.6 [+0.9, +2.3]
writer_affect_y	29.5 / 43.2 / 56.3	0.922	[0.909, 0.935]	4.175e-29	***	-0.2 [-0.7, +0.4]	-2.7 [-3.1, -2.4]	-5.3 [-5.9, -4.7]

Table S200: Distortion by human baseline for 0-100 scale attributes in Study 1 (“edited” subset, 4,503 paragraph pairs, N=66,647 paragraph-reader observations per attribute after excluding baseline ratings). Mixed-effects beta regressions with reader random intercepts and a paragraph type $\times$ baseline interaction. Baseline is the mean rating of the human-written paragraph in the held-out half of its ratings, shown at the 10th, 50th, and 90th percentile (P10 / P50 / P90) of its distribution across paragraph pairs. Interaction OR is the odds ratio for the interaction per standard deviation of the baseline; values below 1 indicate weaker AI distortion for higher baselines. AME is the average marginal effect of AI-written vs. human-written paragraphs (in points on the 0-100 scale) at the given baseline percentile, with 95% confidence intervals. * $p < .05$, ** $p < .01$, *** $p < .001$ for the interaction, after Bonferroni correction across all 25 scale and ordinal attributes.

Attribute	Baseline P10 / P50 / P90	Interaction OR	95% CI (OR)	p	Sig.	AME at P10	AME at P50	AME at P90
writer_knowledge	35.5 / 57.7 / 74.5	0.768	[0.755, 0.781]	2.042e-208	***	+25.9 [+25.3, +26.6]	+15.4 [+15.0, +15.8]	+7.7 [+7.1, +8.2]
writer_importance	45.4 / 67.2 / 83.5	0.804	[0.791, 0.818]	9.720e-136	***	+17.0 [+16.4, +17.7]	+7.8 [+7.5, +8.2]	+2.6 [+2.2, +3.0]
writer_confidence	48.0 / 73.6 / 87.7	0.785	[0.771, 0.800]	2.825e-148	***	+17.7 [+17.0, +18.5]	+6.5 [+6.2, +6.8]	+2.3 [+1.9, +2.6]
writer_stance_polarity	28.8 / 68.4 / 91.3	0.845	[0.828, 0.863]	1.964e-55	***	+11.8 [+10.9, +12.8]	+4.1 [+3.6, +4.5]	+0.4 [-0.0, +0.9]
paragraph_formality	22.5 / 44.6 / 65.4	0.646	[0.635, 0.658]	0	***	+41.4 [+40.8, +42.0]	+28.3 [+27.8, +28.7]	+13.7 [+13.0, +14.3]
paragraph_clarity	48.0 / 72.2 / 88.2	0.783	[0.768, 0.798]	9.034e-138	***	+18.2 [+17.4, +18.9]	+7.9 [+7.6, +8.3]	+3.0 [+2.6, +3.4]
paragraph_informativeness	21.6 / 43.0 / 64.0	0.741	[0.727, 0.755]	2.124e-218	***	+30.1 [+29.4, +30.8]	+21.7 [+21.2, +22.1]	+11.5 [+10.7, +12.3]
paragraph_originality	19.4 / 35.4 / 52.4	0.860	[0.845, 0.876]	3.532e-63	***	+15.4 [+14.8, +16.0]	+12.0 [+11.6, +12.4]	+7.6 [+6.9, +8.3]
paragraph_relevance	55.0 / 74.6 / 87.8	0.785	[0.771, 0.800]	3.807e-148	***	+15.6 [+15.0, +16.2]	+7.4 [+7.1, +7.7]	+3.2 [+2.8, +3.6]
writer_optimism	25.0 / 44.6 / 62.6	0.846	[0.833, 0.860]	3.241e-88	***	+13.3 [+12.7, +13.9]	+9.0 [+8.6, +9.4]	+3.7 [+3.1, +4.4]
writer_community	39.2 / 59.2 / 75.7	0.886	[0.870, 0.902]	1.205e-38	***	+11.4 [+10.7, +12.2]	+6.8 [+6.4, +7.2]	+3.1 [+2.5, +3.7]
writer_friendliness	36.6 / 53.4 / 66.0	0.834	[0.822, 0.846]	6.531e-135	***	+10.1 [+9.5, +10.7]	+3.7 [+3.4, +4.1]	-1.0 [-1.6, -0.5]
writer_openness	15.4 / 33.4 / 56.2	0.857	[0.841, 0.874]	1.850e-57	***	+2.3 [+1.8, +2.8]	-0.8 [-1.3, -0.4]	-6.5 [-7.3, -5.7]
paragraph_hope	19.2 / 38.8 / 60.0	0.858	[0.843, 0.874]	2.795e-62	***	+11.2 [+10.6, +11.8]	+8.6 [+8.2, +9.0]	+3.8 [+3.1, +4.6]
paragraph_excitement	12.2 / 25.2 / 42.0	0.957	[0.940, 0.974]	2.158e-6	***	+3.9 [+3.5, +4.4]	+3.7 [+3.4, +4.0]	+3.1 [+2.5, +3.7]
paragraph_fear	14.8 / 32.8 / 57.2	0.871	[0.855, 0.887]	2.620e-47	***	+1.7 [+1.3, +2.2]	-0.9 [-1.3, -0.6]	-6.1 [-6.9, -5.4]
paragraph_disgust	8.4 / 25.8 / 54.2	0.775	[0.761, 0.791]	6.681e-147	***	+0.8 [+0.4, +1.1]	-3.0 [-3.4, -2.7]	-13.4 [-14.2, -12.6]
paragraph_anger	9.3 / 28.6 / 58.8	0.743	[0.729, 0.758]	5.926e-200	***	+1.5 [+1.1, +1.8]	-3.3 [-3.7, -3.0]	-15.9 [-16.7, -15.0]
writer_affect_x	21.4 / 40.0 / 61.3	0.896	[0.883, 0.910]	9.765e-47	***	+7.6 [+7.1, +8.1]	+5.2 [+4.9, +5.6]	+1.5 [+0.8, +2.1]
writer_affect_y	29.4 / 43.1 / 56.2	0.921	[0.909, 0.933]	7.924e-36	***	+0.2 [-0.3, +0.7]	-2.5 [-2.8, -2.2]	-5.1 [-5.6, -4.6]

Table S201: Distortion by human baseline for 0-100 scale attributes in Study 1 (“unedited” subset, 4,503 paragraph pairs, N=66,614 paragraph-reader observations per attribute after excluding baseline ratings). Mixed-effects beta regressions with reader random intercepts and a paragraph type $\times$ baseline interaction. Baseline is the mean rating of the human-written paragraph in the held-out half of its ratings, shown at the 10th, 50th, and 90th percentile (P10 / P50 / P90) of its distribution across paragraph pairs. Interaction OR is the odds ratio for the interaction per standard deviation of the baseline; values below 1 indicate weaker AI distortion for higher baselines. AME is the average marginal effect of AI-written vs. human-written paragraphs (in points on the 0-100 scale) at the given baseline percentile, with 95% confidence intervals. * $p < .05$, ** $p < .01$, *** $p < .001$ for the interaction, after Bonferroni correction across all 25 scale and ordinal attributes.

Attribute	Baseline P10 / P50 / P90	Interaction OR	95% CI (OR)	p	Sig.	AME at P10	AME at P50	AME at P90
writer_knowledge	35.5 / 57.7 / 74.5	0.753	[0.741, 0.766]	3.473e-240	***	+27.1 [+26.4, +27.7]	+15.8 [+15.4, +16.2]	+7.7 [+7.2, +8.2]
writer_importance	45.4 / 67.2 / 83.5	0.798	[0.784, 0.811]	1.655e-147	***	+17.6 [+16.9, +18.3]	+8.1 [+7.7, +8.4]	+2.6 [+2.2, +3.0]
writer_confidence	48.0 / 73.6 / 87.7	0.778	[0.764, 0.792]	9.576e-160	***	+18.1 [+17.4, +18.9]	+6.6 [+6.2, +6.9]	+2.3 [+1.9, +2.6]
writer_stance_polarity	28.8 / 68.4 / 91.3	0.838	[0.820, 0.855]	2.414e-61	***	+12.2 [+11.3, +13.2]	+4.1 [+3.6, +4.5]	+0.3 [-0.2, +0.8]
paragraph_formality	22.5 / 44.6 / 65.4	0.638	[0.627, 0.650]	0	***	+42.5 [+41.9, +43.1]	+28.9 [+28.5, +29.3]	+14.0 [+13.3, +14.6]
paragraph_clarity	48.0 / 72.2 / 88.2	0.785	[0.771, 0.801]	4.432e-135	***	+18.4 [+17.6, +19.1]	+8.1 [+7.8, +8.5]	+3.3 [+2.9, +3.7]
paragraph_informativeness	21.6 / 43.0 / 64.0	0.734	[0.720, 0.747]	7.766e-235	***	+31.3 [+30.6, +31.9]	+22.5 [+22.1, +23.0]	+12.0 [+11.3, +12.8]
paragraph_originality	19.4 / 35.4 / 52.4	0.860	[0.845, 0.875]	9.273e-64	***	+16.0 [+15.4, +16.6]	+12.6 [+12.2, +13.0]	+8.3 [+7.6, +9.0]
paragraph_relevance	55.0 / 74.6 / 87.8	0.775	[0.761, 0.789]	3.567e-167	***	+16.1 [+15.5, +16.7]	+7.5 [+7.2, +7.8]	+3.2 [+2.8, +3.6]
writer_optimism	25.0 / 44.6 / 62.6	0.839	[0.825, 0.853]	7.618e-97	***	+13.9 [+13.3, +14.5]	+9.3 [+8.9, +9.7]	+3.8 [+3.1, +4.4]
writer_community	39.2 / 59.2 / 75.7	0.868	[0.852, 0.884]	1.075e-52	***	+12.4 [+11.7, +13.2]	+7.0 [+6.6, +7.5]	+2.8 [+2.2, +3.4]
writer_friendliness	36.6 / 53.4 / 66.0	0.815	[0.804, 0.827]	1.322e-170	***	+11.2 [+10.6, +11.8]	+4.0 [+3.6, +4.4]	-1.3 [-1.9, -0.8]
writer_openness	15.4 / 33.4 / 56.2	0.853	[0.837, 0.869]	2.434e-61	***	+2.5 [+2.0, +2.9]	-0.8 [-1.2, -0.4]	-6.6 [-7.4, -5.8]
paragraph_hope	19.2 / 38.8 / 60.0	0.844	[0.829, 0.860]	4.199e-76	***	+12.1 [+11.5, +12.7]	+9.1 [+8.7, +9.5]	+3.8 [+3.0, +4.5]
paragraph_excitement	12.2 / 25.2 / 42.0	0.959	[0.942, 0.977]	7.184e-6	***	+4.1 [+3.6, +4.5]	+3.9 [+3.6, +4.2]	+3.4 [+2.8, +4.0]
paragraph_fear	14.8 / 32.8 / 57.2	0.861	[0.845, 0.878]	9.818e-55	***	+2.1 [+1.6, +2.5]	-0.7 [-1.1, -0.4]	-6.3 [-7.1, -5.5]
paragraph_disgust	8.4 / 25.8 / 54.2	0.767	[0.752, 0.782]	1.560e-159	***	+0.9 [+0.5, +1.2]	-3.1 [-3.4, -2.8]	-13.9 [-14.7, -13.1]
paragraph_anger	9.3 / 28.6 / 58.8	0.736	[0.722, 0.751]	9.860e-213	***	+1.5 [+1.2, +1.9]	-3.4 [-3.7, -3.1]	-16.3 [-17.1, -15.5]
writer_affect_x	21.4 / 40.0 / 61.3	0.891	[0.878, 0.905]	2.540e-51	***	+8.0 [+7.5, +8.5]	+5.5 [+5.1, +5.9]	+1.5 [+0.9, +2.2]
writer_affect_y	29.4 / 43.1 / 56.2	0.919	[0.907, 0.931]	3.137e-37	***	+0.1 [-0.4, +0.6]	-2.6 [-2.9, -2.3]	-5.3 [-5.8, -4.8]

Table S202: Distortion by human baseline for ordinal attributes in Study 1 (“preferred” subset, 3,795 paragraph pairs, N=55,578–56,166 paragraph-reader observations per attribute after excluding baseline ratings). Ordinal logistic regressions (cumulative logit) with a paragraph type $\times$ baseline interaction and reader-clustered robust standard errors. Baseline is the mean category code (1 = lowest category) of the human-written paragraph in the held-out half of its ratings, shown at the 10th, 50th, and 90th percentile (P10 / P50 / P90) of its distribution across paragraph pairs. For writer_english_first, the baseline ranges from 1 (all “No”) to 2 (all “Yes”), and P50 and P90 coincide because most paragraph pairs have a baseline of 2. Interaction OR is the odds ratio for the interaction per standard deviation of the baseline; values below 1 indicate weaker AI distortion for higher baselines. OR at P10 / P50 / P90 is the odds ratio of AI-written vs. human-written paragraphs at the given baseline percentile, with 95% confidence intervals. * $p < .05$, ** $p < .01$, *** $p < .001$ for the interaction, after Bonferroni correction across all 25 scale and ordinal attributes.

Attribute	Baseline P10 / P50 / P90	Interaction OR	95% CI (OR)	p	Sig.	OR at P10	OR at P50	OR at P90
writer_education	1.80 / 3.00 / 4.00	0.655	[0.634, 0.678]	5.011e-134	***	8.09 [7.60, 8.62]	4.29 [4.13, 4.45]	2.53 [2.39, 2.67]
writer_english_skills	2.20 / 2.88 / 3.60	0.633	[0.610, 0.656]	4.065e-135	***	7.87 [7.36, 8.41]	4.27 [4.10, 4.44]	2.21 [2.08, 2.35]
writer_income	2.33 / 3.20 / 4.00	0.743	[0.719, 0.768]	6.659e-71	***	4.91 [4.64, 5.19]	3.35 [3.23, 3.47]	2.36 [2.24, 2.48]
writer_age_binned	1.20 / 2.00 / 2.83	0.783	[0.755, 0.812]	1.588e-39	***	2.12 [2.00, 2.25]	1.56 [1.51, 1.62]	1.14 [1.07, 1.21]
writer_english_first	1.60 / 2.00 / 2.00	0.678	[0.637, 0.721]	1.095e-34	***	6.53 [5.77, 7.39]	2.53 [2.30, 2.78]	2.53 [2.30, 2.78]

Table S203: Distortion by human baseline for ordinal attributes in Study 1 (“edited” subset, 4,503 paragraph pairs, N=65,942–66,647 paragraph-reader observations per attribute after excluding baseline ratings). Ordinal logistic regressions (cumulative logit) with a paragraph type $\times$ baseline interaction and reader-clustered robust standard errors. Baseline is the mean category code (1 = lowest category) of the human-written paragraph in the held-out half of its ratings, shown at the 10th, 50th, and 90th percentile (P10 / P50 / P90) of its distribution across paragraph pairs. For writer_english_first, the baseline ranges from 1 (all “No”) to 2 (all “Yes”), and P50 and P90 coincide because most paragraph pairs have a baseline of 2. Interaction OR is the odds ratio for the interaction per standard deviation of the baseline; values below 1 indicate weaker AI distortion for higher baselines. OR at P10 / P50 / P90 is the odds ratio of AI-written vs. human-written paragraphs at the given baseline percentile, with 95% confidence intervals. * $p < .05$, ** $p < .01$, *** $p < .001$ for the interaction, after Bonferroni correction across all 25 scale and ordinal attributes.

Attribute	Baseline P10 / P50 / P90	Interaction OR	95% CI (OR)	p	Sig.	OR at P10	OR at P50	OR at P90
writer_education	1.83 / 3.00 / 4.00	0.649	[0.629, 0.669]	2.550e-164	***	7.81 [7.37, 8.28]	4.16 [4.02, 4.31]	2.43 [2.31, 2.55]
writer_english_skills	2.20 / 3.00 / 3.60	0.632	[0.612, 0.654]	4.828e-161	***	7.62 [7.16, 8.11]	3.66 [3.53, 3.79]	2.11 [2.00, 2.23]
writer_income	2.40 / 3.20 / 4.00	0.742	[0.720, 0.764]	1.214e-84	***	4.68 [4.45, 4.91]	3.28 [3.17, 3.39]	2.30 [2.19, 2.41]
writer_age_binned	1.20 / 2.00 / 2.83	0.786	[0.761, 0.813]	1.721e-45	***	2.07 [1.96, 2.19]	1.53 [1.49, 1.58]	1.12 [1.06, 1.18]
writer_english_first	1.60 / 2.00 / 2.00	0.684	[0.646, 0.724]	3.160e-39	***	6.18 [5.52, 6.93]	2.43 [2.22, 2.65]	2.43 [2.22, 2.65]

Table S204: Distortion by human baseline for ordinal attributes in Study 1 (“unedited” subset, 4,503 paragraph pairs, N=65,919–66,614 paragraph-reader observations per attribute after excluding baseline ratings). Ordinal logistic regressions (cumulative logit) with a paragraph type $\times$ baseline interaction and reader-clustered robust standard errors. Baseline is the mean category code (1 = lowest category) of the human-written paragraph in the held-out half of its ratings, shown at the 10th, 50th, and 90th percentile (P10 / P50 / P90) of its distribution across paragraph pairs. For writer_english_first, the baseline ranges from 1 (all “No”) to 2 (all “Yes”), and P50 and P90 coincide because most paragraph pairs have a baseline of 2. Interaction OR is the odds ratio for the interaction per standard deviation of the baseline; values below 1 indicate weaker AI distortion for higher baselines. OR at P10 / P50 / P90 is the odds ratio of AI-written vs. human-written paragraphs at the given baseline percentile, with 95% confidence intervals. * $p < .05$, ** $p < .01$, *** $p < .001$ for the interaction, after Bonferroni correction across all 25 scale and ordinal attributes.

Attribute	Baseline P10 / P50 / P90	Interaction OR	95% CI (OR)	p	Sig.	OR at P10	OR at P50	OR at P90
writer_education	1.83 / 3.00 / 4.00	0.643	[0.623, 0.663]	7.635e-171	***	8.20 [7.74, 8.69]	4.31 [4.17, 4.47]	2.49 [2.36, 2.62]
writer_english_skills	2.20 / 3.00 / 3.60	0.624	[0.603, 0.645]	2.422e-168	***	8.04 [7.55, 8.57]	3.78 [3.64, 3.92]	2.14 [2.03, 2.27]
writer_income	2.40 / 3.20 / 4.00	0.737	[0.715, 0.759]	1.410e-88	***	4.81 [4.57, 5.05]	3.35 [3.24, 3.46]	2.33 [2.22, 2.44]
writer_age_binned	1.20 / 2.00 / 2.83	0.780	[0.754, 0.806]	3.227e-48	***	2.10 [1.99, 2.22]	1.54 [1.49, 1.59]	1.11 [1.05, 1.17]
writer_english_first	1.60 / 2.00 / 2.00	0.671	[0.633, 0.710]	2.998e-42	***	6.62 [5.89, 7.44]	2.48 [2.27, 2.70]	2.48 [2.27, 2.70]

5.13 Robustness Checks: Rating Protocol

Each reader in Study 1 rated ten paragraphs and answered all 29 rating questions for each paragraph in the same fixed order (SI:2.4). This protocol, chosen for reasons of efficiency, may introduce fatigue, carryover between consecutive paragraphs, and careless responding. All of these would affect human-written and AI-written paragraphs alike, since question order was identical for both and paragraph order was randomised. They should therefore add noise to our estimates rather than bias the comparison between human-written and AI-written paragraphs. To test this empirically, we re-estimate the effects of AI writing assistance from SI:5.7 under alternative specifications. We focus on the 20 rating attributes measured on a 0-100 scale and on the “preferred” subset (75,972 observations). Inter-rater agreement is reported in SI:5.6, and correlations between rating attributes are reported in SI:5.5.

For each rating, we recorded the position of the paragraph in the reader’s sequence (1–10) and the time between the reader opening the paragraph and submitting the last rating question. Paragraph type was balanced across positions, as intended by our randomisation: AI-written paragraphs made up between 49.4% and 50.7% of human-written and unedited AI-written paragraphs at each position ($\chi^2(9) = 8.41$, $p = .49$). Figure S41 shows that readers spent most time on their first paragraph (median 248 seconds) and became faster with each subsequent paragraph (median 127 seconds for the tenth paragraph). The fastest 1% of ratings still took 65 seconds or more, i.e. more than two seconds per rating question.

For each rating attribute, we fit the mixed-effects beta regression model from SI:5.7, with reader random intercepts only, to the following variants of the data:

- **First paragraph only** (7,608 observations): only ratings of the first paragraph each reader rated. These ratings are unaffected by fatigue and carryover, so this is the cleanest test of our results, at the cost of statistical power.
- **Paragraphs 1–3** (22,850 observations) and **paragraphs 6–10** (37,961 observations): ratings from early and late positions in each reader’s sequence.
- **Control for previous paragraph** (75,972 observations): we add the type of the paragraph the reader rated immediately before (human-written, AI-written, or none) as a fixed effect, to test for carryover between consecutive paragraphs.
- **Excluding survey speeders** (68,340 observations): we exclude all ratings from readers in the bottom 10% of median time per paragraph (≤ 95.7 seconds), as well as any individual rating completed in under 60 seconds.
- **Excluding survey straightliners** (72,173 observations): we exclude ratings in the bottom 5% of within-paragraph standard deviation across the 20 scale rating attributes (≤ 15.4 points), i.e. ratings where readers gave very similar answers to all questions.

In addition, we fit a model to all observations with an interaction between paragraph type and paragraph position (1–10, linear), and report AMEs of AI writing assistance at each position.

Table S205 and Figure S42 show AMEs of AI writing assistance for each specification. Controlling for the previous paragraph, excluding speeders and excluding straightliners all left AMEs essentially unchanged (all within 0.6 points of the full-sample estimates). Across all other specifications, the direction and significance of AMEs were also unchanged, with two exceptions, both for attributes with AMEs close to zero in the full sample. For `writer_openness` (full sample -0.7), the AME was not significant in the first-paragraph subset ($+0.1$) or the paragraphs 1–3 subset (-0.1). For `paragraph_fear` (full sample -0.6), the AME had the same sign but was not significant in the first-paragraph, paragraphs 1–3 and paragraphs 6–10 subsets.

AMEs for several attributes related to paragraph quality and writer competence were smaller for the first paragraph than in the full sample, though still large and significant. For example, the AME for `writer_knowledge` was $+11.6$ for the first paragraph compared to $+17.0$ in the full sample, and the AME for `paragraph_formality` was $+20.6$ compared to $+28.6$. Figure S43 shows that, for these attributes, the AMEs increased gradually with paragraph position, while AMEs for most other attributes were stable. Since excluding speeders and straightliners did not change these AMEs, this pattern is unlikely to reflect careless responding due to fatigue. A more plausible explanation is that readers calibrated their judgements as they were exposed to a wider range of paragraphs, making

differences between human-written and AI-written paragraphs more visible to them. Either way, the first-paragraph estimates, which are unaffected by the rating protocol, confirm the direction and significance of our main results.

This analysis was not pre-registered.

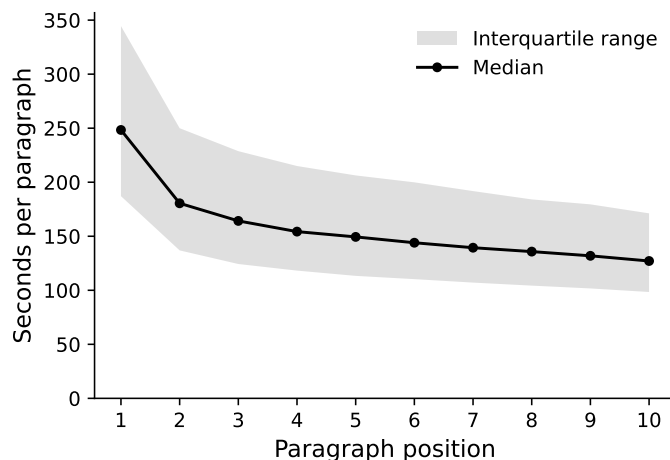


Figure S41: Time spent per paragraph by paragraph position in Study 1 (all 100,124 paragraph-reader observations). The line shows the median time between a reader opening a paragraph and submitting the last of the 29 rating questions for that paragraph; the shaded area shows the interquartile range.

Table S205: Average marginal effects (AMEs) of AI writing assistance on reader perceptions under alternative specifications (Study 1, “preferred” subset). Each cell shows the AME of AI-written vs. human-written paragraphs (in points on the 0-100 scale) from a mixed-effects beta regression with reader random effects, with 95% confidence intervals. Number of observations: full sample and previous-paragraph control N=75,972; first paragraph only N=7,608; paragraphs 1–3 N=22,850; paragraphs 6–10 N=37,961; excluding speeders N=68,340; excluding straightliners N=72,173. [†] 95% confidence interval includes zero.

Attribute	Full sample	First paragraph	Paragraphs 1–3	Paragraphs 6–10	Prev. paragraph control	Excl. speeders	Excl. straightliners
writer_knowledge	+17.0 [+16.6, +17.3]	+11.6 [+10.5, +12.6]	+15.5 [+14.9, +16.1]	+17.9 [+17.4, +18.4]	+16.9 [+16.6, +17.3]	+17.3 [+16.9, +17.7]	+17.2 [+16.9, +17.6]
writer_importance	+8.6 [+8.2, +8.9]	+5.2 [+4.2, +6.2]	+7.3 [+6.7, +7.9]	+10.0 [+9.5, +10.5]	+8.6 [+8.3, +8.9]	+8.7 [+8.4, +9.1]	+8.5 [+8.1, +8.8]
writer_confidence	+7.4 [+7.1, +7.7]	+4.8 [+3.9, +5.7]	+6.5 [+6.0, +7.1]	+8.3 [+7.8, +8.8]	+7.4 [+7.1, +7.7]	+7.5 [+7.1, +7.8]	+7.3 [+7.0, +7.6]
writer_stance_polarity	+4.3 [+3.8, +4.7]	+3.2 [+1.9, +4.5]	+3.9 [+3.2, +4.7]	+4.6 [+4.0, +5.2]	+4.2 [+3.8, +4.7]	+4.3 [+3.9, +4.8]	+4.0 [+3.6, +4.4]
paragraph_formality	+28.6 [+28.2, +29.0]	+20.6 [+19.4, +21.8]	+26.4 [+25.7, +27.1]	+30.4 [+29.9, +30.9]	+28.5 [+28.1, +28.8]	+29.2 [+28.8, +29.6]	+29.2 [+28.8, +29.6]
paragraph_clarity	+9.0 [+8.7, +9.3]	+4.0 [+3.3, +4.6]	+7.2 [+6.7, +7.7]	+10.7 [+10.2, +11.2]	+9.1 [+8.8, +9.4]	+9.0 [+8.7, +9.3]	+8.8 [+8.5, +9.2]
paragraph_informativeness	+22.7 [+22.3, +23.1]	+16.2 [+14.8, +17.6]	+21.6 [+20.8, +22.3]	+23.9 [+23.3, +24.5]	+22.7 [+22.3, +23.1]	+23.2 [+22.8, +23.6]	+23.2 [+22.8, +23.6]
paragraph_originality	+12.5 [+12.1, +12.8]	+7.1 [+5.8, +8.4]	+10.6 [+9.9, +11.3]	+14.1 [+13.5, +14.6]	+12.4 [+12.0, +12.8]	+12.7 [+12.3, +13.1]	+12.8 [+12.4, +13.1]
paragraph_relevance	+8.3 [+8.0, +8.6]	+4.7 [+4.0, +5.4]	+7.2 [+6.7, +7.6]	+9.6 [+9.2, +10.0]	+8.4 [+8.1, +8.7]	+8.3 [+8.1, +8.6]	+8.3 [+8.0, +8.5]
writer_optimism	+9.5 [+9.1, +9.8]	+8.3 [+7.1, +9.6]	+8.9 [+8.2, +9.5]	+9.4 [+8.9, +10.0]	+9.4 [+9.1, +9.8]	+9.6 [+9.2, +10.0]	+9.7 [+9.3, +10.1]
writer_community	+7.4 [+7.0, +7.8]	+6.3 [+5.1, +7.6]	+7.4 [+6.7, +8.1]	+7.4 [+6.9, +8.0]	+7.4 [+7.0, +7.8]	+7.5 [+7.1, +7.9]	+7.4 [+7.0, +7.8]
writer_friendliness	+4.5 [+4.2, +4.8]	+2.7 [+1.7, +3.8]	+3.9 [+3.3, +4.6]	+4.5 [+4.0, +4.9]	+4.5 [+4.2, +4.9]	+4.6 [+4.2, +4.9]	+4.6 [+4.2, +4.9]
writer_openness	−0.7 [−1.1, −0.4]	+0.1 [†] [−1.2, +1.5]	−0.1 [†] [−0.8, +0.6]	−1.2 [−1.7, −0.7]	−0.7 [−1.1, −0.3]	−0.9 [−1.3, −0.5]	−0.7 [−1.1, −0.3]
paragraph_hope	+8.9 [+8.5, +9.3]	+9.4 [+8.0, +10.8]	+9.3 [+8.5, +10.0]	+9.2 [+8.6, +9.7]	+8.9 [+8.5, +9.3]	+9.2 [+8.8, +9.6]	+9.1 [+8.7, +9.5]
paragraph_excitement	+4.1 [+3.8, +4.3]	+3.5 [+2.4, +4.5]	+4.0 [+3.5, +4.6]	+4.1 [+3.7, +4.5]	+4.0 [+3.7, +4.3]	+4.0 [+3.7, +4.3]	+4.2 [+3.9, +4.5]
paragraph_fear	−0.6 [−0.9, −0.3]	−1.3 [†] [−2.6, +0.1]	−0.7 [†] [−1.4, +0.0]	−0.5 [†] [−1.0, +0.0]	−0.5 [−0.9, −0.2]	−0.6 [−1.0, −0.3]	−0.6 [−0.9, −0.2]
paragraph_disgust	−3.1 [−3.4, −2.8]	−4.2 [−5.3, −3.2]	−4.4 [−5.0, −3.7]	−2.9 [−3.4, −2.5]	−3.1 [−3.4, −2.8]	−3.1 [−3.4, −2.8]	−3.1 [−3.4, −2.8]
paragraph_anger	−3.2 [−3.5, −2.9]	−5.3 [−6.4, −4.1]	−4.9 [−5.6, −4.2]	−2.5 [−2.9, −2.0]	−3.2 [−3.6, −2.9]	−3.3 [−3.6, −3.0]	−3.2 [−3.6, −2.9]
writer_affect_x	+5.1 [+4.7, +5.4]	+4.8 [+3.7, +5.8]	+4.9 [+4.2, +5.5]	+5.0 [+4.6, +5.5]	+5.0 [+4.7, +5.4]	+5.1 [+4.8, +5.5]	+5.2 [+4.8, +5.5]
writer_affect_y	−2.6 [−2.9, −2.3]	−1.3 [−2.1, −0.5]	−1.8 [−2.3, −1.3]	−3.3 [−3.7, −2.9]	−2.6 [−2.9, −2.3]	−2.5 [−2.8, −2.2]	−2.7 [−3.0, −2.4]

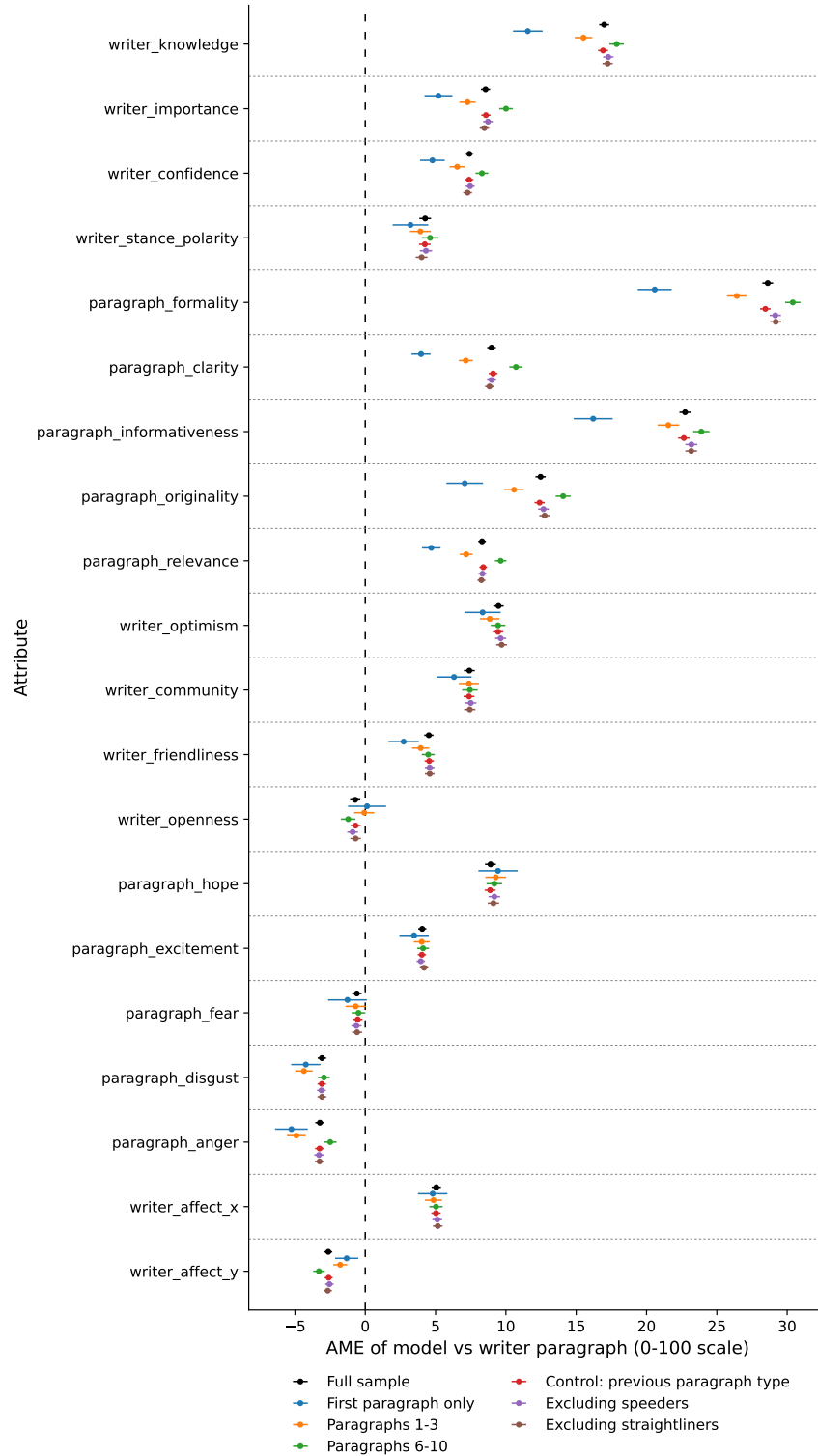


Figure S42: Average marginal effects (AMEs) of AI writing assistance on reader perceptions under alternative specifications (Study 1, “preferred” subset). Each point shows the AME from a mixed-effects beta regression comparing AI-written to human-written paragraphs, with reader random effects, fitted to the full sample (black) or to one of the robustness specifications described in SI:5.13. Error bars show 95% confidence intervals.

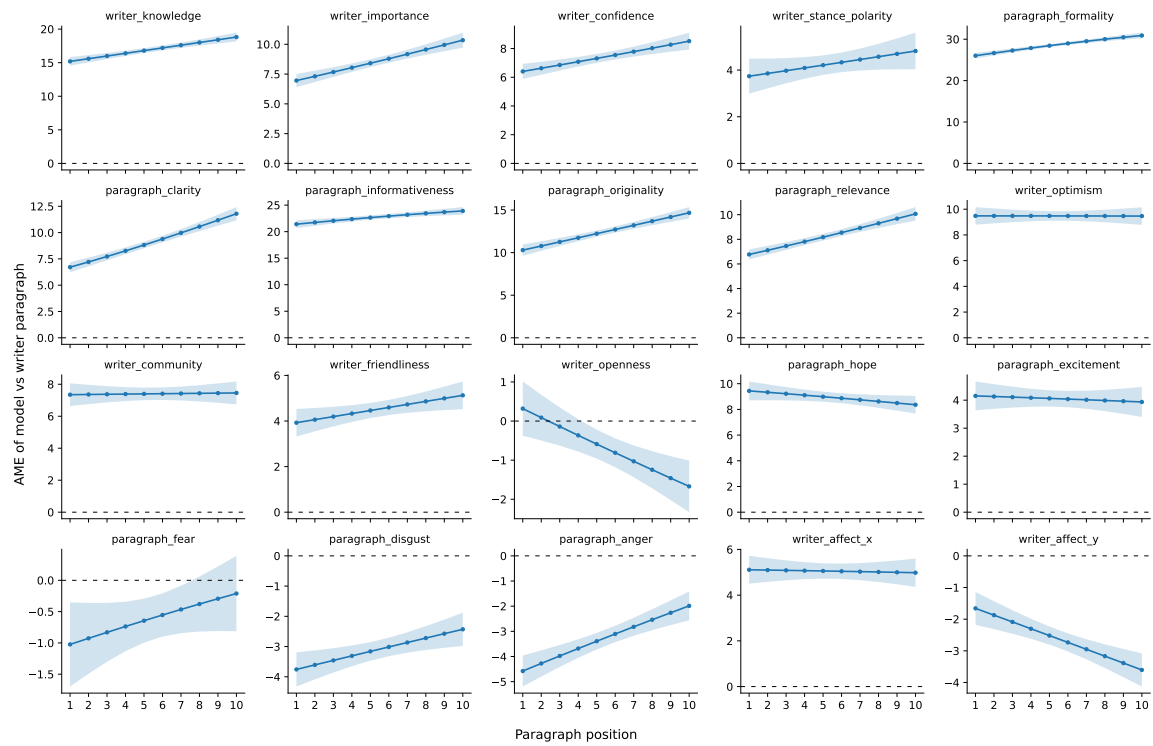


Figure S43: Average marginal effects (AMEs) of AI writing assistance on reader perceptions by paragraph position (Study 1, “preferred” subset). AMEs are estimated from mixed-effects beta regressions with reader random effects and an interaction between paragraph type and paragraph position (1–10, linear). Shaded areas show 95% confidence intervals.

5.14 Robustness Checks: Writer Engagement

In Study 1, propositions were randomly assigned to writers. Writers reported moderate-to-high engagement with their assigned propositions on average (SI:3), but a minority reported low issue knowledge, issue importance, or confidence in their opinion. If writers who were disengaged from an issue wrote less convincing paragraphs, AI writing assistance might improve their paragraphs more than it would improve paragraphs by engaged writers. Our estimates of how AI writing assistance distorts reader perceptions would then partly reflect the random assignment of propositions, rather than how people would use AI writing assistance for issues they choose to write about. To test this, we re-estimate the effects of AI writing assistance from SI:5.7 after excluding writing on issues that writers reported knowing or caring little about.

Writers reported their issue knowledge, issue importance, and confidence in their opinion on 0-100 scales for each of their assigned propositions, after writing their own paragraph and before seeing any AI-written paragraph (SI:2.3). We use these self-reports to identify low-engagement writer-proposition pairs. For each of the three measures, we exclude all writer-proposition pairs where the writer’s self-report was below the scale midpoint of 50 (i.e. below “Some understanding” for issue knowledge). We exclude both the human-written and the AI-written paragraphs for each excluded pair, so that the remaining human-written and AI-written paragraphs are still matched by writer and proposition. We focus on the 20 rating attributes measured on a 0-100 scale and on the “preferred” subset (3,795 writer-proposition pairs, 75,972 observations). This gives the following variants of the data:

- **Excluding low issue knowledge** (48,735 observations): we exclude 1,362 writer-proposition pairs (35.9%) with self-reported issue knowledge below 50.
- **Excluding low issue importance** (54,377 observations): we exclude 1,080 writer-proposition pairs (28.5%) with self-reported issue importance below 50.
- **Excluding low confidence in opinion** (62,105 observations): we exclude 691 writer-proposition pairs (18.2%) with self-reported confidence in opinion below 50.

For each rating attribute and each variant of the data, we fit the mixed-effects beta regression model from SI:5.7, with crossed random intercepts for readers, writers, and propositions. Table S206 and Figure S44 show AMEs of AI writing assistance for each specification. Excluding low-engagement writer-proposition pairs did not change the direction or significance of any AME. All AMEs were within 1.7 points of the full-sample estimates, and 52 of the 60 re-estimated AMEs were within 1 point (median absolute difference 0.4 points).

The largest changes were for attributes related to writer competence and paragraph quality, and when excluding writers with low issue knowledge. For these attributes, AMEs were slightly smaller after exclusion, but still large and significant. For example, the AME for `writer_knowledge` was +15.8 after excluding low issue knowledge, compared to +17.4 in the full sample, and the AME for `paragraph_formality` was +28.3 compared to +29.5. This suggests that AI writing assistance makes writers appear somewhat more knowledgeable and competent when they know little about an issue, but that the distortions we report are not driven by these writers. AMEs for negative emotions (`paragraph_disgust`, `paragraph_anger`, and `paragraph_fear`) became slightly more negative after exclusion (by 0.3 to 0.6 points), while AMEs for most other attributes were essentially unchanged. Overall, our main results hold for writers who were knowledgeable about, cared about, and were confident in their opinions on the issues they wrote about.

This analysis was not pre-registered.

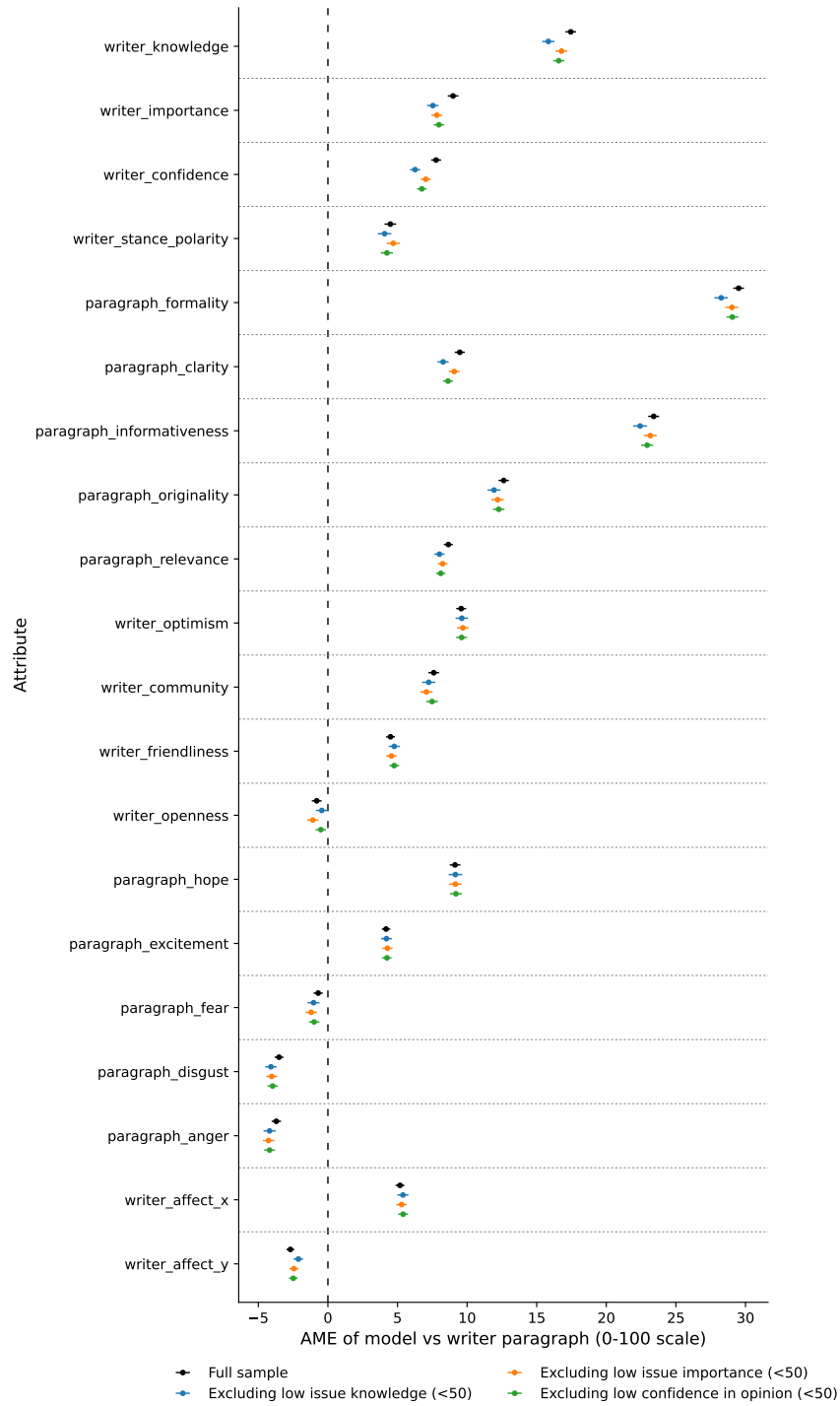


Figure S44: Average marginal effects (AMEs) of AI writing assistance on reader perceptions after excluding low-engagement writers (Study 1, “preferred” subset). Each point shows the AME from a mixed-effects beta regression comparing AI-written to human-written paragraphs, with crossed reader, writer, and proposition random effects, fitted to the full sample (black) or after excluding writer-proposition pairs where the writer’s self-reported issue knowledge, issue importance, or confidence in opinion was below 50, as described in SI:5.14. Error bars show 95% confidence intervals.

Table S206: Average marginal effects (AMEs) of AI writing assistance on reader perceptions after excluding low-engagement writers (Study 1, “preferred” subset). Each cell shows the AME of AI-written vs. human-written paragraphs (in points on the 0-100 scale) from a mixed-effects beta regression with crossed reader, writer, and proposition random effects, with 95% confidence intervals. Exclusions remove writer-proposition pairs where the writer’s self-reported measure was below 50. Number of observations: full sample N=75,972; excluding low issue knowledge N=48,735; excluding low issue importance N=54,377; excluding low confidence in opinion N=62,105. No 95% confidence interval includes zero.

Attribute	Full sample	Excl. low knowledge	Excl. low importance	Excl. low confidence
writer_knowledge	+17.4 [+17.1, +17.8]	+15.8 [+15.4, +16.3]	+16.8 [+16.4, +17.2]	+16.6 [+16.2, +17.0]
writer_importance	+9.0 [+8.6, +9.4]	+7.5 [+7.1, +7.9]	+7.8 [+7.4, +8.2]	+8.0 [+7.6, +8.4]
writer_confidence	+7.8 [+7.4, +8.1]	+6.3 [+5.9, +6.6]	+7.0 [+6.7, +7.4]	+6.7 [+6.4, +7.1]
writer_stance_polarity	+4.5 [+4.1, +4.9]	+4.1 [+3.6, +4.6]	+4.7 [+4.2, +5.2]	+4.2 [+3.8, +4.7]
paragraph_formality	+29.5 [+29.1, +29.9]	+28.3 [+27.8, +28.7]	+29.0 [+28.6, +29.5]	+29.1 [+28.6, +29.5]
paragraph_clarity	+9.5 [+9.1, +9.8]	+8.3 [+7.9, +8.7]	+9.1 [+8.7, +9.5]	+8.6 [+8.3, +9.0]
paragraph_informativeness	+23.4 [+23.0, +23.8]	+22.4 [+21.9, +22.9]	+23.2 [+22.7, +23.6]	+22.9 [+22.5, +23.4]
paragraph_originality	+12.6 [+12.3, +13.0]	+11.9 [+11.5, +12.4]	+12.2 [+11.8, +12.6]	+12.3 [+11.9, +12.7]
paragraph_relevance	+8.7 [+8.3, +9.0]	+8.0 [+7.6, +8.4]	+8.2 [+7.9, +8.6]	+8.1 [+7.8, +8.4]
writer_optimism	+9.6 [+9.2, +9.9]	+9.6 [+9.2, +10.1]	+9.7 [+9.3, +10.1]	+9.6 [+9.2, +10.0]
writer_community	+7.6 [+7.2, +8.0]	+7.2 [+6.8, +7.7]	+7.1 [+6.6, +7.5]	+7.5 [+7.1, +7.9]
writer_friendlyness	+4.5 [+4.2, +4.8]	+4.8 [+4.4, +5.2]	+4.6 [+4.2, +4.9]	+4.8 [+4.4, +5.1]
writer_openness	−0.8 [−1.2, −0.5]	−0.4 [−0.9, −0.0]	−1.1 [−1.5, −0.7]	−0.5 [−0.9, −0.1]
paragraph_hope	+9.1 [+8.7, +9.5]	+9.2 [+8.7, +9.6]	+9.2 [+8.7, +9.6]	+9.2 [+8.8, +9.6]
paragraph_excitement	+4.2 [+3.9, +4.5]	+4.2 [+3.8, +4.6]	+4.3 [+3.9, +4.6]	+4.2 [+3.9, +4.6]
paragraph_fear	−0.7 [−1.0, −0.4]	−1.0 [−1.5, −0.6]	−1.2 [−1.6, −0.8]	−1.0 [−1.4, −0.6]
paragraph_disgust	−3.5 [−3.8, −3.2]	−4.1 [−4.5, −3.7]	−4.0 [−4.4, −3.6]	−4.0 [−4.3, −3.6]
paragraph_anger	−3.7 [−4.0, −3.4]	−4.2 [−4.6, −3.8]	−4.3 [−4.7, −3.9]	−4.2 [−4.6, −3.8]
writer_affect_x	+5.2 [+4.9, +5.5]	+5.4 [+5.0, +5.8]	+5.3 [+4.9, +5.7]	+5.4 [+5.1, +5.8]
writer_affect_y	−2.7 [−3.0, −2.4]	−2.1 [−2.5, −1.8]	−2.4 [−2.8, −2.1]	−2.5 [−2.8, −2.2]

5.15 Statistical Analysis: Reader Perceptions vs Writer Self-Reports

Our distortion analyses compare how readers perceived human-written and AI-written paragraphs by the same writers. They show that AI writing assistance shifts reader perceptions, but not whether these perceptions were accurate to begin with, or whether AI writing assistance moves them closer to or further away from how writers describe themselves. In Study 1, writers reported their own stance, knowledge, importance, confidence, and affect for each proposition (SI:2.3), as well as their demographics (SI:2.1). Readers answered the same questions about the authors of the paragraphs they read (SI:2.4). Here, we compare reader perceptions to writer self-reports, separately for human-written and AI-written paragraphs.

For each paragraph-reader observation, we match the reader’s perception to the corresponding self-report by the writer of the paragraph:

- **Proposition-level attributes**, i.e. `writer_stance`, `writer_knowledge`, `writer_importance`, `writer_confidence`, `writer_affect_x`, and `writer_affect_y` are matched to the writer’s self-report on the same proposition. Writer and reader questions used identical 0-100 scales and anchors. For `writer_stance`, we use `writer_stance_post`, which writers reported directly after writing their own paragraph and before seeing any AI-written paragraph. We report results for `writer_stance_pre` and `writer_stance_final` as a robustness check below. For `writer_stance_polarity`, we apply the transformation from SI:5.3 to the self-reported stance.
- **Ordinal demographics**, i.e. `writer_age_binned`, `writer_education`, `writer_income`, `writer_english_skills`, and `writer_politicalIdeology` are matched to the writer’s self-reported category, with self-reported age binned as in SI:5.3.
- **Nominal demographics**, i.e. `writer_english_first`, `writer_gender`, `writer_race`, and `writer_politicalParty` are matched to the writer’s self-reported category. Writers could answer “Bilingual from birth” to the question about English as a first language, whereas readers could only answer “Yes” or “No”. We therefore count “Bilingual from birth” as “Yes”.
- Writers who answered “Prefer not to say” are excluded from the analysis of the corresponding attribute.

For the seven attributes measured on a 0-100 scale, we report three **accuracy metrics**:

- The **Pearson correlation** between perceived and self-reported values measures how well reader perceptions track differences between writers. It is unaffected by readers and writers using the scale differently overall, e.g. if readers systematically rate all writers as more knowledgeable than writers rate themselves.
- The **mean signed error** (perceived minus self-reported) measures bias, i.e. whether readers systematically over- or underestimate writers on a given attribute.
- The **mean absolute error** measures overall deviation of perceptions from self-reports.

For the nine categorical demographic attributes, we report the share of reader perceptions that match the writer’s self-reported category (**accuracy**). Raw accuracy depends heavily on how often readers choose each category. For example, if most writers are native English speakers and readers mostly guess that writers are native English speakers, accuracy is high even if readers cannot tell writers apart. We therefore also report **chance accuracy**, i.e. the accuracy expected if reader perceptions were independent of self-reports, given the observed distributions of both, and **Cohen’s κ** , which measures accuracy above chance ($\kappa = 0$ at chance, $\kappa = 1$ for perfect agreement). For the five ordinal attributes, we additionally report the **Pearson correlation of ordinal codes** (1 for the lowest category upwards, excluding “Other”). Unlike κ , this credits perceptions that are close to the self-reported category without matching it.

We compute all metrics at two levels: 1) for individual paragraph-reader observations, and 2) for paragraph-level perceptions aggregated across the approximately 10 readers of each paragraph, using the mean for scale attributes and ordinal codes and the modal category for categorical attributes. Aggregated perceptions average out reader-specific noise and indicate how much information about the writer a paragraph conveys to readers collectively. We compute metrics separately for human-written and AI-written paragraphs in each of the three data subsets (SI:5.2). We estimate 95% confidence intervals for each metric, and for the difference between AI-written and human-written paragraphs,

with a cluster bootstrap that resamples writers with replacement (1,000 resamples), using percentile intervals. Resampling writers accounts for the fact that each writer contributes three propositions, two paragraph types, and about 10 readers per paragraph, and keeps human-written and AI-written paragraphs by the same writer paired in each resample. The bootstrap does not account for repeated ratings by the same reader (each reader rated ten paragraphs).

Tables S207 and S208 and Figures S45 and S46 show results for the seven **scale attributes**.

For human-written paragraphs, reader perceptions tracked self-reported stance well ($r = 0.66$ for individual readers, $r = 0.80$ aggregated across readers). They tracked self-reported stance polarity moderately ($r = 0.37 / 0.56$), and knowledge, importance, confidence and affective valence (writer_affect_x) weakly ($r = 0.18$ to $0.28 / 0.33$ to 0.46). Reader perceptions barely tracked self-reported arousal (writer_affect_y, $r = 0.06 / 0.14$).

AI writing assistance slightly improved tracking of self-reported stance for individual readers (+0.026), but not when aggregating across readers (+0.005, CI includes zero). This suggests that AI-written paragraphs conveyed the writer’s stance slightly more consistently to individual readers, but not more informatively overall. Tracking of stance polarity improved very slightly at both levels (+0.043 / +0.028). For all five other scale attributes, AI writing assistance reduced tracking at both levels, and all 95% confidence intervals exclude zero. For example, the correlation for knowledge fell from 0.18 to 0.11 for individual readers and from 0.33 to 0.20 aggregated across readers. In other words, AI-written paragraphs conveyed less information about how knowledgeable, invested and confident writers were, and how they felt, according to their own self-reports.

For human-written paragraphs, reader perceptions were on average close to writer self-reports for stance (−0.6 points), knowledge (+2.2), importance (+3.9) and confidence (+0.7). Readers overestimated stance polarity (+5.2) and underestimated affective valence (−5.1). The direction of the shift from AI writing assistance relative to this baseline differed across attributes: For knowledge, importance, confidence and stance polarity, AI writing assistance introduced or increased overestimation. Readers overestimated writers’ knowledge by 17.7 points, importance by 12.8 points, confidence by 9.0 points and stance polarity by 10.0 points for AI-written paragraphs. For affective valence, AI writing assistance removed the underestimation (−5.1 to 0.0). For arousal, slight overestimation (+1.5) turned into slight underestimation (−1.2).

Changes in mean absolute error were small by comparison (−0.6 to +3.1 points for individual readers; −0.2 to +5.1 points aggregated across readers), because individual perceptions deviated substantially from self-reports for both paragraph types (mean absolute errors of 20 to 29 points for individual readers). The largest increase was for knowledge (23.7 to 26.8 for individual readers; 17.6 to 22.7 aggregated across readers).

Table S207: Accuracy of reader perceptions relative to writer self-reports for scale attributes in Study 1 (“preferred” subset, individual paragraph-reader observations; $N=37,995$ human-written and 37,977 AI-written observations from 1,438 writers). Correlation is the Pearson correlation between perceived and self-reported values. Bias is the mean signed error (perceived minus self-reported), in points on the 0-100 scale. Δ is the difference between AI-written and human-written paragraphs. 95% confidence intervals from a cluster bootstrap over writers (1,000 resamples). [†] 95% confidence interval includes zero.

Attribute	Correlation: Human	Correlation: AI	Δ Correlation	Bias: Human	Bias: AI	Δ Bias
writer_stance	0.658 [0.644, 0.672]	0.684 [0.669, 0.699]	+0.026 [+0.015, +0.036]	−0.6 [−1.3, +0.2]	+1.2 [+0.5, +1.9]	+1.8 [+1.3, +2.3]
writer_stance_polarity	0.369 [0.350, 0.388]	0.412 [0.393, 0.431]	+0.043 [+0.027, +0.062]	+5.2 [+4.2, +6.2]	+10.0 [+9.0, +11.0]	+4.8 [+4.2, +5.4]
writer_knowledge	0.184 [0.164, 0.207]	0.113 [0.093, 0.133]	−0.072 [−0.094, −0.051]	+2.2 [+1.4, +3.1]	+17.7 [+16.7, +18.5]	+15.4 [+14.9, +16.0]
writer_importance	0.264 [0.246, 0.285]	0.220 [0.200, 0.237]	−0.045 [−0.062, −0.026]	+3.9 [+2.9, +4.8]	+12.8 [+11.9, +13.7]	+8.9 [+8.5, +9.3]
writer_confidence	0.284 [0.262, 0.306]	0.232 [0.211, 0.253]	−0.052 [−0.072, −0.032]	+0.7 [−0.2, +1.6]	+9.0 [+8.0, +10.0]	+8.3 [+7.9, +8.8]
writer_affect_x	0.212 [0.192, 0.232]	0.167 [0.147, 0.187]	−0.044 [−0.061, −0.027]	−5.1 [−6.0, −4.1]	−0.0 [−0.9, +0.9]	+5.0 [+4.6, +5.4]
writer_affect_y	0.063 [0.049, 0.078]	0.039 [0.026, 0.053]	−0.024 [−0.039, −0.010]	+1.5 [+0.6, +2.4]	−1.2 [−2.1, −0.3]	−2.7 [−3.0, −2.4]

Table S208: Accuracy of paragraph-level reader perceptions relative to writer self-reports for scale attributes in Study 1 (“preferred” subset, perceptions averaged across the readers of each paragraph; N=3,795 human-written and 3,795 AI-written paragraphs from 1,438 writers). Columns as in Table S207. 95% confidence intervals from a cluster bootstrap over writers (1,000 resamples). [†] 95% confidence interval includes zero.

Attribute	Correlation: Human	Correlation: AI	Δ Correlation	Bias: Human	Bias: AI	Δ Bias
writer_stance	0.797 [0.779, 0.814]	0.802 [0.785, 0.819]	+0.005 [−0.006, +0.014] [†]	−0.7 [−1.3, +0.1]	+1.2 [+0.5, +2.0]	+1.9 [+1.4, +2.4]
writer_stance_polarity	0.560 [0.534, 0.586]	0.588 [0.563, 0.610]	+0.028 [+0.003, +0.050]	+5.3 [+4.3, +6.3]	+10.0 [+9.1, +11.0]	+4.7 [+4.1, +5.3]
writer_knowledge	0.325 [0.290, 0.361]	0.199 [0.163, 0.234]	−0.126 [−0.164, −0.091]	+2.3 [+1.4, +3.2]	+17.7 [+16.8, +18.6]	+15.4 [+14.9, +15.8]
writer_importance	0.443 [0.413, 0.472]	0.370 [0.342, 0.398]	−0.072 [−0.101, −0.045]	+3.9 [+3.0, +4.8]	+12.8 [+11.9, +13.7]	+8.9 [+8.5, +9.3]
writer_confidence	0.455 [0.424, 0.486]	0.368 [0.337, 0.398]	−0.086 [−0.115, −0.059]	+0.7 [−0.2, +1.6]	+9.0 [+8.1, +9.9]	+8.3 [+7.9, +8.8]
writer_affect_x	0.355 [0.323, 0.391]	0.293 [0.260, 0.328]	−0.062 [−0.090, −0.035]	−5.0 [−5.9, −4.1]	−0.0 [−1.0, +1.0]	+5.0 [+4.6, +5.4]
writer_affect_y	0.142 [0.107, 0.178]	0.095 [0.064, 0.128]	−0.047 [−0.083, −0.015]	+1.5 [+0.6, +2.3]	−1.1 [−2.0, −0.3]	−2.6 [−3.0, −2.3]

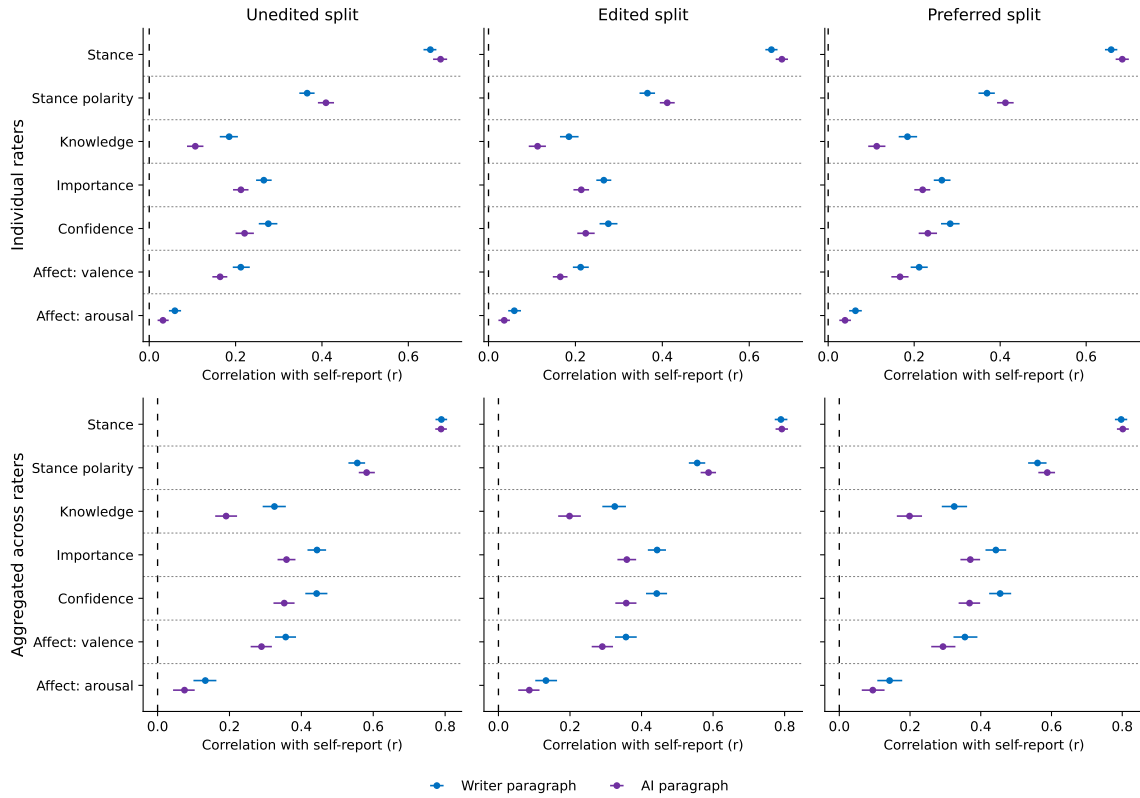


Figure S45: Correlation between reader perceptions and writer self-reports for scale attributes in Study 1, for human-written (blue) and AI-written (purple) paragraphs. Columns show the three data subsets. The top row shows correlations for individual paragraph-reader observations, and the bottom row for paragraph-level perceptions averaged across readers. Error bars show 95% confidence intervals from a cluster bootstrap over writers.

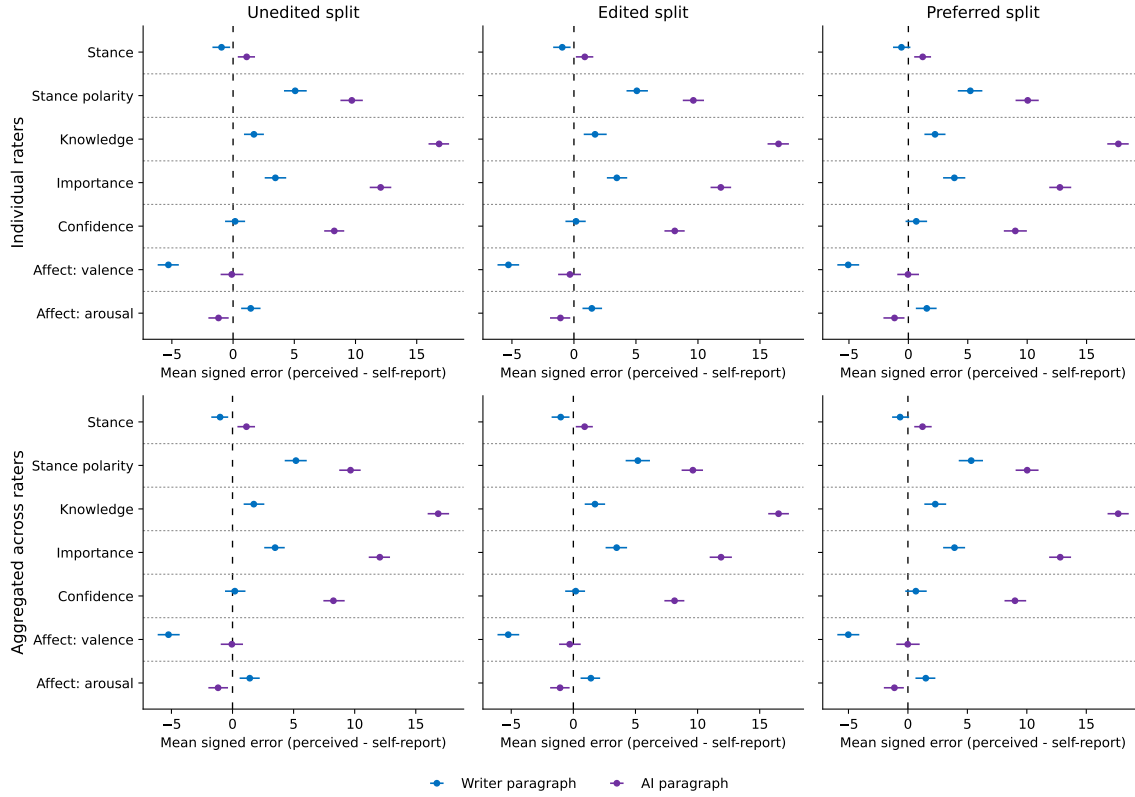


Figure S46: Mean signed error of reader perceptions relative to writer self-reports for scale attributes (perceived minus self-reported, in points on the 0-100 scale) in Study 1, for human-written (blue) and AI-written (purple) paragraphs. Positive values indicate that readers overestimated writers relative to their self-reports. Columns show the three data subsets. The top row shows individual paragraph-reader observations, and the bottom row shows paragraph-level perceptions averaged across readers. Error bars show 95% confidence intervals from a cluster bootstrap over writers.

Table S209 and Figure S47 show results for the nine **demographic attributes**.

Even for human-written paragraphs, reader perceptions of writer demographics were close to chance, with κ between 0.007 (income) and 0.066 (gender). Ordinal correlations were also low, from 0.029 (English skills) to 0.159 (political ideology). AI writing assistance reduced κ further for age group, education, English as a first language, gender and ethnicity (95% CIs exclude zero), as well as ordinal correlations for age group, education and political ideology. All other differences were small, and their confidence intervals included zero.

Raw accuracy can be misleading for these attributes. For example, accuracy for English skills rose from 31.3% for human-written paragraphs to 53.4% for AI-written paragraphs, but chance accuracy rose by the same amount (30.7% to 53.0%), so that κ remained close to zero for both. This is illustrated by Figure S48, which shows that readers perceived about 60% of AI-written paragraphs as written by an “Expert” English speaker, regardless of the writer’s self-reported skills. Since most writers described their English skills as “Expert”, this raised accuracy without any increase in readers’ ability to distinguish writers. Similarly, readers perceived about 43% of AI-written paragraphs as written by someone with an undergraduate degree, regardless of the writer’s self-reported education. This is consistent with the homogenisation reported in SI:5.11.

Figure S48 also shows that readers perceived writers as much younger than writers reported being. For human-written paragraphs, readers placed 43.8% of writers in the 18-29 age group and 12.9% in age groups of 50 or older, whereas 17.8% and 49.0% of writers, respectively, reported being in these age groups. AI writing assistance shifted perceived age slightly upwards (30.4% in the 18-29 age group and 14.4% aged 50 or older), but further reduced readers’ ability to distinguish older from younger writers (ordinal correlation 0.093 to 0.035).

Table S209: Accuracy of reader perceptions relative to writer self-reports for demographic attributes in Study 1 (“preferred” subset, individual paragraph-reader observations; N=35,658 to 37,995 observations per paragraph type from 1,348 to 1,438 writers, depending on the number of writers who answered “Prefer not to say”). Accuracy is the share of perceptions matching the self-reported category, and chance is the accuracy expected if perceptions were independent of self-reports. κ is Cohen’s κ . r_{ord} is the Pearson correlation of ordinal codes, excluding “Other” (ordinal attributes only). Δ is the difference between AI-written and human-written paragraphs. 95% confidence intervals from a cluster bootstrap over writers (1,000 resamples). † 95% confidence interval includes zero.

Attribute	Accuracy / chance: Human	Accuracy / chance: AI	κ : Human	κ : AI	$\Delta\kappa$	r_{ord} : Human	r_{ord} : AI	Δr_{ord}
writer_age_binned	0.198 / 0.180	0.185 / 0.180	0.022 [0.016, 0.028]	0.006 [0.001, 0.011]	-0.016 [-0.022, -0.009]	0.093 [0.075, 0.107]	0.035 [0.020, 0.049]	-0.058 [-0.075, -0.040]
writer_education	0.233 / 0.213	0.260 / 0.254	0.025 [0.018, 0.032]	0.009 [0.003, 0.015]	-0.016 [-0.024, -0.008]	0.097 [0.078, 0.117]	0.046 [0.032, 0.059]	-0.052 [-0.072, -0.032]
writer_income	0.197 / 0.191	0.170 / 0.170	0.007 [0.002, 0.013]	0.000 [-0.005, 0.005]	-0.007 [-0.014, +0.000] [†]	0.040 [0.021, 0.061]	0.024 [0.008, 0.039]	-0.017 [-0.036, +0.004] [†]
writer_english_skills	0.313 / 0.307	0.534 / 0.530	0.009 [0.003, 0.014]	0.007 [-0.002, 0.015]	-0.002 [-0.011, +0.008] [†]	0.029 [0.008, 0.049]	0.009 [-0.005, 0.021]	-0.020 [-0.041, +0.001] [†]
writer_politicalIdeology	0.271 / 0.239	0.271 / 0.246	0.042 [0.034, 0.050]	0.034 [0.027, 0.042]	-0.008 [-0.016, +0.000] [†]	0.159 [0.140, 0.179]	0.131 [0.111, 0.149]	-0.029 [-0.044, -0.013]
writer_english_first	0.852 / 0.847	0.918 / 0.918	0.031 [0.008, 0.053]	0.004 [-0.008, 0.015]	-0.027 [-0.052, -0.001]	-	-	-
writer_gender	0.521 / 0.487	0.506 / 0.486	0.066 [0.051, 0.082]	0.039 [0.025, 0.055]	-0.027 [-0.042, -0.012]	-	-	-
writer_race	0.642 / 0.633	0.648 / 0.645	0.026 [0.018, 0.035]	0.008 [0.000, 0.015]	-0.019 [-0.028, -0.009]	-	-	-
writer_politicalParty	0.233 / 0.208	0.236 / 0.215	0.031 [0.024, 0.038]	0.026 [0.019, 0.033]	-0.005 [-0.012, +0.002] [†]	-	-	-

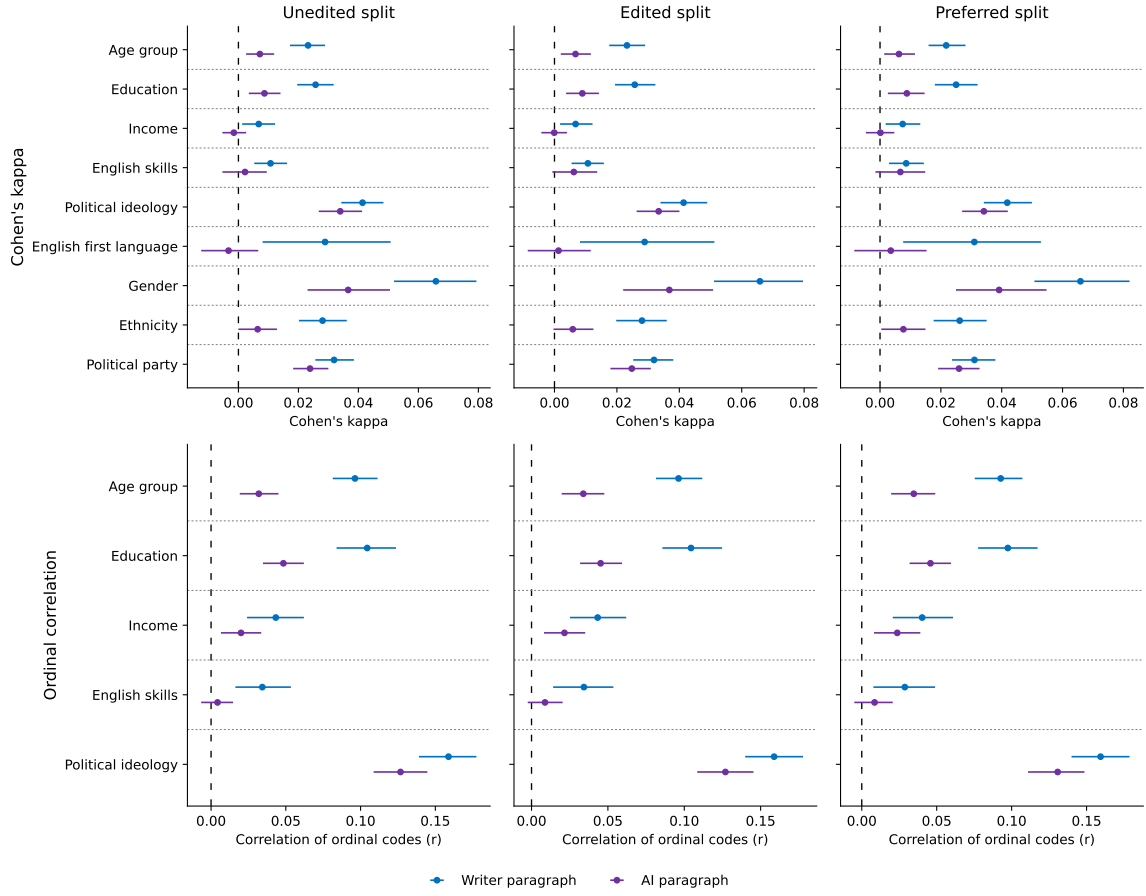


Figure S47: Accuracy of reader perceptions of writer demographics relative to writer self-reports in Study 1, for human-written (blue) and AI-written (purple) paragraphs. The top row shows Cohen's κ (accuracy above chance) for all nine demographic attributes, and the bottom row shows the Pearson correlation of ordinal codes for the five ordinal attributes. Columns show the three data subsets. All metrics are computed for individual paragraph-reader observations. Error bars show 95% confidence intervals from a cluster bootstrap over writers.

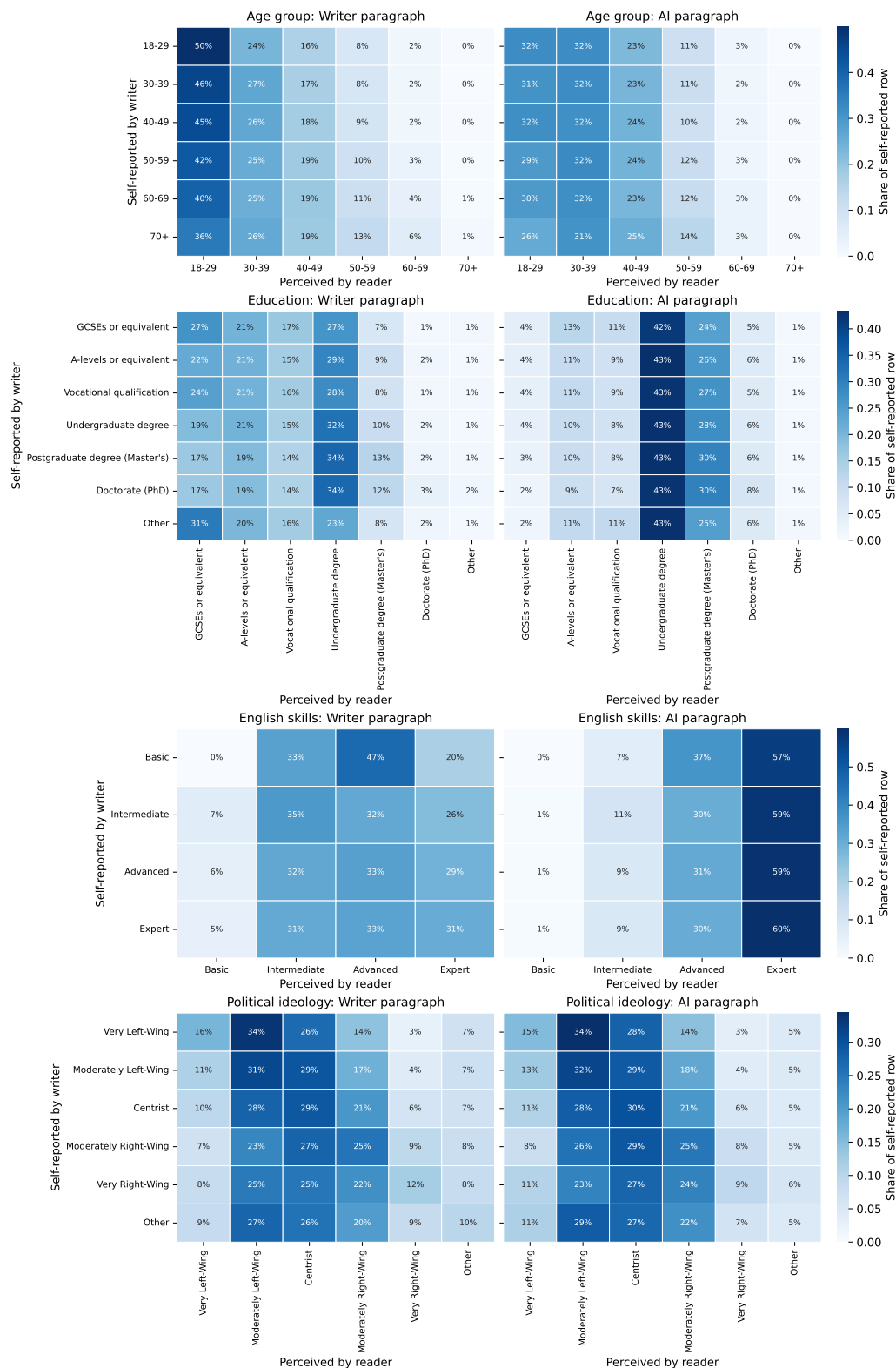


Figure S48: Reader perceptions of writer demographics by writer self-report in Study 1 (“preferred” subset, individual paragraph-reader observations), for human-written (left) and AI-written (right) paragraphs. Each row shows how readers perceived writers who reported a given category, as a share of all perceptions for that row. Accurate perceptions fall on the diagonal.

Table S210 and Figure S49 show **differences between AI-written and human-written paragraphs by AI input condition** (SI:2.11), for the “unedited” subset, where each AI-written paragraph is generated under exactly one input condition.

In the “stance-based” condition, the AI model saw only the writer’s numeric stance rating and not their text. Reader perceptions of these AI-written paragraphs therefore cannot contain information about the writer beyond their stance. This serves as a sanity check for our analysis. As expected, tracking of self-reported knowledge fell to $r = 0.02$ (compared to $r = 0.18$ for human-written paragraphs by the same writers), and the ordinal correlation for age group to $r = 0.00$ (0.10). Tracking of stance (+0.067) and stance polarity (+0.160) improved substantially in this condition, plausibly because the AI model received the writer’s stance rating directly. The “bullets-based” condition showed similar losses in tracking. In the “rewrite” and “improve” conditions, in which the AI model received the writer’s full paragraph, tracking was largely preserved, and most confidence intervals included zero. Overestimation of knowledge, importance and confidence occurred in all four conditions, but was about half as large in the “rewrite” and “improve” conditions (e.g. +8.6 and +10.7 points for knowledge) as in the “stance-based” and “bullets-based” conditions (+18.4 and +17.4 points).

Results were consistent across the “unedited”, “edited” and “preferred” data subsets (Figures S45 to S47). Differences between AI-written and human-written paragraphs in stance tracking and stance bias were also robust to the choice of self-reported stance. In the “preferred” subset, the difference in correlation was +0.043 (pre), +0.026 (post) and +0.033 (final) for stance, and +0.063, +0.043 and +0.059 for stance polarity. The difference in mean signed error was +1.8 points for stance and +4.8 points for stance polarity with all three self-reports. By contrast, the mean signed error for stance polarity in human-written paragraphs depends on which self-report is used: +8.1 (pre), +5.2 (post) and −0.7 (final), with corresponding values of +12.9, +10.0 and +4.1 for AI-written paragraphs. The absolute level of polarity bias should therefore be interpreted with caution, whereas the additional overestimation of polarity from AI writing assistance is robust.

Table S210: Difference in perception accuracy between AI-written and human-written paragraphs for scale attributes, by AI input condition in Study 1 (“unedited” subset, individual paragraph-reader observations; 1,501 writers in the “stance-based” and “bullets-based” conditions, 740 in “rewrite” and 761 in “improve”). Each cell shows the difference between AI-written paragraphs generated under the given input condition and the human-written paragraphs on the same propositions by the same writers. The top panel shows differences in the correlation between perceived and self-reported values, and the bottom panel differences in mean signed error (bias), in points on the 0-100 scale. 95% confidence intervals from a cluster bootstrap over writers (1,000 resamples). † 95% confidence interval includes zero.

Attribute	Stance-based	Bullets-based	Rewrite	Improve
<i>Δ Correlation</i>				
writer_stance	+0.067 [+0.046, +0.085]	+0.004 [−0.015, +0.022]†	+0.000 [−0.019, +0.018]†	+0.008 [−0.010, +0.026]†
writer_stance_polarity	+0.160 [+0.128, +0.193]	−0.020 [−0.053, +0.012]†	−0.006 [−0.032, +0.019]†	−0.017 [−0.046, +0.012]†
writer_knowledge	−0.155 [−0.190, −0.117]	−0.101 [−0.138, −0.065]	−0.005 [−0.040, +0.031]†	−0.004 [−0.035, +0.027]†
writer_importance	−0.094 [−0.124, −0.060]	−0.067 [−0.099, −0.037]	+0.015 [−0.012, +0.045]†	−0.022 [−0.053, +0.007]†
writer_confidence	−0.060 [−0.097, −0.025]	−0.089 [−0.123, −0.052]	−0.035 [−0.067, −0.006]	−0.015 [−0.049, +0.016]†
writer_affect_x	−0.076 [−0.103, −0.047]	−0.062 [−0.088, −0.035]	+0.002 [−0.024, +0.028]†	−0.004 [−0.035, +0.027]†
writer_affect_y	−0.038 [−0.064, −0.010]	−0.027 [−0.050, −0.001]	−0.017 [−0.048, +0.012]†	−0.019 [−0.050, +0.011]†
<i>Δ Bias</i>				
writer_stance	+2.1 [+1.2, +3.1]	+3.1 [+2.2, +4.0]	+0.7 [−0.1, +1.4]†	+1.3 [+0.5, +2.1]
writer_stance_polarity	+1.8 [+0.7, +3.0]	+9.7 [+8.7, +10.7]	+1.9 [+0.9, +2.9]	+2.7 [+1.7, +3.6]
writer_knowledge	+18.4 [+17.6, +19.1]	+17.4 [+16.7, +18.0]	+8.6 [+7.8, +9.4]	+10.7 [+9.9, +11.5]
writer_importance	+9.7 [+8.9, +10.4]	+10.9 [+10.3, +11.6]	+4.3 [+3.6, +5.0]	+6.0 [+5.3, +6.7]
writer_confidence	+8.1 [+7.4, +8.9]	+11.0 [+10.3, +11.6]	+4.4 [+3.6, +5.1]	+6.0 [+5.3, +6.7]
writer_affect_x	+8.0 [+7.3, +8.6]	+5.4 [+4.8, +6.1]	+1.7 [+1.0, +2.4]	+2.5 [+1.8, +3.2]
writer_affect_y	−2.5 [−3.1, −2.0]	−3.7 [−4.2, −3.2]	−1.3 [−2.0, −0.7]	−1.9 [−2.6, −1.3]

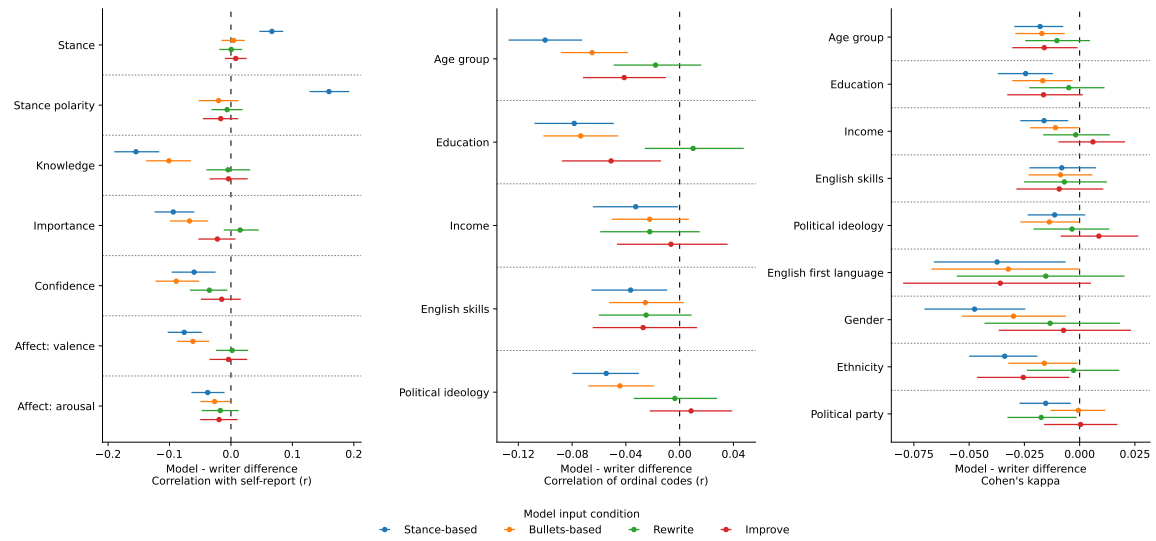


Figure S49: Difference in perception accuracy between AI-written and human-written paragraphs by AI input condition in Study 1 (“unedited” subset, individual paragraph-reader observations). Each AI-written paragraph is compared to the human-written paragraphs on the same propositions by the same writers. The left panel shows differences in the correlation with self-reports for scale attributes, the middle panel differences in the correlation of ordinal codes for ordinal demographics, and the right panel differences in Cohen’s κ for all demographics. Negative values indicate lower accuracy for AI-written paragraphs. Error bars show 95% confidence intervals from a cluster bootstrap over writers.

5.16 Statistical Analysis: Partisan Bias of AI Writing Assistance

To quantify variation in distortions across left- and right-leaning propositions (N=46 each), we fit separate **mixed-effects regression models** to each proposition-specific subset of our data, which were specified as follows:

$$\text{rating_attribute} \sim \text{paragraph_type} + (1 \mid \text{reader_id})$$

where `rating_attribute`, `paragraph_type`, and `(1 | reader_id)` are defined as in SI:5.7. The measurement scale of each rating attribute again determined the appropriate link function, following the same scheme as in SI:5.7: beta regression for 0-100 scale attributes, multinomial logistic regression for nominal categorical attributes, and ordinal logistic regression for ordinal categorical attributes.

Having confirmed that results are highly consistent across data subsets (SI:5.7), we fit each regression model to the “preferred” data subset as defined in SI:5.2, where 35,049 observations correspond to left-leaning propositions and 35,094 observations correspond to right-leaning propositions. Figures S50, S51, and S52 summarise results for scale, ordinal, and nominal variables. Tables S211 to S258 show results for each of the 29 rating attributes in our study. For this analysis only, we also show results for `writer_stance` rather than the transformed `writer_stance_polarity` because the direction of stance shifts is substantively meaningful when conditioning on proposition leaning: a shift from 40 to 60 on a left-leaning proposition (toward agreement) has a different political interpretation than the same shift on a right-leaning proposition, yet both would appear identical under the polarity transformation.

This analysis was not pre-registered

Please note that while we constructed equal numbers of left- and right-leaning propositions, these sets are not parallel. They address different political issues, vary in salience and extremity, and may elicit different levels of engagement from writers and readers. Any differences in distortion patterns between left- and right-leaning propositions therefore cannot be attributed solely to ideological direction, as they may also reflect differences in topic content, writer familiarity, or the strength of opinions elicited. Results from this analysis should be interpreted with this limitation in mind.

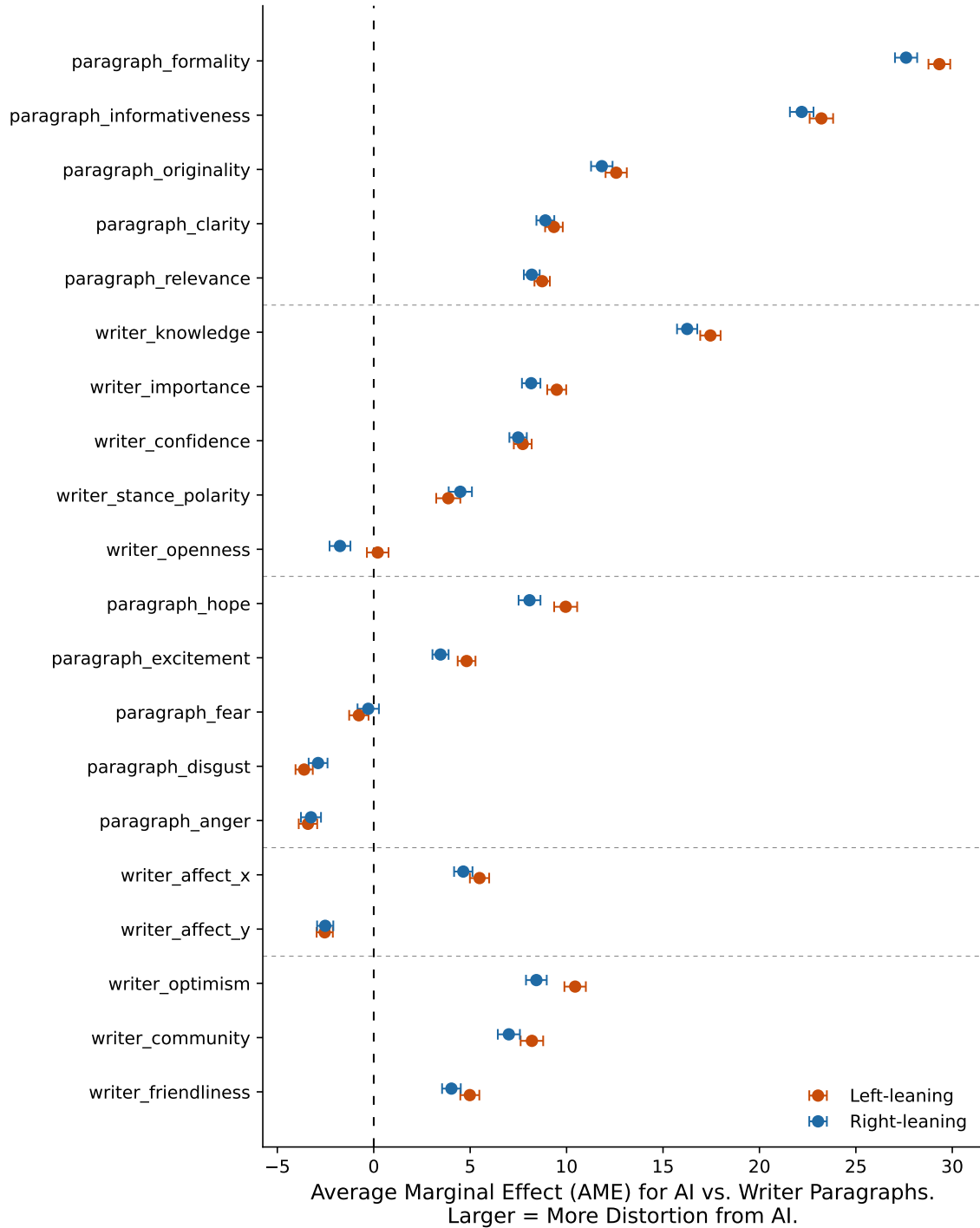


Figure S50: Average marginal effects (AMEs) of AI writing assistance on reader perceptions across all 0-100 scale rating attributes in Study 1. Each point shows the AME from a mixed-effects beta regression comparing AI-written to human-written paragraphs, with reader random effects. Results shown separately for left- and right- leaning propositions (n=46 each). Error bars show 95% confidence intervals.

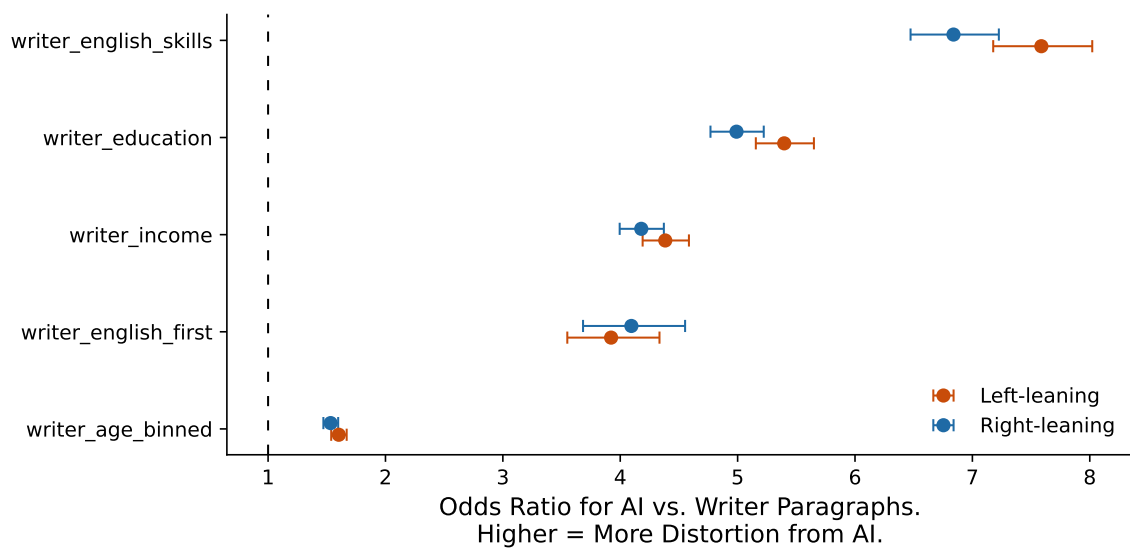


Figure S51: Odds ratios (ORs) of AI writing assistance on reader perceptions across all ordinal rating attributes in Study 1. Each point shows the odds ratio from a mixed-effects ordinal logistic regression comparing AI-written to human-written paragraphs, with reader random effects. Results shown separately for left- and right- leaning propositions (n=46 each). Error bars show 95% confidence intervals.

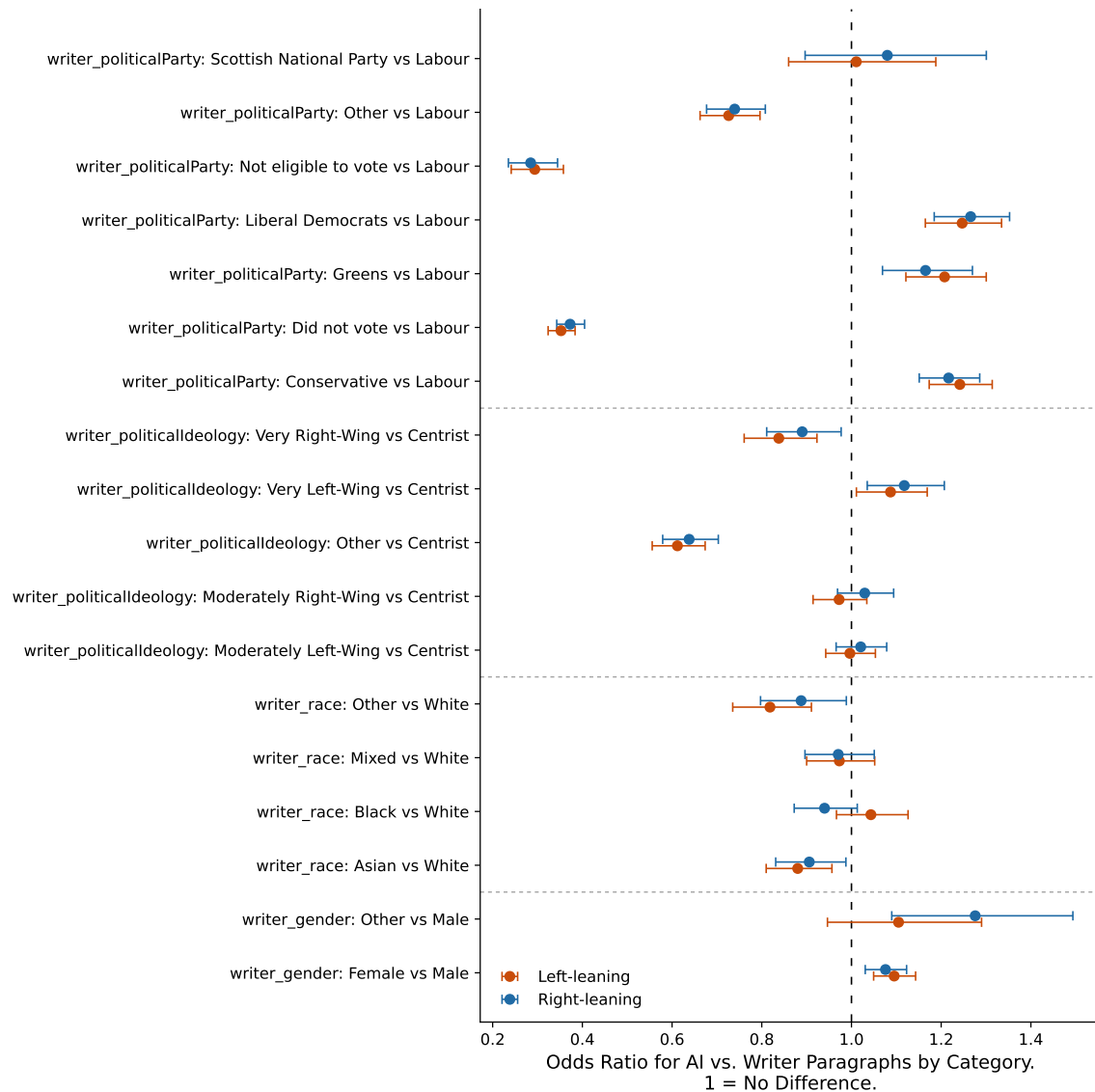


Figure S52: Odds ratios (ORs) of AI writing assistance on reader perceptions across all nominal rating attributes in Study 1. Each point shows the odds ratio of a specific category vs a reference category from a mixed-effects multinomial logistic regression comparing AI-written to human-written paragraphs, with reader random effects. Results shown separately for left- and right-leaning propositions (n=46 each). Error bars show 95% confidence intervals.

Table S211: Beta regression predicting perceived paragraph formality in Study 1 (“preferred” subset, left-leaning propositions, N=35,049). Outcome: “paragraph_formality”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	3.402	[3.318, 3.488]	96.155	0	29.329	[28.765, 29.893]

Table S212: Beta regression predicting perceived paragraph formality in Study 1 (“preferred” subset, right-leaning propositions, N=35,094). Outcome: “paragraph_formality”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	3.155	[3.076, 3.235]	89.261	0	27.604	[27.028, 28.179]

Table S213: Beta regression predicting perceived paragraph clarity in Study 1 (“preferred” subset, left-leaning propositions, N=35,049). Outcome: “paragraph_clarity”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.710	[1.667, 1.755]	40.540	0	9.344	[8.886, 9.801]

Table S214: Beta regression predicting perceived paragraph clarity in Study 1 (“preferred” subset, right-leaning propositions, N=35,094). Outcome: “paragraph_clarity”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.664	[1.621, 1.708]	38.051	8.427e-317	8.899	[8.436, 9.362]

Table S215: Beta regression predicting perceived paragraph informativeness in Study 1 (“preferred” subset, left-leaning propositions, N=35,049). Outcome: “paragraph_informativeness”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	2.575	[2.509, 2.642]	72.069	0	23.212	[22.604, 23.820]

Table S216: Beta regression predicting perceived paragraph informativeness in Study 1 (“preferred” subset, right-leaning propositions, N=35,094). Outcome: “paragraph_informativeness”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	2.466	[2.404, 2.531]	68.727	0	22.192	[21.580, 22.804]

Table S217: Beta regression predicting perceived paragraph originality in Study 1 (“preferred” subset, left-leaning propositions, N=35,049). Outcome: “paragraph_originality”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.717	[1.677, 1.758]	44.613	0	12.572	[12.021, 13.122]

Table S218: Beta regression predicting perceived paragraph originality in Study 1 (“preferred” subset, right-leaning propositions, N=35,094). Outcome: “paragraph_originality”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.660	[1.621, 1.700]	41.457	0	11.826	[11.269, 12.383]

Table S219: Beta regression predicting perceived paragraph relevance in Study 1 (“preferred” subset, left-leaning propositions, N=35,049). Outcome: “paragraph_relevance”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.766	[1.722, 1.811]	44.173	0	8.734	[8.334, 9.133]

Table S220: Beta regression predicting perceived paragraph relevance in Study 1 (“preferred” subset, right-leaning propositions, N=35,094). Outcome: “paragraph_relevance”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.686	[1.644, 1.730]	40.059	0	8.193	[7.783, 8.603]

Table S221: Beta regression predicting perceived writer knowledge in Study 1 (“preferred” subset, left-leaning propositions, N=35,049). Outcome: “writer_knowledge”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	2.147	[2.097, 2.197]	63.998	0	17.458	[16.930, 17.985]

Table S222: Beta regression predicting perceived writer knowledge in Study 1 (“preferred” subset, right-leaning propositions, N=35,094). Outcome: “writer_knowledge”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	2.036	[1.989, 2.084]	60.080	0	16.251	[15.727, 16.775]

Table S223: Beta regression predicting perceived writer importance in Study 1 (“preferred” subset, left-leaning propositions, N=35,049). Outcome: “writer_importance”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.595	[1.557, 1.634]	37.904	2.172e-314	9.489	[8.997, 9.980]

Table S224: Beta regression predicting perceived writer importance in Study 1 (“preferred” subset, right-leaning propositions, N=35,094). Outcome: “writer_importance”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.513	[1.476, 1.550]	33.524	2.167e-246	8.164	[7.685, 8.642]

Table S225: Beta regression predicting perceived writer confidence in Study 1 (“preferred” subset, left-leaning propositions, N=35,049). Outcome: “writer_confidence”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.530	[1.492, 1.570]	32.875	4.935e-237	7.727	[7.263, 8.190]

Table S226: Beta regression predicting perceived writer confidence in Study 1 (“preferred” subset, right-leaning propositions, N=35,094). Outcome: “writer_confidence”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.527	[1.489, 1.567]	32.778	1.224e-235	7.485	[7.034, 7.936]

Table S227: Beta regression predicting perceived writer stance in Study 1 (“preferred” subset, left-leaning propositions, N=35,049). Outcome: “writer_stance”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.148	[1.115, 1.181]	9.434	3.938e-21	3.437	[2.724, 4.151]

Table S228: Beta regression predicting perceived writer stance in Study 1 (“preferred” subset, right-leaning propositions, N=35,094). Outcome: “writer_stance”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.016	[0.988, 1.046]	1.103	0.270	0.396	[-0.308, 1.101]

Table S229: Beta regression predicting perceived writer stance polarity in Study 1 (“preferred” subset, left-leaning propositions, N=35,049). Outcome: “writer_stance_polarity”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.189	[1.156, 1.223]	12.112	9.104e-34	3.865	[3.239, 4.490]

Table S230: Beta regression predicting perceived writer stance polarity in Study 1 (“preferred” subset, right-leaning propositions, N=35,094). Outcome: “writer_stance_polarity”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.230	[1.197, 1.265]	14.608	2.511e-48	4.489	[3.887, 5.091]

Table S231: Beta regression predicting perceived paragraph hope in Study 1 (“preferred” subset, left-leaning propositions, N=35,049). Outcome: “paragraph_hope”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.510	[1.473, 1.548]	32.538	3.078e-232	9.952	[9.356, 10.548]

Table S232: Beta regression predicting perceived paragraph hope in Study 1 (“preferred” subset, right-leaning propositions, N=35,094). Outcome: “paragraph_hope”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.410	[1.376, 1.444]	27.854	9.511e-171	8.078	[7.511, 8.645]

Table S233: Beta regression predicting perceived paragraph excitement in Study 1 (“preferred” subset, left-leaning propositions, N=35,049). Outcome: “paragraph_excitement”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.298	[1.266, 1.331]	20.559	6.445e-94	4.815	[4.353, 5.276]

Table S234: Beta regression predicting perceived paragraph excitement in Study 1 (“preferred” subset, right-leaning propositions, N=35,094). Outcome: “paragraph_excitement”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.225	[1.195, 1.255]	16.335	5.602e-60	3.463	[3.045, 3.880]

Table S235: Beta regression predicting perceived paragraph fear in Study 1 (“preferred” subset, left-leaning propositions, N=35,049). Outcome: “paragraph_fear”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	0.962	[0.937, 0.986]	−3.013	0.003	−0.772	[−1.274, −0.270]

Table S236: Beta regression predicting perceived paragraph fear in Study 1 (“preferred” subset, right-leaning propositions, N=35,094). Outcome: “paragraph_fear”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	0.987	[0.962, 1.013]	−1.009	0.313	−0.286	[−0.841, 0.269]

Table S237: Beta regression predicting perceived paragraph disgust in Study 1 (“preferred” subset, left-leaning propositions, N=35,049). Outcome: “paragraph_disgust”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	0.809	[0.788, 0.830]	−15.872	9.920e-57	−3.608	[−4.055, −3.161]

Table S238: Beta regression predicting perceived paragraph disgust in Study 1 (“preferred” subset, right-leaning propositions, N=35,094). Outcome: “paragraph_disgust”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	0.855	[0.833, 0.878]	−11.578	5.329e-31	−2.886	[−3.375, −2.397]

Table S239: Beta regression predicting perceived paragraph anger in Study 1 (“preferred” subset, left-leaning propositions, N=35,049). Outcome: “paragraph_anger”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	0.829	[0.807, 0.851]	−13.989	1.815e-44	−3.411	[−3.889, −2.932]

Table S240: Beta regression predicting perceived paragraph anger in Study 1 (“preferred” subset, right-leaning propositions, N=35,094). Outcome: “paragraph_anger”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	0.847	[0.825, 0.870]	−12.282	1.129e-34	−3.255	[−3.775, −2.735]

Table S241: Beta regression predicting perceived writer affect x in Study 1 (“preferred” subset, left-leaning propositions, N=35,049). Outcome: “writer_affect_x”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.249	[1.224, 1.274]	21.468	3.108e-102	5.485	[4.986, 5.985]

Table S242: Beta regression predicting perceived writer affect x in Study 1 (“preferred” subset, right-leaning propositions, N=35,094). Outcome: “writer_affect_x”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.212	[1.189, 1.237]	19.057	5.747e-81	4.643	[4.166, 5.120]

Table S243: Beta regression predicting perceived writer affect y in Study 1 (“preferred” subset, left-leaning propositions, N=35,049). Outcome: “writer_affect_y”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	0.901	[0.885, 0.916]	−11.832	2.671e-32	−2.541	[−2.962, −2.120]

Table S244: Beta regression predicting perceived writer affect y in Study 1 (“preferred” subset, right-leaning propositions, N=35,094). Outcome: “writer_affect_y”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	0.901	[0.886, 0.917]	−11.727	9.296e-32	−2.517	[−2.938, −2.097]

Table S245: Beta regression predicting perceived writer optimism in Study 1 (“preferred” subset, left-leaning propositions, N=35,049). Outcome: “writer_optimism”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.522	[1.488, 1.556]	36.573	7.596e-293	10.443	[9.887, 10.998]

Table S246: Beta regression predicting perceived writer optimism in Study 1 (“preferred” subset, right-leaning propositions, N=35,094). Outcome: “writer_optimism”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.406	[1.376, 1.437]	30.644	3.191e-206	8.432	[7.896, 8.969]

Table S247: Beta regression predicting perceived writer community orientation in Study 1 (“preferred” subset, left-leaning propositions, N=35,049). Outcome: “writer_community”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.413	[1.379, 1.449]	27.403	2.510e-165	8.201	[7.617, 8.786]

Table S248: Beta regression predicting perceived writer community orientation in Study 1 (“preferred” subset, right-leaning propositions, N=35,094). Outcome: “writer_community”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.347	[1.314, 1.380]	23.923	1.750e-126	7.006	[6.433, 7.579]

Table S249: Beta regression predicting perceived writer friendliness in Study 1 (“preferred” subset, left-leaning propositions, N=35,049). Outcome: “writer_friendliness”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.223	[1.199, 1.247]	19.785	4.043e-87	4.982	[4.490, 5.475]

Table S250: Beta regression predicting perceived writer friendliness in Study 1 (“preferred” subset, right-leaning propositions, N=35,094). Outcome: “writer_friendliness”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.176	[1.153, 1.199]	16.286	1.242e-59	4.028	[3.544, 4.512]

Table S251: Beta regression predicting perceived writer openness in Study 1 (“preferred” subset, left-leaning propositions, N=35,049). Outcome: “writer_openness”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	1.009	[0.984, 1.035]	0.720	0.472	0.206	[-0.355, 0.766]

Table S252: Beta regression predicting perceived writer openness in Study 1 (“preferred” subset, right-leaning propositions, N=35,094). Outcome: “writer_openness”. Reference category: “paragraph_type_writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
paragraph_type_model	0.921	[0.898, 0.945]	-6.338	2.320e-10	-1.750	[-2.291, -1.209]

Table S253: Multinomial logistic regression predicting perceived writer race in Study 1 (“preferred” subset, left-leaning propositions, N=35,049). Outcome: “writer_race”. Predictor reference category: “paragraph_type_writer”. Outcome reference level: “White”. OR is odds ratio.

Term	Target level	Reference level	OR	95% CI (OR)	statistic	p
paragraph_type_model	Asian	White	0.880	[0.810, 0.956]	−3.009	0.003
paragraph_type_model	Black	White	1.043	[0.967, 1.126]	1.087	0.277
paragraph_type_model	Mixed	White	0.973	[0.900, 1.052]	−0.690	0.490
paragraph_type_model	Other	White	0.818	[0.735, 0.911]	−3.673	2.395e-4

Table S254: Multinomial logistic regression predicting perceived writer race in Study 1 (“preferred” subset, right-leaning propositions, N=35,094). Outcome: “writer_race”. Predictor reference category: “paragraph_type_writer”. Outcome reference level: “White”. OR is odds ratio.

Term	Target level	Reference level	OR	95% CI (OR)	statistic	p
paragraph_type_model	Asian	White	0.906	[0.831, 0.988]	−2.243	0.025
paragraph_type_model	Black	White	0.940	[0.872, 1.013]	−1.623	0.105
paragraph_type_model	Mixed	White	0.971	[0.896, 1.051]	−0.738	0.461
paragraph_type_model	Other	White	0.888	[0.797, 0.989]	−2.170	0.030

Table S255: Multinomial logistic regression predicting perceived writer gender in Study 1 (“preferred” subset, left-leaning propositions, N=35,049). Outcome: “writer_gender”. Predictor reference category: “paragraph_type_writer”. Outcome reference level: “Male”. OR is odds ratio.

Term	Target level	Reference level	OR	95% CI (OR)	statistic	p
paragraph_type_model	Female	Male	1.095	[1.049, 1.143]	4.173	3.001e-5
paragraph_type_model	Other	Male	1.105	[0.946, 1.290]	1.263	0.206

Table S256: Multinomial logistic regression predicting perceived writer gender in Study 1 (“preferred” subset, right-leaning propositions, N=35,094). Outcome: “writer_gender”. Predictor reference category: “paragraph_type_writer”. Outcome reference level: “Male”. OR is odds ratio.

Term	Target level	Reference level	OR	95% CI (OR)	statistic	p
paragraph_type_model	Female	Male	1.076	[1.031, 1.123]	3.343	8.293e-4
paragraph_type_model	Other	Male	1.276	[1.090, 1.494]	3.030	0.002

Table S257: Multinomial logistic regression predicting perceived writer political party in Study 1 (“preferred” subset, left-leaning propositions, N=35,049). Outcome: “writer_politicalParty”. Predictor reference category: “paragraph_type_writer”. Outcome reference level: “Labour”. OR is odds ratio.

Term	Target level	Reference level	OR	95% CI (OR)	statistic	p
paragraph_type_model	Conservative	Labour	1.242	[1.173, 1.314]	7.498	6.483e-14
paragraph_type_model	Did not vote	Labour	0.352	[0.324, 0.384]	−23.987	3.818e-127
paragraph_type_model	Greens	Labour	1.208	[1.121, 1.300]	4.989	6.072e-7
paragraph_type_model	Liberal Democrats	Labour	1.247	[1.165, 1.335]	6.348	2.186e-10
paragraph_type_model	Not eligible to vote	Labour	0.294	[0.241, 0.358]	−12.197	3.229e-34
paragraph_type_model	Other	Labour	0.726	[0.662, 0.796]	−6.824	8.841e-12
paragraph_type_model	Scottish National Party	Labour	1.011	[0.860, 1.188]	0.130	0.897

Table S258: Multinomial logistic regression predicting perceived writer political party in Study 1 (“preferred” subset, right-leaning propositions, N=35,094). Outcome: “writer_politicalParty”. Predictor reference category: “paragraph_type_writer”. Outcome reference level: “Labour”. OR is odds ratio.

Term	Target level	Reference level	OR	95% CI (OR)	statistic	p
paragraph_type_model	Conservative	Labour	1.217	[1.151, 1.286]	6.937	3.993e-12
paragraph_type_model	Did not vote	Labour	0.373	[0.343, 0.405]	−23.258	1.172e-119
paragraph_type_model	Greens	Labour	1.165	[1.069, 1.270]	3.485	4.926e-4
paragraph_type_model	Liberal Democrats	Labour	1.266	[1.185, 1.353]	6.964	3.308e-12
paragraph_type_model	Not eligible to vote	Labour	0.285	[0.235, 0.345]	−12.877	6.030e-38
paragraph_type_model	Other	Labour	0.740	[0.677, 0.808]	−6.685	2.305e-11
paragraph_type_model	Scottish National Party	Labour	1.080	[0.897, 1.301]	0.810	0.418

Table S259: Multinomial logistic regression predicting perceived writer political ideology in Study 1 (“preferred” subset, left-leaning propositions, N=35,049). Outcome: “writer_politicalIdeology”. Predictor reference category: “paragraph_type_writer”. Outcome reference level: “Centrist”. OR is odds ratio.

Term	Target level	Reference level	OR	95% CI (OR)	statistic	p
paragraph_type_model	Moderately Left-Wing	Centrist	0.996	[0.943, 1.053]	−0.125	0.900
paragraph_type_model	Moderately Right-Wing	Centrist	0.972	[0.914, 1.034]	−0.889	0.374
paragraph_type_model	Other	Centrist	0.612	[0.556, 0.674]	−9.994	1.626e-23
paragraph_type_model	Very Left-Wing	Centrist	1.087	[1.011, 1.169]	2.255	0.024
paragraph_type_model	Very Right-Wing	Centrist	0.838	[0.761, 0.923]	−3.583	3.392e-4

Table S260: Multinomial logistic regression predicting perceived writer political ideology in Study 1 (“preferred” subset, right-leaning propositions, N=35,094). Outcome: “writer_politicalIdeology”. Predictor reference category: “paragraph_type_writer”. Outcome reference level: “Centrist”. OR is odds ratio.

Term	Target level	Reference level	OR	95% CI (OR)	statistic	p
paragraph_type_model	Moderately Left-Wing	Centrist	1.021	[0.966, 1.079]	0.724	0.469
paragraph_type_model	Moderately Right-Wing	Centrist	1.030	[0.969, 1.094]	0.941	0.347
paragraph_type_model	Other	Centrist	0.638	[0.579, 0.703]	−9.076	1.126e-19
paragraph_type_model	Very Left-Wing	Centrist	1.118	[1.035, 1.207]	2.842	0.004
paragraph_type_model	Very Right-Wing	Centrist	0.890	[0.811, 0.977]	−2.450	0.014

Table S261: Ordinal logistic regression predicting perceived writer English first-language status in Study 1 (“preferred” subset, left-leaning propositions, N=35,049). Outcome: “writer_english_first”. Reference category: “paragraph_type_writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
paragraph_type_model	3.922	[3.549, 4.334]	26.801	3.166e-158

Table S262: Ordinal logistic regression predicting perceived writer English first-language status in Study 1 (“preferred” subset, right-leaning propositions, N=35,094). Outcome: “writer_english_first”. Reference category: “paragraph_type_writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
paragraph_type_model	4.095	[3.683, 4.553]	26.078	6.522e-150

Table S263: Ordinal logistic regression predicting perceived writer education level in Study 1 (“preferred” subset, left-leaning propositions, N=35,049). Outcome: “writer_education”. Reference category: “paragraph_type_writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
paragraph_type_model	5.397	[5.155, 5.650]	72.098	0

Table S264: Ordinal logistic regression predicting perceived writer education level in Study 1 (“preferred” subset, right-leaning propositions, N=35,094). Outcome: “writer_education”. Reference category: “paragraph_type_writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
paragraph_type_model	4.991	[4.769, 5.223]	69.326	0

Table S265: Ordinal logistic regression predicting perceived writer English skill level in Study 1 (“preferred” subset, left-leaning propositions, N=35,049). Outcome: “writer_english_skills”. Reference category: “paragraph_type_writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
paragraph_type_model	7.588	[7.177, 8.021]	71.471	0

Table S266: Ordinal logistic regression predicting perceived writer English skill level in Study 1 (“preferred” subset, right-leaning propositions, N=35,094). Outcome: “writer_english_skills”. Reference category: “paragraph_type_writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
paragraph_type_model	6.839	[6.473, 7.225]	68.562	0

Table S267: Ordinal logistic regression predicting perceived writer income level in Study 1 (“preferred” subset, left-leaning propositions, N=35,049). Outcome: “writer_income”. Reference category: “paragraph_type_writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
paragraph_type_model	4.384	[4.191, 4.585]	64.408	0

Table S268: Ordinal logistic regression predicting perceived writer income level in Study 1 (“preferred” subset, right-leaning propositions, N=35,094). Outcome: “writer_income”. Reference category: “paragraph_type_writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
paragraph_type_model	4.179	[3.995, 4.372]	62.265	0

Table S269: Ordinal logistic regression predicting perceived writer age group in Study 1 (“preferred” subset, left-leaning propositions, N=35,049). Outcome: “writer_age_binned”. Reference category: “paragraph_type_writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
paragraph_type_model	1.602	[1.538, 1.669]	22.485	5.843e-112

Table S270: Ordinal logistic regression predicting perceived writer age group in Study 1 (“preferred” subset, right-leaning propositions, N=35,094). Outcome: “writer_age_binned”. Reference category: “paragraph_type_writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
paragraph_type_model	1.533	[1.471, 1.597]	20.372	2.946e-92

5.17 Statistical Analysis: Linguistic Properties of Human and AI-Assisted Writing

To complement our results on perceptual distortion, we compare the linguistic properties of human-written and AI-written paragraphs in Study 1. We analyse all 4,503 human-written paragraphs and the 4,503 AI-written paragraphs generated for the same writer-proposition pairs, before any writer edits (as in the “unedited” subset; see SI:5.2). We additionally report results for the 1,002 AI-written paragraphs that writers edited, in their edited form. We compute four groups of features for each paragraph.

1) Errors. We check each paragraph with LanguageTool 6.8⁴⁵, an open-source rule-based spelling and grammar checker. Since writers were UK-based, we check paragraphs against British English. To avoid penalising American English spellings, which were common in AI-written paragraphs (e.g. “recognize”), we only count a spelling error if it is also flagged by the American English spell checker, and we disable rules that concern the choice between British and American variants. We do not count spelling errors for acronym-like tokens with two or more capital letters (e.g. “UBI”), which are mostly correct but not in the dictionary. We group errors by LanguageTool category into spelling, grammar, punctuation, and casing errors, as well as whitespace issues (e.g. double spaces) and other style issues. Our main error measure is the number of spelling, grammar, punctuation, and casing errors per 100 words, which excludes whitespace and style issues.

2) Length and readability. We count words and sentences, and compute the mean number of words per sentence and of characters per word. We compute the Flesch-Kincaid grade level⁴⁶, which estimates the US school grade needed to understand a text based on words per sentence and syllables per word. We count syllables using the pyphen hyphenation dictionary.

3) Lexical diversity. We compute the Measure of Textual Lexical Diversity MTLD;⁴⁷, which is the mean length of word sequences that maintain a type-token ratio above 0.72. Higher values indicate more diverse vocabulary. Unlike the raw type-token ratio, MTLD is largely independent of text length. We average MTLD over forward and backward passes through each paragraph.

4) Pronoun use. We count first-person singular pronouns (e.g. “I”, “my”) and first-person plural pronouns (e.g. “we”, “our”) per 100 words. Pronoun use is a well-established marker of how writers position themselves socially⁴⁸.

For each paragraph type, we report mean feature values with 95% confidence intervals from a cluster bootstrap (2,000 resamples), resampling writers to account for each writer contributing up to three paragraphs. To compare human-written and AI-written paragraphs directly, we compute the difference between the AI-written paragraph and the human-written paragraph for each writer-proposition pair (AI – human). We report the mean difference with a writer-clustered bootstrap 95% confidence interval, and test for significance with a Wilcoxon signed-rank test, applying Bonferroni correction across all 16 features. The paired comparison uses AI-written paragraphs as generated, before any writer edits.

Table S271 shows results for all features. AI-written paragraphs contained almost no errors. Human-written paragraphs contained 1.64 errors per 100 words on average, compared to 0.04 for AI-written paragraphs, and 67.3% of human-written paragraphs contained at least one error, compared to 3.8% of AI-written paragraphs. AI-written paragraphs had fewer errors than the corresponding human-written paragraph in 66.9% of writer-proposition pairs and more errors in 1.3%, with the remaining pairs tied. Writer edits reintroduced some errors (0.19 errors per 100 words in edited AI-written paragraphs), but far fewer than in human-written paragraphs.

AI-written paragraphs also used more complex and more diverse vocabulary. Words in AI-written paragraphs were 1.23 characters longer on average, and MTLD was 73% higher than in human-written paragraphs (154.0 vs. 89.2). As a result, Flesch-Kincaid grade levels were three grades higher for AI-written paragraphs (13.9 vs. 10.8), even though AI-written sentences were shorter (21.7 vs. 24.8 words). AI-written paragraphs used fewer first-person pronouns, and in particular less than half as many first-person plural pronouns as human-written paragraphs (0.51 vs. 1.15 per 100 words). AI-written paragraphs were slightly longer than human-written paragraphs (119.0 vs. 115.8 words). Note that paragraph length was constrained by study design: writers had to write at least 100 words, and AI models were instructed to write 100-150 words or to preserve the length of the human-written paragraph (SI:2.11).

All differences are significant at $p < .001$ after Bonferroni correction. Differences point in the same direction for all three AI models, except for paragraph length (DeepSeek V3 paragraphs were 3.2 words shorter than human-written paragraphs on average), with some variation in magnitude. For example, Flesch-Kincaid grade levels ranged from 12.9 (DeepSeek V3) to 15.6 (Claude Sonnet 4), and MTLT from 138.8 (ChatGPT-4o) to 164.9 (Claude Sonnet 4). Across input conditions, AI-written paragraphs contained slightly more errors when the AI model was given the full human-written paragraph to rewrite (0.06 errors per 100 words) or improve (0.07) than when it was given only the writer’s stance (0.02) or bullet points (0.03), consistent with some errors being carried over from the human-written paragraph. Full results by AI model and input condition are available in our code repository.

Table S271: Text features of human-written vs. AI-written paragraphs in Study 1. Values are means with 95% confidence intervals from a writer-clustered bootstrap (2,000 resamples), for 4,503 human-written paragraphs, 4,503 AI-written paragraphs, and 1,002 AI-written paragraphs edited by writers. Δ is the mean paired difference between the AI-written (unedited) and human-written paragraph for the same writer-proposition pair (AI – human; 4,503 pairs). Errors are counted with LanguageTool; “all errors” comprises spelling, grammar, punctuation, and casing errors, but not whitespace and style issues. Significance from Wilcoxon signed-rank tests on paired differences: * $p < .05$, ** $p < .01$, *** $p < .001$, after Bonferroni correction across all 16 features.

Feature	Human-written	AI-written	AI-written (edited)	Δ	95% CI (Δ)	Sig.
<i>Errors</i>						
All errors per 100 words	1.64 [1.55, 1.73]	0.04 [0.03, 0.04]	0.19 [0.13, 0.28]	−1.60	[−1.70, −1.52]	***
% paragraphs with ≥ 1 error	67.3 [65.6, 68.9]	3.8 [3.2, 4.3]	14.4 [12.2, 16.7]	−63.6	[−65.2, −61.7]	***
Spelling errors per 100 words	0.80 [0.73, 0.87]	0.02 [0.01, 0.02]	0.10 [0.06, 0.17]	−0.78	[−0.85, −0.72]	***
Grammar errors per 100 words	0.24 [0.22, 0.26]	0.01 [0.01, 0.01]	0.03 [0.02, 0.04]	−0.23	[−0.25, −0.21]	***
Punctuation errors per 100 words	0.38 [0.36, 0.40]	0.01 [0.01, 0.01]	0.04 [0.03, 0.05]	−0.37	[−0.39, −0.35]	***
Casing errors per 100 words	0.22 [0.19, 0.25]	0.00 [0.00, 0.00]	0.02 [0.00, 0.04]	−0.22	[−0.26, −0.19]	***
Whitespace issues per 100 words	0.32 [0.27, 0.37]	0.00 [0.00, 0.00]	0.07 [0.05, 0.10]	−0.32	[−0.37, −0.27]	***
Style issues per 100 words	0.29 [0.28, 0.31]	0.03 [0.02, 0.03]	0.05 [0.03, 0.06]	−0.27	[−0.28, −0.25]	***
<i>Length and readability</i>						
Words	115.8 [114.9, 116.7]	119.0 [118.5, 119.4]	120.3 [118.2, 122.2]	+3.19	[+2.34, +4.00]	***
Sentences	5.31 [5.23, 5.39]	5.59 [5.56, 5.62]	5.70 [5.59, 5.80]	+0.28	[+0.21, +0.35]	***
Words per sentence	24.8 [24.2, 25.4]	21.7 [21.6, 21.8]	21.8 [21.4, 22.1]	−3.13	[−3.73, −2.60]	***
Characters per word	4.74 [4.73, 4.76]	5.97 [5.96, 5.98]	5.90 [5.87, 5.94]	+1.23	[+1.21, +1.24]	***
Flesch-Kincaid grade level	10.8 [10.6, 11.1]	13.9 [13.8, 13.9]	13.6 [13.5, 13.8]	+3.04	[+2.81, +3.25]	***
<i>Lexical diversity</i>						
MTLD	89.2 [88.2, 90.2]	154.0 [152.9, 155.2]	151.5 [148.8, 154.6]	+64.8	[+63.5, +66.2]	***
<i>Personal voice</i>						
First-person singular per 100 words	1.43 [1.36, 1.50]	1.15 [1.10, 1.19]	1.38 [1.26, 1.50]	−0.29	[−0.34, −0.24]	***
First-person plural per 100 words	1.15 [1.08, 1.21]	0.51 [0.47, 0.54]	0.51 [0.43, 0.58]	−0.64	[−0.69, −0.59]	***

To identify words that distinguish AI-written from human-written paragraphs, we compute log-odds ratios with an informative Dirichlet prior⁴⁹, using the pooled word counts of all human-written and AI-written (unedited) paragraphs as the prior. We lowercase all words and only consider words that occur at least 20 times across both corpora. We rank words by the z-score of their log-odds ratio, where positive values indicate over-representation in AI-written paragraphs and negative values over-representation in human-written paragraphs.

Table S272 shows the 25 most distinctive words in each direction. Words over-represented in AI-written paragraphs were mostly abstract nouns, formal verbs, and discourse markers (e.g. “approach”, “ensuring”, “challenges”, “fostering”, “additionally”, “crucial”). This is broadly consistent with prior evidence of characteristic vocabulary in AI-written text⁵⁰, although many cliché words are absent, likely because in most input conditions writers provided more of their own writing as input to the AI assistant. Words over-represented in human-written paragraphs were mostly function words, informal verbs, and vague nouns (e.g. “it”, “they”, “people”, “think”, “get”, “things”), reflecting a more conversational register. The human-written paragraphs also used “we” more than three times as often as AI-written paragraphs, in line with the pronoun results in Table S271.

Table S272: Most distinctive words in AI-written vs. human-written paragraphs in Study 1. Words are ranked by the z-score of their log-odds ratio with an informative Dirichlet prior⁴⁹, comparing 4,503 AI-written (unedited) and 4,503 human-written paragraphs. Rates are occurrences per 10,000 words. Words are lowercased; only words with at least 20 occurrences across both corpora are included.

Over-represented in AI-written				Over-represented in human-written			
Word	AI	Human	z	Word	AI	Human	z
while	58.4	6.3	26.7	it	53.9	144.0	-27.1
approach	19.1	2.5	14.8	be	72.5	161.8	-24.8
rather	28.5	7.5	14.6	we	19.6	73.0	-23.4
ensuring	17.1	2.1	14.2	is	82.4	168.9	-23.2
essential	19.0	3.2	13.9	people	20.1	72.0	-22.8
economic	23.1	5.5	13.7	they	21.5	73.9	-22.6
toward	12.1	0.6	13.3	will	6.6	40.9	-20.8
significant	15.8	2.2	13.3	so	4.8	35.4	-20.2
concerns	15.5	2.2	13.1	do	5.8	36.5	-19.7
challenges	13.2	1.4	12.7	if	11.1	46.4	-19.5
risks	14.3	2.0	12.6	there	12.5	47.7	-19.0
UK's	14.9	2.3	12.6	have	31.0	77.5	-18.8
recognize	9.8	0.2	12.5	to	263.1	369.9	-18.4
public	33.4	13.5	12.1	think	6.2	33.5	-18.2
environmental	15.6	3.0	12.1	not	43.5	91.9	-17.6
maintaining	10.6	0.8	12.0	are	49.0	98.8	-17.4
and	394.6	322.8	11.7	this	52.9	97.0	-15.2
fostering	8.5	0.4	11.2	then	1.1	16.3	-15.1
global	19.0	5.7	11.1	very	1.9	16.4	-14.2
long-term	9.2	0.7	11.1	get	0.6	13.2	-14.1
potential	15.1	3.6	11.0	being	5.2	22.5	-13.8
address	9.8	1.1	10.8	what	5.8	23.5	-13.7
individuals	17.0	4.9	10.8	able	0.6	12.0	-13.3
additionally	11.5	1.9	10.8	things	0.4	11.2	-13.1
crucial	9.4	1.0	10.7	much	1.9	14.5	-12.9

6 Study 1 (Main Study): Writer Distortion Tolerance Analysis

Table [S273](#) shows the distribution of ratings from writers in Study 1 (N=1,501) for each distortion tolerance question, alongside mean and median ratings. Figure [S53](#) shows the same mean ratings sorted in ascending order with 95% bootstrapped confidence intervals.

Table S273: Writer tolerance for different types of distortion from AI writing assistance in Study 1. Writers (N=1,501) indicated agreement with different statements about distortion (e.g. “I am okay with AI writing assistance making [me seem more optimistic].”, see §2.3) on a 0-100 scale, from 0 = “Strongly disagree” to 100 = “Strongly agree”. For visualising rating distributions, we split the scale into five equal-sized bins and colour the bins from dark red (strong agree) to dark green (strong disagree).

Type of Distortion	Rating Distribution	Mean	Median
“[...] my opinion on an issue seem more extreme [...]”		34.97 ●	30.00 ●
“[...] my opinion on an issue seem more moderate [...]”		50.70 ●	54.00 ●
“[...] me seem more confident in my opinion [...]”		66.43 ●	72.00 ●
“[...] me seem less confident in my opinion [...]”		20.48 ●	16.00 ●
“[...] me seem more knowledgeable about an issue [...]”		63.64 ●	69.00 ●
“[...] me seem less knowledgeable about an issue [...]”		17.01 ●	10.00 ●
“[...] an issue seem more important to me [...]”		48.30 ●	51.00 ●
“[...] an issue seem less important to me [...]”		21.60 ●	17.00 ●
“[...] my writing seem more relevant or on-topic [...]”		71.73 ●	75.00 ●
“[...] my writing seem less relevant or on-topic [...]”		16.18 ●	10.00 ●
“[...] my opinion clearer to others [...]”		80.76 ●	86.00 ●
“[...] my opinion less clear others [...]”		11.38 ●	4.00 ●
“[...] my writing seem more formal [...]”		65.26 ●	68.00 ●
“[...] my writing seem less formal [...]”		34.68 ●	35.00 ●
“[...] my writing seem more informative or educational [...]”		72.46 ●	76.00 ●
“[...] my writing seem less informative or educational [...]”		18.66 ●	13.00 ●
“[...] my writing seem more original or novel [...]”		60.29 ●	62.00 ●
“[...] my writing seem less original or novel [...]”		22.63 ●	19.00 ●
“[...] me seem more friendly [...]”		59.90 ●	62.00 ●
“[...] me seem less friendly [...]”		24.79 ●	21.00 ●
“[...] me seem more optimistic [...]”		52.42 ●	53.00 ●
“[...] me seem less optimistic [...]”		28.92 ●	30.00 ●
“[...] me seem more community-oriented [...]”		53.48 ●	53.00 ●
“[...] me seem less community-oriented [...]”		27.79 ●	28.00 ●
“[...] me seem more open to reconsidering my views [...]”		52.89 ●	56.00 ●
“[...] me seem less open to reconsidering my views [...]”		28.64 ●	27.00 ●
“[...] my writing express more hope [...]”		55.19 ●	59.00 ●
“[...] my writing express less hope [...]”		26.56 ●	26.00 ●
“[...] my writing express more excitement [...]”		54.12 ●	57.00 ●
“[...] my writing express less excitement [...]”		29.71 ●	30.00 ●
“[...] my writing express more fear [...]”		28.18 ●	25.00 ●
“[...] my writing express less fear [...]”		38.70 ●	43.00 ●
“[...] my writing express more disgust [...]”		26.18 ●	21.00 ●
“[...] my writing express less disgust [...]”		37.58 ●	39.00 ●
“[...] my writing express more anger [...]”		27.62 ●	22.00 ●
“[...] my writing express less anger [...]”		39.13 ●	41.00 ●
“[...] me seem older [...]”		37.33 ●	46.00 ●
“[...] me seem younger [...]”		34.73 ●	39.00 ●
“[...] my gender seem different [...]”		27.89 ●	21.00 ●
“[...] my race or ethnicity seem different [...]”		24.19 ●	15.00 ●
“[...] my English language skills seem better [...]”		66.95 ●	73.00 ●
“[...] my English language skills seem worse [...]”		12.22 ●	4.00 ●
“[...] me seem more educated [...]”		63.32 ●	69.00 ●
“[...] me seem less educated [...]”		14.10 ●	6.00 ●
“[...] my income seem higher [...]”		39.79 ●	49.00 ●
“[...] my income seem lower [...]”		25.90 ●	22.00 ●
“[...] it seem like I voted for a different political party [...]”		21.06 ●	10.00 ●
“[...] me seem more politically extreme [...]”		17.65 ●	8.00 ●
“[...] me seem more politically moderate / centrist [...]”		33.47 ●	36.00 ●

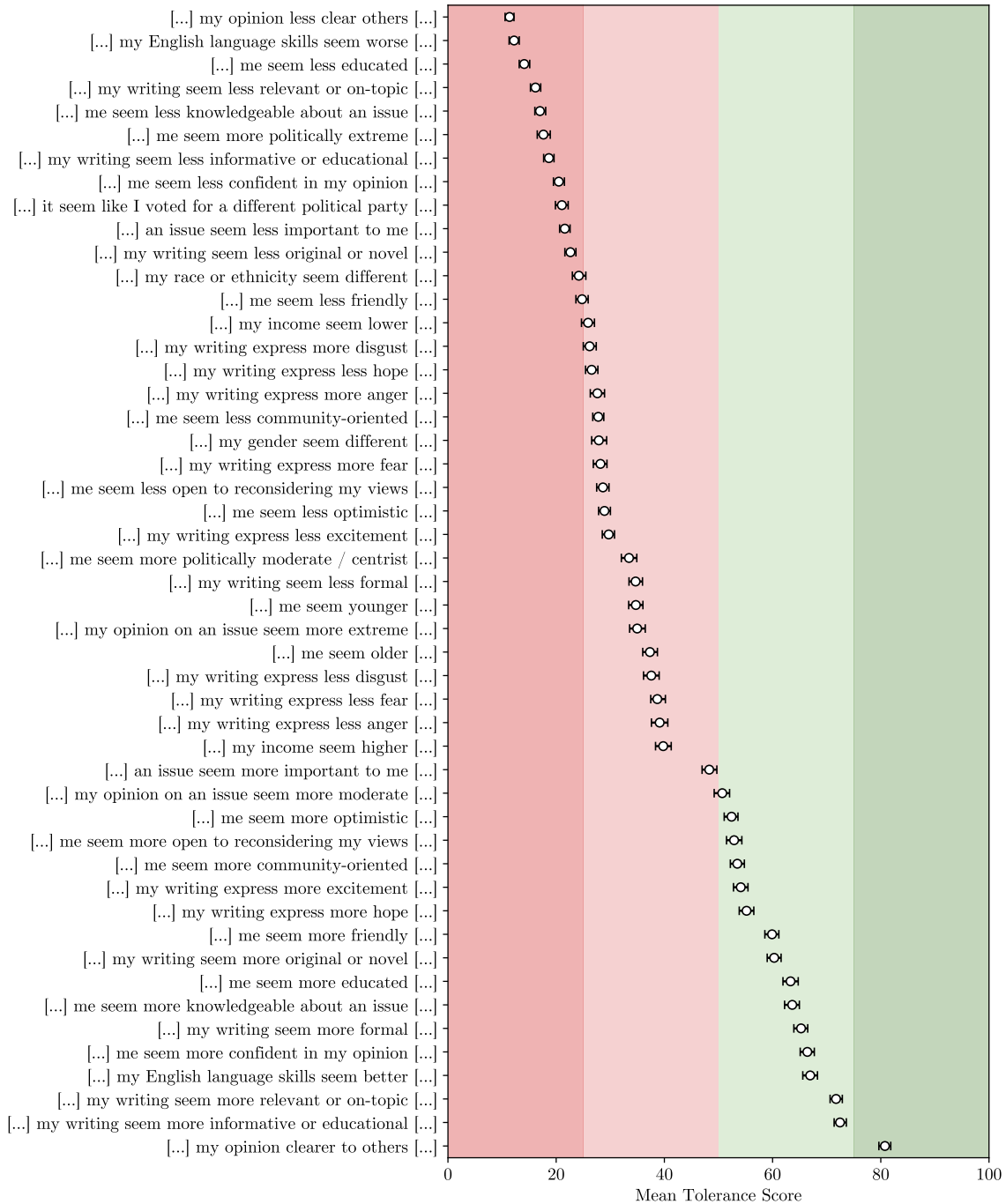


Figure S53: Average writer tolerance for different types of distortion from AI writing assistance in Study 1. Writers (N=1,501) indicated agreement with different statements about distortion (e.g. “I am okay with AI writing assistance making [me seem more optimistic].”, see §2.3) on a 0-100 scale, from 0 = “Strongly disagree” to 100 = “Strongly agree”. Sorted by average tolerance score, shown with 95% bootstrapped confidence intervals.

7 Study 2 (Disclaimer Study)

7.1 Disclaimer Design

In Study 2, we showed writers a pop-up disclaimer that they had to acknowledge before they could edit the AI paragraph, and again before they chose which paragraph (AI or their own) they preferred for communicating their opinion. The content of the disclaimer was randomly assigned at the writer level across three disclaimer conditions and a no-disclaimer control.

Disclaimer design was based on the writer tolerance measures collected in Study 1. The first disclaimer, which we refer to as the ‘liked’ disclaimer, highlighted the two observed AI distortions that, on average, writers were most accepting of: AI making writing seem clearer and more relevant to the issue that the paragraph was responding to. The second disclaimer, which we refer to as the ‘disliked’ disclaimer, highlighted the two observed AI distortions that, on average, writers were least accepting of: AI making writers seem more extreme in their political opinion and less open to changing their views. The third disclaimer, which we refer to as the ‘full’ disclaimer, highlighted all four of these observed AI distortions. The order in which distortions were highlighted within each disclaimer was randomised every time a disclaimer was shown.

Figure S54 shows a screenshot of our study interface, in which a writer assigned to the third ‘full’ disclaimer condition is shown a disclaimer on the paragraph edit screen. This particular writer would have been shown the same disclaimer on every edit and paragraph preference screen, with only the order of the four bullet points randomised.

Consider this statement: Statement: 1 of 3 Step: 2 of 3 Screen: 2 of 4

The UK should mandate the inclusion of feminist perspectives in the national literature curriculum.

Please **edit the AI-written paragraph** below until you feel that it reflects your opinion about the **statement**. If you feel that the paragraph already reflects your opinion, you do not have to edit it. Otherwise please edit it until it does.

PLEASE NOTE:

Research has found:

- AI tends to make you seem **less open** to changing your views
- AI tends to make you seem **more extreme** in your political opinion
- AI tends to make your opinion **clearer** to others
- AI tends to make your writing seem **more informative**

☐ I have read this note.

I believe the UK should mandate feminist perspectives in literature curriculum because diverse viewpoints enrich students' understanding of human experience. Historically, women's voices have been underrepresented in academic study, creating gaps in our cultural knowledge. Including feminist texts and analyses would provide students with more complete perspectives on society, relationships, and power structures. This inclusion would help develop critical thinking skills by exposing students to different interpretations of classic works and contemporary issues. Furthermore, mandatory inclusion ensures consistent implementation across all schools, preventing inequality in educational opportunities. Students of all genders benefit from understanding how literature reflects and shapes social attitudes toward women and marginalized groups.

Continue

Figure S54: Screenshot of the disclaimer shown to writers on the paragraph edit screen of Study 2. Each writer was randomly assigned to one of three disclaimer conditions or a no-disclaimer control. Shown here is the ‘full’ disclaimer that highlights distortions that writers were most and least accepting of in Study 1 (see §4.1).

7.2 Descriptive Results

Figures S55, S56, and S57 show writers' edit and preference rates across disclaimer conditions with 95% bootstrapped confidence intervals.

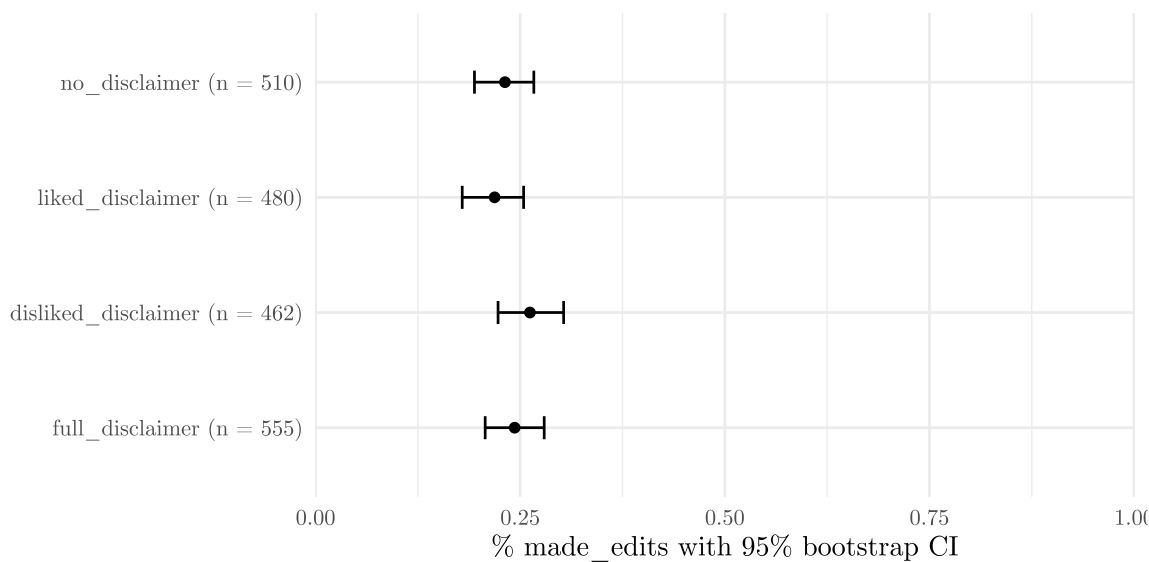


Figure S55: AI paragraph edit rates in Study 2 (N=669 writers, each writing about 3 propositions) across disclaimer conditions.

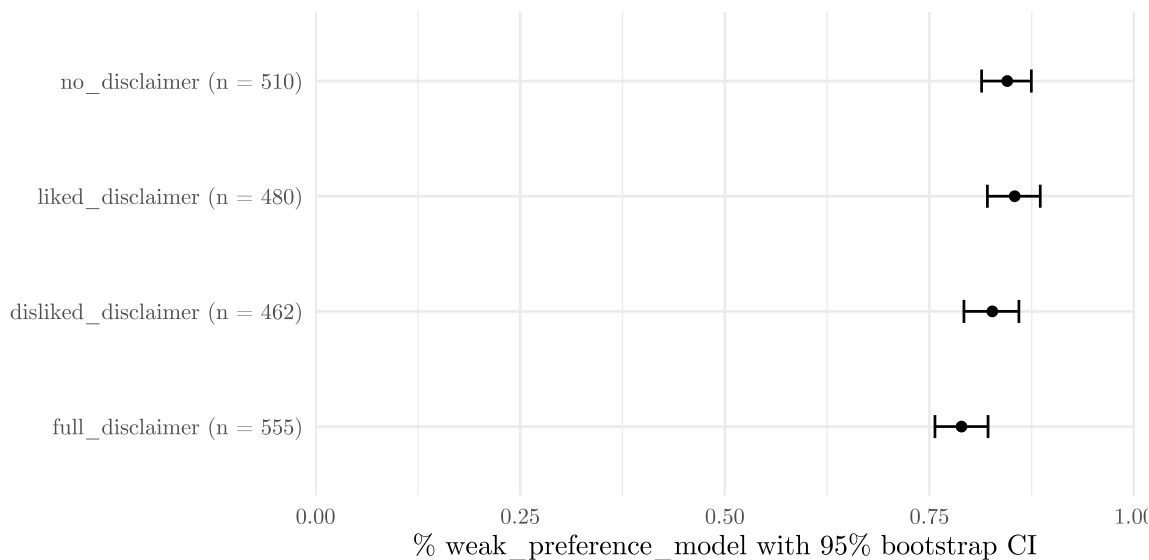


Figure S56: Weak preference for AI-assisted paragraphs over own writing in Study 2 (N=669 writers, each writing about 3 propositions) across disclaimer conditions. AI-written paragraphs include any edits made.

Unlike in Study 1, all writers in Study 2 who strictly preferred one paragraph over the other had to explain their preference in free text (see §2.5). This gave 1,135 free-text reasons for strict preferences for the AI-assisted paragraph and 346 for strict preferences for the human-written paragraph. We coded these responses using the same codebook and LLM-based coding procedure as in Study 1 (§4.3, Table S34). Responses received 2.2 codes on average, and 72.9% of responses received more

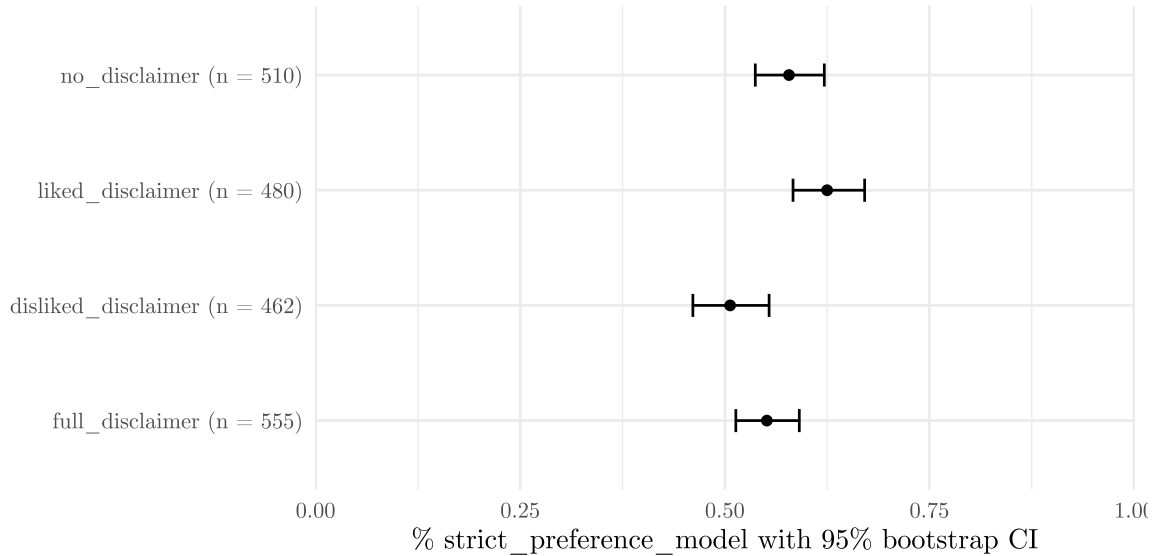


Figure S57: Strict preference for AI-assisted paragraphs over own writing in Study 2 (N=669 writers, each writing about 3 propositions) across disclaimer conditions. AI-written paragraphs include any edits made.

than one code. Table S274 and Figure S58 show the share of responses mentioning each reason, and Table S275 breaks this down by disclaimer condition.

Writers who strictly preferred the AI-written paragraph most often said that this was because it was better written (40.1%), clearer (31.3%), more informative (28.5%) or better reflected their opinion (27.8%), followed by concision (15.9%), persuasiveness (13.8%), and structure (12.9%). Writers who strictly preferred their own paragraph most often said that this was because it better reflected their opinion (35.3%) or sounded more human or more like their own voice (29.8%), followed by clarity (15.6%), informativeness (14.2%), and a general dislike of AI (12.4%).

Some reasons varied across disclaimer conditions in ways that mirror the content of the disclaimers. The ‘liked’ and ‘full’ disclaimers highlighted that AI made writing clearer: among writers who preferred the AI-written paragraph, 40.0% (‘liked’) and 35.3% (‘full’) said that it was clearer, compared to 26.4% without a disclaimer. The ‘disliked’ and ‘full’ disclaimers highlighted that AI made writers seem more extreme: among writers who preferred their own paragraph, 11.2% (‘disliked’) and 7.7% (‘full’) said that this was because their own paragraph made their opinion seem less extreme, compared to 1.3% without a disclaimer. Note that these comparisons are based on small numbers of writers per condition, particularly for writers who preferred their own paragraph (70 to 117 responses per condition).

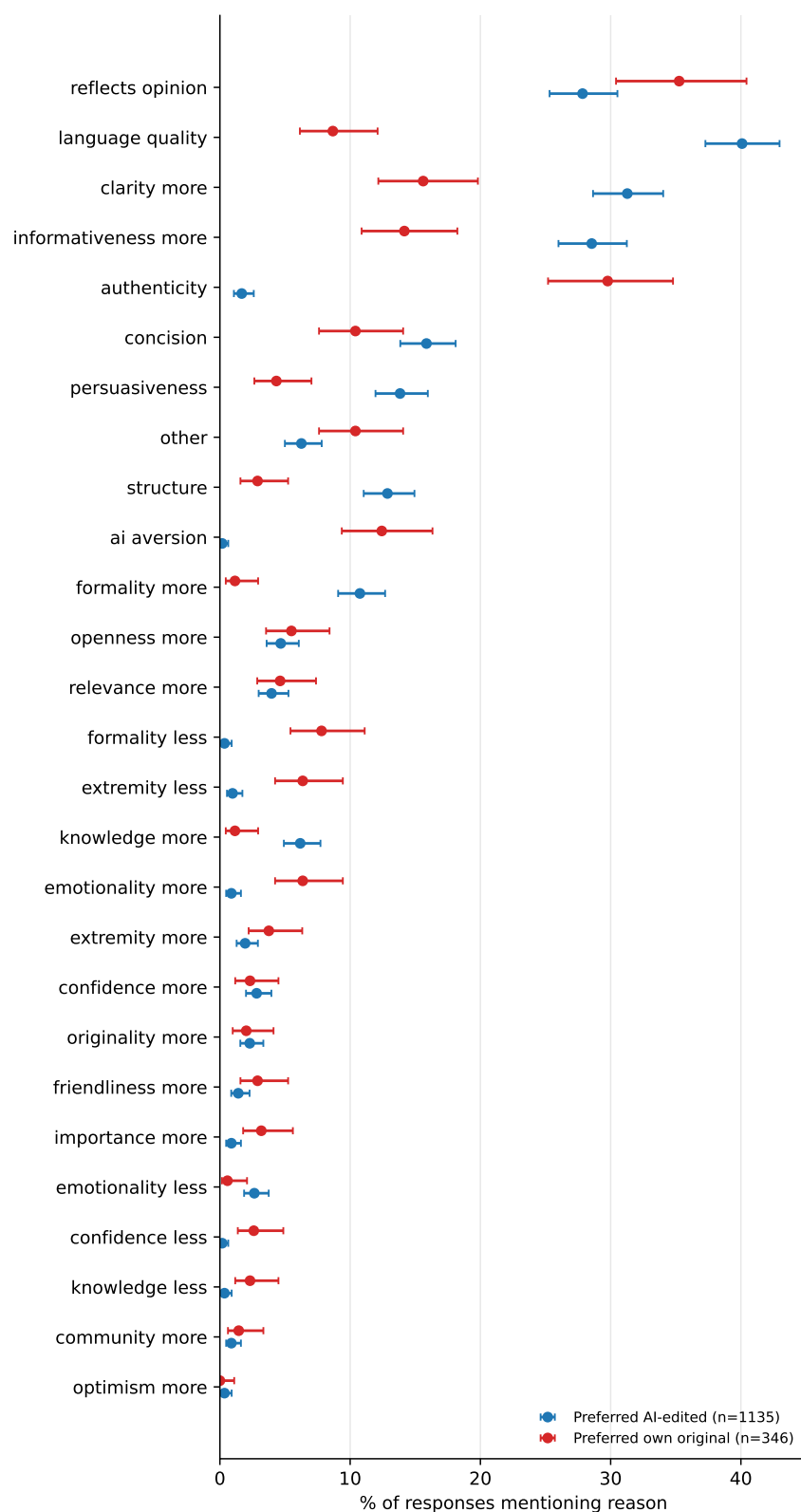


Figure S58: Free-text reasons for strict paragraph preference in Study 2 (n=1,135 strict preferences for the AI-assisted paragraph, n=346 for the human-written paragraph). Points show the share of responses mentioning each reason, with 95% Wilson CIs. See Table S34 for code definitions.

Table S274: Free-text reasons for strict paragraph preference in Study 2 (n=1,135 strict preferences for the AI-assisted paragraph, n=346 for the human-written paragraph). Responses can receive multiple codes, so percentages do not sum to 100%. See Table S34 for code definitions.

Code	AI-written preferred		Human-written preferred	
	n	%	n	%
reflects_opinion	316	■ 27.8%	122	■ 35.3%
extremity_more	22	1.9%	13	3.8%
extremity_less	11	1.0%	22	6.4%
confidence_more	32	2.8%	8	2.3%
confidence_less	2	0.2%	9	2.6%
knowledge_more	70	6.2%	4	1.2%
knowledge_less	4	0.4%	8	2.3%
importance_more	10	0.9%	11	3.2%
relevance_more	45	4.0%	16	4.6%
clarity_more	355	■ 31.3%	54	■ 15.6%
formality_more	122	■ 10.7%	4	1.2%
formality_less	4	0.4%	27	■ 7.8%
informativeness_more	324	■ 28.5%	49	■ 14.2%
originality_more	26	2.3%	7	2.0%
friendliness_more	16	1.4%	10	2.9%
optimism_more	4	0.4%	0	0.0%
community_more	10	0.9%	5	1.4%
openness_more	53	4.7%	19	■ 5.5%
emotionality_more	10	0.9%	22	■ 6.4%
emotionality_less	30	2.6%	2	0.6%
language_quality	455	■ 40.1%	30	■ 8.7%
concision	180	■ 15.9%	36	■ 10.4%
structure	146	■ 12.9%	10	2.9%
persuasiveness	157	■ 13.8%	15	4.3%
authenticity	19	1.7%	103	■ 29.8%
ai_aversion	2	0.2%	43	■ 12.4%
other	71	■ 6.3%	36	■ 10.4%

Table S275: Free-text reasons for strict paragraph preference by disclaimer condition in Study 2. Cells show the percentage of responses mentioning each reason, with the number of responses per condition in the column header. See Table S34 for code definitions.

Code	AI-written preferred				Human-written preferred			
	None (295)	Liked (300)	Disliked (234)	Full (306)	None (79)	Liked (70)	Disliked (80)	Full (117)
reflects_opinion	25.8	24.3	32.1	30.1	30.4	40.0	40.0	32.5
extremity_more	2.4	1.7	1.7	2.0	0.0	5.7	3.8	5.1
extremity_less	0.3	1.0	1.3	1.3	1.3	4.3	11.2	7.7
confidence_more	2.4	2.7	2.6	3.6	0.0	1.4	2.5	4.3
confidence_less	0.3	0.0	0.4	0.0	0.0	1.4	7.5	1.7
knowledge_more	6.8	7.3	3.8	6.2	0.0	1.4	0.0	2.6
knowledge_less	0.0	0.7	0.4	0.3	1.3	2.9	3.8	1.7
importance_more	1.7	0.0	0.9	1.0	1.3	2.9	3.8	4.3
relevance_more	3.7	3.3	3.8	4.9	12.7	0.0	2.5	3.4
clarity_more	26.4	40.0	20.9	35.3	12.7	18.6	17.5	14.5
formality_more	9.8	12.3	11.5	9.5	0.0	4.3	0.0	0.9
formality_less	0.7	0.3	0.4	0.0	5.1	5.7	8.8	10.3
informativeness_more	26.4	30.7	28.6	28.4	16.5	11.4	15.0	13.7
originality_more	3.4	1.3	1.7	2.6	5.1	1.4	0.0	1.7
friendliness_more	1.0	1.0	2.1	1.6	2.5	0.0	3.8	4.3
optimism_more	0.3	0.7	0.0	0.3	0.0	0.0	0.0	0.0
community_more	1.7	0.3	0.9	0.7	1.3	2.9	1.2	0.9
openness_more	5.4	3.7	5.1	4.6	6.3	2.9	8.8	4.3
emotionality_more	1.7	0.0	0.9	1.0	6.3	8.6	8.8	3.4
emotionality_less	3.1	3.3	1.7	2.3	2.5	0.0	0.0	0.0
language_quality	45.8	36.0	38.5	39.9	6.3	7.1	8.8	11.1
concision	18.0	15.3	16.7	13.7	15.2	10.0	5.0	11.1
structure	10.8	12.3	16.2	12.7	3.8	1.4	5.0	1.7
persuasiveness	14.6	13.7	12.8	14.1	6.3	1.4	3.8	5.1
authenticity	1.4	0.3	1.7	3.3	26.6	25.7	31.2	33.3
ai_aversion	0.3	0.0	0.0	0.3	15.2	7.1	10.0	15.4
other	7.5	6.3	7.3	4.2	12.7	10.0	6.2	12.0

Table S276: Writer tolerance for different types of distortion from AI writing assistance in Study 2. Writers (N=669) rated different statements about distortion (e.g. “How would you feel about AI making [your opinion clearer]”, see §2.5) on a 0-100 scale, from 0 = “Strongly dislike” to 100 = “Strongly like”. In Study 2, we collected tolerance ratings for eight distortion statements.

Type of Distortion	Mean	Median
“[...] my opinion on an issue seem more extreme [...]”	25.03 ●	19.00 ●
“[...] my opinion on an issue seem more moderate [...]”	42.26 ●	48.00 ●
“[...] my writing seem more informative or educational [...]”	73.16 ●	77.00 ●
“[...] my writing seem less informative or educational [...]”	14.68 ●	9.00 ●
“[...] my opinion clearer to others [...]”	79.22 ●	82.00 ●
“[...] my opinion less clear others [...]”	9.33 ●	3.00 ●
“[...] me seem more open to reconsidering my views [...]”	46.67 ●	50.00 ●
“[...] me seem less open to reconsidering my views [...]”	26.54 ●	24.00 ●

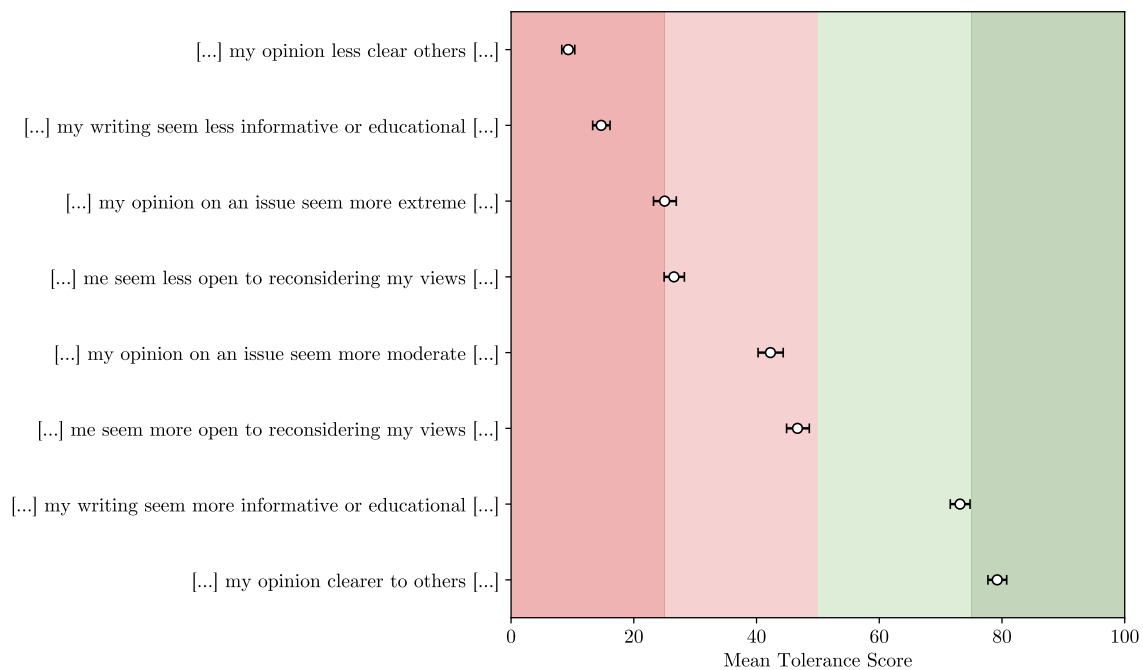


Figure S59: Average writer tolerance for different types of distortion from AI writing assistance in Study 2. Writers (N=669) rated different statements about distortion (e.g. “How would you feel about AI making [your opinion clearer]”, see §2.5) on a 0-100 scale, from 0 = “Strongly dislike” to 100 = “Strongly like”. This study included eight distortion statements. Sorted by average tolerance score, shown with 95% bootstrapped confidence intervals.

7.3 Statistical Analysis

We fit logistic regression models with writer random effects to measure the effect of disclaimers on writer preference for AI writing. Matching our main analysis, we used three binary outcome variables observed for each writer-proposition pair: whether the writer made any edits to the AI-written paragraph (“made_edits”), whether the writer weakly preferred the AI-written paragraph (“weak_preference_model”), and whether the writer strictly preferred the AI-written paragraph (“strict_preference_model”). In addition to a regressor for the assigned disclaimer condition, we included controls for model and input condition. Regression models were specified as follows:

```
made_edits ~ disclaimer_condition +
model + input_condition + (1 | writer_id)
```

```
weak_preference_model ~ disclaimer_condition +
model + input_condition + (1 | writer_id)
```

```
strict_preference_model ~ disclaimer_condition +
model + input_condition + (1 | writer_id)
```

Table S277 reports results for the made_edits outcome. No disclaimer has a significant effect compared to the no-disclaimer control. Paragraphs generated by ChatGPT-4o are less likely to be edited than those generated by Claude Sonnet 4 (OR=0.617, $p<.001$). Paragraphs generated based on the writer’s numeric issue stance (OR=1.553, $p=.016$) or based on rewriting the writer’s full paragraph (OR=1.643, $p<.001$) were significantly more likely to be edited than those generated based on bullet point inputs.

Table S277: Logistic regression predicting writer editing of AI writing in Study 2 (N=669 writers, each writing about 3 propositions). Binary outcome: “made_edits”. Reference categories: “model_anthropic/claude-sonnet-4”, “input_condition_bullets-based”, and “disclaimer_condition_none”

Term	Estimate	SE	OR [95% CI]	z	p
disclaimer_condition_disliked_disclaimer	0.199	0.215	1.220 [0.800, 1.859]	0.924	.355
disclaimer_condition_full_disclaimer	0.079	0.207	1.082 [0.722, 1.622]	0.381	.703
disclaimer_condition_liked_disclaimer	−0.100	0.217	0.905 [0.592, 1.384]	−0.460	.646
model_deepseek/deepseek-chat-v3-0324	−0.254	0.141	0.776 [0.589, 1.022]	−1.803	.071
model_openai/chatgpt-4o-latest	−0.483	0.144	0.617 [0.465, 0.819]	−3.341	< .001
input_condition_improve	0.203	0.182	1.225 [0.857, 1.751]	1.115	.265
input_condition_rewrite	0.440	0.182	1.553 [1.087, 2.220]	2.419	.016
input_condition_stance-based	0.497	0.145	1.643 [1.236, 2.184]	3.420	< .001

Table S278 reports results for the weak_preference_model outcome. Only the ‘full’ disclaimer has a significant effect compared to the no-disclaimer control, reducing weak preference for AI-written paragraphs (OR=0.605, p=.036). Weak preference rate does not differ significantly across AI models. Paragraphs generated based on the writer’s numeric issue stance were significantly less likely to be weakly preferred (OR=0.453, p<.001) than those generated based on bullet point inputs.

Table S278: Logistic regression predicting weak writer preference for AI writing in Study 2 (N=669 writers, each writing about 3 propositions). Binary outcome: “weak_preference_model”. Reference categories: “model_anthropic/claude-sonnet-4”, “input_condition_bullets-based”, and “disclaimer_condition_none”

Term	Estimate	SE	OR [95% CI]	z	p
disclaimer_condition_disliked_disclaimer	−0.185	0.255	0.831 [0.505, 1.369]	−0.727	.467
disclaimer_condition_full_disclaimer	−0.502	0.240	0.605 [0.378, 0.968]	−2.094	.036
disclaimer_condition_liked_disclaimer	0.069	0.257	1.071 [0.647, 1.772]	0.267	.789
model_deepseek/deepseek-chat-v3-0324	−0.273	0.163	0.761 [0.553, 1.048]	−1.670	.095
model_openai/chatgpt-4o-latest	−0.113	0.166	0.893 [0.646, 1.236]	−0.681	.496
input_condition_improve	−0.356	0.210	0.701 [0.464, 1.058]	−1.694	.090
input_condition_rewrite	−0.349	0.214	0.705 [0.464, 1.072]	−1.634	.102
input_condition_stance-based	−0.792	0.167	0.453 [0.327, 0.629]	−4.738	< .001

Table S279 reports results for the strict_preference_model outcome. No disclaimer condition has a significant effect compared to the no-disclaimer control, though the ‘disliked’ disclaimer approaches significance (OR=0.671, p=.057). Strict preference does not differ significantly across AI models. All non-bullet-point input conditions significantly reduce strict preference for AI writing: paragraphs generated by improving the writer’s paragraph (OR=0.485, p<.001), by rewriting it (OR=0.433, p<.001), or based on the writer’s numeric issue stance (OR=0.624, p<.001)

Table S279: Logistic regression predicting strict writer preference for AI writing in Study 2 (N=669 writers, each writing about 3 propositions). Binary outcome: “strict_preference_model”. Reference categories: “model_anthropic/claude-sonnet-4”, “input_condition_bullets-based”, and “disclaimer_condition_none”

Term	Estimate	SE	OR [95% CI]	z	p
disclaimer_condition_disliked_disclaimer	−0.398	0.210	0.671 [0.445, 1.013]	−1.900	.057
disclaimer_condition_full_disclaimer	−0.153	0.200	0.858 [0.580, 1.270]	−0.765	.444
disclaimer_condition_liked_disclaimer	0.265	0.209	1.304 [0.865, 1.966]	1.268	.205
model_deepseek/deepseek-chat-v3-0324	−0.249	0.130	0.780 [0.604, 1.006]	−1.911	.056
model_openai/chatgpt-4o-latest	−0.126	0.130	0.881 [0.683, 1.138]	−0.969	.332
input_condition_improve	−0.725	0.165	0.485 [0.351, 0.669]	−4.405	< .001
input_condition_rewrite	−0.837	0.167	0.433 [0.312, 0.601]	−5.005	< .001
input_condition_stance-based	−0.472	0.132	0.624 [0.482, 0.807]	−3.585	< .001

8 Study 3 (Mitigation Study)

In Study 3, we piloted two model-level interventions to reduce distortions from AI writing assistance. Study 3 matched Study 1 design, except for how paragraphs were generated by AI, where we assigned a model-level intervention (Prompting, Reranking) or a no-intervention control. Assignment was such that each writer, across the three propositions assigned to them, experienced the two interventions and control in random order. We targeted the polarising effect of AI with our interventions, i.e. AI writing assistance making writers seem more extreme in their political opinion, but our interventions are equally compatible with targeting other distortions. We restricted this study to the “bullet-based” input condition, where the polarising effect of AI was particularly pronounced.

8.1 Prompting Intervention Design

For the Prompting intervention, we appended the text below to each AI paragraph generation prompt. The findings mentioned in the text are those of Study 1. Writer dislike is highlighted to match the design of our Reranking mitigation, detailed in 8.2.

“Prompting” mitigation, appended to each AI generation prompt

Research has shown that AI writing assistance makes writers seem more extreme in their issue stance than their own independent writing, on average by 4.0 points on a 0-100 scale.

The same research also asked a representative sample of writers to which extent they were okay with this type of distortion:
68.8% dislike AI writing assistance making them seem more extreme in their issue stance.
44.8% dislike AI writing assistance making them seem more moderate in their issue stance.

When you generate the paragraph, please avoid distorting the writer’s issue stance.
When weighing the different types of distortion against each other, use weights proportional to how many people dislike each type of distortion.

8.2 Reranking Intervention Design

For the Reranking intervention, we fine-tuned two reward models (RMs) on data collected in Study 1. For this purpose, we randomly split the 1,501 writers in Study 1 into a training set of 1,126 writers, a validation set of 75 writers, and a test set of 300 writers.

A first **Paragraph RM**, f_{para} , was fine-tuned on paragraph-rating pairs, to predict third-party perception of new paragraphs. Specifically, each training instance was a paragraph (written by a human or AI) and the average perceived stance of the writer of the paragraph (`writer_stance`) across the 10 ratings that the paragraph received in Study 1. For fine-tuning, each training instance was inserted into the template below:

Paragraph RM scoring prompt used in fine-tuning

You will be shown an OPINION PARAGRAPH concerning a PROPOSITION. We showed this paragraph to 10 independent raters and asked a RATING QUESTION. For this rating question, your task is to predict the MEAN rating value across all raters.

RATING QUESTION:

`writer_stance`: What is the author’s level of agreement with the proposition? [0-100 scale, where 0 = "Strongly disagree", 50 =

```
"Ambivalent", and 100 = "Strongly agree"]
```

```
PROPOSITION: proposition
```

```
OPINION PARAGRAPH: paragraph
```

Since each writer wrote three paragraphs, and for each human-written paragraph there was one AI-written counterpart (excluding human edits, since these are only made after generation), there were a total of 6,756 training instances available. We fine-tuned different sizes of GPT-4.1 on different amounts of training data using the OpenAI fine-tuning API, finding clear benefits to scaling model and training data size (Table S280). For our study, we chose the best-performing model based on mean absolute error on the 0-100 rating scale, which was GPT-4.1 fine-tuned on all available training data.

Note that we also evaluated settings in which the Paragraph RM made predictions for all 20 rating attributes that were measured on a 0-100 scale, but found that this degraded performance relative to only predicting writer issue stance.

Table S280: Paragraph RM performance across different base models and amounts of training. MAE is Mean Absolute Error on the 0-100 rating scale. Best / chosen model highlighted in **bold**.

Base Model	Training Data Size	MAE ↓
GPT-4.1 nano	100	10.17
GPT-4.1 nano	1,000	7.10
GPT-4.1 mini	100	7.91
GPT-4.1 mini	1,000	6.62
GPT-4.1	100	6.90
GPT-4.1	1,000	6.40
GPT-4.1	6,756	6.15

A second **Bullet RM**, f_{bullet} , was fine-tuned on bullet-rating pairs, i.e. bullet points provided by writers and third-party ratings for the corresponding writer paragraphs. Specifically, each training instance consisted of a writer's bullet point reasons for their opinion on an assigned proposition plus the average perceived stance of the writer's full opinion paragraph (`writer_stance`) across the 10 ratings that the paragraph received in Study 1. For fine-tuning, each training instance was inserted into the template below:

Bullet RM scoring prompt used in fine-tuning

```
You will be shown BULLET POINTS written by a person to explain their position on a PROPOSITION. The person who wrote these bullet points expanded them into a full paragraph. We showed this paragraph to 10 independent raters and asked a RATING QUESTION. For this rating question, your task is to predict the MEAN rating value across all raters.
```

```
RATING QUESTION:
```

```
writer_stance: What is the author's level of agreement with the proposition? [0-100 scale, where 0 = "Strongly disagree", 50 = "Ambivalent", and 100 = "Strongly agree"]
```

```
PROPOSITION: proposition
```

```
BULLET POINTS: writer_bullets
```

Since each writer expressed their opinion on three propositions, there were a total of 3,378 training instances available. Again, we fine-tuned different sizes of GPT-4.1 on different amounts of training data using the OpenAI fine-tuning API, finding clear benefits to scaling model and training data size (Table S281). For our study, we chose GPT-4.1 fine-tuned on all available training data, which performed roughly on par with GPT-4.1 models fine-tuned on less training data.

Table S281: Bullet RM performance across different base models and amounts of training. MAE is Mean Absolute Error on the 0-100 rating scale. Chosen model highlighted in **bold**.

Base Model	Training Data Size	MAE ↓
GPT-4.1 nano	100	14.97
GPT-4.1 nano	1,000	10.07
GPT-4.1 mini	100	10.66
GPT-4.1 mini	1,000	10.25
GPT-4.1	100	9.03
GPT-4.1	1,000	9.59
GPT-4.1	3,378	9.28

All RM fine-tuning was done asynchronously, before launch of Study 3.

During Study 3, at generation time under the Reranking condition, we used verbalised sampling⁵¹ to generate two sets of five AI paragraph candidates, $y_1, \dots, y_k$, from the assigned AI model, using the prompt below appended to the standard generation prompt. We extracted individual paragraph candidates from the model responses using a custom regex function. When fewer than ten candidates were successfully extracted, selection proceeded over the available set.

Verbalised sampling prompt, appended to generation prompt in “Reranking”

Generate 5 independent paragraphs with their corresponding probabilities, sampled from the full distribution.

Follow this format: "[probability] paragraph text".

Each paragraph should account for all the bullet points.

Reply only with the generated paragraphs, nothing else.

Then, simultaneously, we used the Paragraph RM to predict the perceived issue stance of each AI-written candidate, $s_k = f_{\text{para}}(y_k)$, and the Bullet RM to predict the perceived issue stance of the human-written paragraph based on the writer’s bullet points, $s^{\text{target}} = f_{\text{bullet}}(b)$.

Finally, we selected the AI-written paragraph candidate that minimised stance discrepancy, with directional weighting:

$$y^* = y_{\arg \min_k w(\delta_k) \cdot |\delta_k|} \quad (1)$$

where $\delta_k = |f_{\text{para}}(y_k) - 50| - |f_{\text{bullet}}(b) - 50|$ measures how much more extreme ($\delta_k > 0$) or more moderate ($\delta_k \leq 0$) candidate k is predicted to appear relative to the writer’s own text, and $w(\delta) = 0.688$ if $\delta > 0$ and $w(\delta) = 0.448$ otherwise, so that polarising distortions are penalised more heavily than moderating ones, in proportion to the share of writers in Study 1 who objected to each direction of distortion (stated tolerance < 50).

8.3 Reader Assignment

In Study 3, readers were each assigned ten paragraphs to rate. Few readers (<0.4%) were affected by technical issues in the study interface that resulted in them rating just nine paragraphs, as shown in Table S282.

Table S282: Distribution of paragraphs rated per reader in Study 3. Overall, 2,543 readers rated 5,016 paragraphs in Study 3.

Paragraphs Rated	Readers	Proportion
9	8	0.3%
10	2,535	 99.7%
Total	2,543	

We targeted collecting five sets of ratings for every paragraph. Paragraphs were randomly assigned to readers, with those paragraphs that had received five ratings or more periodically removed from the assignment pool. Table S283 shows the distribution of ratings received per paragraph. Most paragraphs received five ratings. No paragraph received fewer than four ratings.

Table S283: Distribution of ratings received per paragraph in Study 3. Overall, 2,543 readers rated 5,016 paragraphs in Study 3.

Ratings Received	Paragraphs	Proportion
4	1,145	 22.8%
5	2,829	 56.4%
6	754	 15.0%
7	193	3.9%
8	58	1.2%
9	25	0.5%
10	4	0.1%
11	4	0.1%
12	3	0.1%
13	1	0.0%
Total	5,016	

8.4 Descriptive Results: Inter-Rater Agreement

Tables S284 and S285 show inter-rater agreement for reader perceptions in Study 3. Note that our goal was to elicit subjective reader perceptions of writers and their opinions, which we expected to differ across readers, and that we included reader random effects in our analyses below to account for this.

Table S284: Inter-rater agreement (Krippendorff’s α) for all 29 dimensions of reader perception in Study 3 (5,016 paragraphs: 2,307 human-written, 2,307 AI-written, and 402 writer-edited AI-written; 25,422 paragraph-reader observations; about 5 readers per paragraph). Units are paragraphs, rated by different readers. α uses the interval metric for scale attributes, the ordinal metric for ordinal attributes, and the nominal metric for nominal attributes. $\alpha = 1$ indicates perfect agreement and $\alpha = 0$ indicates agreement at chance level. Columns report α computed across all paragraphs and separately by paragraph type. “writer_education” responses of “Other” are treated as missing.

Attribute	Type	All paragraphs	Human-written	AI-written	AI-written (edited)
writer_knowledge	scale	0.385	0.314	0.303	0.356
writer_importance	scale	0.364	0.350	0.311	0.393
writer_confidence	scale	0.391	0.372	0.353	0.369
writer_stance_polarity	scale	0.436	0.403	0.453	0.473
paragraph_formality	scale	0.354	0.212	0.119	0.129
paragraph_clarity	scale	0.279	0.257	0.194	0.228
paragraph_informativeness	scale	0.288	0.200	0.153	0.218
paragraph_originality	scale	0.117	0.087	0.061	0.099
paragraph_relevance	scale	0.216	0.199	0.134	0.120
writer_optimism	scale	0.233	0.239	0.188	0.167
writer_community	scale	0.184	0.181	0.157	0.146
writer_friendliness	scale	0.134	0.166	0.090	0.117
writer_openness	scale	0.273	0.274	0.261	0.321
paragraph_hope	scale	0.227	0.227	0.194	0.180
paragraph_excitement	scale	0.127	0.109	0.126	0.144
paragraph_fear	scale	0.183	0.190	0.176	0.179
paragraph_disgust	scale	0.226	0.267	0.171	0.201
paragraph_anger	scale	0.262	0.303	0.211	0.232
writer_affect_x	scale	0.265	0.270	0.242	0.267
writer_affect_y	scale	0.118	0.131	0.087	0.136
writer_education	ordinal	0.257	0.178	0.089	0.107
writer_english_skills	ordinal	0.202	0.130	0.035	0.078
writer_income	ordinal	0.195	0.135	0.060	0.079
writer_age_binned	ordinal	0.125	0.113	0.103	0.101
writer_english_first	ordinal	0.085	0.080	0.032	0.003
writer_politicalParty	nominal	0.083	0.072	0.083	0.087
writer_politicalIdeology	nominal	0.078	0.079	0.076	0.089
writer_gender	nominal	0.151	0.152	0.146	0.172
writer_race	nominal	0.029	0.032	0.023	0.035

Table S285: Mean within-paragraph standard deviation of reader ratings for scale attributes in Study 3 (5,016 paragraphs: 2,307 human-written, 2,307 AI-written, and 402 writer-edited AI-written; 25,422 paragraph-reader observations; about 5 readers per paragraph). For each paragraph, we compute the standard deviation of the ratings it received, then average across paragraphs. Values are in rating points on the 0–100 scale. Columns report the mean SD across all paragraphs and separately by paragraph type.

Attribute	All paragraphs	Human-written	AI-written	AI-written (edited)
writer_knowledge	17.34	19.19	15.70	16.13
writer_importance	16.69	18.00	15.59	15.47
writer_confidence	16.51	18.18	15.02	15.51
writer_stance_polarity	22.47	23.89	21.25	21.36
paragraph_formality	19.89	21.86	18.09	18.97
paragraph_clarity	18.25	20.66	16.05	17.14
paragraph_informativeness	21.70	22.93	20.70	20.35
paragraph_originality	22.65	22.51	22.83	22.41
paragraph_relevance	17.53	19.75	15.49	16.56
writer_optimism	17.95	18.01	17.83	18.28
writer_community	19.27	19.83	18.84	18.56
writer_friendliness	16.02	16.35	15.65	16.26
writer_openness	20.13	19.99	20.30	19.93
paragraph_hope	20.27	20.18	20.25	20.97
paragraph_excitement	20.50	19.91	20.98	21.12
paragraph_fear	22.21	22.36	21.98	22.63
paragraph_disgust	21.06	21.30	20.86	20.79
paragraph_anger	21.10	21.20	21.03	21.02
writer_affect_x	17.79	17.80	17.81	17.63
writer_affect_y	16.52	16.86	16.24	16.15

8.5 Statistical Analysis: Distortions

To quantify variation in distortions across mitigation conditions, we fit **mixed-effects regression models** specified as follows:

$$\text{rating_attribute} \sim \text{mitigation_condition} + (1 \mid \text{reader_id})$$

where `rating_attribute` and `(1|reader_id)` are defined as in SI:5.7, and `mitigation_condition` is a categorical fixed effect taking four values: one for each of the three mitigation conditions (Prompting, Reranking, None) and one for human-written paragraphs, which serves as the reference category. The measurement scale of each rating attribute again determined the appropriate link function, following the same scheme as in SI:5.7: beta regression for 0-100 scale attributes, multinomial logistic regression for nominal categorical attributes, and ordinal logistic regression for ordinal categorical attributes.

Having confirmed that results are highly consistent across data subsets in Study 1 (SI:5.7), we fit each regression model to Study 3 equivalent of the “preferred” data subset as defined in SI:5.2, which contained 19,102 observations. Figures S60, S61, and S62 summarise results for scale, ordinal, and nominal variables. Tables S286 to S314 show results for each of the 29 rating attributes in our study.

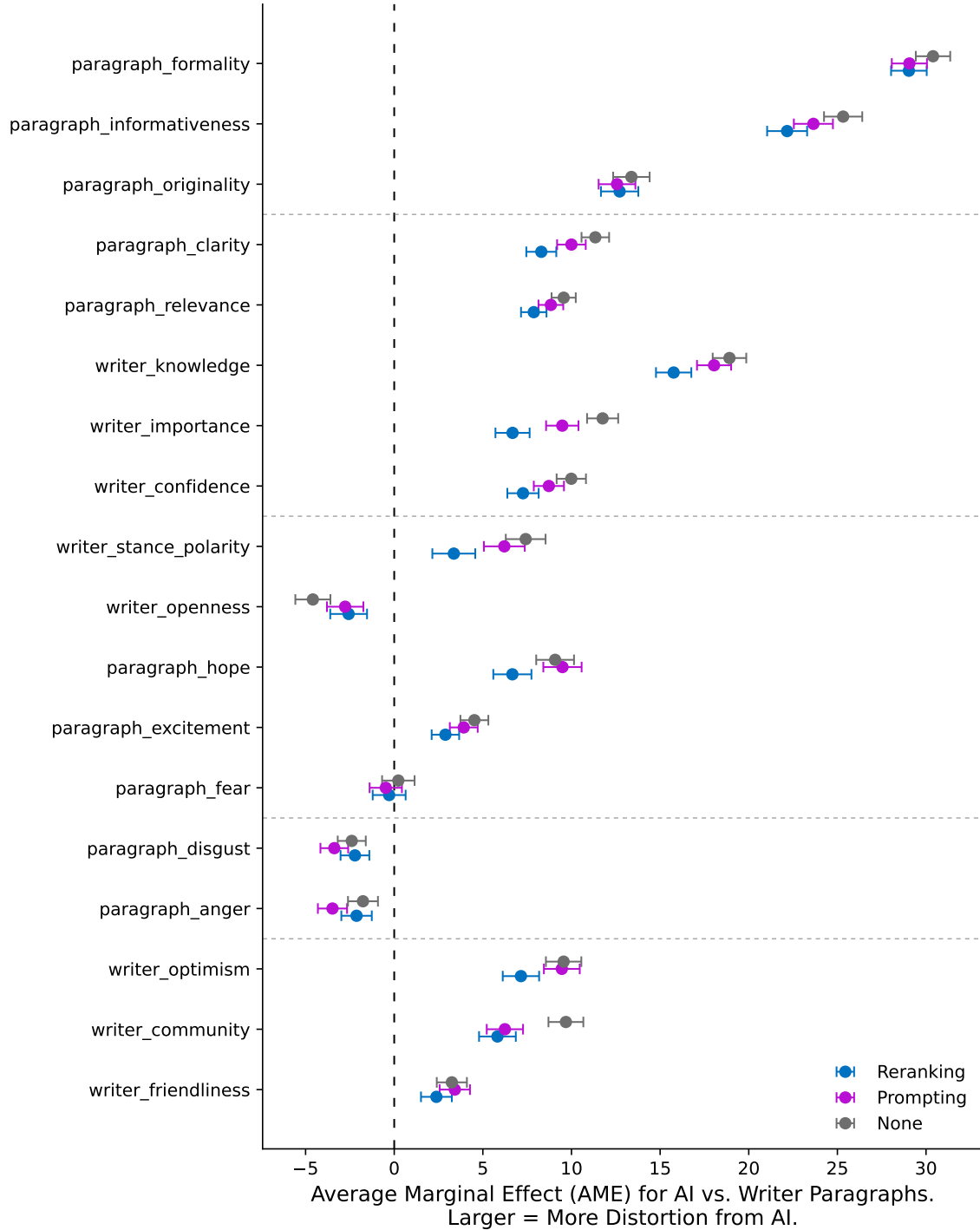


Figure S60: Average marginal effects (AMEs) of AI writing assistance on reader perceptions across all 0-100 scale rating attributes in Study 3. Each point shows the AME from a mixed-effects beta regression comparing AI-written to human-written paragraphs, with reader random effects. Results shown separately for each mitigation condition. Error bars show 95% confidence intervals.

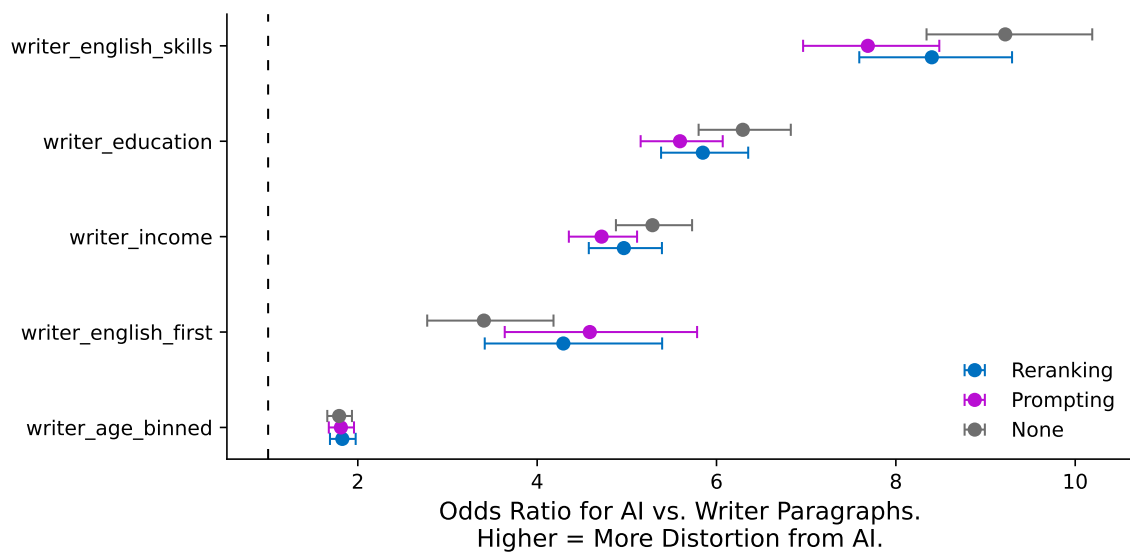


Figure S61: Odds ratios (ORs) of AI writing assistance on reader perceptions across all ordinal rating attributes in Study 3. Each point shows the odds ratio from a mixed-effects ordinal logistic regression comparing AI-written to human-written paragraphs, with reader random effects. Results shown separately for each mitigation condition. Error bars show 95% confidence intervals.

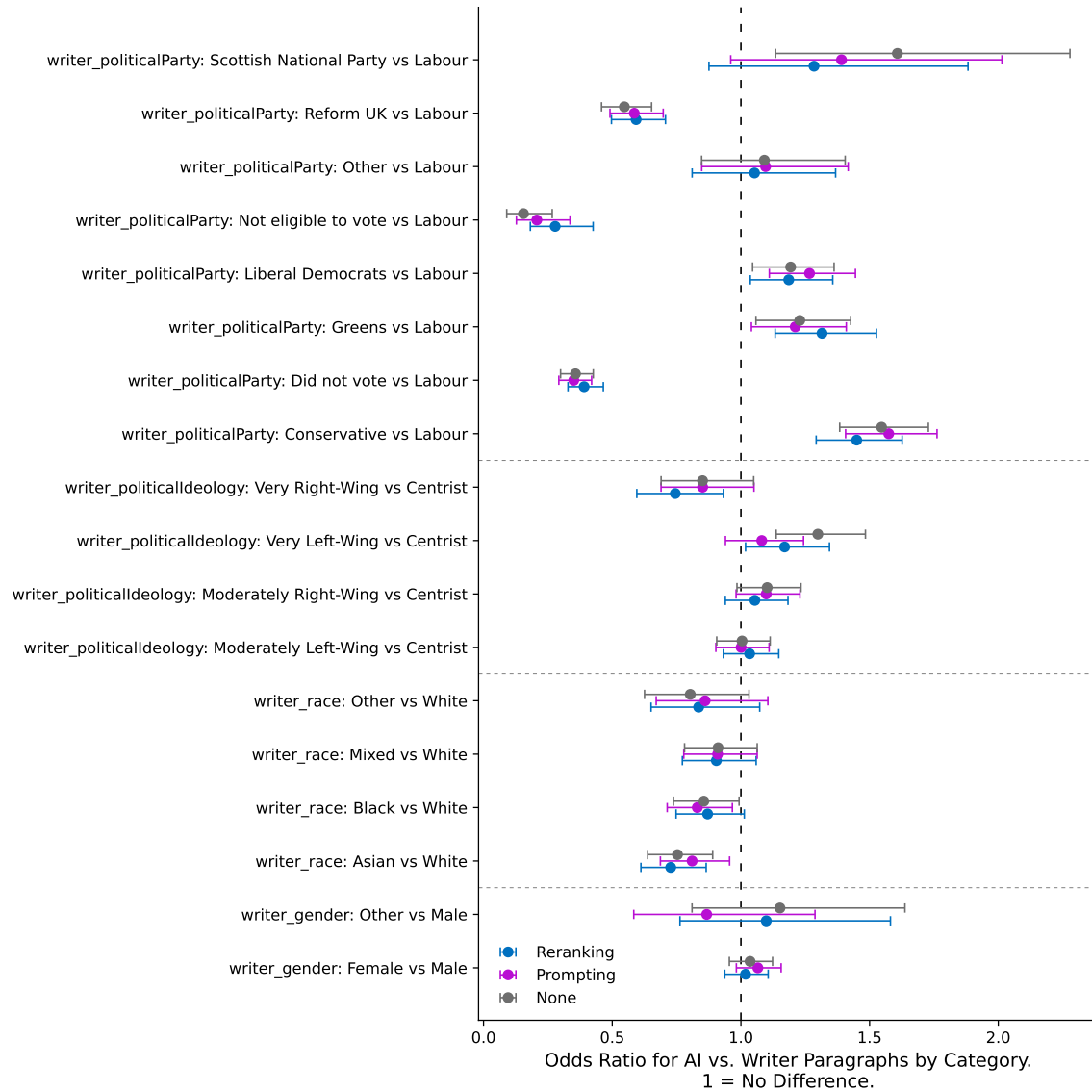


Figure S62: Odds ratios (ORs) of AI writing assistance on reader perceptions across all nominal rating attributes in Study 3. Each point shows the odds ratio of a specific category vs a reference category from a mixed-effects multinomial logistic regression comparing AI-written to human-written paragraphs, with reader random effects. Results shown separately for each mitigation condition. Error bars show 95% confidence intervals.

Table S286: Beta regression predicting paragraph formality by mitigation condition in Study 3 (“preferred” subset, N=19,102). Outcome: “paragraph_formality”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
mitigation_condition_none	3.587	[3.428, 3.752]	55.535	2.691e-672	30.394	[29.423, 31.365]
mitigation_condition_prompting	3.364	[3.215, 3.519]	52.506	3.805e-601	29.062	[28.069, 30.054]
mitigation_condition_reranking	3.359	[3.209, 3.517]	51.733	1.262e-583	29.035	[28.030, 30.040]

Table S287: Beta regression predicting paragraph clarity by mitigation condition in Study 3 (“preferred” subset, N=19,102). Outcome: “paragraph_clarity”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
mitigation_condition_none	1.903	[1.816, 1.995]	26.902	2.077e-159	11.344	[10.566, 12.122]
mitigation_condition_prompting	1.740	[1.659, 1.824]	22.935	2.065e-116	9.996	[9.189, 10.803]
mitigation_condition_reranking	1.564	[1.491, 1.640]	18.296	8.924e-75	8.297	[7.452, 9.143]

Table S288: Beta regression predicting paragraph informativeness by mitigation condition in Study 3 (“preferred” subset, N=19,102). Outcome: “paragraph_informativeness”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
mitigation_condition_none	2.817	[2.690, 2.951]	43.864	3.219e-420	25.324	[24.246, 26.403]
mitigation_condition_prompting	2.622	[2.503, 2.748]	40.487	2.478e-358	23.647	[22.545, 24.749]
mitigation_condition_reranking	2.464	[2.350, 2.583]	37.401	3.753e-306	22.164	[21.037, 23.292]

Table S289: Beta regression predicting paragraph originality by mitigation condition in Study 3 (“preferred” subset, N=19,102). Outcome: “paragraph_originality”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
mitigation_condition_none	1.771	[1.697, 1.848]	26.213	1.868e-151	13.380	[12.352, 14.409]
mitigation_condition_prompting	1.714	[1.641, 1.789]	24.442	6.091e-132	12.568	[11.532, 13.605]
mitigation_condition_reranking	1.724	[1.650, 1.801]	24.335	8.354e-131	12.716	[11.661, 13.770]

Table S290: Beta regression predicting paragraph relevance by mitigation condition in Study 3 (“preferred” subset, N=19,102). Outcome: “paragraph_relevance”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
mitigation_condition_none	1.828	[1.747, 1.912]	26.221	1.527e-151	9.559	[8.877, 10.241]
mitigation_condition_prompting	1.731	[1.654, 1.812]	23.634	1.737e-123	8.837	[8.138, 9.535]
mitigation_condition_reranking	1.614	[1.542, 1.690]	20.460	4.887e-93	7.872	[7.153, 8.590]

Table S291: Beta regression predicting perceived writer knowledge by mitigation condition in Study 3 (“preferred” subset, N=19,102). Outcome: “writer_knowledge”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
mitigation_condition_none	2.264	[2.167, 2.365]	36.596	3.257e-293	18.913	[17.965, 19.860]
mitigation_condition_prompting	2.170	[2.076, 2.267]	34.521	3.938e-261	18.043	[17.081, 19.006]
mitigation_condition_reranking	1.948	[1.864, 2.037]	29.480	5.273e-191	15.763	[14.767, 16.760]

Table S292: Beta regression predicting perceived writer importance by mitigation condition in Study 3 (“preferred” subset, N=19,102). Outcome: “writer_importance”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
mitigation_condition_none	1.764	[1.686, 1.844]	24.856	2.215e-136	11.756	[10.879, 12.634]
mitigation_condition_prompting	1.562	[1.494, 1.634]	19.489	1.374e-84	9.480	[8.566, 10.393]
mitigation_condition_reranking	1.358	[1.297, 1.421]	13.192	9.733e-40	6.675	[5.711, 7.638]

Table S293: Beta regression predicting perceived writer confidence by mitigation condition in Study 3 (“preferred” subset, N=19,102). Outcome: “writer_confidence”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
mitigation_condition_none	1.705	[1.627, 1.786]	22.407	3.344e-111	9.989	[9.162, 10.816]
mitigation_condition_prompting	1.580	[1.507, 1.655]	19.141	1.156e-81	8.724	[7.873, 9.575]
mitigation_condition_reranking	1.452	[1.385, 1.522]	15.467	5.778e-54	7.265	[6.381, 8.148]

Table S294: Beta regression predicting perceived writer stance polarity by mitigation condition in Study 3 (“preferred” subset, N=19,102). Outcome: “writer_stance_polarity”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
mitigation_condition_none	1.397	[1.326, 1.472]	12.584	2.574e-36	7.416	[6.294, 8.538]
mitigation_condition_prompting	1.320	[1.252, 1.391]	10.313	6.185e-25	6.211	[5.058, 7.364]
mitigation_condition_reranking	1.159	[1.098, 1.222]	5.380	7.433e-8	3.362	[2.151, 4.573]

Table S295: Beta regression predicting paragraph hope by mitigation condition in Study 3 (“preferred” subset, N=19,102). Outcome: “paragraph_hope”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
mitigation_condition_none	1.462	[1.399, 1.528]	16.934	2.535e-64	9.076	[8.010, 10.143]
mitigation_condition_prompting	1.487	[1.423, 1.555]	17.500	1.440e-68	9.492	[8.412, 10.571]
mitigation_condition_reranking	1.325	[1.267, 1.386]	12.323	6.780e-35	6.665	[5.589, 7.740]

Table S296: Beta regression predicting paragraph excitement by mitigation condition in Study 3 (“preferred” subset, N=19,102). Outcome: “paragraph_excitement”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
mitigation_condition_none	1.290	[1.236, 1.346]	11.710	1.135e-31	4.522	[3.734, 5.310]
mitigation_condition_prompting	1.249	[1.197, 1.304]	10.133	3.954e-24	3.922	[3.136, 4.708]
mitigation_condition_reranking	1.181	[1.131, 1.234]	7.494	6.706e-14	2.887	[2.112, 3.663]

Table S297: Beta regression predicting paragraph fear by mitigation condition in Study 3 (“preferred” subset, N=19,102). Outcome: “paragraph_fear”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
mitigation_condition_none	1.012	[0.967, 1.059]	0.496	.620	0.232	[−0.685, 1.149]
mitigation_condition_prompting	0.976	[0.933, 1.022]	−1.024	.306	−0.477	[−1.388, 0.434]
mitigation_condition_reranking	0.986	[0.941, 1.033]	−0.602	.547	−0.286	[−1.214, 0.643]

Table S298: Beta regression predicting paragraph disgust by mitigation condition in Study 3 (“preferred” subset, N=19,102). Outcome: “paragraph_disgust”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
mitigation_condition_none	0.870	[0.831, 0.912]	−5.834	5.408e-9	−2.396	[−3.188, −1.605]
mitigation_condition_prompting	0.819	[0.782, 0.859]	−8.270	1.340e-16	−3.380	[−4.163, −2.598]
mitigation_condition_reranking	0.880	[0.839, 0.923]	−5.263	1.420e-7	−2.215	[−3.027, −1.404]

Table S299: Beta regression predicting paragraph anger by mitigation condition in Study 3 (“preferred” subset, N=19,102). Outcome: “paragraph_anger”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
mitigation_condition_none	0.908	[0.867, 0.952]	−4.043	5.272e-5	−1.764	[−2.609, −0.918]
mitigation_condition_prompting	0.823	[0.785, 0.863]	−8.113	4.924e-16	−3.484	[−4.306, −2.661]
mitigation_condition_reranking	0.890	[0.848, 0.934]	−4.777	1.775e-6	−2.125	[−2.984, −1.266]

Table S300: Beta regression predicting perceived writer affect x by mitigation condition in Study 3 (“preferred” subset, N=19,102). Outcome: “writer_affect_x”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
mitigation_condition_none	1.189	[1.147, 1.233]	9.353	8.550e-21	4.257	[3.361, 5.153]
mitigation_condition_prompting	1.248	[1.203, 1.295]	11.854	2.049e-32	5.458	[4.551, 6.365]
mitigation_condition_reranking	1.153	[1.111, 1.196]	7.508	6.001e-14	3.487	[2.573, 4.401]

Table S301: Beta regression predicting perceived writer affect y by mitigation condition in Study 3 (“preferred” subset, N=19,102). Outcome: “writer_affect_y”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
mitigation_condition_none	0.814	[0.788, 0.840]	−12.821	1.247e-37	−4.986	[−5.740, −4.232]
mitigation_condition_prompting	0.870	[0.842, 0.897]	−8.667	4.435e-18	−3.400	[−4.164, −2.636]
mitigation_condition_reranking	0.917	[0.888, 0.947]	−5.307	1.113e-7	−2.116	[−2.894, −1.337]

Table S302: Beta regression predicting perceived writer optimism by mitigation condition in Study 3 (“preferred” subset, N=19,102). Outcome: “writer_optimism”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
mitigation_condition_none	1.468	[1.410, 1.529]	18.708	4.221e-78	9.556	[8.557, 10.555]
mitigation_condition_prompting	1.462	[1.404, 1.523]	18.298	8.546e-75	9.455	[8.444, 10.466]
mitigation_condition_reranking	1.333	[1.280, 1.390]	13.675	1.433e-42	7.148	[6.122, 8.174]

Table S303: Beta regression predicting perceived writer community by mitigation condition in Study 3 (“preferred” subset, N=19,102). Outcome: “writer_community”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
mitigation_condition_none	1.521	[1.456, 1.590]	18.689	6.121e-78	9.685	[8.701, 10.669]
mitigation_condition_prompting	1.303	[1.247, 1.362]	11.751	7.002e-32	6.243	[5.220, 7.267]
mitigation_condition_reranking	1.279	[1.224, 1.338]	10.827	2.569e-27	5.822	[4.785, 6.859]

Table S304: Beta regression predicting perceived writer friendliness by mitigation condition in Study 3 (“preferred” subset, N=19,102). Outcome: “writer_friendliness”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
mitigation_condition_none	1.140	[1.102, 1.180]	7.469	8.074e-14	3.250	[2.400, 4.100]
mitigation_condition_prompting	1.148	[1.109, 1.188]	7.793	6.547e-15	3.417	[2.561, 4.273]
mitigation_condition_reranking	1.101	[1.063, 1.140]	5.351	8.735e-8	2.380	[1.510, 3.249]

Table S305: Beta regression predicting perceived writer openness by mitigation condition in Study 3 (“preferred” subset, N=19,102). Outcome: “writer_openness”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
mitigation_condition_none	0.811	[0.775, 0.849]	−8.940	3.908e-19	−4.589	[−5.578, −3.601]
mitigation_condition_prompting	0.883	[0.843, 0.925]	−5.229	1.705e-7	−2.769	[−3.796, −1.742]
mitigation_condition_reranking	0.891	[0.850, 0.934]	−4.807	1.529e-6	−2.572	[−3.610, −1.534]

Table S306: Ordinal logistic regression predicting perceived writer first-language status by mitigation condition in Study 3 (“preferred” subset, N=19,102). Outcome: “writer_english_first”. Reference category: “writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
mitigation_condition_none	3.406	[2.774, 4.182]	11.701	1.263e-31
mitigation_condition_prompting	4.587	[3.638, 5.784]	12.884	5.554e-38
mitigation_condition_reranking	4.292	[3.416, 5.394]	12.498	7.689e-36

Table S307: Ordinal logistic regression predicting perceived writer education by mitigation condition in Study 3 (“preferred” subset, N=19,102). Outcome: “writer_education”. Reference category: “writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
mitigation_condition_none	6.294	[5.801, 6.828]	44.199	1.191e-426
mitigation_condition_prompting	5.593	[5.154, 6.069]	41.280	1.905e-372
mitigation_condition_reranking	5.848	[5.382, 6.353]	41.710	3.358e-380

Table S308: Ordinal logistic regression predicting perceived writer English skills by mitigation condition in Study 3 (“preferred” subset, N=19,102). Outcome: “writer_english_skills”. Reference category: “writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
mitigation_condition_none	9.220	[8.342, 10.190]	43.518	1.154e-413
mitigation_condition_prompting	7.688	[6.966, 8.484]	40.536	3.548e-359
mitigation_condition_reranking	8.401	[7.592, 9.296]	41.210	3.873e-321

Table S309: Ordinal logistic regression predicting perceived writer income by mitigation condition in Study 3 (“preferred” subset, N=19,102). Outcome: “writer_income”. Reference category: “writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
mitigation_condition_none	5.286	[4.878, 5.728]	40.666	1.810e-361
mitigation_condition_prompting	4.719	[4.353, 5.114]	37.759	5.301e-312
mitigation_condition_reranking	4.967	[4.576, 5.391]	38.312	3.873e-321

Table S310: Ordinal logistic regression predicting perceived writer age by mitigation condition in Study 3 (“preferred” subset, N=19,102). Outcome: “writer_age_binned”. Reference category: “writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
mitigation_condition_none	1.792	[1.660, 1.935]	14.910	2.833e-50
mitigation_condition_prompting	1.812	[1.677, 1.958]	15.033	4.490e-51
mitigation_condition_reranking	1.827	[1.690, 1.976]	15.083	2.103e-51

Table S311: Multinomial logistic regression predicting perceived writer race by mitigation condition in Study 3 (“preferred” subset, N=19,102). Outcome: “writer_race”. Predictor reference category: “writer”. Outcome reference level: “White”. OR is odds ratio.

Term	Target level	Reference level	OR	95% CI (OR)	statistic	p
mitigation_condition_none	Asian	White	0.753	[0.638, 0.890]	−3.325	8.835e-4
mitigation_condition_none	Black	White	0.856	[0.738, 0.993]	−2.054	.040
mitigation_condition_none	Mixed	White	0.911	[0.781, 1.063]	−1.180	.238
mitigation_condition_none	Other	White	0.803	[0.626, 1.031]	−1.717	.086
mitigation_condition_prompting	Asian	White	0.810	[0.687, 0.956]	−2.495	.013
mitigation_condition_prompting	Black	White	0.831	[0.714, 0.967]	−2.396	.017
mitigation_condition_prompting	Mixed	White	0.910	[0.778, 1.063]	−1.191	.234
mitigation_condition_prompting	Other	White	0.861	[0.671, 1.105]	−1.177	.239
mitigation_condition_reranking	Asian	White	0.727	[0.611, 0.865]	−3.598	3.203e-4
mitigation_condition_reranking	Black	White	0.871	[0.748, 1.013]	−1.792	.073
mitigation_condition_reranking	Mixed	White	0.905	[0.772, 1.059]	−1.246	.213
mitigation_condition_reranking	Other	White	0.836	[0.651, 1.073]	−1.407	.159

Table S312: Multinomial logistic regression predicting perceived writer gender by mitigation condition in Study 3 (“preferred” subset, N=19,102). Outcome: “writer_gender”. Predictor reference category: “writer”. Outcome reference level: “Male”. OR is odds ratio.

Term	Target level	Reference level	OR	95% CI (OR)	statistic	p
mitigation_condition_none	Female	Male	1.036	[0.955, 1.123]	0.847	.397
mitigation_condition_none	Other	Male	1.152	[0.810, 1.637]	0.787	.432
mitigation_condition_prompting	Female	Male	1.066	[0.982, 1.156]	1.530	.126
mitigation_condition_prompting	Other	Male	0.867	[0.584, 1.288]	−0.707	.480
mitigation_condition_reranking	Female	Male	1.018	[0.937, 1.106]	0.424	.671
mitigation_condition_reranking	Other	Male	1.099	[0.763, 1.581]	0.506	.613

Table S313: Multinomial logistic regression predicting perceived writer political party by mitigation condition in Study 3 (“preferred” subset, N=19,102). Outcome: “writer_politicalParty”. Predictor reference category: “writer”. Outcome reference level: “Labour”. OR is odds ratio.

Term	Target level	Reference level	OR	95% CI (OR)	statistic	p
mitigation_condition_none	Conservative	Labour	1.547	[1.384, 1.728]	7.688	1.500e-14
mitigation_condition_none	Did not vote	Labour	0.357	[0.299, 0.426]	-11.412	3.640e-30
mitigation_condition_none	Greens	Labour	1.229	[1.058, 1.426]	2.706	.007
mitigation_condition_none	Liberal Democrats	Labour	1.193	[1.045, 1.362]	2.618	.009
mitigation_condition_none	Not eligible to vote	Labour	0.155	[0.090, 0.267]	-6.728	1.723e-11
mitigation_condition_none	Other	Labour	1.091	[0.847, 1.405]	0.676	.499
mitigation_condition_none	Reform UK	Labour	0.547	[0.458, 0.653]	-6.684	2.323e-11
mitigation_condition_none	Scottish National Party	Labour	1.608	[1.135, 2.278]	2.671	.008
mitigation_condition_prompting	Conservative	Labour	1.574	[1.407, 1.762]	7.902	2.744e-15
mitigation_condition_prompting	Did not vote	Labour	0.351	[0.293, 0.420]	-11.375	5.550e-30
mitigation_condition_prompting	Greens	Labour	1.211	[1.041, 1.409]	2.477	.013
mitigation_condition_prompting	Liberal Democrats	Labour	1.267	[1.110, 1.445]	3.518	4.347e-4
mitigation_condition_prompting	Not eligible to vote	Labour	0.207	[0.128, 0.336]	-6.380	1.773e-10
mitigation_condition_prompting	Other	Labour	1.096	[0.848, 1.417]	0.699	.484
mitigation_condition_prompting	Reform UK	Labour	0.586	[0.491, 0.698]	-5.971	2.360e-9
mitigation_condition_prompting	Scottish National Party	Labour	1.391	[0.960, 2.014]	1.746	.081
mitigation_condition_reranking	Conservative	Labour	1.450	[1.292, 1.626]	6.337	2.348e-10
mitigation_condition_reranking	Did not vote	Labour	0.391	[0.328, 0.465]	-10.540	5.660e-26
mitigation_condition_reranking	Greens	Labour	1.315	[1.133, 1.527]	3.601	3.171e-4
mitigation_condition_reranking	Liberal Democrats	Labour	1.186	[1.036, 1.357]	2.481	.013
mitigation_condition_reranking	Not eligible to vote	Labour	0.278	[0.182, 0.426]	-5.896	3.730e-9
mitigation_condition_reranking	Other	Labour	1.053	[0.810, 1.368]	0.386	.699
mitigation_condition_reranking	Reform UK	Labour	0.593	[0.497, 0.707]	-5.818	5.959e-9
mitigation_condition_reranking	Scottish National Party	Labour	1.284	[0.876, 1.882]	1.281	.200

Table S314: Multinomial logistic regression predicting perceived writer political ideology by mitigation condition in Study 3 (“preferred” subset, N=19,102). Outcome: “writer_politicalIdeology”. Predictor reference category: “writer”. Outcome reference level: “Centrist”. OR is odds ratio.

Term	Target level	Reference level	OR	95% CI (OR)	statistic	p
mitigation_condition_none	Moderately Left-Wing	Centrist	1.004	[0.906, 1.114]	0.085	.933
mitigation_condition_none	Moderately Right-Wing	Centrist	1.102	[0.985, 1.233]	1.692	.091
mitigation_condition_none	Very Left-Wing	Centrist	1.299	[1.137, 1.484]	3.850	1.180e-4
mitigation_condition_none	Very Right-Wing	Centrist	0.851	[0.690, 1.050]	-1.507	.132
mitigation_condition_prompting	Moderately Left-Wing	Centrist	1.001	[0.903, 1.110]	0.015	.988
mitigation_condition_prompting	Moderately Right-Wing	Centrist	1.098	[0.981, 1.229]	1.631	.103
mitigation_condition_prompting	Very Left-Wing	Centrist	1.081	[0.940, 1.243]	1.091	.275
mitigation_condition_prompting	Very Right-Wing	Centrist	0.851	[0.690, 1.051]	-1.499	.134
mitigation_condition_reranking	Moderately Left-Wing	Centrist	1.034	[0.932, 1.147]	0.627	.531
mitigation_condition_reranking	Moderately Right-Wing	Centrist	1.054	[0.939, 1.183]	0.897	.370
mitigation_condition_reranking	Very Left-Wing	Centrist	1.170	[1.018, 1.344]	2.212	.027
mitigation_condition_reranking	Very Right-Wing	Centrist	0.745	[0.596, 0.932]	-2.579	.010

Finally, to quantify variation in distortions across specific model-mitigation combinations, we fit **mixed-effects regression models** specified as follows:

$$\text{rating_attribute} \sim \text{model_and_mitigation} + (1 \mid \text{reader_id})$$

where `rating_attribute` and `(1|reader_id)` are defined as in SI:5.7, and `model_and_mitigation` is a categorical fixed effect taking ten values: one for human-written paragraphs, which serves as the reference category, and nine corresponding to the 3×3 combinations of model identity and mitigation condition (Prompting, Reranking, None, crossed with Claude Sonnet 4, DeepSeek Chat V3, and ChatGPT-4o).

We used this combined-factor specification in order to estimate the distortion associated with each exact model-mitigation combination directly relative to human-written text, without imposing the assumption that mitigation effects are additive and constant across models. This parameterisation is also more natural than a standard model-by-mitigation interaction for our design, because the human-written baseline does not populate the full crossed model $\times$ mitigation structure. The combined factor therefore yields direct, interpretable contrasts for each treatment condition while avoiding unnecessary extrapolation.

The measurement scale of each rating attribute again determined the appropriate link function, following the same scheme as in SI:5.7: beta regression for 0–100 scale attributes, multinomial logistic regression for nominal categorical attributes, and ordinal logistic regression for ordinal categorical attributes.

As above, we fit each regression model to the Study 3 equivalent of the “preferred” data subset as defined in SI:5.2, which contained 19,102 observations. Tables S315 to S343 show results for each of the 29 rating attributes in our study.

This analysis was not pre-registered, and we note that by subsetting the data we are operating with relatively small per-group sample sizes for some model-mitigation combinations. As a result, estimates for these conditions may be noisier and more sensitive to idiosyncrasies in the sampled paragraphs and readers, and should therefore be interpreted with appropriate caution.

Table S315: Beta regression predicting perceived paragraph formality in Study 3 (“preferred” subset, N=19,102). Outcome: “paragraph_formality”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
model_and_mitigation_anthropic/claude-sonnet-4_none	4.260	[3.957, 4.586]	38.535	0	33.792	[32.351, 35.233]
model_and_mitigation_anthropic/claude-sonnet-4_prompting	4.541	[4.206, 4.902]	38.745	0	34.984	[33.530, 36.438]
model_and_mitigation_anthropic/claude-sonnet-4_reranking	4.900	[4.556, 5.269]	42.873	0	36.351	[35.010, 37.693]
model_and_mitigation_deepseek/deepseek-chat-v3-0324_none	3.323	[3.095, 3.568]	33.119	1.577e-240	28.808	[27.280, 30.335]
model_and_mitigation_deepseek/deepseek-chat-v3-0324_prompting	3.269	[3.047, 3.508]	32.958	3.201e-238	28.461	[26.939, 29.984]
model_and_mitigation_deepseek/deepseek-chat-v3-0324_reranking	2.551	[2.370, 2.746]	24.936	3.011e-137	22.941	[21.233, 24.649]
model_and_mitigation_openai/chatgpt-4o-latest_none	3.308	[3.081, 3.552]	32.998	8.651e-239	28.710	[27.181, 30.240]
model_and_mitigation_openai/chatgpt-4o-latest_prompting	2.609	[2.430, 2.801]	26.505	8.502e-155	23.461	[21.824, 25.098]
model_and_mitigation_openai/chatgpt-4o-latest_reranking	2.855	[2.648, 3.078]	27.312	3.037e-164	25.500	[23.805, 27.195]

Table S316: Beta regression predicting perceived paragraph clarity in Study 3 (“preferred” subset, N=19,102). Outcome: “paragraph_clarity”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
model_and_mitigation_anthropic/claude-sonnet-4_none	1.953	[1.807, 2.111]	16.877	6.607e-64	11.713	[10.537, 12.889]
model_and_mitigation_anthropic/claude-sonnet-4_prompting	1.954	[1.804, 2.117]	16.433	1.101e-60	11.722	[10.516, 12.928]
model_and_mitigation_anthropic/claude-sonnet-4_reranking	1.581	[1.466, 1.704]	11.912	1.026e-32	8.472	[7.203, 9.742]
model_and_mitigation_deepseek/deepseek-chat-v3-0324_none	1.919	[1.781, 2.067]	17.138	7.669e-66	11.461	[10.318, 12.604]
model_and_mitigation_deepseek/deepseek-chat-v3-0324_prompting	1.796	[1.667, 1.934]	15.441	8.726e-54	10.477	[9.299, 11.655]
model_and_mitigation_deepseek/deepseek-chat-v3-0324_reranking	1.535	[1.420, 1.661]	10.731	7.313e-27	7.995	[6.659, 9.331]
model_and_mitigation_openai/chatgpt-4o-latest_none	1.845	[1.712, 1.988]	16.096	2.720e-58	10.883	[9.717, 12.048]
model_and_mitigation_openai/chatgpt-4o-latest_prompting	1.496	[1.388, 1.613]	10.484	1.025e-25	7.561	[6.258, 8.864]
model_and_mitigation_openai/chatgpt-4o-latest_reranking	1.580	[1.459, 1.712]	11.192	4.441e-29	8.469	[7.123, 9.815]

Table S317: Beta regression predicting perceived paragraph informativeness in Study 3 (“preferred” subset, N=19,102). Outcome: “paragraph_informativeness”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
model_and_mitigation_anthropic/claude-sonnet-4__none	2.893	[2.683, 3.120]	27.586	1.645e-167	25.939	[24.196, 27.682]
model_and_mitigation_anthropic/claude-sonnet-4__prompting	3.472	[3.207, 3.759]	30.719	3.189e-207	30.028	[28.288, 31.767]
model_and_mitigation_anthropic/claude-sonnet-4__reranking	3.193	[2.963, 3.441]	30.417	3.280e-203	28.181	[26.498, 29.863]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__none	2.668	[2.478, 2.872]	26.102	3.445e-150	24.052	[22.318, 25.787]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__prompting	2.434	[2.263, 2.617]	23.964	6.624e-127	21.867	[20.123, 23.611]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__reranking	1.914	[1.773, 2.067]	16.608	6.134e-62	15.991	[14.096, 17.886]
model_and_mitigation_openai/chatgpt-4o-latest__none	2.923	[2.715, 3.148]	28.451	4.680e-178	26.179	[24.477, 27.881]
model_and_mitigation_openai/chatgpt-4o-latest__prompting	2.172	[2.018, 2.338]	20.655	8.808e-95	19.109	[17.313, 20.905]
model_and_mitigation_openai/chatgpt-4o-latest__reranking	2.364	[2.185, 2.557]	21.423	8.127e-102	21.165	[19.269, 23.062]

Table S318: Beta regression predicting perceived paragraph originality in Study 3 (“preferred” subset, N=19,102). Outcome: “paragraph_originality”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
model_and_mitigation_anthropic/claude-sonnet-4__none	1.730	[1.611, 1.858]	15.105	1.502e-51	12.805	[11.066, 14.544]
model_and_mitigation_anthropic/claude-sonnet-4__prompting	2.075	[1.927, 2.234]	19.375	1.263e-83	17.323	[15.495, 19.150]
model_and_mitigation_anthropic/claude-sonnet-4__reranking	2.080	[1.939, 2.233]	20.329	7.053e-92	17.387	[15.640, 19.133]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__none	1.756	[1.639, 1.881]	16.076	3.774e-58	13.169	[11.490, 14.848]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__prompting	1.569	[1.466, 1.680]	12.942	2.598e-38	10.411	[8.766, 12.056]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__reranking	1.501	[1.397, 1.613]	11.068	1.785e-28	9.338	[7.614, 11.062]
model_and_mitigation_openai/chatgpt-4o-latest__none	1.838	[1.716, 1.969]	17.387	1.044e-67	14.300	[12.615, 15.985]
model_and_mitigation_openai/chatgpt-4o-latest__prompting	1.560	[1.456, 1.671]	12.635	1.358e-36	10.259	[8.598, 11.920]
model_and_mitigation_openai/chatgpt-4o-latest__reranking	1.601	[1.489, 1.721]	12.696	6.233e-37	10.896	[9.138, 12.655]

Table S319: Beta regression predicting perceived paragraph relevance in Study 3 (“preferred” subset, N=19,102). Outcome: “paragraph_relevance”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
model_and_mitigation_anthropic/claude-sonnet-4__none	1.853	[1.718, 1.998]	16.012	1.060e-57	9.730	[8.696, 10.764]
model_and_mitigation_anthropic/claude-sonnet-4__prompting	1.895	[1.753, 2.048]	16.086	3.218e-58	10.019	[8.968, 11.071]
model_and_mitigation_anthropic/claude-sonnet-4__reranking	1.687	[1.568, 1.815]	13.994	1.686e-44	8.481	[7.424, 9.537]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__none	1.868	[1.738, 2.008]	16.955	1.767e-64	9.836	[8.844, 10.827]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__prompting	1.763	[1.641, 1.895]	15.484	4.419e-54	9.083	[8.067, 10.098]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__reranking	1.478	[1.371, 1.592]	10.241	1.305e-24	6.580	[5.427, 7.733]
model_and_mitigation_openai/chatgpt-4o-latest__none	1.769	[1.646, 1.901]	15.564	1.282e-54	9.126	[8.112, 10.140]
model_and_mitigation_openai/chatgpt-4o-latest__prompting	1.553	[1.444, 1.670]	11.896	1.248e-32	7.313	[6.220, 8.406]
model_and_mitigation_openai/chatgpt-4o-latest__reranking	1.684	[1.559, 1.819]	13.216	7.131e-40	8.458	[7.347, 9.569]

Table S320: Beta regression predicting perceived writer knowledge in Study 3 (“preferred” subset, N=19,102). Outcome: “writer_knowledge”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
model_and_mitigation_anthropic/claude-sonnet-4__none	2.248	[2.093, 2.415]	22.135	1.448e-108	18.771	[17.273, 20.268]
model_and_mitigation_anthropic/claude-sonnet-4__prompting	2.695	[2.502, 2.904]	26.085	5.332e-150	22.322	[20.876, 23.768]
model_and_mitigation_anthropic/claude-sonnet-4__reranking	2.339	[2.181, 2.509]	23.835	1.448e-125	19.573	[18.133, 21.013]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__none	2.202	[2.054, 2.361]	22.213	2.593e-109	18.343	[16.876, 19.811]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__prompting	2.150	[2.007, 2.304]	21.803	2.160e-105	17.855	[16.392, 19.318]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__reranking	1.664	[1.549, 1.789]	13.878	8.644e-44	12.268	[10.615, 13.921]
model_and_mitigation_openai/chatgpt-4o-latest__none	2.355	[2.196, 2.525]	24.072	4.878e-128	19.703	[18.270, 21.137]
model_and_mitigation_openai/chatgpt-4o-latest__prompting	1.770	[1.651, 1.898]	16.048	5.945e-58	13.654	[12.078, 15.231]
model_and_mitigation_openai/chatgpt-4o-latest__reranking	1.844	[1.713, 1.985]	16.236	2.825e-59	14.563	[12.916, 16.211]

Table S321: Beta regression predicting perceived writer importance in Study 3 (“preferred” subset, N=19,102). Outcome: “writer_importance”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
model_and_mitigation_anthropic/claude-sonnet-4__none	1.854	[1.722, 1.996]	16.340	5.103e-60	12.650	[11.294, 14.006]
model_and_mitigation_anthropic/claude-sonnet-4__prompting	1.887	[1.749, 2.036]	16.353	4.159e-60	12.962	[11.582, 14.343]
model_and_mitigation_anthropic/claude-sonnet-4__reranking	1.485	[1.382, 1.596]	10.794	3.673e-27	8.487	[7.041, 9.934]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__none	1.803	[1.679, 1.937]	16.165	8.923e-59	12.156	[10.827, 13.485]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__prompting	1.618	[1.507, 1.736]	13.344	1.287e-40	10.148	[8.771, 11.524]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__reranking	1.201	[1.115, 1.292]	4.865	1.144e-6	4.077	[2.475, 5.678]
model_and_mitigation_openai/chatgpt-4o-latest__none	1.646	[1.532, 1.767]	13.696	1.066e-42	10.470	[9.092, 11.848]
model_and_mitigation_openai/chatgpt-4o-latest__prompting	1.245	[1.159, 1.337]	6.010	1.850e-9	4.853	[3.319, 6.388]
model_and_mitigation_openai/chatgpt-4o-latest__reranking	1.392	[1.290, 1.502]	8.516	1.651e-17	7.189	[5.618, 8.759]

Table S322: Beta regression predicting perceived writer confidence in Study 3 (“preferred” subset, N=19,102). Outcome: “writer_confidence”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
model_and_mitigation_anthropic/claude-sonnet-4__none	1.791	[1.659, 1.934]	14.898	3.419e-50	10.771	[9.516, 12.026]
model_and_mitigation_anthropic/claude-sonnet-4__prompting	1.853	[1.713, 2.006]	15.313	6.300e-53	11.304	[10.036, 12.572]
model_and_mitigation_anthropic/claude-sonnet-4__reranking	1.607	[1.492, 1.731]	12.524	5.547e-36	9.012	[7.729, 10.296]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__none	1.714	[1.591, 1.847]	14.144	2.029e-45	10.069	[8.818, 11.321]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__prompting	1.626	[1.511, 1.750]	12.950	2.363e-38	9.207	[7.940, 10.473]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__reranking	1.316	[1.218, 1.422]	6.973	3.106e-12	5.473	[4.012, 6.935]
model_and_mitigation_openai/chatgpt-4o-latest__none	1.617	[1.501, 1.742]	12.673	8.331e-37	9.114	[7.833, 10.396]
model_and_mitigation_openai/chatgpt-4o-latest__prompting	1.300	[1.207, 1.401]	6.888	5.658e-12	5.242	[3.820, 6.663]
model_and_mitigation_openai/chatgpt-4o-latest__reranking	1.424	[1.315, 1.541]	8.761	1.939e-18	6.909	[5.467, 8.351]

Table S323: Beta regression predicting perceived writer stance polarity in Study 3 (“preferred” subset, N=19,102). Outcome: “writer_stance_polarity”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
model_and_mitigation_anthropic/claude-sonnet-4__none	1.546	[1.422, 1.682]	10.161	2.951e-24	9.481	[7.772, 11.191]
model_and_mitigation_anthropic/claude-sonnet-4__prompting	1.600	[1.467, 1.745]	10.605	2.811e-26	10.158	[8.418, 11.899]
model_and_mitigation_anthropic/claude-sonnet-4__reranking	1.176	[1.083, 1.277]	3.856	1.153e-4	3.697	[1.855, 5.539]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__none	1.356	[1.248, 1.473]	7.212	5.527e-13	6.786	[5.018, 8.553]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__prompting	1.244	[1.145, 1.351]	5.174	2.287e-7	4.928	[3.113, 6.743]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__reranking	1.127	[1.032, 1.230]	2.674	0.008	2.743	[0.761, 4.725]
model_and_mitigation_openai/chatgpt-4o-latest__none	1.299	[1.195, 1.411]	6.173	6.689e-10	5.867	[4.069, 7.665]
model_and_mitigation_openai/chatgpt-4o-latest__prompting	1.156	[1.063, 1.258]	3.388	0.001	3.318	[1.432, 5.203]
model_and_mitigation_openai/chatgpt-4o-latest__reranking	1.176	[1.076, 1.286]	3.566	3.626e-4	3.703	[1.709, 5.696]

Table S324: Beta regression predicting perceived paragraph hope in Study 3 (“preferred” subset, N=19,102). Outcome: “paragraph_hope”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
model_and_mitigation_anthropic/claude-sonnet-4__none	1.313	[1.221, 1.412]	7.313	2.606e-13	6.436	[4.670, 8.201]
model_and_mitigation_anthropic/claude-sonnet-4__prompting	1.470	[1.363, 1.585]	9.998	1.552e-23	9.203	[7.349, 11.056]
model_and_mitigation_anthropic/claude-sonnet-4__reranking	1.264	[1.177, 1.358]	6.439	1.201e-10	5.515	[3.801, 7.229]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__none	1.511	[1.408, 1.621]	11.456	2.183e-30	9.879	[8.144, 11.615]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__prompting	1.537	[1.433, 1.649]	11.983	4.346e-33	10.313	[8.581, 12.045]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__reranking	1.352	[1.256, 1.454]	8.090	5.989e-16	7.142	[5.369, 8.915]
model_and_mitigation_openai/chatgpt-4o-latest__none	1.583	[1.475, 1.699]	12.736	3.739e-37	11.036	[9.292, 12.780]
model_and_mitigation_openai/chatgpt-4o-latest__prompting	1.457	[1.357, 1.564]	10.369	3.444e-25	8.985	[7.241, 10.728]
model_and_mitigation_openai/chatgpt-4o-latest__reranking	1.372	[1.273, 1.479]	8.264	1.411e-16	7.512	[5.685, 9.339]

Table S325: Beta regression predicting perceived paragraph excitement in Study 3 (“preferred” subset, N=19,102). Outcome: “paragraph_excitement”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
model_and_mitigation_anthropic/claude-sonnet-4__none	1.235	[1.149, 1.327]	5.730	1.006e-8	3.703	[2.374, 5.032]
model_and_mitigation_anthropic/claude-sonnet-4__prompting	1.246	[1.157, 1.341]	5.809	6.281e-9	3.867	[2.495, 5.239]
model_and_mitigation_anthropic/claude-sonnet-4__reranking	1.223	[1.140, 1.311]	5.620	1.905e-8	3.521	[2.236, 4.806]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__none	1.283	[1.198, 1.373]	7.153	8.491e-13	4.417	[3.139, 5.696]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__prompting	1.269	[1.185, 1.359]	6.846	7.582e-12	4.216	[2.944, 5.488]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__reranking	1.123	[1.045, 1.206]	3.179	0.001	1.984	[0.727, 3.240]
model_and_mitigation_openai/chatgpt-4o-latest__none	1.356	[1.266, 1.452]	8.720	2.773e-18	5.481	[4.165, 6.797]
model_and_mitigation_openai/chatgpt-4o-latest__prompting	1.233	[1.150, 1.321]	5.923	3.152e-9	3.674	[2.400, 4.948]
model_and_mitigation_openai/chatgpt-4o-latest__reranking	1.193	[1.109, 1.283]	4.746	2.079e-6	3.073	[1.751, 4.395]

Table S326: Beta regression predicting perceived paragraph fear in Study 3 (“preferred” subset, N=19,102). Outcome: “paragraph_fear”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
model_and_mitigation_anthropic/claude-sonnet-4__none	1.072	[0.992, 1.157]	1.767	0.077	1.414	[−0.173, 3.000]
model_and_mitigation_anthropic/claude-sonnet-4__prompting	1.038	[0.961, 1.122]	0.948	0.343	0.764	[−0.826, 2.355]
model_and_mitigation_anthropic/claude-sonnet-4__reranking	0.979	[0.909, 1.054]	−0.568	0.570	−0.431	[−1.911, 1.049]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__none	1.003	[0.932, 1.079]	0.074	0.941	0.055	[−1.413, 1.524]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__prompting	1.002	[0.932, 1.077]	0.052	0.958	0.039	[−1.423, 1.502]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__reranking	1.012	[0.938, 1.092]	0.309	0.757	0.242	[−1.297, 1.781]
model_and_mitigation_openai/chatgpt-4o-latest__none	0.965	[0.897, 1.038]	−0.964	0.335	−0.713	[−2.153, 0.727]
model_and_mitigation_openai/chatgpt-4o-latest__prompting	0.897	[0.833, 0.965]	−2.924	0.003	−2.143	[−3.550, −0.736]
model_and_mitigation_openai/chatgpt-4o-latest__reranking	0.970	[0.897, 1.049]	−0.753	0.452	−0.601	[−2.157, 0.955]

Table S327: Beta regression predicting perceived paragraph disgust in Study 3 (“preferred” subset, N=19,102). Outcome: “paragraph_disgust”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
model_and_mitigation_anthropic/claude-sonnet-4__none	1.000	[0.925, 1.082]	0.009	0.993	0.006	[−1.397, 1.410]
model_and_mitigation_anthropic/claude-sonnet-4__prompting	0.881	[0.813, 0.956]	−3.064	0.002	−2.186	[−3.544, −0.827]
model_and_mitigation_anthropic/claude-sonnet-4__reranking	0.942	[0.873, 1.016]	−1.553	0.121	−1.061	[−2.383, 0.261]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__none	0.860	[0.798, 0.926]	−3.960	7.482e-5	−2.603	[−3.849, −1.357]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__prompting	0.845	[0.784, 0.910]	−4.460	8.189e-6	−2.890	[−4.114, −1.666]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__reranking	0.853	[0.789, 0.923]	−3.963	7.412e-5	−2.725	[−4.025, −1.425]
model_and_mitigation_openai/chatgpt-4o-latest__none	0.769	[0.714, 0.829]	−6.881	5.930e-12	−4.373	[−5.548, −3.199]
model_and_mitigation_openai/chatgpt-4o-latest__prompting	0.738	[0.684, 0.796]	−7.850	4.158e-15	−4.998	[−6.163, −3.833]
model_and_mitigation_openai/chatgpt-4o-latest__reranking	0.840	[0.776, 0.910]	−4.261	2.034e-5	−2.973	[−4.287, −1.659]

Table S328: Beta regression predicting perceived paragraph anger in Study 3 (“preferred” subset, N=19,102). Outcome: “paragraph_anger”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
model_and_mitigation_anthropic/claude-sonnet-4__none	1.083	[1.001, 1.171]	1.992	0.046	1.527	[0.000, 3.054]
model_and_mitigation_anthropic/claude-sonnet-4__prompting	0.947	[0.873, 1.026]	−1.335	0.182	−1.016	[−2.491, 0.458]
model_and_mitigation_anthropic/claude-sonnet-4__reranking	0.956	[0.886, 1.032]	−1.158	0.247	−0.836	[−2.237, 0.566]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__none	0.881	[0.817, 0.949]	−3.317	9.113e-4	−2.305	[−3.632, −0.979]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__prompting	0.826	[0.766, 0.889]	−5.063	4.118e-7	−3.427	[−4.702, −2.153]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__reranking	0.877	[0.810, 0.948]	−3.282	0.001	−2.389	[−3.777, −1.002]
model_and_mitigation_openai/chatgpt-4o-latest__none	0.790	[0.733, 0.851]	−6.204	5.512e-10	−4.170	[−5.423, −2.917]
model_and_mitigation_openai/chatgpt-4o-latest__prompting	0.717	[0.665, 0.773]	−8.674	4.156e-18	−5.732	[−6.937, −4.527]
model_and_mitigation_openai/chatgpt-4o-latest__reranking	0.834	[0.770, 0.904]	−4.419	9.921e-6	−3.247	[−4.631, −1.863]

Table S329: Beta regression predicting perceived writer affect x in Study 3 (“preferred” subset, N=19,102). Outcome: “writer_affect_x”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
model_and_mitigation_anthropic/claude-sonnet-4_none	1.009	[0.951, 1.070]	0.301	0.763	0.220	[-1.213, 1.653]
model_and_mitigation_anthropic/claude-sonnet-4_prompting	1.154	[1.086, 1.225]	4.641	3.466e-6	3.504	[2.014, 4.995]
model_and_mitigation_anthropic/claude-sonnet-4_reranking	1.071	[1.012, 1.134]	2.356	0.018	1.673	[0.276, 3.070]
model_and_mitigation_deepseek/deepseek-chat-v3-0324_none	1.295	[1.223, 1.372]	8.805	1.310e-18	6.387	[4.953, 7.820]
model_and_mitigation_deepseek/deepseek-chat-v3-0324_prompting	1.236	[1.167, 1.309]	7.241	4.438e-13	5.214	[3.791, 6.637]
model_and_mitigation_deepseek/deepseek-chat-v3-0324_reranking	1.181	[1.112, 1.254]	5.414	6.173e-8	4.086	[2.595, 5.577]
model_and_mitigation_openai/chatgpt-4o-latest_none	1.288	[1.216, 1.365]	8.635	5.852e-18	6.249	[4.818, 7.679]
model_and_mitigation_openai/chatgpt-4o-latest_prompting	1.363	[1.286, 1.444]	10.428	1.850e-25	7.653	[6.203, 9.102]
model_and_mitigation_openai/chatgpt-4o-latest_reranking	1.228	[1.154, 1.306]	6.521	6.987e-11	5.053	[3.521, 6.585]

Table S330: Beta regression predicting perceived writer affect y in Study 3 (“preferred” subset, N=19,102). Outcome: “writer_affect_y”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
model_and_mitigation_anthropic/claude-sonnet-4_none	0.771	[0.732, 0.812]	-9.870	5.601e-23	-6.257	[-7.471, -5.043]
model_and_mitigation_anthropic/claude-sonnet-4_prompting	0.798	[0.757, 0.842]	-8.355	6.569e-17	-5.432	[-6.682, -4.182]
model_and_mitigation_anthropic/claude-sonnet-4_reranking	0.876	[0.833, 0.920]	-5.239	1.617e-7	-3.226	[-4.422, -2.030]
model_and_mitigation_deepseek/deepseek-chat-v3-0324_none	0.810	[0.771, 0.852]	-8.186	2.711e-16	-5.082	[-6.278, -3.885]
model_and_mitigation_deepseek/deepseek-chat-v3-0324_prompting	0.827	[0.786, 0.869]	-7.497	6.515e-14	-4.610	[-5.797, -3.422]
model_and_mitigation_deepseek/deepseek-chat-v3-0324_reranking	0.953	[0.905, 1.004]	-1.796	0.072	-1.168	[-2.440, 0.103]
model_and_mitigation_openai/chatgpt-4o-latest_none	0.861	[0.819, 0.905]	-5.870	4.354e-9	-3.632	[-4.832, -2.433]
model_and_mitigation_openai/chatgpt-4o-latest_prompting	0.996	[0.947, 1.047]	-0.158	0.874	-0.100	[-1.331, 1.132]
model_and_mitigation_openai/chatgpt-4o-latest_reranking	0.930	[0.881, 0.981]	-2.670	0.008	-1.780	[-3.081, -0.479]

Table S331: Beta regression predicting perceived writer optimism in Study 3 (“preferred” subset, N=19,102). Outcome: “writer_optimism”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
model_and_mitigation_anthropic/claude-sonnet-4_none	1.336	[1.250, 1.427]	8.564	1.087e-17	7.192	[5.539, 8.846]
model_and_mitigation_anthropic/claude-sonnet-4_prompting	1.460	[1.363, 1.564]	10.809	3.106e-27	9.413	[7.705, 11.122]
model_and_mitigation_anthropic/claude-sonnet-4_reranking	1.318	[1.236, 1.406]	8.369	5.806e-17	6.861	[5.247, 8.476]
model_and_mitigation_deepseek/deepseek-chat-v3-0324_none	1.493	[1.400, 1.592]	12.195	3.311e-34	9.966	[8.365, 11.568]
model_and_mitigation_deepseek/deepseek-chat-v3-0324_prompting	1.458	[1.367, 1.554]	11.561	6.489e-31	9.372	[7.781, 10.962]
model_and_mitigation_deepseek/deepseek-chat-v3-0324_reranking	1.288	[1.204, 1.377]	7.373	1.668e-13	6.276	[4.598, 7.953]
model_and_mitigation_openai/chatgpt-4o-latest_none	1.590	[1.491, 1.696]	14.118	2.961e-45	11.537	[9.943, 13.132]
model_and_mitigation_openai/chatgpt-4o-latest_prompting	1.472	[1.379, 1.572]	11.628	2.977e-31	9.625	[8.002, 11.249]
model_and_mitigation_openai/chatgpt-4o-latest_reranking	1.401	[1.308, 1.502]	9.563	1.140e-21	8.390	[6.665, 10.115]

Table S332: Beta regression predicting perceived writer community orientation in Study 3 (“preferred” subset, N=19,102). Outcome: “writer_community”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
model_and_mitigation_anthropic/claude-sonnet-4_none	1.457	[1.355, 1.567]	10.126	4.241e-24	8.742	[7.124, 10.360]
model_and_mitigation_anthropic/claude-sonnet-4_prompting	1.360	[1.262, 1.465]	8.066	7.239e-16	7.209	[5.516, 8.901]
model_and_mitigation_anthropic/claude-sonnet-4_reranking	1.352	[1.260, 1.450]	8.426	3.582e-17	7.079	[5.485, 8.673]
model_and_mitigation_deepseek/deepseek-chat-v3-0324_none	1.512	[1.410, 1.623]	11.501	1.299e-30	9.560	[8.011, 11.109]
model_and_mitigation_deepseek/deepseek-chat-v3-0324_prompting	1.277	[1.191, 1.368]	6.910	4.834e-12	5.774	[4.176, 7.372]
model_and_mitigation_deepseek/deepseek-chat-v3-0324_reranking	1.194	[1.111, 1.284]	4.801	1.577e-6	4.235	[2.535, 5.935]
model_and_mitigation_openai/chatgpt-4o-latest_none	1.595	[1.487, 1.710]	13.071	4.796e-39	10.698	[9.186, 12.210]
model_and_mitigation_openai/chatgpt-4o-latest_prompting	1.278	[1.191, 1.371]	6.812	9.641e-12	5.795	[4.169, 7.422]
model_and_mitigation_openai/chatgpt-4o-latest_reranking	1.285	[1.192, 1.385]	6.571	4.982e-11	5.923	[4.202, 7.644]

Table S333: Beta regression predicting perceived writer friendliness in Study 3 (“preferred” subset, N=19,102). Outcome: “writer_friendliness”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
model_and_mitigation_anthropic/claude-sonnet-4__none	1.024	[0.967, 1.085]	0.809	0.418	0.591	[−0.839, 2.020]
model_and_mitigation_anthropic/claude-sonnet-4__prompting	1.105	[1.042, 1.172]	3.337	8.471e-4	2.479	[1.028, 3.929]
model_and_mitigation_anthropic/claude-sonnet-4__reranking	1.009	[0.955, 1.067]	0.326	0.744	0.230	[−1.151, 1.612]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__none	1.178	[1.115, 1.245]	5.823	5.764e-9	4.054	[2.700, 5.408]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__prompting	1.120	[1.061, 1.183]	4.076	4.586e-5	2.820	[1.470, 4.170]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__reranking	1.133	[1.070, 1.200]	4.249	2.143e-5	3.099	[1.677, 4.521]
model_and_mitigation_openai/chatgpt-4o-latest__none	1.229	[1.163, 1.298]	7.314	2.592e-13	5.082	[3.735, 6.429]
model_and_mitigation_openai/chatgpt-4o-latest__prompting	1.225	[1.159, 1.296]	7.127	1.029e-12	5.020	[3.654, 6.386]
model_and_mitigation_openai/chatgpt-4o-latest__reranking	1.184	[1.117, 1.256]	5.623	1.874e-8	4.187	[2.740, 5.635]

Table S334: Beta regression predicting perceived writer openness in Study 3 (“preferred” subset, N=19,102). Outcome: “writer_openness”. Reference category: “writer”. OR is odds ratio. AME is average marginal effect.

Term	OR	95% CI (OR)	z	p	AME	95% CI (AME)
model_and_mitigation_anthropic/claude-sonnet-4__none	0.777	[0.720, 0.838]	−6.539	6.207e-11	−5.490	[−7.074, −3.906]
model_and_mitigation_anthropic/claude-sonnet-4__prompting	0.798	[0.737, 0.864]	−5.590	2.275e-8	−4.930	[−6.601, −3.260]
model_and_mitigation_anthropic/claude-sonnet-4__reranking	0.825	[0.765, 0.889]	−5.065	4.079e-7	−4.231	[−5.823, −2.639]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__none	0.779	[0.724, 0.839]	−6.679	2.409e-11	−5.415	[−6.947, −3.884]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__prompting	0.854	[0.793, 0.919]	−4.230	2.335e-5	−3.499	[−5.085, −1.913]
model_and_mitigation_deepseek/deepseek-chat-v3-0324__reranking	1.029	[0.954, 1.111]	0.743	0.458	0.661	[−1.089, 2.412]
model_and_mitigation_openai/chatgpt-4o-latest__none	0.883	[0.820, 0.951]	−3.303	0.001	−2.760	[−4.371, −1.150]
model_and_mitigation_openai/chatgpt-4o-latest__prompting	1.012	[0.939, 1.090]	0.312	0.755	0.269	[−1.425, 1.963]
model_and_mitigation_openai/chatgpt-4o-latest__reranking	0.839	[0.776, 0.908]	−4.352	1.346e-5	−3.856	[−5.549, −2.164]

Table S335: Ordinal logistic regression predicting perceived writer English first-language status in Study 3 (“preferred” subset, N=19,102). Outcome: “writer_english_first”. Reference category: “writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
model_and_mitigation_anthropic/claude-sonnet-4__none	3.708	[2.614, 5.260]	7.349	2.000e-13
model_and_mitigation_anthropic/claude-sonnet-4__prompting	3.307	[2.338, 4.677]	6.763	1.355e-11
model_and_mitigation_anthropic/claude-sonnet-4__reranking	4.637	[3.216, 6.686]	8.215	2.120e-16
model_and_mitigation_deepseek/deepseek-chat-v3-0324__none	3.795	[2.650, 5.434]	7.280	3.346e-13
model_and_mitigation_deepseek/deepseek-chat-v3-0324__prompting	5.277	[3.518, 7.915]	8.043	8.797e-16
model_and_mitigation_deepseek/deepseek-chat-v3-0324__reranking	3.399	[2.370, 4.874]	6.653	2.868e-11
model_and_mitigation_openai/chatgpt-4o-latest__none	2.858	[2.074, 3.937]	6.421	1.353e-10
model_and_mitigation_openai/chatgpt-4o-latest__prompting	6.051	[3.908, 9.368]	8.072	6.914e-16
model_and_mitigation_openai/chatgpt-4o-latest__reranking	5.170	[3.335, 8.014]	7.345	2.059e-13

Table S336: Ordinal logistic regression predicting perceived writer education level in Study 3 (“preferred” subset, N=19,102). Outcome: “writer_education”. Reference category: “writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
model_and_mitigation_anthropic/claude-sonnet-4__none	7.283	[6.387, 8.303]	29.671	1.831e-193
model_and_mitigation_anthropic/claude-sonnet-4__prompting	8.209	[7.169, 9.401]	30.442	1.527e-203
model_and_mitigation_anthropic/claude-sonnet-4__reranking	9.309	[8.184, 10.590]	33.936	1.968e-252
model_and_mitigation_deepseek/deepseek-chat-v3-0324__none	6.265	[5.521, 7.109]	28.456	4.067e-178
model_and_mitigation_deepseek/deepseek-chat-v3-0324__prompting	5.354	[4.726, 6.066]	26.348	5.376e-153
model_and_mitigation_deepseek/deepseek-chat-v3-0324__reranking	3.824	[3.358, 4.356]	20.207	8.503e-91
model_and_mitigation_openai/chatgpt-4o-latest__none	5.627	[4.965, 6.377]	27.057	3.145e-161
model_and_mitigation_openai/chatgpt-4o-latest__prompting	4.161	[3.670, 4.717]	22.278	6.040e-110
model_and_mitigation_openai/chatgpt-4o-latest__reranking	5.403	[4.727, 6.175]	24.751	3.007e-135

Table S337: Ordinal logistic regression predicting perceived writer English skill level in Study 3 (“preferred” subset, N=19,102). Outcome: “writer_english_skills”. Reference category: “writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
model_and_mitigation_anthropic/claude-sonnet-4__none	9.986	[8.493, 11.742]	27.846	1.193e-170
model_and_mitigation_anthropic/claude-sonnet-4__prompting	10.210	[8.643, 12.063]	27.318	2.597e-164
model_and_mitigation_anthropic/claude-sonnet-4__reranking	12.998	[11.048, 15.293]	30.919	6.724e-210
model_and_mitigation_deepseek/deepseek-chat-v3-0324__none	9.956	[8.474, 11.697]	27.945	7.532e-172
model_and_mitigation_deepseek/deepseek-chat-v3-0324__prompting	8.218	[7.038, 9.597]	26.620	3.976e-156
model_and_mitigation_deepseek/deepseek-chat-v3-0324__reranking	5.519	[4.718, 6.457]	21.338	5.068e-101
model_and_mitigation_openai/chatgpt-4o-latest__none	8.141	[6.969, 9.510]	26.443	4.370e-154
model_and_mitigation_openai/chatgpt-4o-latest__prompting	5.582	[4.800, 6.491]	22.340	1.525e-110
model_and_mitigation_openai/chatgpt-4o-latest__reranking	8.019	[6.783, 9.480]	24.373	3.286e-131

Table S338: Ordinal logistic regression predicting perceived writer income level in Study 3 (“preferred” subset, N=19,102). Outcome: “writer_income”. Reference category: “writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
model_and_mitigation_anthropic/claude-sonnet-4__none	5.415	[4.751, 6.171]	25.325	1.705e-141
model_and_mitigation_anthropic/claude-sonnet-4__prompting	5.555	[4.857, 6.354]	25.023	3.462e-138
model_and_mitigation_anthropic/claude-sonnet-4__reranking	7.203	[6.333, 8.193]	30.064	1.428e-198
model_and_mitigation_deepseek/deepseek-chat-v3-0324__none	5.146	[4.542, 5.829]	25.749	3.318e-146
model_and_mitigation_deepseek/deepseek-chat-v3-0324__prompting	5.186	[4.582, 5.870]	26.053	1.230e-149
model_and_mitigation_deepseek/deepseek-chat-v3-0324__reranking	3.763	[3.305, 4.284]	20.021	3.646e-89
model_and_mitigation_openai/chatgpt-4o-latest__none	5.374	[4.745, 6.087]	26.458	2.949e-154
model_and_mitigation_openai/chatgpt-4o-latest__prompting	3.688	[3.254, 4.179]	20.446	6.543e-93
model_and_mitigation_openai/chatgpt-4o-latest__reranking	4.335	[3.796, 4.951]	21.646	6.666e-104

Table S339: Ordinal logistic regression predicting perceived writer age group in Study 3 (“preferred” subset, N=19,102). Outcome: “writer_age_binned”. Reference category: “writer”. OR is odds ratio.

Term	OR	95% CI (OR)	statistic	p
model_and_mitigation_anthropic/claude-sonnet-4__none	1.733	[1.529, 1.965]	8.613	7.111e-18
model_and_mitigation_anthropic/claude-sonnet-4__prompting	1.586	[1.394, 1.804]	7.008	2.410e-12
model_and_mitigation_anthropic/claude-sonnet-4__reranking	2.071	[1.838, 2.334]	11.951	6.434e-33
model_and_mitigation_deepseek/deepseek-chat-v3-0324__none	1.747	[1.549, 1.970]	9.079	1.100e-19
model_and_mitigation_deepseek/deepseek-chat-v3-0324__prompting	2.043	[1.813, 2.302]	11.709	1.141e-31
model_and_mitigation_deepseek/deepseek-chat-v3-0324__reranking	1.609	[1.417, 1.826]	7.348	2.019e-13
model_and_mitigation_openai/chatgpt-4o-latest__none	1.899	[1.686, 2.140]	10.531	6.198e-26
model_and_mitigation_openai/chatgpt-4o-latest__prompting	1.812	[1.605, 2.046]	9.584	9.317e-22
model_and_mitigation_openai/chatgpt-4o-latest__reranking	1.777	[1.561, 2.022]	8.697	3.398e-18

Table S340: Multinomial logistic regression predicting perceived writer gender in Study 3 (“preferred” subset, N=19,102). Outcome: “writer_gender”. Predictor reference category: “writer”. Outcome reference level: “Male”. OR is odds ratio.

Term	Target level	Reference level	OR	95% CI (OR)	statistic	p
model_and_mitigation_anthropic/claude-sonnet-4__none	Female	Male	1.058	[0.931, 1.203]	0.864	0.388
model_and_mitigation_anthropic/claude-sonnet-4__none	Other	Male	1.197	[0.694, 2.064]	0.647	0.517
model_and_mitigation_anthropic/claude-sonnet-4__prompting	Female	Male	1.020	[0.894, 1.164]	0.294	0.769
model_and_mitigation_anthropic/claude-sonnet-4__prompting	Other	Male	0.835	[0.435, 1.603]	-0.541	0.588
model_and_mitigation_anthropic/claude-sonnet-4__reranking	Female	Male	0.919	[0.811, 1.041]	-1.326	0.185
model_and_mitigation_anthropic/claude-sonnet-4__reranking	Other	Male	1.323	[0.809, 2.164]	1.114	0.265
model_and_mitigation_deepseek/deepseek-chat-v3-0324__none	Female	Male	1.135	[0.999, 1.289]	1.946	0.052
model_and_mitigation_deepseek/deepseek-chat-v3-0324__none	Other	Male	1.227	[0.712, 2.117]	0.737	0.461
model_and_mitigation_deepseek/deepseek-chat-v3-0324__prompting	Female	Male	0.926	[0.814, 1.052]	-1.183	0.237
model_and_mitigation_deepseek/deepseek-chat-v3-0324__prompting	Other	Male	1.034	[0.590, 1.812]	0.117	0.907
model_and_mitigation_deepseek/deepseek-chat-v3-0324__reranking	Female	Male	1.108	[0.970, 1.266]	1.512	0.130
model_and_mitigation_deepseek/deepseek-chat-v3-0324__reranking	Other	Male	0.979	[0.524, 1.830]	-0.065	0.948
model_and_mitigation_openai/chatgpt-4o-latest__none	Female	Male	0.925	[0.813, 1.052]	-1.193	0.233
model_and_mitigation_openai/chatgpt-4o-latest__none	Other	Male	1.040	[0.594, 1.823]	0.138	0.890
model_and_mitigation_openai/chatgpt-4o-latest__prompting	Female	Male	1.280	[1.127, 1.454]	3.799	1.452e-4
model_and_mitigation_openai/chatgpt-4o-latest__prompting	Other	Male	0.702	[0.341, 1.445]	-0.960	0.337
model_and_mitigation_openai/chatgpt-4o-latest__reranking	Female	Male	1.056	[0.921, 1.211]	0.786	0.432
model_and_mitigation_openai/chatgpt-4o-latest__reranking	Other	Male	0.924	[0.481, 1.775]	-0.236	0.813

Table S341: Multinomial logistic regression predicting perceived writer race in Study 3 (“preferred” subset, N=19,102). Outcome: “writer_race”. Predictor reference category: “writer”. Outcome reference level: “White”. OR is odds ratio.

Term	Target level	Reference level	OR	95% CI (OR)	statistic	p
model_and_mitigation_anthropic/claude-sonnet-4__none	Asian	White	0.623	[0.464, 0.837]	-3.144	0.002
model_and_mitigation_anthropic/claude-sonnet-4__none	Black	White	0.989	[0.785, 1.246]	-0.095	0.924
model_and_mitigation_anthropic/claude-sonnet-4__none	Mixed	White	1.004	[0.788, 1.280]	0.035	0.972
model_and_mitigation_anthropic/claude-sonnet-4__none	Other	White	0.926	[0.637, 1.347]	-0.401	0.688
model_and_mitigation_anthropic/claude-sonnet-4__prompting	Asian	White	0.859	[0.656, 1.125]	-1.106	0.269
model_and_mitigation_anthropic/claude-sonnet-4__prompting	Black	White	0.819	[0.634, 1.060]	-1.517	0.129
model_and_mitigation_anthropic/claude-sonnet-4__prompting	Mixed	White	0.950	[0.735, 1.226]	-0.396	0.692
model_and_mitigation_anthropic/claude-sonnet-4__prompting	Other	White	1.239	[0.873, 1.760]	1.200	0.230
model_and_mitigation_anthropic/claude-sonnet-4__reranking	Asian	White	0.807	[0.624, 1.044]	-1.629	0.103
model_and_mitigation_anthropic/claude-sonnet-4__reranking	Black	White	0.825	[0.650, 1.047]	-1.583	0.114
model_and_mitigation_anthropic/claude-sonnet-4__reranking	Mixed	White	0.910	[0.714, 1.159]	-0.765	0.445
model_and_mitigation_anthropic/claude-sonnet-4__reranking	Other	White	1.054	[0.752, 1.479]	0.306	0.760
model_and_mitigation_deepseek/deepseek-chat-v3-0324__none	Asian	White	0.721	[0.551, 0.944]	-2.377	0.017
model_and_mitigation_deepseek/deepseek-chat-v3-0324__none	Black	White	0.709	[0.551, 0.913]	-2.669	0.008
model_and_mitigation_deepseek/deepseek-chat-v3-0324__none	Mixed	White	0.890	[0.695, 1.138]	-0.932	0.351
model_and_mitigation_deepseek/deepseek-chat-v3-0324__none	Other	White	0.707	[0.462, 1.081]	-1.601	0.109
model_and_mitigation_deepseek/deepseek-chat-v3-0324__prompting	Asian	White	0.755	[0.578, 0.987]	-2.055	0.040
model_and_mitigation_deepseek/deepseek-chat-v3-0324__prompting	Black	White	0.907	[0.719, 1.143]	-0.827	0.408
model_and_mitigation_deepseek/deepseek-chat-v3-0324__prompting	Mixed	White	0.927	[0.726, 1.184]	-0.607	0.544
model_and_mitigation_deepseek/deepseek-chat-v3-0324__prompting	Other	White	0.914	[0.622, 1.344]	-0.457	0.648
model_and_mitigation_deepseek/deepseek-chat-v3-0324__reranking	Asian	White	0.624	[0.463, 0.841]	-3.096	0.002
model_and_mitigation_deepseek/deepseek-chat-v3-0324__reranking	Black	White	0.883	[0.693, 1.126]	-1.002	0.317
model_and_mitigation_deepseek/deepseek-chat-v3-0324__reranking	Mixed	White	0.792	[0.606, 1.036]	-1.704	0.088
model_and_mitigation_deepseek/deepseek-chat-v3-0324__reranking	Other	White	0.809	[0.530, 1.235]	-0.983	0.326
model_and_mitigation_openai/chatgpt-4o-latest__none	Asian	White	0.914	[0.713, 1.173]	-0.706	0.480
model_and_mitigation_openai/chatgpt-4o-latest__none	Black	White	0.877	[0.695, 1.107]	-1.106	0.269
model_and_mitigation_openai/chatgpt-4o-latest__none	Mixed	White	0.843	[0.655, 1.084]	-1.330	0.183
model_and_mitigation_openai/chatgpt-4o-latest__none	Other	White	0.768	[0.510, 1.156]	-1.264	0.206
model_and_mitigation_openai/chatgpt-4o-latest__prompting	Asian	White	0.822	[0.634, 1.065]	-1.485	0.138
model_and_mitigation_openai/chatgpt-4o-latest__prompting	Black	White	0.765	[0.598, 0.977]	-2.144	0.032
model_and_mitigation_openai/chatgpt-4o-latest__prompting	Mixed	White	0.855	[0.666, 1.098]	-1.228	0.219
model_and_mitigation_openai/chatgpt-4o-latest__prompting	Other	White	0.433	[0.253, 0.740]	-3.064	0.002
model_and_mitigation_openai/chatgpt-4o-latest__reranking	Asian	White	0.742	[0.553, 0.996]	-1.988	0.047
model_and_mitigation_openai/chatgpt-4o-latest__reranking	Black	White	0.915	[0.712, 1.176]	-0.695	0.487
model_and_mitigation_openai/chatgpt-4o-latest__reranking	Mixed	White	1.023	[0.795, 1.317]	0.178	0.859
model_and_mitigation_openai/chatgpt-4o-latest__reranking	Other	White	0.568	[0.344, 0.938]	-2.212	0.027

Table S342: Multinomial logistic regression predicting perceived writer political party in Study 3 (“preferred” subset, N=19,102). Outcome: “writer_politicalParty”. Predictor reference category: “writer”. Outcome reference level: “Labour”. OR is odds ratio.

Term	Target level	Reference level	OR	95% CI (OR)	statistic	p
model_and_mitigation_anthropic/claude-sonnet-4_none	Conservative	Labour	1.395	[1.165, 1.670]	3.626	2.883e-4
model_and_mitigation_anthropic/claude-sonnet-4_none	Did not vote	Labour	0.415	[0.314, 0.548]	-6.183	6.309e-10
model_and_mitigation_anthropic/claude-sonnet-4_none	Greens	Labour	1.345	[1.067, 1.695]	2.508	0.012
model_and_mitigation_anthropic/claude-sonnet-4_none	Liberal Democrats	Labour	1.228	[0.995, 1.515]	1.911	0.056
model_and_mitigation_anthropic/claude-sonnet-4_none	Not eligible to vote	Labour	0.165	[0.067, 0.403]	-3.954	7.688e-5
model_and_mitigation_anthropic/claude-sonnet-4_none	Other	Labour	1.219	[0.834, 1.783]	1.023	0.306
model_and_mitigation_anthropic/claude-sonnet-4_none	Reform UK	Labour	0.652	[0.496, 0.858]	-3.060	0.002
model_and_mitigation_anthropic/claude-sonnet-4_none	Scottish National Party	Labour	1.705	[1.013, 2.869]	2.008	0.045
model_and_mitigation_anthropic/claude-sonnet-4_prompting	Conservative	Labour	1.547	[1.287, 1.860]	4.647	3.362e-6
model_and_mitigation_anthropic/claude-sonnet-4_prompting	Did not vote	Labour	0.330	[0.240, 0.453]	-6.842	7.796e-12
model_and_mitigation_anthropic/claude-sonnet-4_prompting	Greens	Labour	1.387	[1.093, 1.760]	2.689	0.007
model_and_mitigation_anthropic/claude-sonnet-4_prompting	Liberal Democrats	Labour	1.421	[1.151, 1.754]	3.273	0.001
model_and_mitigation_anthropic/claude-sonnet-4_prompting	Not eligible to vote	Labour	0.334	[0.170, 0.657]	-3.177	0.001
model_and_mitigation_anthropic/claude-sonnet-4_prompting	Other	Labour	1.303	[0.882, 1.926]	1.330	0.184
model_and_mitigation_anthropic/claude-sonnet-4_prompting	Reform UK	Labour	0.629	[0.470, 0.843]	-3.110	0.002
model_and_mitigation_anthropic/claude-sonnet-4_prompting	Scottish National Party	Labour	1.377	[0.759, 2.497]	1.052	0.293
model_and_mitigation_anthropic/claude-sonnet-4_reranking	Conservative	Labour	1.665	[1.403, 1.976]	5.834	5.408e-9
model_and_mitigation_anthropic/claude-sonnet-4_reranking	Did not vote	Labour	0.396	[0.300, 0.522]	-6.568	5.112e-11
model_and_mitigation_anthropic/claude-sonnet-4_reranking	Greens	Labour	1.375	[1.099, 1.720]	2.789	0.005
model_and_mitigation_anthropic/claude-sonnet-4_reranking	Liberal Democrats	Labour	1.313	[1.073, 1.607]	2.642	0.008
model_and_mitigation_anthropic/claude-sonnet-4_reranking	Not eligible to vote	Labour	0.190	[0.084, 0.430]	-3.980	6.894e-5
model_and_mitigation_anthropic/claude-sonnet-4_reranking	Other	Labour	1.334	[0.926, 1.922]	1.549	0.121
model_and_mitigation_anthropic/claude-sonnet-4_reranking	Reform UK	Labour	0.538	[0.402, 0.719]	-4.183	2.883e-5
model_and_mitigation_anthropic/claude-sonnet-4_reranking	Scottish National Party	Labour	1.457	[0.845, 2.513]	1.353	0.176
model_and_mitigation_deepseek/deepseek-chat-v3-0324_none	Conservative	Labour	1.487	[1.249, 1.770]	4.458	8.256e-6
model_and_mitigation_deepseek/deepseek-chat-v3-0324_none	Did not vote	Labour	0.327	[0.242, 0.442]	-7.255	4.005e-13
model_and_mitigation_deepseek/deepseek-chat-v3-0324_none	Greens	Labour	1.323	[1.055, 1.659]	2.427	0.015
model_and_mitigation_deepseek/deepseek-chat-v3-0324_none	Liberal Democrats	Labour	1.178	[0.956, 1.450]	1.540	0.123
model_and_mitigation_deepseek/deepseek-chat-v3-0324_none	Not eligible to vote	Labour	0.202	[0.089, 0.458]	-3.829	1.287e-4
model_and_mitigation_deepseek/deepseek-chat-v3-0324_none	Other	Labour	1.115	[0.750, 1.658]	0.539	0.590
model_and_mitigation_deepseek/deepseek-chat-v3-0324_none	Reform UK	Labour	0.545	[0.409, 0.727]	-4.127	3.677e-5
model_and_mitigation_deepseek/deepseek-chat-v3-0324_none	Scottish National Party	Labour	1.952	[1.197, 3.184]	2.679	0.007
model_and_mitigation_deepseek/deepseek-chat-v3-0324_prompting	Conservative	Labour	1.620	[1.364, 1.923]	5.506	3.677e-8
model_and_mitigation_deepseek/deepseek-chat-v3-0324_prompting	Did not vote	Labour	0.347	[0.258, 0.466]	-7.027	2.104e-12
model_and_mitigation_deepseek/deepseek-chat-v3-0324_prompting	Greens	Labour	1.214	[0.960, 1.535]	1.617	0.106
model_and_mitigation_deepseek/deepseek-chat-v3-0324_prompting	Liberal Democrats	Labour	1.290	[1.054, 1.579]	2.466	0.014
model_and_mitigation_deepseek/deepseek-chat-v3-0324_prompting	Not eligible to vote	Labour	0.101	[0.032, 0.317]	-3.926	8.642e-5
model_and_mitigation_deepseek/deepseek-chat-v3-0324_prompting	Other	Labour	0.880	[0.570, 1.358]	-0.579	0.563
model_and_mitigation_deepseek/deepseek-chat-v3-0324_prompting	Reform UK	Labour	0.639	[0.487, 0.839]	-3.229	0.001
model_and_mitigation_deepseek/deepseek-chat-v3-0324_prompting	Scottish National Party	Labour	1.131	[0.612, 2.089]	0.393	0.695
model_and_mitigation_deepseek/deepseek-chat-v3-0324_reranking	Conservative	Labour	1.438	[1.197, 1.728]	3.882	1.037e-4
model_and_mitigation_deepseek/deepseek-chat-v3-0324_reranking	Did not vote	Labour	0.427	[0.320, 0.568]	-5.832	5.486e-9
model_and_mitigation_deepseek/deepseek-chat-v3-0324_reranking	Greens	Labour	1.311	[1.033, 1.664]	2.229	0.026
model_and_mitigation_deepseek/deepseek-chat-v3-0324_reranking	Liberal Democrats	Labour	1.077	[0.863, 1.345]	0.657	0.511
model_and_mitigation_deepseek/deepseek-chat-v3-0324_reranking	Not eligible to vote	Labour	0.332	[0.169, 0.653]	-3.197	0.001
model_and_mitigation_deepseek/deepseek-chat-v3-0324_reranking	Other	Labour	0.877	[0.556, 1.385]	-0.561	0.574
model_and_mitigation_deepseek/deepseek-chat-v3-0324_reranking	Reform UK	Labour	0.726	[0.554, 0.952]	-2.312	0.021
model_and_mitigation_deepseek/deepseek-chat-v3-0324_reranking	Scottish National Party	Labour	1.141	[0.603, 2.160]	0.405	0.685
model_and_mitigation_openai/chatgpt-4o-latest_none	Conservative	Labour	1.755	[1.484, 2.075]	6.582	4.647e-11
model_and_mitigation_openai/chatgpt-4o-latest_none	Did not vote	Labour	0.331	[0.246, 0.447]	-7.234	4.678e-13
model_and_mitigation_openai/chatgpt-4o-latest_none	Greens	Labour	1.023	[0.800, 1.307]	0.178	0.859
model_and_mitigation_openai/chatgpt-4o-latest_none	Liberal Democrats	Labour	1.175	[0.956, 1.444]	1.529	0.126
model_and_mitigation_openai/chatgpt-4o-latest_none	Not eligible to vote	Labour	0.099	[0.032, 0.311]	-3.959	7.518e-5
model_and_mitigation_openai/chatgpt-4o-latest_none	Other	Labour	0.946	[0.625, 1.432]	-0.260	0.795
model_and_mitigation_openai/chatgpt-4o-latest_none	Reform UK	Labour	0.448	[0.329, 0.610]	-5.088	3.612e-7
model_and_mitigation_openai/chatgpt-4o-latest_none	Scottish National Party	Labour	1.181	[0.652, 2.137]	0.549	0.583
model_and_mitigation_openai/chatgpt-4o-latest_prompting	Conservative	Labour	1.555	[1.307, 1.849]	4.983	6.246e-7
model_and_mitigation_openai/chatgpt-4o-latest_prompting	Did not vote	Labour	0.374	[0.281, 0.499]	-6.704	2.026e-11
model_and_mitigation_openai/chatgpt-4o-latest_prompting	Greens	Labour	1.055	[0.828, 1.345]	0.434	0.664
model_and_mitigation_openai/chatgpt-4o-latest_prompting	Liberal Democrats	Labour	1.109	[0.899, 1.368]	0.969	0.332
model_and_mitigation_openai/chatgpt-4o-latest_prompting	Not eligible to vote	Labour	0.198	[0.088, 0.450]	-3.875	1.068e-4
model_and_mitigation_openai/chatgpt-4o-latest_prompting	Other	Labour	1.129	[0.763, 1.670]	0.606	0.544
model_and_mitigation_openai/chatgpt-4o-latest_prompting	Reform UK	Labour	0.495	[0.368, 0.667]	-4.624	3.759e-6
model_and_mitigation_openai/chatgpt-4o-latest_prompting	Scottish National Party	Labour	1.658	[0.986, 2.789]	1.907	0.057
model_and_mitigation_openai/chatgpt-4o-latest_reranking	Conservative	Labour	1.220	[1.008, 1.476]	2.040	0.041
model_and_mitigation_openai/chatgpt-4o-latest_reranking	Did not vote	Labour	0.351	[0.258, 0.477]	-6.697	2.123e-11
model_and_mitigation_openai/chatgpt-4o-latest_reranking	Greens	Labour	1.256	[0.987, 1.597]	1.857	0.063
model_and_mitigation_openai/chatgpt-4o-latest_reranking	Liberal Democrats	Labour	1.156	[0.930, 1.436]	1.305	0.192
model_and_mitigation_openai/chatgpt-4o-latest_reranking	Not eligible to vote	Labour	0.329	[0.167, 0.646]	-3.224	0.001
model_and_mitigation_openai/chatgpt-4o-latest_reranking	Other	Labour	0.914	[0.588, 1.423]	-0.397	0.692
model_and_mitigation_openai/chatgpt-4o-latest_reranking	Reform UK	Labour	0.523	[0.386, 0.709]	-4.179	2.933e-5
model_and_mitigation_openai/chatgpt-4o-latest_reranking	Scottish National Party	Labour	1.234	[0.668, 2.282]	0.671	0.502

Table S343: Multinomial logistic regression predicting perceived writer political ideology in Study 3 (“preferred” subset, N=19,102). Outcome: “writer_politicalIdeology”. Predictor reference category: “writer”. Outcome reference level: “Centrist”. OR is odds ratio.

Term	Target level	Reference level	OR	95% CI (OR)	statistic	p
model_and_mitigation_anthropic/claude-sonnet-4__none	Moderately Left-Wing	Centrist	1.102	[0.930, 1.305]	1.125	0.261
model_and_mitigation_anthropic/claude-sonnet-4__none	Moderately Right-Wing	Centrist	1.173	[0.976, 1.411]	1.700	0.089
model_and_mitigation_anthropic/claude-sonnet-4__none	Very Left-Wing	Centrist	1.745	[1.426, 2.135]	5.407	6.418e-8
model_and_mitigation_anthropic/claude-sonnet-4__none	Very Right-Wing	Centrist	1.207	[0.886, 1.646]	1.193	0.233
model_and_mitigation_anthropic/claude-sonnet-4__prompting	Moderately Left-Wing	Centrist	1.102	[0.931, 1.304]	1.130	0.258
model_and_mitigation_anthropic/claude-sonnet-4__prompting	Moderately Right-Wing	Centrist	1.073	[0.889, 1.295]	0.732	0.464
model_and_mitigation_anthropic/claude-sonnet-4__prompting	Very Left-Wing	Centrist	1.280	[1.029, 1.594]	2.213	0.027
model_and_mitigation_anthropic/claude-sonnet-4__prompting	Very Right-Wing	Centrist	0.979	[0.701, 1.368]	−0.124	0.901
model_and_mitigation_anthropic/claude-sonnet-4__reranking	Moderately Left-Wing	Centrist	1.034	[0.884, 1.211]	0.422	0.673
model_and_mitigation_anthropic/claude-sonnet-4__reranking	Moderately Right-Wing	Centrist	1.141	[0.963, 1.353]	1.520	0.128
model_and_mitigation_anthropic/claude-sonnet-4__reranking	Very Left-Wing	Centrist	1.023	[0.824, 1.270]	0.203	0.839
model_and_mitigation_anthropic/claude-sonnet-4__reranking	Very Right-Wing	Centrist	0.774	[0.552, 1.085]	−1.487	0.137
model_and_mitigation_deepseek/deepseek-chat-v3-0324__none	Moderately Left-Wing	Centrist	1.026	[0.872, 1.206]	0.306	0.760
model_and_mitigation_deepseek/deepseek-chat-v3-0324__none	Moderately Right-Wing	Centrist	1.001	[0.835, 1.201]	0.015	0.988
model_and_mitigation_deepseek/deepseek-chat-v3-0324__none	Very Left-Wing	Centrist	1.255	[1.019, 1.547]	2.137	0.033
model_and_mitigation_deepseek/deepseek-chat-v3-0324__none	Very Right-Wing	Centrist	0.857	[0.615, 1.194]	−0.913	0.361
model_and_mitigation_deepseek/deepseek-chat-v3-0324__prompting	Moderately Left-Wing	Centrist	0.863	[0.733, 1.015]	−1.778	0.075
model_and_mitigation_deepseek/deepseek-chat-v3-0324__prompting	Moderately Right-Wing	Centrist	1.035	[0.870, 1.232]	0.389	0.697
model_and_mitigation_deepseek/deepseek-chat-v3-0324__prompting	Very Left-Wing	Centrist	1.073	[0.867, 1.327]	0.647	0.518
model_and_mitigation_deepseek/deepseek-chat-v3-0324__prompting	Very Right-Wing	Centrist	0.965	[0.710, 1.313]	−0.225	0.822
model_and_mitigation_deepseek/deepseek-chat-v3-0324__reranking	Moderately Left-Wing	Centrist	1.074	[0.905, 1.273]	0.816	0.415
model_and_mitigation_deepseek/deepseek-chat-v3-0324__reranking	Moderately Right-Wing	Centrist	1.160	[0.963, 1.397]	1.565	0.118
model_and_mitigation_deepseek/deepseek-chat-v3-0324__reranking	Very Left-Wing	Centrist	1.254	[1.004, 1.568]	1.992	0.046
model_and_mitigation_deepseek/deepseek-chat-v3-0324__reranking	Very Right-Wing	Centrist	0.844	[0.590, 1.207]	−0.929	0.353
model_and_mitigation_openai/chatgpt-4o-latest__none	Moderately Left-Wing	Centrist	0.904	[0.769, 1.063]	−1.221	0.222
model_and_mitigation_openai/chatgpt-4o-latest__none	Moderately Right-Wing	Centrist	1.138	[0.960, 1.349]	1.485	0.138
model_and_mitigation_openai/chatgpt-4o-latest__none	Very Left-Wing	Centrist	0.975	[0.783, 1.214]	−0.228	0.819
model_and_mitigation_openai/chatgpt-4o-latest__none	Very Right-Wing	Centrist	0.552	[0.375, 0.812]	−3.011	0.003
model_and_mitigation_openai/chatgpt-4o-latest__prompting	Moderately Left-Wing	Centrist	1.060	[0.901, 1.246]	0.704	0.482
model_and_mitigation_openai/chatgpt-4o-latest__prompting	Moderately Right-Wing	Centrist	1.191	[1.000, 1.418]	1.962	0.050
model_and_mitigation_openai/chatgpt-4o-latest__prompting	Very Left-Wing	Centrist	0.906	[0.716, 1.146]	−0.825	0.409
model_and_mitigation_openai/chatgpt-4o-latest__prompting	Very Right-Wing	Centrist	0.607	[0.412, 0.895]	−2.518	0.012
model_and_mitigation_openai/chatgpt-4o-latest__reranking	Moderately Left-Wing	Centrist	0.995	[0.839, 1.180]	−0.059	0.953
model_and_mitigation_openai/chatgpt-4o-latest__reranking	Moderately Right-Wing	Centrist	0.848	[0.695, 1.035]	−1.623	0.105
model_and_mitigation_openai/chatgpt-4o-latest__reranking	Very Left-Wing	Centrist	1.269	[1.019, 1.581]	2.127	0.033
model_and_mitigation_openai/chatgpt-4o-latest__reranking	Very Right-Wing	Centrist	0.616	[0.412, 0.920]	−2.366	0.018

8.6 Statistical Analysis: Writer Preference

Figures S63, S64, and S65 show writers' edit and preference rates across mitigation conditions with 95% bootstrapped confidence intervals.

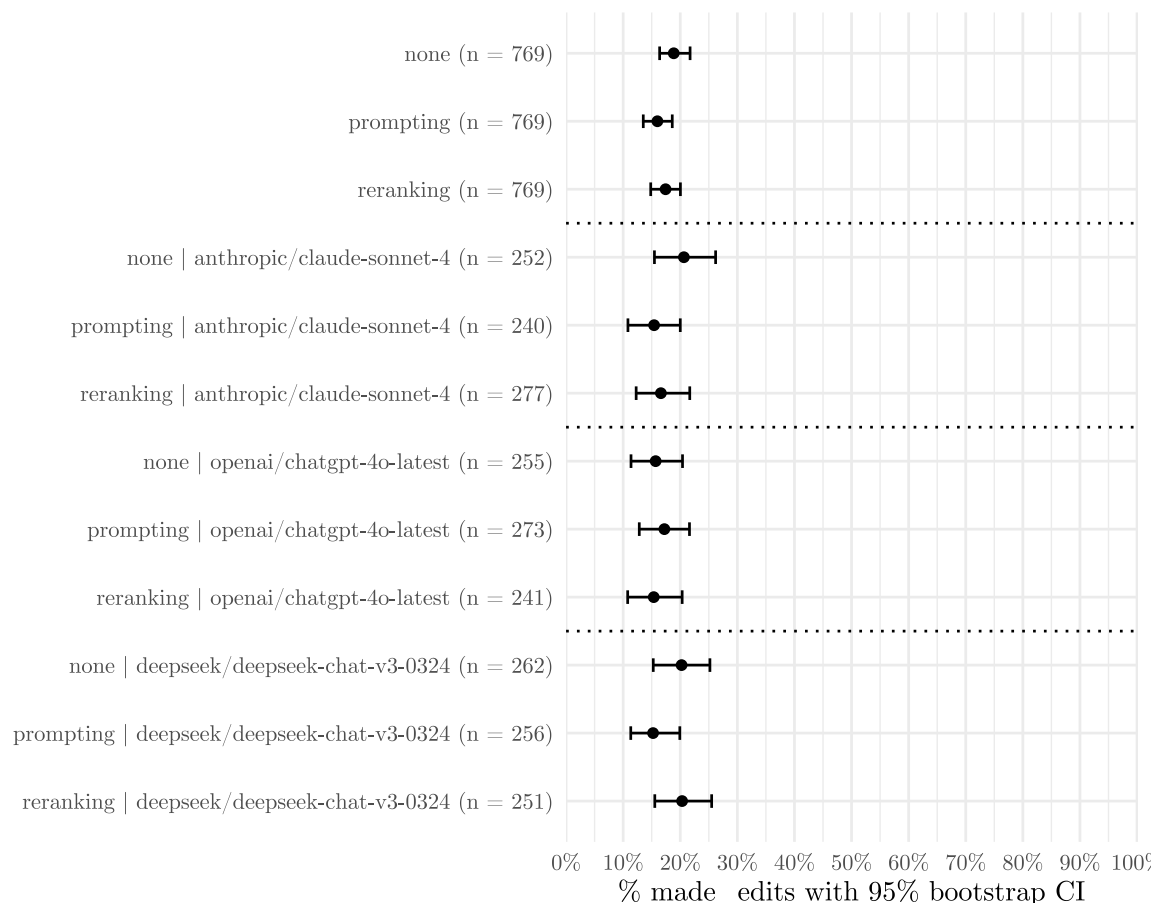


Figure S63: AI paragraph edit rates in Study 3 (N=769 writers, each writing about 3 propositions) across mitigation conditions.

As in Study 2, all writers who strictly preferred one paragraph over the other explained their preference in free text (see §2.6). This gave 1,255 free-text reasons for strict preferences for the AI-assisted paragraph and 423 for strict preferences for the human-written paragraph. We coded these responses using the same codebook and LLM-based coding procedure as in Study 1 (§4.3, Table S34). Responses received 2.1 codes on average, and 69.6% of responses received more than one code. Table S344 and Figure S66 show the share of responses mentioning each reason, and Table S345 breaks this down by mitigation condition.

Reasons were very similar to Study 2. Writers who strictly preferred the AI-written paragraph most often said that that this was because it was better written (43.1%), more informative (29.2%), clearer (27.8%) or better reflected their opinion (26.4%), followed by persuasiveness (13.8%), structure (13.4%), formality (12.7%) and concision (12.4%). Writers who strictly preferred their own paragraph most often said that this was because it sounded more human or more like their own voice (31.2%) or better reflected their opinion (27.7%), followed by informativeness (19.6%), clarity (17.7%), concision (13.0%) and a general dislike of AI (12.3%).

Reasons for preferring the AI-written paragraph were largely stable across mitigation conditions. Among writers who preferred their own paragraph, fewer said that their own paragraph better reflected their opinion in the reranking condition (19.3%) than without mitigation (33.1%) or with prompting (33.1%), and more said that their own paragraph was clearer (26.5%, compared to 14.9%

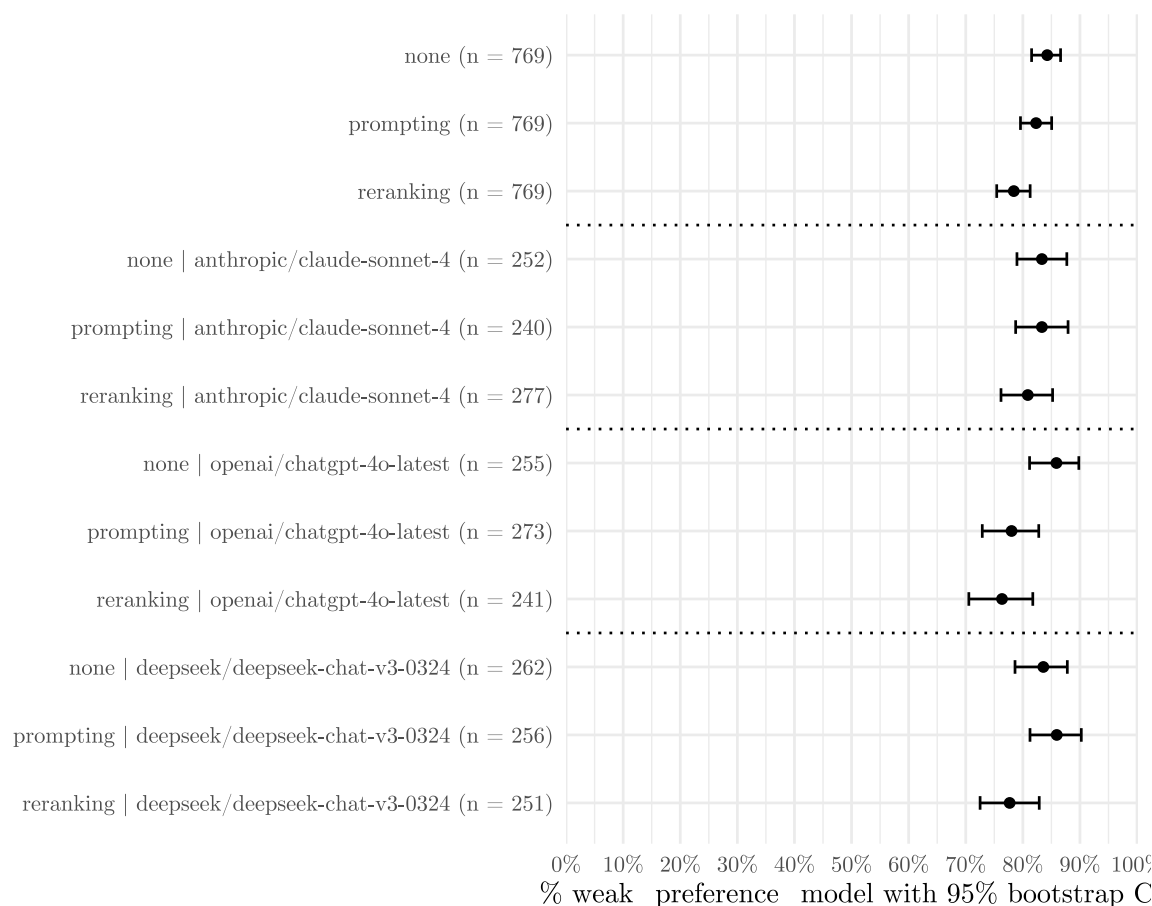


Figure S64: Weak preference for AI-assisted paragraphs over own writing in Study 3 (N=769 writers, each writing about 3 propositions) across mitigation conditions. AI-written paragraphs include any edits made.

without mitigation and 9.6% with prompting). A general dislike of AI was mentioned less often with prompting (11.8%) and reranking (10.2%) than without mitigation (15.7%). Note that these comparisons are based on small numbers of writers per condition (121 to 166 responses per condition for writers who preferred their own paragraph).

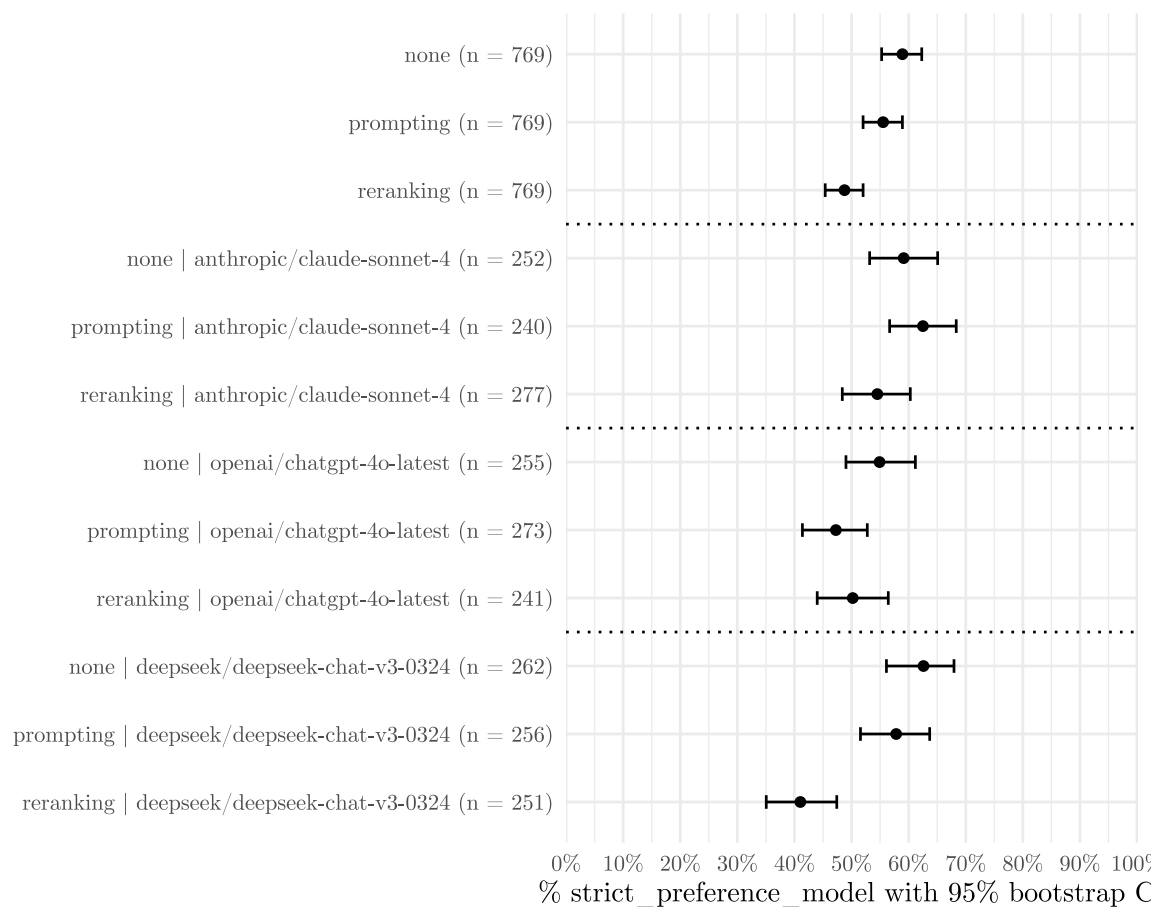


Figure S65: Strict preference for AI-assisted paragraphs over own writing in Study 3 (N=769 writers, each writing about 3 propositions) across mitigation conditions. AI-written paragraphs include any edits made.

Table S344: Free-text reasons for strict paragraph preference in Study 3 (n=1,255 strict preferences for the AI-assisted paragraph, n=423 for the human-written paragraph). Responses can receive multiple codes, so percentages do not sum to 100%. See Table S34 for code definitions.

Code	AI-written preferred		Human-written preferred	
	n	%	n	%
reflects_opinion	331	■ 26.4%	117	■ 27.7%
extremity_more	12	1.0%	17	4.0%
extremity_less	10	0.8%	6	1.4%
confidence_more	37	2.9%	7	1.7%
confidence_less	1	0.1%	4	0.9%
knowledge_more	93	■ 7.4%	5	1.2%
knowledge_less	0	0.0%	6	1.4%
importance_more	13	1.0%	5	1.2%
relevance_more	29	2.3%	21	5.0%
clarity_more	349	■ 27.8%	75	■ 17.7%
formality_more	159	■ 12.7%	3	0.7%
formality_less	3	0.2%	36	■ 8.5%
informativeness_more	366	■ 29.2%	83	■ 19.6%
originality_more	16	1.3%	9	2.1%
friendliness_more	12	1.0%	6	1.4%
optimism_more	3	0.2%	1	0.2%
community_more	4	0.3%	4	0.9%
openness_more	32	2.5%	13	3.1%
emotionality_more	20	1.6%	33	■ 7.8%
emotionality_less	28	2.2%	5	1.2%
language_quality	541	■ 43.1%	36	■ 8.5%
concision	156	■ 12.4%	55	■ 13.0%
structure	168	■ 13.4%	9	2.1%
persuasiveness	173	■ 13.8%	13	3.1%
authenticity	17	1.4%	132	■ 31.2%
ai_aversion	2	0.2%	52	■ 12.3%
other	72	■ 5.7%	32	■ 7.6%

Table S345: Free-text reasons for strict paragraph preference by mitigation condition in Study 3. Cells show the percentage of responses mentioning each reason, with the number of responses per condition in the column header. See Table S34 for code definitions.

Code	AI-written preferred			Human-written preferred		
	None (453)	Prompting (427)	Reranking (375)	None (121)	Prompting (136)	Reranking (166)
reflects_opinion	27.8	25.8	25.3	33.1	33.1	19.3
extremity_more	0.9	0.7	1.3	3.3	5.1	3.6
extremity_less	0.7	1.2	0.5	3.3	0.7	0.6
confidence_more	3.1	2.8	2.9	0.8	2.2	1.8
confidence_less	0.2	0.0	0.0	0.0	2.2	0.6
knowledge_more	7.3	7.3	7.7	0.0	0.7	2.4
knowledge_less	0.0	0.0	0.0	0.0	2.2	1.8
importance_more	1.1	0.9	1.1	1.7	1.5	0.6
relevance_more	3.1	1.4	2.4	5.0	5.1	4.8
clarity_more	25.6	30.9	26.9	14.9	9.6	26.5
formality_more	13.7	11.0	13.3	0.0	2.2	0.0
formality_less	0.2	0.2	0.3	10.7	6.6	8.4
informativeness_more	32.2	26.9	28.0	16.5	19.9	21.7
originality_more	1.8	0.7	1.3	1.7	2.2	2.4
friendliness_more	0.9	1.2	0.8	1.7	0.7	1.8
optimism_more	0.4	0.2	0.0	0.0	0.7	0.0
community_more	0.2	0.2	0.5	0.0	1.5	1.2
openness_more	1.5	4.0	2.1	3.3	3.7	2.4
emotionality_more	1.5	0.9	2.4	6.6	9.6	7.2
emotionality_less	2.2	3.0	1.3	0.8	0.7	1.8
language_quality	41.7	46.4	41.1	5.0	11.0	9.0
concision	9.3	11.0	17.9	11.6	14.0	13.3
structure	11.9	14.8	13.6	3.3	2.2	1.2
persuasiveness	15.5	13.6	12.0	4.1	2.2	3.0
authenticity	1.1	0.7	2.4	30.6	33.8	29.5
ai_aversion	0.4	0.0	0.0	15.7	11.8	10.2
other	4.4	6.1	6.9	9.1	7.4	6.6

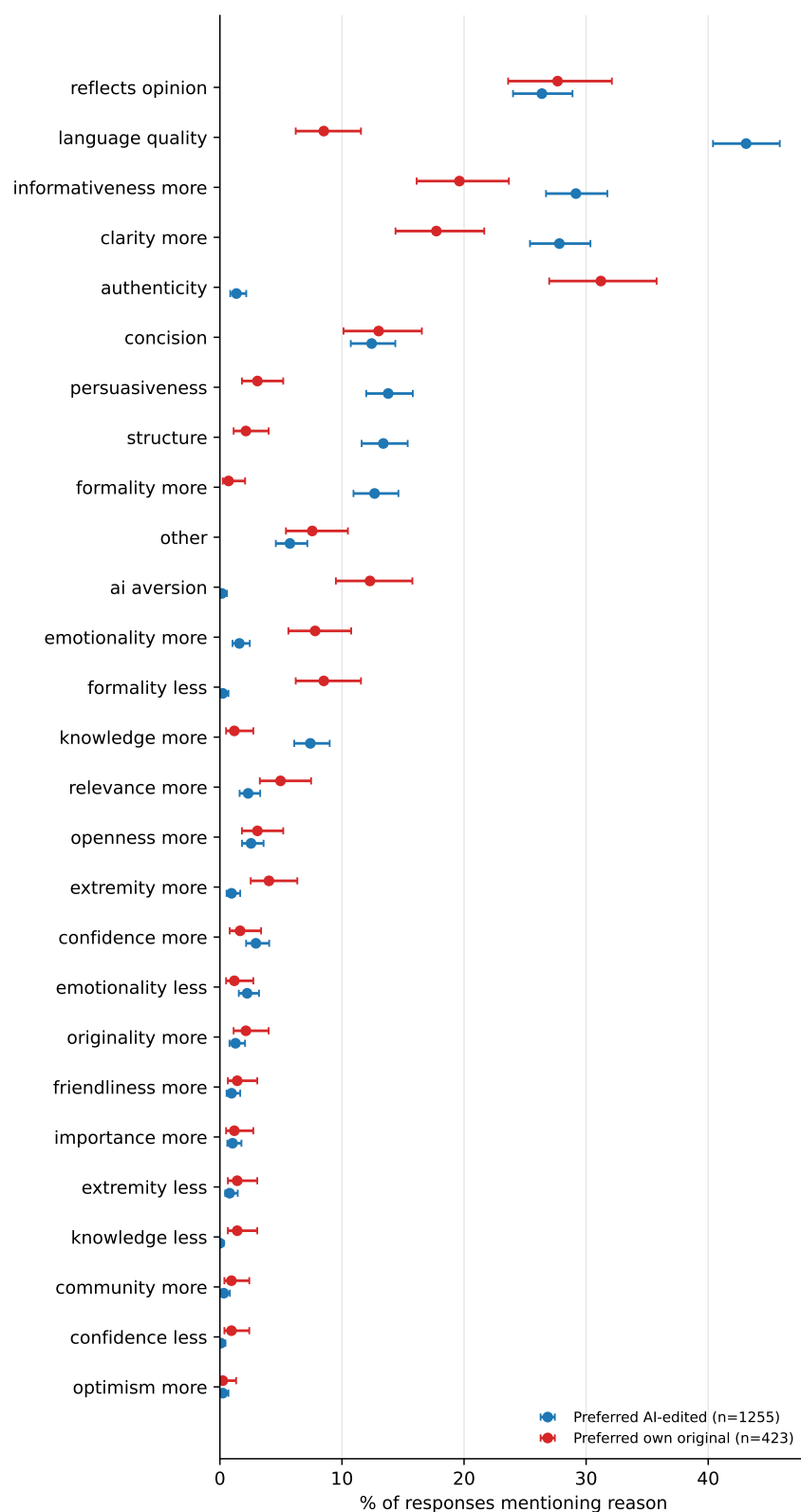


Figure S66: Free-text reasons for strict paragraph preference in Study 3 (n=1,255 strict preferences for the AI-assisted paragraph, n=423 for the human-written paragraph). Points show the share of responses mentioning each reason, with 95% Wilson CIs. See Table S34 for code definitions.

Next, we fit logistic regression models with writer random effects to measure the effect of mitigations on writer preference for AI writing. Matching our main analysis, we used three binary outcome variables observed for each writer-proposition pair: whether the writer made any edits to the AI-written paragraph (“made_edits”), whether the writer weakly preferred the AI-written paragraph (“weak_preference_model”), and whether the writer strictly preferred the AI-written paragraph (“strict_preference_model”). Regression models were specified as follows:

$$\text{made_edits} \sim \text{mitigation_condition} + (1 \mid \text{writer_id})$$

$$\text{weak_preference_model} \sim \text{mitigation_condition} + (1 \mid \text{writer_id})$$

$$\text{strict_preference_model} \sim \text{mitigation_condition} + (1 \mid \text{writer_id})$$

Table S346 reports results for the made_edits outcome. Neither mitigation condition has a significant effect compared to the no-mitigation control. Prompting shows a marginal, non-significant reduction in the likelihood that writers edited the AI paragraph (OR=0.780, p=.099), while Reranking has no significant effect on editing behaviour (OR=0.886, p=.415).

Table S346: Logistic regression predicting writer editing of AI writing in Study 3 (N=769 writers, each writing about 3 propositions). Binary outcome: “made_edits”. Reference category: “mitigation_condition_none”

Term	Estimate	SE	OR [95% CI]	z	p
mitigation_condition_prompting	−0.248	0.151	0.780 [0.581, 1.048]	−1.648	.099
mitigation_condition_reranking	−0.121	0.148	0.886 [0.663, 1.185]	−0.814	.415

Table S347 reports results for the weak_preference_model outcome. Only Reranking has a significant effect compared to the no-mitigation control, reducing weak preference for AI-written paragraphs (OR=0.598, $p<.001$). Prompting does not significantly affect weak preference for AI writing (OR=0.832, $p=.241$).

Table S347: Logistic regression predicting weak writer preference for AI writing in Study 3 (N=769 writers, each writing about 3 propositions). Binary outcome: “weak_preference_model”. Reference category: “mitigation_condition_none”

Term	Estimate	SE	OR [95% CI]	z	p
mitigation_condition_prompting	−0.184	0.157	0.832 [0.612, 1.132]	−1.172	.241
mitigation_condition_reranking	−0.514	0.153	0.598 [0.443, 0.807]	−3.358	< .001

Table S348 reports results for the strict_preference_model outcome. Only Reranking has a significant effect compared to the no-mitigation control, reducing strict preference for AI-written paragraphs (OR=0.537, $p<.001$). Prompting does not significantly affect strict preference for AI writing, though the effect approaches significance (OR=0.811, $p=.100$).

Table S348: Logistic regression predicting strict writer preference for AI writing in Study 3 (N=769 writers, each writing about 3 propositions). Binary outcome: “strict_preference_model”. Reference category: “mitigation_condition_none”

Term	Estimate	SE	OR [95% CI]	z	p
mitigation_condition_prompting	−0.209	0.127	0.811 [0.632, 1.041]	−1.646	.100
mitigation_condition_reranking	−0.622	0.128	0.537 [0.417, 0.690]	−4.846	< .001

Finally, we fit logistic regression models with writer random effects to measure the effect of mitigations on writer preference for AI writing while adjusting for the AI model used to generate the paragraph:

$$\text{made_edits} \sim \text{mitigation_condition} + \text{model} + (1 \mid \text{writer_id})$$

$$\text{weak_preference_model} \sim \text{mitigation_condition} + \text{model} + (1 \mid \text{writer_id})$$

$$\text{strict_preference_model} \sim \text{mitigation_condition} + \text{model} + (1 \mid \text{writer_id})$$

Table S349 reports results for the `made_edits` outcome. After adjusting for model, neither mitigation condition has a significant effect relative to the no-mitigation control. Prompting shows a non-significant reduction in the likelihood that writers edited the AI paragraph (OR = 0.785, $p = .109$), while Reranking remains non-significant (OR = 0.887, $p = .418$). There are also no significant differences between Claude and either DeepSeek (OR = 1.093, $p = .552$) or ChatGPT-4o (OR = 0.888, $p = .434$) on this outcome.

Table S349: Logistic regression predicting writer editing of AI writing in Study 3 (N = 769 writers, each writing about 3 propositions). Binary outcome: “made_edits”. Reference categories: “mitigation_condition_none” and Claude Sonnet 4 for model.

Term	Estimate	SE	OR [95% CI]	z	p
mitigation_condition_prompting	−0.242	0.151	0.785 [0.584, 1.056]	−1.602	.109
mitigation_condition_reranking	−0.120	0.149	0.887 [0.663, 1.186]	−0.811	.418
DeepSeek (vs Claude)	0.088	0.149	1.093 [0.816, 1.462]	0.595	.552
ChatGPT-4o (vs Claude)	−0.119	0.152	0.888 [0.659, 1.197]	−0.782	.434

Table S350 reports results for the `weak_preference_model` outcome. Only Reranking has a significant effect compared to the no-mitigation control, reducing weak preference for AI-written paragraphs (OR = 0.597, $p < .001$). Prompting does not significantly affect weak preference for AI writing (OR = 0.841, $p = .269$). Neither DeepSeek nor ChatGPT-4o differs significantly from Claude on this outcome after controlling for mitigation condition.

Table S350: Logistic regression predicting weak writer preference for AI writing in Study 3 (N = 769 writers, each writing about 3 propositions). Binary outcome: “`weak_preference_model`”. Reference categories: “`mitigation_condition_none`” and Claude Sonnet 4 for model.

Term	Estimate	SE	OR [95% CI]	z	p
<code>mitigation_condition_prompting</code>	-0.174	0.157	0.841 [0.618, 1.143]	-1.106	.269
<code>mitigation_condition_reranking</code>	-0.515	0.153	0.597 [0.442, 0.807]	-3.362	< .001
DeepSeek (vs Claude)	-0.011	0.154	0.989 [0.731, 1.339]	-0.070	.944
ChatGPT-4o (vs Claude)	-0.215	0.152	0.806 [0.599, 1.086]	-1.419	.156

Table S351 reports results for the `strict_preference_model` outcome. Only Reranking has a significant effect compared to the no-mitigation control, reducing strict preference for AI-written paragraphs (OR = 0.527, $p < .001$). Prompting does not significantly affect strict preference for AI writing (OR = 0.817, $p = .115$). By contrast, model fixed effects are significant here: relative to Claude, both DeepSeek (OR = 0.743, $p = .020$) and ChatGPT-4o (OR = 0.604, $p < .001$) are associated with lower odds of strict writer preference for the AI-written paragraph.

Table S351: Logistic regression predicting strict writer preference for AI writing in Study 3 (N = 769 writers, each writing about 3 propositions). Binary outcome: “`strict_preference_model`”. Reference categories: “`mitigation_condition_none`” and Claude Sonnet 4 for model.

Term	Estimate	SE	OR [95% CI]	z	p
<code>mitigation_condition_prompting</code>	-0.202	0.128	0.817 [0.636, 1.050]	-1.576	.115
<code>mitigation_condition_reranking</code>	-0.641	0.129	0.527 [0.409, 0.679]	-4.956	< .001
DeepSeek (vs Claude)	-0.297	0.128	0.743 [0.578, 0.955]	-2.318	.020
ChatGPT-4o (vs Claude)	-0.504	0.129	0.604 [0.469, 0.777]	-3.918	< .001

9 Study 4 (Trust Study)

9.1 Paragraph Selection

Each reader in Study 4 was assigned four paragraphs drawn from a fixed pool constructed from the paragraphs collected in Study 1. To construct this pool, we began with all 1,501 paragraph pairs from Study 1 for which the writer’s input to the AI was a set of bullet points. Each pair consists of a human-written and an AI-assisted paragraph (including any writer edits) from the same writer on the same proposition. We restricted the pool to bullet-point pairs so that generation condition did not vary across the pool. Otherwise, the low- vs. high-distortion selection described below could systematically draw on different input formats, introducing generation condition as a confounder of the distortion contrast. We then removed the pairs for which writers strictly preferred their own writing to the AI-assisted paragraph.

From the remaining pairs, we selected equally many “low-distortion” and “high-distortion” pairs based on reader perceptions in Study 1. Specifically, for each pair we computed the average absolute distortion across the 20 scale rating attributes (§5.7), and selected the 200 pairs with the highest average and the 200 pairs with the lowest average. We additionally balanced the selected pairs across the three AI models that produced the AI-assisted paragraphs, selecting 66 or 67 paragraphs from each model within both the low- and high-distortion sets, so that the producing model did not confound the distortion contrast. This resulted in 800 paragraphs, with 200 in each of the four cells defined by authorship (human vs. AI) $\times$ distortion (low vs. high). Note that our goal was not to test paragraph-specific effects: the 200 paragraphs per cell were intended to provide sufficient diversity within each cell, not to support paragraph-level inference.

9.2 Reader Assignment

Study 4 used a **within-subjects** design. Each reader was assigned one paragraph from each of the four authorship $\times$ distortion cells, drawn at random from the corresponding pool subject to the constraint that no reader was ever assigned more than one paragraph per proposition. The presentation order of the four assigned paragraphs was independently randomised for each reader, so that presentation order was orthogonal to `paragraph_type` in expectation. All randomisation was performed at the reader level.

Of the 856 participants who opened the study, 804 completed it, and 802 remained after list-wise deletion of incomplete cases (Table S22). Readers who failed the Trust Game comprehension check (§2.12) were screened out before the game began and are not counted among these participants. The 802 analysable readers each contributed four observations, yielding 3,208 reader-paragraph observations with complete data on all variables used in the analyses below.

Our target sample size of 800 analysable readers was determined in advance and pre-registered (osf.io/38sg7). It was based on a simulation-based power analysis under conservative assumptions, primarily driven by the interaction test (whether the AI advantage scales with distortion), since the main effect of paragraph type is well powered at every sample size considered. We simulated data under the design described above and fitted the pre-registered models, treating the `paragraph_type` $\times$ `distortion_bin` interaction as the binding test. Assuming a small AI advantage in the low-distortion bin (8.0p vs. 7.5p average allocation) that widens in the high-distortion bin (8.5p vs. 7.0p), with a between-reader SD of 0.60 on the logit scale and beta precision $\phi = 3$, 800 readers yield 84.2% power for the interaction and effectively 100% power for the main effect at $\alpha = .05$ (for comparison, 600 readers yield 71.8% power for the interaction, and 1,000 readers raise it to 89.6%). 800 is the smallest grid point clearing the conventional 80% threshold on the binding test. The realised sample of 802 analysable readers meets this target.

9.3 Descriptive Results

Across all 3,208 observations, readers allocated an average of 8.72p (SD = 5.44) of their 20p endowment to paragraph authors. Table S352 reports mean allocations broken down by the four cells. Readers allocated more to the authors of AI-assisted paragraphs than to the authors of human-written paragraphs overall (9.26p vs. 8.18p, a raw difference of +1.08p). This raw AI advantage was concentrated in the high-distortion cells: when distortion was low, human and AI authors received similar allocations (9.03p vs. 9.34p, +0.31p), whereas when distortion was high, allocations to human

authors dropped markedly (to 7.33p) while allocations to AI authors held steady (9.18p), widening the raw gap to +1.85p. The regression analyses below formalise these patterns while accounting for reader-level heterogeneity.

Table S352: Mean Trust Game allocations by authorship and distortion in Study 4 (N = 802 readers, 3,208 reader-paragraph observations). Cells report the mean allocation in pence (0–20p) with standard deviation in parentheses, and n observations. The final column reports the raw AI – human difference in mean allocation.

Distortion	Human	AI	n per cell	AI – Human
Low distortion	9.03 (5.23)	9.34 (5.51)	802	+0.31
High distortion	7.33 (5.38)	9.18 (5.40)	802	+1.85
Overall	8.18 (5.38)	9.26 (5.45)	1,604	+1.08

9.4 Statistical Analysis

Our primary outcome was the reader’s allocation amount (0–20p) per paragraph, which serves as a behavioural measure of the interpersonal trust placed in the author. Statistical analysis followed the approach used for the 0–100 scale rating attributes in Study 1 (§5.7), applied to this outcome. For all models below, we fit **mixed-effects beta regression models** with a logit link. Because beta regression requires the response to lie on the open interval (0, 1), allocations were first rescaled from 0–20p to [0, 1] and then transformed via $y' = \frac{y(n-1)+0.5}{n}$, where n is the number of observations, to compress the boundary values 0 and 1 into the open interval. All models included reader random intercepts to account for individual differences in baseline allocations. We report effects as average marginal effects (AMEs) on the original 0–20p scale, alongside the model coefficients as odds ratios (ORs), each with 95% confidence intervals.

To test whether AI writing assistance affects the interpersonal trust that readers place in writers, we fit the following primary model:

$$\text{trust_allocation} \sim \text{paragraph_type} + (1 \mid \text{reader_id})$$

where `paragraph_type` is a binary predictor taking the values *human* (reference) and *AI*. The quantity of interest is the AME of *AI* vs. *human* authorship on trust allocations. Readers allocated significantly more to the authors of AI-assisted paragraphs than to the authors of human-written paragraphs, with an AME of +1.56p (95% CI [1.16, 1.95], $p < .001$; Table S353, Figure S67).

This effect was robust to two pre-registered secondary specifications (Table S354):

1) Controlling for `stance_distance`, the absolute distance between the reader’s stated agreement with the proposition and the paragraph’s perceived stance (using the average perceived paragraph stance from reader ratings in Study 1), left the AI advantage essentially unchanged (AME = +1.53p, [1.15, 1.91], $p < .001$). As expected, greater stance disagreement was itself associated with lower allocations (−0.062p per 1-point increase in stance distance, $p < .001$).

2) Controlling for `presentation_order` (the position of the paragraph in the reader’s randomised sequence) also left the AI advantage unchanged (AME = +1.56p, [1.16, 1.95], $p < .001$), with a small positive order effect (+0.21p per position, $p = .019$).

Next, to test whether the effect of AI writing assistance on trust scales with the magnitude of distortion, we fit an interaction model using the pre-registered binary distortion bin:

$$\text{trust_allocation} \sim \text{paragraph_type} \times \text{distortion_bin} + (1 \mid \text{reader_id})$$

where `distortion_bin` takes the values *low_distortion* (reference) and *high_distortion*, fixed at the paragraph-pair level (§9.1). The AI advantage was significantly larger in the high-distortion bin

Table S353: Beta regression predicting Trust Game allocation in Study 4 (N = 802 readers, 3,208 observations). Outcome: “trust_allocation_pence” (0–20p). Reference category: “paragraph_type_human”. OR is odds ratio. AME is average marginal effect, in pence.

Term	Estimate	OR [95% CI]	z	p	AME	95% CI (AME)
(Intercept)	−0.280	0.756 [0.696, 0.820]	−6.695	< .001	—	—
paragraph_type_ai	0.313	1.368 [1.264, 1.481]	7.742	< .001	+1.557	[1.165, 1.950]

Table S354: Robustness of the paragraph type effect to control variables in Study 4. Each block reports a separate beta regression adding one control to the primary model in Table S353. Outcome: “trust_allocation_pence” (0–20p). Reference category: “paragraph_type_human”. AME is average marginal effect, in pence (per 1-point increase for continuous controls).

Model / Term	AME	[95% CI]	p
<i>+ stance_distance</i>			
paragraph_type_ai	+1.529	[1.151, 1.906]	< .001
stance_distance	−0.062	[−0.071, −0.054]	< .001
<i>+ presentation_order</i>			
paragraph_type_ai	+1.556	[1.164, 1.948]	< .001
presentation_order	+0.209	[0.034, 0.384]	.019

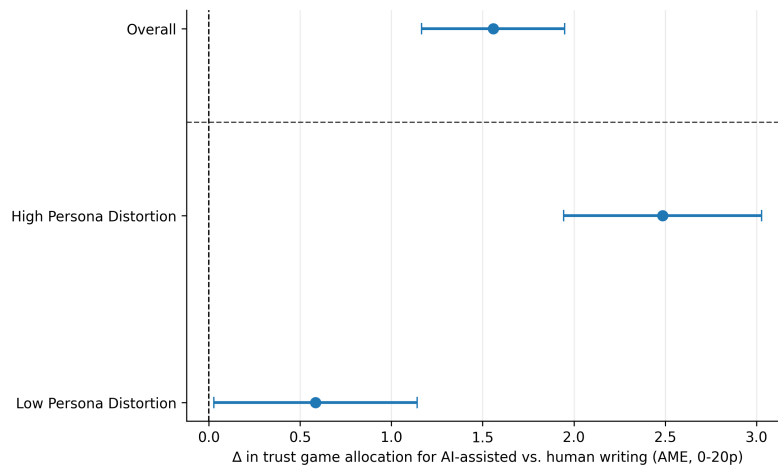


Figure S67: Average marginal effects (AMEs) of AI writing assistance on Trust Game allocation in Study 4. Points show the AME of AI vs. human authorship (in pence, on the 0–20p allocation scale) from mixed-effects beta regressions with reader random effects: the overall effect, and the conditional effects within the low- and high-distortion bins. Error bars show 95% confidence intervals.

than in the low-distortion bin (paragraph_type $\times$ distortion_bin interaction OR = 1.476, [1.261, 1.727], $z = 4.85$, $p < .001$; Table S355). Decomposing this interaction, the AI advantage was small in the low-distortion bin (AME = +0.58p, [0.03, 1.14], $p = .040$) but more than four times as large in the high-distortion bin (AME = +2.49p, [1.94, 3.03], $p < .001$; Figure S67).

As a secondary analysis, we re-fit the model with continuous distortion magnitude (the average absolute distortion across the 20 scale attributes) in place of the binary bin. This confirmed a clear and significant dose-response relationship. The AI advantage grew by +0.122p per 1-unit increase in distortion magnitude ([0.074, 0.170], $p < .001$), equivalent to +0.98p per 1 SD increase in distortion ([0.60, 1.37], $p < .001$).

Table S355: Beta regression predicting Trust Game allocation, with distortion as a moderator in Study 4 (N = 802 readers, 3,208 observations). Outcome: “trust_allocation_pence” (0–20p). Reference categories: “paragraph_type_human” and “distortion_bin_low_distortion”. OR is odds ratio.

Term	Estimate	SE	OR [95% CI]	z	p
(Intercept)	−0.074	0.051	0.929 [0.841, 1.026]	−1.461	.144
paragraph_type_ai	0.117	0.057	1.124 [1.006, 1.257]	2.057	.040
distortion_bin_high	−0.409	0.057	0.665 [0.594, 0.743]	−7.173	< .001
paragraph_type_ai $\times$ distortion_bin_high	0.389	0.080	1.476 [1.261, 1.727]	4.851	< .001

The analyses above treat distortion from AI writing assistance as a single aggregate magnitude. As an exploratory extension, we examined whether specific paragraph attributes are associated with trust allocations, and in which direction. For each of the 20 scale rating attributes from Study 1 (e.g. perceived writer knowledgeability, friendliness, formality), we used the average rating that paragraph received from readers in Study 1 as a predictor, fitting a separate model of the form of the primary model (i.e. including a paragraph type control) with that attribute’s rating added as a predictor. Because we fit 20 such models, we apply a Bonferroni correction across the 20 tests ($\alpha = .05/20 = .0025$).

Of the 20 attributes, 13 were significantly and positively associated with trust allocations after Bonferroni correction (Table S356, Figure S68). The strongest associations were for perceived writer friendliness (+0.082p per 1-point increase in the attribute rating), writer community-orientation (+0.081p), paragraph relevance (+0.073p), and paragraph informativeness (+0.060p). The negative-emotion attributes (perceived paragraph fear, disgust, and anger) showed small negative associations that were not significant after correction.

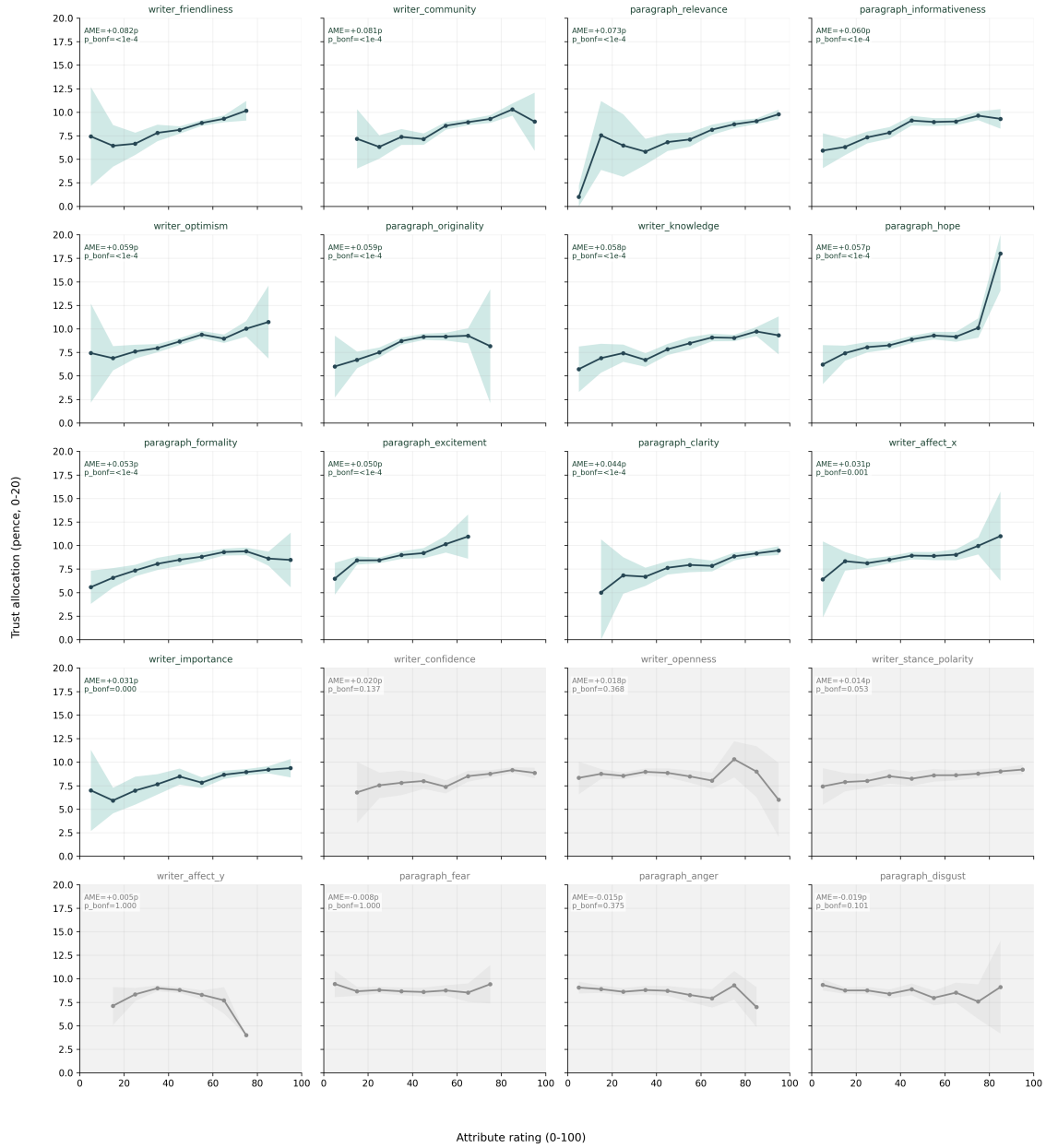


Figure S68: Relationship between paragraph attributes and Trust Game allocation in Study 4. Each panel plots mean trust allocation (0–20p) against binned values of one of the 20 scale rating attributes (0–100, the paragraph’s average rating in Study 1). Panels are ordered by effect size from beta regressions (Table S356). Each panel reports the per-attribute AME and Bonferroni-corrected p -value, and is highlighted when significant at $\alpha = .0025$.

Table S356: Per-attribute associations with Trust Game allocation in Study 4 (N = 802 readers, 3,208 observations). Each row reports a separate beta regression adding the paragraph’s average Study 1 rating on that attribute to the primary model. AME is the average marginal effect on allocation (in pence) per 1-point increase in the attribute rating. p_{Bonf} is the Bonferroni-corrected p -value across the 20 tests. Effects significant at $\alpha = .0025$ are marked ✓.

Attribute	AME	95% CI (AME)	p_{Bonf}
writer_friendliness	+0.082	[0.063, 0.102]	< .001 ✓
writer_community	+0.081	[0.065, 0.096]	< .001 ✓
paragraph_relevance	+0.073	[0.056, 0.090]	< .001 ✓
paragraph_informativeness	+0.060	[0.047, 0.073]	< .001 ✓
writer_optimism	+0.059	[0.044, 0.075]	< .001 ✓
paragraph_originality	+0.059	[0.040, 0.077]	< .001 ✓
writer_knowledge	+0.058	[0.045, 0.072]	< .001 ✓
paragraph_hope	+0.057	[0.043, 0.072]	< .001 ✓
paragraph_formality	+0.053	[0.039, 0.068]	< .001 ✓
paragraph_excitement	+0.050	[0.031, 0.069]	< .001 ✓
paragraph_clarity	+0.044	[0.029, 0.059]	< .001 ✓
writer_affect_x	+0.031	[0.017, 0.046]	< .001 ✓
writer_importance	+0.031	[0.017, 0.045]	< .001 ✓
writer_confidence	+0.020	[0.005, 0.034]	.137
writer_openness	+0.018	[0.003, 0.032]	.368
writer_stance_polarity	+0.014	[0.005, 0.024]	.053
writer_affect_y	+0.005	[−0.019, 0.030]	1.000
paragraph_fear	−0.008	[−0.023, 0.007]	1.000
paragraph_anger	−0.015	[−0.028, −0.003]	.375
paragraph_disgust	−0.019	[−0.033, −0.006]	.101

10 Study 5 (Persuasion Study)

10.1 Paragraph Selection

Each reader in Study 5 was assigned five paragraphs, one from each of five conditions: the four *treated* cells defined by authorship (human vs. AI) $\times$ distortion (low vs. high), plus a topically unrelated *control*. The treated pool was constructed exactly as in Study 4 (§9.1). The control condition used informative paragraphs on off-topic, non-political subjects (Table S357) that carry no stance on any of our political propositions. On a control cycle, the proposition rated and the control paragraph shown were drawn independently: the proposition was sampled uniformly at random from the propositions used in the treated conditions, and the control paragraph was sampled uniformly at random from a set of 10 topically distinct control paragraphs.

10.2 Reader Assignment

Study 5 used a **within-subjects** design. Each reader evaluated one paragraph from each of the five conditions ($\{\text{human, AI}\} \times \{\text{low, high}\}$, plus control), on five distinct propositions, with the presentation order randomised independently for each reader. The four treated paragraphs were drawn from four distinct pairs on four distinct propositions, and a reader never saw both members of the same pair, so that each assigned proposition was rated exactly once and no PRE rating was contaminated by prior exposure to the matched paragraph. For each paragraph, readers completed a three-item attitude battery about the corresponding proposition before (PRE) and after (POST) reading it (§2.13). All randomisation was performed at the reader level.

Of the 9,241 participants who opened the study, 8,010 completed it, and 7,996 remained after list-wise deletion of incomplete cases (Table S23). Readers who failed the attention check (§2.2) were screened out before the task began and are not counted among these participants. The 7,996 analysable readers each contributed five observations (four treated, one control), yielding 39,980 reader-paragraph observations, with complete data on all variables used in the analyses below.

Our target sample size of 8,000 analysable readers was determined in advance and pre-registered (osf.io/e48g2). As in Study 4, it was based on a simulation-based power analysis under conservative assumptions, primarily driven by the dose-response contrast (whether the AI advantage scales with distortion), which we treated as the binding test since the main effect of paragraph type is well powered at every sample size considered. We simulated data under the design described above and fitted the same crossed reader/proposition models used in our confirmatory analyses (§10.4). We assumed the off-topic control produces no net drift (0 points), a small AI advantage in the low-distortion bin (2.5 vs. 3.0 points of stance shift) that widens in the high-distortion bin (1.0 vs. 3.0 points) – i.e. an average AI-vs.-human effect of 1.25 points and a distortion interaction of 1.5 points on the 0–100 attitude scale – with a between-reader SD of 10 points and within-reader variability partitioned into a between-proposition SD of 6 points and a residual SD of approximately 22 points. Under these assumptions, 8,000 readers yield 87.8% power for the interaction and effectively 100% power for the main effect at $\alpha = .05$ (for comparison, 6,000 readers yield 74.4% power for the interaction, and 10,000 readers raise it to 92.4%). The larger target than in Study 4 reflects the smaller expected per-observation effect on the 0–100 stance-shift scale and the resulting need for more readers to power the interaction. 8,000 is the smallest grid point clearing the 80% threshold on the binding test with a comfortable margin. The realised sample of 7,996 analysable readers meets this target.

10.3 Descriptive Results

Our primary outcome was `stance_shift`, the signed change in a reader’s policy attitude from PRE to POST, oriented towards the position expressed in the paragraph. We first combined the three attitude-battery items into a single 0–100 `policy_attitude` score, as the mean of the *support* and *good consequences* items and the reverse-scored *bad idea* item, so that higher values indicate greater policy support. For treated cycles, we then oriented the PRE-to-POST change by the sign of the paragraph’s perceived stance relative to the scale midpoint of 50 (using the average perceived stance from reader ratings in Study 1), so that positive `stance_shift` indicates the reader moved *towards* the paragraph’s position. For control cycles, which carry no stance on the rated proposition, the

Topic	Control Paragraph
watercycle	The water cycle describes how water moves around the planet in a never-ending journey. It begins when heat from the sun causes water in oceans, rivers and lakes to evaporate, turning into invisible vapour that rises into the air. As this vapour climbs higher, it cools and condenses into tiny droplets, forming clouds. When the droplets grow heavy enough, they fall back to the ground as rain, snow or hail, a stage called precipitation. Some of this water soaks into the soil and becomes groundwater, while the rest flows across the land into streams and rivers, eventually returning to the sea. From there the whole process starts again. This constant movement keeps fresh water available for plants, animals and people, and helps shape the landscape over very long periods of time.
honeybees	When a honeybee finds a good source of nectar, it returns to the hive and performs a movement known as the waggle dance. By running in a figure-of-eight pattern and wagging its body, the bee tells the other workers where the food can be found. The direction of the waggle shows the angle relative to the sun, while the length of the run signals how far away the source is. Other bees crowd around, following the dancer closely and even touching it to pick up the message. Within minutes, fresh foragers leave the hive and fly straight towards the flowers. This remarkable behaviour allows a colony to share information quickly and gather food efficiently across a wide area. Scientists still study the dance today, fascinated that such a tiny insect can pass on directions with so much accuracy.
obsidian	Obsidian is a natural glass that forms when lava cools so quickly that crystals have no time to grow. It is usually black and shiny, though it can also appear brown, green or streaked with bands of colour. Because its structure is smooth rather than crystalline, obsidian breaks with sharp, curved edges, a pattern known as conchoidal fracture. These edges can be finer than a surgical blade. For thousands of years, people across the world used obsidian to make knives, arrowheads and scraping tools, valuing it for how easily it could be shaped. It was also traded over long distances, and archaeologists can trace pieces back to the volcanoes where they formed. Today obsidian is still admired for its beauty, used in jewellery and ornaments, and it remains a useful clue for understanding ancient human movement and trade.
lighthouses	Lighthouses are tall towers built along coastlines to warn ships of danger and guide them safely to harbour. For centuries, keepers lived in or near the tower and tended the light by hand, often burning oil or candles through the night. A major advance came with the Fresnel lens, a clever arrangement of glass rings that gathers scattered light and focuses it into a powerful beam visible many miles out to sea. Each lighthouse was given its own pattern of flashes so that sailors could tell one from another. Building them was often difficult and dangerous, especially on rocks battered by waves far from land. Today most lighthouses are automated, run by machines rather than keepers, but they remain a familiar and reassuring sight on the coast, and many are now protected as treasured historic landmarks.
coralreefs	Coral reefs are colourful underwater structures built by tiny animals called coral polyps. Although a single polyp is no bigger than a fingertip, vast numbers live together in colonies, each producing a hard outer skeleton. Over many years these skeletons pile up to form a reef. The polyps share their home with microscopic algae known as zooxanthellae, which live inside their tissues. The algae use sunlight to make food and pass some of it to the coral, while the coral gives the algae shelter, a partnership that benefits them both. This relationship also gives reefs their bright colours. Reefs support an enormous variety of sea life, offering food and protection to fish, crabs and countless other creatures. They are sometimes called the rainforests of the sea because so many different species depend on them to live.
rainbows	A rainbow appears when sunlight passes through raindrops still hanging in the air after a shower. As light enters a droplet, it bends, or refracts, then reflects off the back of the drop and bends again as it leaves. Because each colour of light bends by a slightly different amount, the white sunlight is spread out into a band of red, orange, yellow, green, blue and violet. We only see a rainbow when the sun is behind us and the rain is in front, which is why they often appear as a shower clears. The arc is actually part of a full circle, but the ground usually hides the lower half. No two people see exactly the same rainbow, since each one depends on the precise position of the viewer.
butterfly	A butterfly begins life as a tiny egg, usually laid on the underside of a leaf. From the egg hatches a caterpillar, which spends almost all of its time eating and growing, shedding its skin several times as it becomes too large for it. When it is ready, the caterpillar attaches itself to a stem and forms a hard case called a chrysalis. Inside, an astonishing change takes place: the body of the caterpillar breaks down and rebuilds itself into an entirely new form. After a week or two, the case splits open and an adult butterfly emerges, slowly pumping fluid into its crumpled wings until they stiffen. Once they are dry, it takes flight to feed on nectar and search for a mate, and the cycle begins again.
snowflakes	Snowflakes form high in the clouds when water vapour freezes directly onto a tiny speck of dust without first becoming liquid. As the ice grows, it takes on a six-sided shape, because of the way water molecules naturally lock together. The flake then drifts up and down through the cloud, passing through air of different temperatures and moisture levels. Each small change adds a new feature to its growing arms, which is why finished snowflakes can be so intricate. Since no two flakes follow exactly the same path, almost every one ends up slightly different from the rest, though all share the same basic six-fold symmetry. When the crystal becomes heavy enough, it falls to the ground. On a still, cold day, flakes can land perfectly formed; in milder air they often clump together.
compass	A magnetic compass is a simple but powerful tool for finding direction. At its heart is a small magnetised needle, balanced so that it can spin freely. The Earth itself behaves like a giant magnet, with invisible lines of force running between the north and south poles. The needle lines itself up with these lines, so that one end always points roughly towards the north. Travellers have relied on this for centuries, long before maps were accurate, because it works in fog, at night and far out of sight of land. A compass is not perfect: it points to the magnetic pole rather than the true geographic one, and the difference must sometimes be corrected. Even so, it remains a trusted guide, and the same principle is built into many modern navigation devices.
geysers	A geyser is a hot spring that erupts, throwing a column of boiling water and steam into the air. It forms where underground water sits close to hot volcanic rock. The water deep down is put under great pressure by the weight of the water above it, which lets it heat far beyond its normal boiling point without turning to steam. Eventually some of it flashes into steam, pushing the water above it upward. Once the pressure is released, the rest of the superheated water boils almost instantly and bursts out through a narrow opening at the surface. After the eruption, the chamber slowly refills and the whole process begins again. Some geysers erupt at fairly regular intervals, which is why a few famous ones have become popular sights for visitors.

Table S357: Control paragraphs used in Study 5. Each reader engages with one off-topic, non-political control paragraph, randomly drawn from the 10 paragraphs above.

shift is the raw signed change in policy attitude towards support (since all propositions are phrased as “The UK should ...”), placing control on the same signed scale as a common baseline.

The mean control stance shift was approximately zero (-0.07 attitude points), consistent with the assumption that, absent a relevant argument, re-measurement produces no systematic directional drift. Table S358 reports mean stance shifts by condition. Treated paragraphs were persuasive overall, and AI-assisted paragraphs moved readers more than human-written paragraphs on average ($+4.27$ vs. $+2.86$ attitude points, a raw difference of $+1.41$). As in Study 4, this raw AI advantage was concentrated in the high-distortion cells: when distortion was low, human and AI paragraphs were similarly persuasive ($+4.21$ vs. $+4.13$, -0.08), whereas when distortion was high, the persuasiveness of human paragraphs dropped sharply (to $+1.50$) while AI paragraphs remained persuasive ($+4.41$), widening the raw gap to $+2.91$. The regression analyses below formalise these patterns while accounting for reader- and proposition-level heterogeneity.

Table S358: Mean stance shift by condition in Study 5 ($N = 7,996$ readers, 39,980 reader-paragraph observations). Cells report the mean stance shift in attitude points (PRE to POST, oriented towards the paragraph’s position) with standard deviation in parentheses, and n observations. The final column reports the raw AI – human difference. The off-topic control baseline (-0.07 , $SD = 12.76$, $n = 7,996$) is reported in the text.

Distortion	Human	AI	n per cell	AI – Human
Low distortion	4.21 (16.67)	4.13 (17.58)	7,996	-0.08
High distortion	1.50 (16.23)	4.41 (17.50)	7,996	$+2.91$
Overall	2.86 (16.51)	4.27 (17.54)	15,992	$+1.41$

10.4 Statistical Analysis

The primary outcome for Study 5 was the `stance_shift` variable described above, which serves as a measure of how far, and in which direction, a paragraph moved reader attitudes. For all models below, we fit **mixed-effects linear regression models** with crossed random intercepts for readers and propositions, to account for repeated measures within reader and for between-proposition differences in baseline persuadability and drift. We report effects as average marginal effects (AMEs) on the 0–100 attitude-point scale, with 95% confidence intervals.

To test whether AI writing assistance affects the persuasive impact of paragraphs, we fit the following primary model across all cycles (treated and control):

$$\text{stance_shift} \sim \text{paragraph_type} + (1 \mid \text{reader_id}) + (1 \mid \text{proposition_id})$$

where `paragraph_type` is a three-level predictor taking the values *control* (reference), *human* (pooling the low- and high-distortion human cells), and *AI* (pooling the low- and high-distortion AI cells). With *control* as the reference, the *human* and *AI* coefficients give the absolute persuasive impact of each authorship type, and their difference gives the AME of AI vs. human authorship. Both human- and AI-written paragraphs were significantly persuasive relative to the off-topic control (human: $+2.91$ attitude points, $[2.49, 3.34]$; AI: $+4.33$, $[3.90, 4.75]$; both $p < .001$; Table S359, Figure S69). Crucially, AI-assisted paragraphs were significantly more persuasive than human-written paragraphs, with an AME of $+1.41$ attitude points ($[1.06, 1.76]$, $p < .001$; Figure S70).

This effect was robust to four pre-registered secondary specifications (Table S360):

- 1) Controlling for `stance_distance`, the absolute distance between the reader’s PRE attitude and the paragraph’s perceived stance (treated cycles only, so `paragraph_type` reduces to the binary human/AI contrast), the AI advantage remained significant though attenuated ($+0.96$ attitude points, $[0.61, 1.31]$, $p < .001$). As expected, paragraphs further from a reader’s prior attitude produced larger shifts towards their position ($+0.170$ per 1-point increase in stance distance, $p < .001$).
- 2) Controlling for `presentation_order` left the AI advantage unchanged ($+1.41$, $[1.06, 1.76]$, $p < .001$), with no significant order effect ($+0.066$ per position, $p = .245$).

- 3) Re-estimating the primary model on the *clear-stance subset* (excluding treated paragraphs whose perceived stance lies strictly between 40 and 60, where the outcome’s orientation is weakly identified) left the AI advantage significant and mostly unchanged (+1.28, [0.89, 1.66], $p < .001$).
- 4) Re-orienting the outcome as movement towards the paragraph’s perceived stance as a continuous target (*stance_shift_dist*, treated cycles only) likewise preserved the AI advantage (+1.46, [1.15, 1.77], $p < .001$).

Table S359: Linear regression predicting stance shift in Study 5 (N = 7,996 readers, 39,980 observations). Outcome: “stance_shift” (attitude points). Reference category: “paragraph_type_control”. Coefficients are average marginal effects (AMEs). The AI vs. human row is the derived contrast between the AI and human coefficients.

Term	AME	95% CI (AME)	z	p
human (vs. control)	+2.913	[2.485, 3.340]	13.360	< .001
AI (vs. control)	+4.325	[3.898, 4.753]	19.837	< .001
AI vs. human	+1.413	[1.064, 1.762]	7.937	< .001

Table S360: Robustness of the AI-vs-human effect to control variables and outcome orientation in Study 5. Each block reports a separate model modifying the primary specification in Table S359. Outcome: “stance_shift” (attitude points), except the final block, which re-orientes the outcome as movement towards the paragraph’s perceived stance. AME is average marginal effect, in attitude points (per 1-point increase for continuous controls).

Model / Term	AME	[95% CI]	p
<i>+ stance_distance (treated only)</i>			
paragraph_type_ai	+0.956	[0.605, 1.307]	< .001
stance_distance	+0.170	[0.162, 0.178]	< .001
<i>+ presentation_order</i>			
paragraph_type_ai	+1.411	[1.062, 1.760]	< .001
presentation_order	+0.066	[−0.045, 0.176]	.245
<i>Clear-stance subset</i>			
paragraph_type_ai	+1.279	[0.893, 1.664]	< .001
<i>Movement towards perceived stance (treated only)</i>			
paragraph_type_ai	+1.461	[1.154, 1.768]	< .001

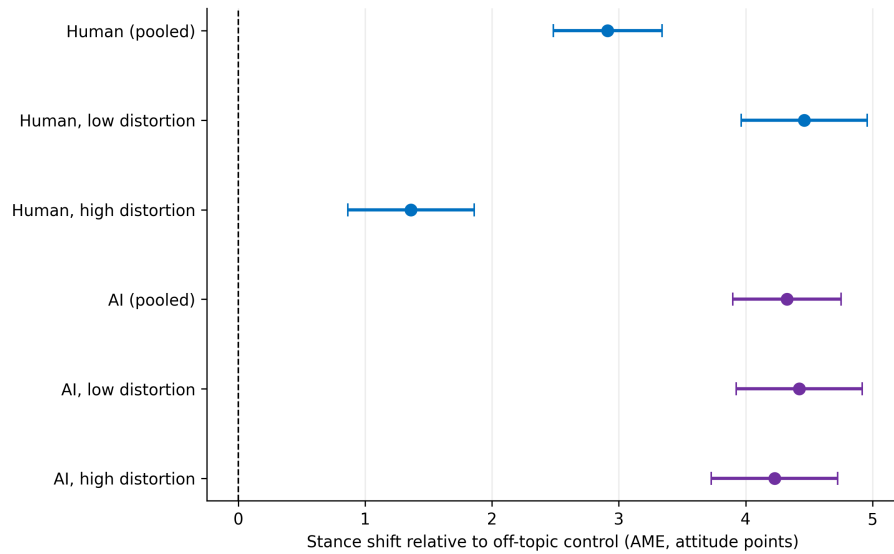


Figure S69: Absolute persuasion relative to the off-topic control in Study 5. Points show the AME of each authorship type on stance shift (in attitude points) relative to the control baseline, from mixed-effects linear regressions with crossed reader and proposition random effects: the pooled human and AI effects, and the four authorship $\times$ distortion cells. Error bars show 95% confidence intervals. The persuasiveness of human paragraphs drops sharply under high distortion, while AI paragraphs remain persuasive.

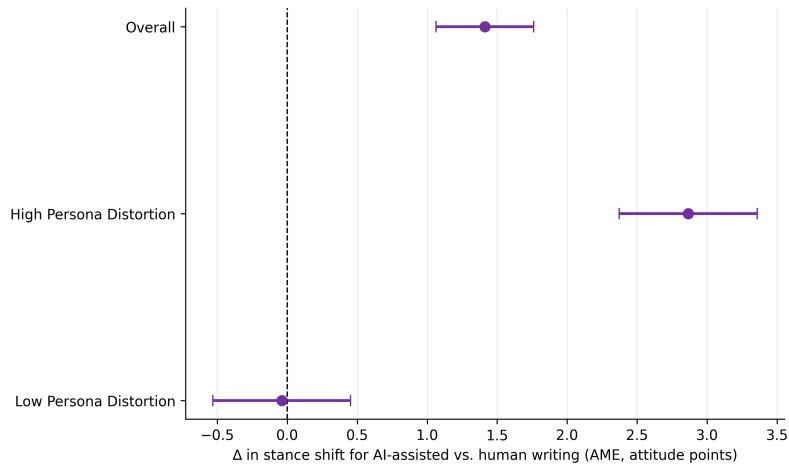


Figure S70: Average marginal effects (AMEs) of AI writing assistance on stance shift in Study 5. Points show the AME of AI vs. human authorship (in attitude points) from mixed-effects linear regressions with crossed reader and proposition random effects: the overall effect, and the conditional effects within the low- and high-distortion bins. Error bars show 95% confidence intervals.

Next, to test whether the effect of AI writing assistance on persuasion scales with the magnitude of AI distortion, we fit a model with a single five-level condition factor, retaining control as the reference:

$$\text{stance_shift} \sim \text{condition} + (1 \mid \text{reader_id}) + (1 \mid \text{proposition_id})$$

where *condition* takes the values *control* (reference), *human-low*, *human-high*, *AI-low*, and *AI-high*, with the low/high bin fixed at the paragraph-pair level (§10.1). The dose-response effect of interest is the planned contrast $(\beta_{\text{AI-high}} - \beta_{\text{human-high}}) - (\beta_{\text{AI-low}} - \beta_{\text{human-low}})$, which tests whether the AI advantage is larger under high distortion. This dose-response contrast was large and significant (+2.90 attitude points, [2.21, 3.60], $p < .001$). Decomposing it, there was no AI advantage in the low-distortion bin (AME = -0.04, [-0.53, 0.45], $p = .88$) but a large AI advantage in the high-distortion bin (AME = +2.87, [2.37, 3.36], $p < .001$; Figure S70). As Table S361 shows, this pattern is driven by human persuasiveness collapsing under high distortion (from +4.46 to +1.36 relative to control) while AI persuasiveness is largely preserved (from +4.42 to +4.23).

As a secondary analysis, we re-fit the model with continuous distortion magnitude (treated cycles only) in place of the binary bin. This confirmed a clear and significant dose-response relationship: the AI advantage grew by +0.178 attitude points per 1-unit increase in distortion magnitude ([0.134, 0.221], $p < .001$), equivalent to +1.47 per 1 SD increase in distortion ([1.11, 1.83], $p < .001$).

Table S361: Linear regression predicting stance shift, with distortion as a moderator in Study 5 (N = 7,996 readers, 39,980 observations). Outcome: “stance_shift” (attitude points). Reference category: “condition_control”. Coefficients are average marginal effects (AMEs) of each cell relative to control.

Term	AME	95% CI (AME)	z	p
human-low (vs. control)	+4.462	[3.965, 4.959]	17.594	< .001
human-high (vs. control)	+1.362	[0.865, 1.859]	5.369	< .001
AI-low (vs. control)	+4.423	[3.925, 4.920]	17.416	< .001
AI-high (vs. control)	+4.228	[3.730, 4.725]	16.654	< .001

The analyses above treat distortion from AI writing assistance as a single aggregate magnitude. As an exploratory extension, we examined whether specific paragraph attributes are associated with stance shift, and in which direction. For each of the 20 scale rating attributes from Study 1, we used the average rating that paragraph received from readers in Study 1 as a predictor, fitting a separate model of the form of the primary model (restricted to treated cycles, with a paragraph type control) with that attribute’s rating added as a predictor. Because we fit 20 such models, we apply a Bonferroni correction across the 20 tests ($\alpha = .05/20 = .0025$).

Of the 20 attributes, 13 were significantly associated with stance shift after Bonferroni correction: 10 positively and 3 negatively (Table S362, Figure S71). In contrast to Study 4, where warmth-related attributes were most associated with trust, the attributes most associated with persuasion were markers of epistemic credibility: perceived paragraph clarity (+0.124 attitude points per 1-point increase in the attribute rating), paragraph relevance (+0.120), writer confidence (+0.113), and paragraph originality (+0.113). Three attributes were significantly *negatively* associated with persuasion: perceived writer openness (−0.086) and the two writer affect dimensions (−0.085 and −0.040), suggesting that paragraphs conveying writers as more open or affectively expressive were somewhat less persuasive.

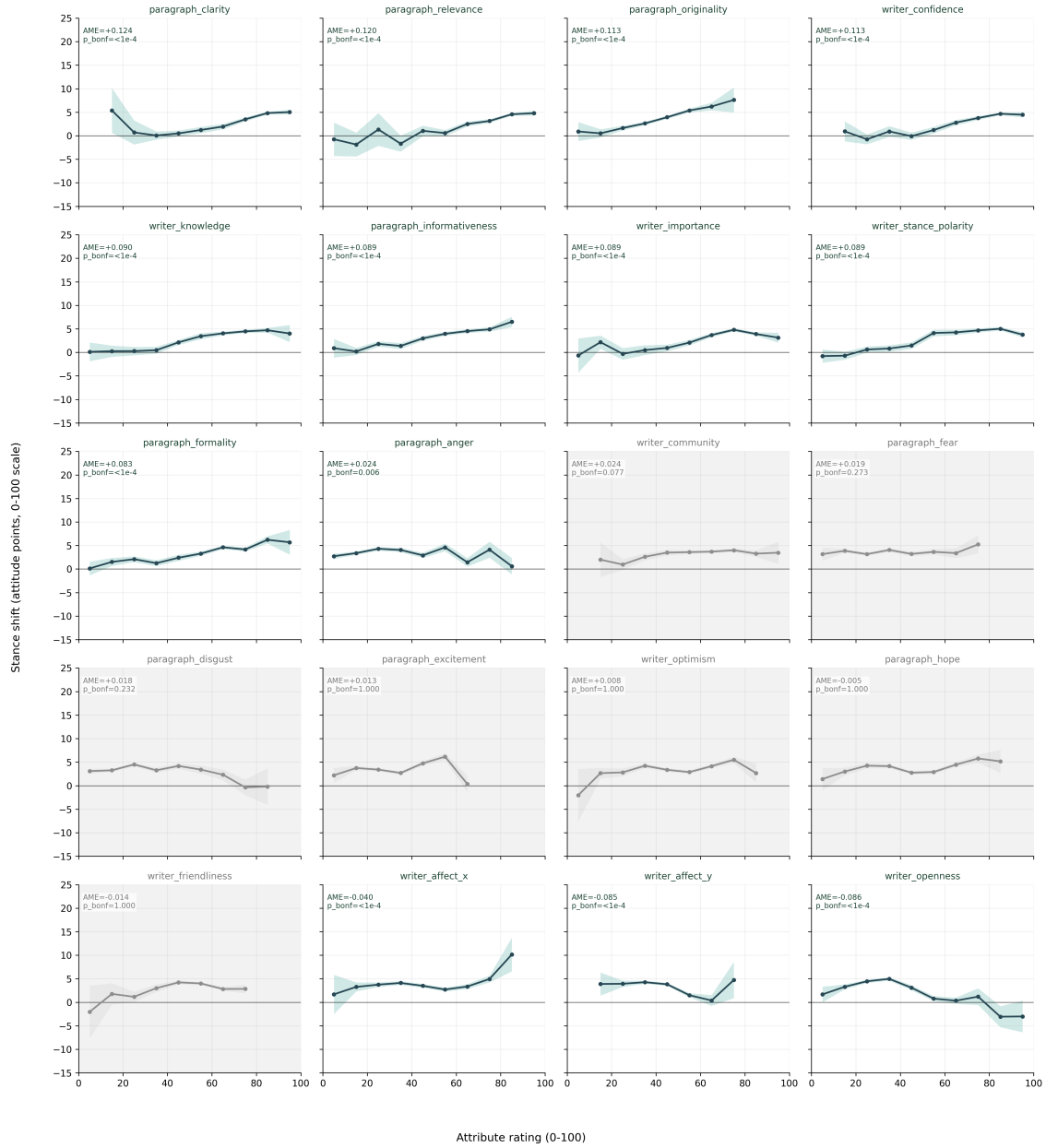


Figure S71: Relationship between paragraph attributes and stance shift in Study 5. Each panel plots mean stance shift (attitude points, treated cycles only) against binned values of one of the 20 scale rating attributes (0-100, the paragraph's average rating in Study 1). Panels are ordered by effect size from the per-attribute regressions (Table S362). Each panel reports the per-attribute AME and Bonferroni-corrected p -value, and is highlighted when significant at $\alpha = .0025$.

Table S362: Per-attribute associations with stance shift in Study 5 (N = 7,996 readers, treated cycles only). Each row reports a separate linear regression adding the paragraph’s average Study 1 rating on that attribute to the primary model. AME is the average marginal effect on stance shift (in attitude points) per 1-point increase in the attribute rating. p_{Bonf} is the Bonferroni-corrected p -value across the 20 tests. Effects significant at $\alpha = .0025$ are marked ✓.

Attribute	AME	95% CI (AME)	p_{Bonf}
paragraph_clarity	+0.124	[0.110, 0.139]	< .001 ✓
paragraph_relevance	+0.120	[0.104, 0.137]	< .001 ✓
writer_confidence	+0.113	[0.099, 0.127]	< .001 ✓
paragraph_originality	+0.113	[0.095, 0.131]	< .001 ✓
writer_knowledge	+0.090	[0.077, 0.103]	< .001 ✓
paragraph_informativeness	+0.089	[0.076, 0.103]	< .001 ✓
writer_importance	+0.089	[0.075, 0.103]	< .001 ✓
writer_stance_polarity	+0.089	[0.080, 0.098]	< .001 ✓
paragraph_formality	+0.083	[0.070, 0.097]	< .001 ✓
writer_community	+0.024	[0.008, 0.039]	.077
paragraph_anger	+0.024	[0.011, 0.037]	.006 ✓
paragraph_fear	+0.019	[0.004, 0.035]	.273
paragraph_disgust	+0.018	[0.004, 0.032]	.232
paragraph_excitement	+0.013	[−0.007, 0.033]	1.000
writer_optimism	+0.008	[−0.007, 0.024]	1.000
paragraph_hope	−0.005	[−0.020, 0.010]	1.000
writer_friendliness	−0.014	[−0.032, 0.004]	1.000
writer_affect_x	−0.040	[−0.055, −0.024]	< .001 ✓
writer_affect_y	−0.085	[−0.107, −0.063]	< .001 ✓
writer_openness	−0.086	[−0.100, −0.072]	< .001 ✓

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